

Artin approximation for polynomial equations and its desingularization proof

GPT-6.1 Sol (OpenAI)
Codex, Ultra setting

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Online course: <https://kokunoyumeto.github.io/open-math-courses/courses/AG-RG/>.

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Artin approximation for polynomial equations and its desingularization proof

Written and self-checked by GPT-6.1 Sol (OpenAI), Codex, Ultra setting, 5 October 2026. This reconstruction draft includes the desingularization proof and its consumed foundations. It retains adapted GNU FDL 1.2 expression.

The purpose of approximation is to retain exact equations while replacing coefficients in a completed local ring by coefficients on an étale neighbourhood. A coefficient need not itself belong to the local ring, and approximating each coefficient separately generally destroys its equations. Instead, retain all equations in one finite system, including those that impose the desired congruence.

The only mathematical source used to construct the new desingularization material is the freely accessible pinned Stacks exposition and its supporting chapters. Earlier programme proofs are used where their exact statements apply. The source is material for the proof written here; a link to that source does not discharge a mathematical obligation. The original theorem below remains the target until all of its consumed dependencies are proved in this lesson or an earlier programme lesson.

1. The exact approximation statement

A map of Noetherian rings is **regular** if it is flat and every fibre is geometrically regular. An algebra over a field is geometrically regular if after every finite field extension its local rings are regular. A **G-ring** is a Noetherian ring whose maps from localizations to their completions are regular. The relevant implication for a local G-ring is therefore immediate from its definition: $R \rightarrow \hat{R}$ is regular. The broader local criterion and permanence statements, including the property for local rings essentially of finite type over $\mathbb{Z}$, are supplied among the foundation arguments below.

Theorem 1.1 (Artin approximation in a pointed étale neighbourhood). Let $(R, \mathfrak{m})$ be a Noetherian local G-ring. Let

$$f_1, \dots, f_q \in R[X_1, \dots, X_n], \quad \hat{a} \in \hat{R}^n, \quad f_j(\hat{a}) = 0.$$

For every positive integer N , there exist a finitely presented étale R -algebra R' , a maximal ideal $\mathfrak{m}' \subset R'$ over $\mathfrak{m}$, and a solution $b \in (R')^n$, such that

$$\kappa(\mathfrak{m}') = \kappa(\mathfrak{m}), \quad f_j(b) = 0, \quad \rho(b_i) - \hat{a}_i \in \mathfrak{m}^N \hat{R}. \quad (1.1)$$

Here $\rho : R' \rightarrow \hat{R}$ is the canonical lift of the specified residue-field point. This is a homomorphism; no assertion that the whole étale algebra embeds into the completion is required. The statement allows arbitrary characteristic, imperfect residue fields, nilpotent base elements, and nonregular local rings.

Lemma 1.2 (the canonical comparison map). A finitely presented étale R -algebra and a specified R -map $R' \rightarrow R/\mathfrak{m}$ have a unique compatible map $R' \rightarrow \widehat{R}$.

Proof. Starting with the specified map to $R/\mathfrak{m}$, lift successively to $R/\mathfrak{m}^r$. The kernel of $R/\mathfrak{m}^{r+1} \rightarrow R/\mathfrak{m}^r$ is square zero for $r \geq 1$, so existence and uniqueness follow from formal étaleness. The lifts are compatible by uniqueness. Taking their inverse limit gives the desired map. A map into an inverse limit of rings is exactly a compatible collection of maps into the constituent rings, which also proves uniqueness. The equality $\widehat{R}/\mathfrak{m}^r \widehat{R} = R/\mathfrak{m}^r$ used here is proved in [AG-CA, Completion, Proposition 2.2 and Theorem 3.3](#). Formal étaleness of an étale algebra is the defining lifting property in [AG-CA, Formally smooth, unramified and étale ring maps, §6](#). $\square$

Lemma 1.3 (a smooth residue-field point has an étale lift). If B is a smooth finitely presented R -algebra and $B \rightarrow \kappa = R/\mathfrak{m}$ is an R -map, there are a finitely presented étale R -algebra R' , a point $R' \rightarrow \kappa$, and an R -map $B \rightarrow R'$ inducing the original point.

Proof. Use the local standard form proved in [AG-CA, Formally smooth, unramified and étale ring maps, Theorem 5.1 and Solution 8.5](#). At the kernel of the given point, localize once to write

$$B_g = R[X_1, \dots, X_s]/(h_1, \dots, h_c), \quad \Delta = \det(\partial h_i / \partial X_j)_{i,j \leq c} \text{ a unit.}$$

This presentation includes any localization variable and its equation. Choose $r_{c+1}, \dots, r_s \in R$ lifting the point's coordinates in the remaining variables. Substitute these elements in the equations, keeping the first c variables. Localize the resulting algebra at the substituted Δ and at any denominators needed for the original presentation. Its number of equations and variables agree, and its square Jacobian determinant is a unit. The direct Newton calculation proving a standard étale presentation is étale appears in the same earlier lesson, Theorem 4.1 and §6: a square-zero error vector is corrected uniquely by the inverse Jacobian matrix. Thus the resulting R' is finitely presented étale. The original residue coordinates define its map to κ , since the inverted elements have nonzero residues. The kernel of that map is maximal and has residue field κ , because the structural map from R already surjects onto κ . Substitution gives $B_g \rightarrow R'$, hence $B \rightarrow R'$. $\square$

Proof of Theorem 1.1. The equality of finite-order quotients of a Noetherian local ring and its completion is proved at the preceding Completion locators. Choose $c_i \in R$ representing $\widehat{a}_i$ modulo $\mathfrak{m}^N \widehat{R}$, and choose finitely many generators $d_1, \dots, d_M$ of $\mathfrak{m}^N$. Write

$$\widehat{a}_i = c_i + \sum_{l=1}^M d_l \widehat{u}_{il}.$$

Put

$$g_j(U) = f_j \left(c_1 + \sum_l d_l U_{1l}, \dots, c_n + \sum_l d_l U_{nl} \right), \quad A = R[U_{il}]/(g_j). \quad (1.2)$$

The tuple $\widehat{u}$ gives an R -map $A \rightarrow \widehat{R}$. Since $R \rightarrow \widehat{R}$ is regular, the desingularization theorem proved by the sequence ending at [Theorem D.38](#), and the elementary factorization criterion proved at [the corresponding foundation lemma](#), give a factorization

$$A \longrightarrow B \longrightarrow \widehat{R} \quad (1.3)$$

with B smooth and finitely presented over R . Compose with the residue map $\widehat{R} \rightarrow \kappa$, then apply Lemma 1.3. The images $u_{il} \in R'$ solve the g_j exactly. Set $b_i = c_i + \sum_l d_l u_{il}$. Substitution in (1.2) gives $f_j(b) = 0$ exactly. Under the canonical map of Lemma 1.2,

$$\rho(b_i) - \widehat{a}_i = \sum_l d_l (\rho(u_{il}) - \widehat{u}_{il}) \in \mathfrak{m}^N \widehat{R}.$$

This proves the prescribed congruence without demanding that the auxiliary u_{il} themselves be close to $\hat{u}_{il}$. The exact written foundations and the earlier programme proofs used in this argument are registered in §6. $\square$

Corollary 1.4 (henselian formulation). If R is henselian, the solution can be chosen in R . In general it gives a solution in the henselization R^h , descending to the pointed étale neighbourhood just constructed.

Proof. The section criterion proved in [AG-CA, Henselian local rings and henselization, Theorem 1.2](#) gives $R' \rightarrow R$ with the specified residue map. Its composition with $R \rightarrow \hat{R}$ is the map ρ by Lemma 1.2, so (1.1) is preserved. For R^h , use the pointed étale colimit and its universal property proved in that lesson, Lemma 4.1 and Theorem 4.2. Compatibility with the map into the complete henselian ring follows from the same uniqueness. Here the principal monic chart consumed by that section-criterion proof is supplied by the full [earlier structural étale proof, Lemma D2.2](#), whose integral closure input is [the affine conductor theorem, Theorem C4.3](#). Its polynomial normality input is proved in [the preliminary coefficient argument, Corollary B1.2](#). To spell out the actual section step, localize the selected étale algebra to $(R[T]/(F))_h$, with F monic and F' nonzero at the given residue root $\bar{a}$. The henselian simple-root property gives $a \in R$ with $F(a) = 0$ and $a \bmod \mathfrak{m} = \bar{a}$. Every denominator has nonzero residue at $\bar{a}$, hence is a unit at a , and evaluation defines the section. The resulting map $R' \rightarrow R$ induces the prescribed residue map; formal étale uniqueness along the quotients $R/\mathfrak{m}^n$ proves its composition with completion is ρ . Thus this corollary uses these written earlier proofs rather than an external conductor proof tag. $\square$

2. Returning from a local model to the original base

Lemma 2.1 (spreading finite presentation data). Let R_0 be Noetherian and $\mathfrak{p} \subset R_0$. A pointed étale algebra, a finite tuple, and finitely many equations over $(R_0)_{\mathfrak{p}}$, as in Theorem 1.1, spread to $(R_0)_s$ for some $s \notin \mathfrak{p}$. Their solution therefore gives a pointed étale neighbourhood over R_0 .

Proof. In the construction of Lemma 1.3, every polynomial coefficient, every fixed nonpivot coordinate, and every chosen inverse has a denominator outside $\mathfrak{p}$. Invert their product in R_0 . The finitely many equations for the tuple and the finitely many witnesses verifying them likewise hold after multiplying by finitely many additional denominators; invert their product too. The standard étale presentation spreads with its invertible Jacobian determinant, so its spread algebra is étale by the same direct Newton calculation. Regard it as an R_0 -algebra through the étale localization $R_0 \rightarrow (R_0)_s$; composing two formally étale, finitely presented maps remains étale because a square-zero lifting problem is solved uniquely in two successive steps. The chosen residue map survives and gives a point over $\mathfrak{p}$ with unchanged residue field. The defining equations remain exact, and localization at the selected point gives the original local comparison. $\square$

This proof uses the specific standard étale algebra produced above and therefore does not invoke an external spreading theorem for an arbitrary presentation. Pulling the resulting neighbourhood back to the original scheme preserves every defining equation and the specified fibre point.

3. The finite system for an embedded torus

The approximation step in the torus lesson begins after the formal deformation and effectivity argument has produced an affine torus $\hat{T}$ over the completed local model and a closed homomorphism $\hat{T} \hookrightarrow G_{\hat{R}}$. The present lesson proves the approximation input; it does not claim to supply that preceding effectivity argument.

Lemma 3.A (finite splitting over a complete local ring). Let $(A, \mathfrak{m}, \kappa)$ be a local ring with

$$A \simeq \varprojlim_{n \geq 1} A/\mathfrak{m}^n.$$

No Noetherian, reducedness, normality or perfection hypothesis is imposed. Suppose a torus T/A has an affine etale splitting chart $U \rightarrow \operatorname{Spec} A$ and a point of its closed fibre with finite separable residue field K/κ . Then a finite free, faithfully flat, etale algebra E/A splits T . We may choose E local and complete, with residue field a finite Galois extension L/κ containing K , and with an action of $\Gamma = \operatorname{Gal}(L/\kappa)$ for which

$$E \otimes_A E \xrightarrow{\sim} \prod_{\sigma \in \Gamma} E, \quad x \otimes y \mapsto (x\sigma(y))_{\sigma}. \quad (3.2)$$

Here a torus is etale locally a product of copies of $\mathbf{G}_m$. Starting instead with the fpqc multiplicative-type definition gives this same condition by [AG-GS-05, Theorem 7.7](#); its finite torsion-free character-group step is proved in [AG-RG-S04, Lemma P0.10](#). Every torus therefore has such a chart and residue point: choose an affine finite-presentation part of an etale splitting cover meeting the closed fibre, and a closed point of that fibre. Its residue field is finite over κ by the finite-type field-algebra argument in [AG-RG-S04, Lemma P0.9](#). The quotient sequence for differentials gives $\Omega_{K/\kappa} = 0$, because the chart is formally unramified. The finite-field differential calculation in [AG-CA-17, Lemma 7.1](#) then proves separability.

Furthermore, a torus T_0/κ , with character lattice split by L , has a torus lift over A split by this same E . Its character lattice and descent action reduce compatibly modulo every $\mathfrak{m}^n$.

Proof. A finite separable extension embeds in a finite normal separable extension: adjoin all the roots of the minimal polynomials of a finite generating list. There are finitely many roots and each adjunction has finite separable degree. Every embedding permutes the root sets, so the resulting extension L/κ is normal and separable. Counting embeddings successively gives $|\Gamma| = [L : \kappa]$. The primitive-element argument of [AG-RG-S02, Lemma 4.2](#), applies also to finite fields, and gives $L = \kappa(\alpha)$. Let $\bar{f} \in \kappa[X]$ be the monic minimal polynomial of α , and lift its coefficients to a monic $f \in A[X]$. Put

$$E = A[t] = A[X]/(f), \quad d = \deg(f).$$

Successively subtracting leading multiples of the monic polynomial f reduces every polynomial to degree less than d . A nonzero multiple qf has degree $\deg(q) + d$, since its leading coefficient is the nonzero leading coefficient of q . Thus the remainder is unique, and $1, t, \dots, t^{d-1}$ is a basis making E free of positive finite rank over A ; it is faithfully flat because tensoring a module with this free module gives a direct sum of d copies. Its quotient by $\mathfrak{m}E$ is L . If $e \in E$ has nonzero residue, multiplication by e on this basis is invertible modulo $\mathfrak{m}$. Its determinant is a unit of A , and the cofactor calculation proved in [Lemma 3.B](#) gives an inverse to multiplication by e . Thus e is a unit, E is local with maximal ideal $\mathfrak{m}E$, and coordinatewise completeness gives

$$E \simeq \varprojlim_n E/\mathfrak{m}^n E.$$

Separability makes $f'(t)$ a unit by the same argument. A root in a quotient by a square-zero ideal lifts uniquely by replacing any lift a with $a - f'(a)^{-1}f(a)$. Polynomial expansion proves existence, and $f(b) - f(a) = (b - a)(f'(a) + (b - a)H)$ proves uniqueness. Iteration gives the corresponding assertion for every nilpotent ideal. This proves formal etaleness of E/A ; its finite presentation makes it etale.

The residue point of U gives an algebra map $\mathcal{O}(U) \rightarrow K \rightarrow L$. Formal etaleness of U/A lifts this map uniquely through $E/\mathfrak{m}^{n+1}E \rightarrow E/\mathfrak{m}^n E$, whose kernel is square-zero for $n \geq 1$. Taking the compatible limit gives a map $\mathcal{O}(U) \rightarrow E$, hence a section of $U_E \rightarrow \operatorname{Spec} E$. Pulling back the splitting on U therefore splits T_E .

We now construct (3.2), including its automorphisms. For every $\sigma \in \Gamma$, the residue $\sigma(\alpha) \in L$ is a simple root of $\bar{f}$. In the complete local ring E , Newton's recursion

$$a_{j+1} = a_j - f'(a_j)^{-1} f(a_j)$$

starting from any lift of this residue gives $f(a_j) \in (\mathfrak{m}E)^{2^j}$ and $a_{j+1} - a_j \in (\mathfrak{m}E)^{2^j}$. Completeness and separatedness give a root a_σ , and the difference formula above gives its uniqueness with that residue. Distinct a_σ have unit differences. If a polynomial P vanishes at all these roots, monic division by $X - a_1$ gives $P = (X - a_1)Q$, since the remainder is $P(a_1) = 0$. At each other root a_σ , the factor $a_\sigma - a_1$ is a unit, so $Q(a_\sigma) = 0$. Repeating the division shows that $\prod_\sigma (X - a_\sigma)$ divides P . If $\deg(P) < d$, monicity forces $P = 0$. The difference between f and this product has degree less than d and vanishes at every root, so

$$f(X) = \prod_{\sigma \in \Gamma} (X - a_\sigma).$$

The map $t \mapsto a_\sigma$ defines an A -algebra endomorphism g_σ of E . The two roots $g_\sigma g_\tau(t)$ and $a_{\sigma\tau}$ have the same residue, so uniqueness gives $g_\sigma g_\tau = g_{\sigma\tau}$. Also $a_1 = t$; hence these maps are automorphisms and lift the Galois action on L .

The isomorphism $E \otimes_A E = E[X]/(f)$ and evaluation at the a_σ now give (3.2). Explicitly, the polynomials

$$\ell_\sigma(X) = \prod_{\tau \neq \sigma} \frac{X - a_\tau}{a_\sigma - a_\tau}$$

give the inverse to evaluation: a tuple $(b_\sigma)_\sigma$ is represented by $\sum_\sigma b_\sigma \ell_\sigma(X)$. Thus the isomorphism uses no division by $|\Gamma|$. Its compatibility on triple products is exactly $g_\sigma g_\tau = g_{\sigma\tau}$. This is the required finite étale Galois torsor.

For the final assertion choose the finite Galois splitting field of T_0 supplied by AG-GS-05, Lemma 5.1, and let M be its free finite-rank character lattice, with the resulting Γ -action. On $E[M]$ put

$$\sigma(ae^m) = g_\sigma(a)e^{\sigma m}. \quad (3.3)$$

The group law and (3.2) make this a descent datum. Multiplication and every Hopf operation commute with it, since addition in M commutes with its action. The actual module and algebra effectivity proof in AG-RG-S04, Lemmas A1.1–A1.2, gives the invariant Hopf algebra

$$C = E[M]^\Gamma, \quad E \otimes_A C \simeq E[M]. \quad (3.4)$$

Tensor products and Hopf identities descend by this isomorphism and faithful flatness. Finite presentation descends here by a finite-generator argument. Express the finitely many Laurent algebra generators over E as finite E -linear combinations of elements of C . Those elements generate C , since the cokernel of their generated subalgebra becomes zero over E . For a resulting finite polynomial surjection $A[Z] \rightarrow C$, its kernel becomes a finite ideal after tensoring with E , because $E[M]$ is finitely presented. To see that finiteness for this particular surjection, compare it with a finite presentation of $E[M]$: express each presentation's generators in the other, and substitute these expressions in the finitely many relations and composite-generator identities. They generate the new kernel. Write a finite list of its generators as finite sums of elements of the original kernel with coefficients in E . Their finitely many kernel components generate the original kernel by faithful flatness. Thus C is finitely presented. It defines a torus T_A , because (3.4) splits it on the finite étale faithfully flat cover.

Put $A_n = A/\mathfrak{m}^n$ and $E_n = E \otimes_A A_n$. Tensoring (3.2)–(3.4) with A_n gives the same torsor and the same character action over E_n :

$$E_n \otimes_{A_n} (C \otimes_A A_n) \simeq E_n[M]. \quad (3.5)$$

The algebra on the left descends the datum (3.3) modulo $\mathfrak{m}^n$; uniqueness in Lemma A1.2 proves that it is its effective descent. This proves compatibility for every n , even though $A \rightarrow A_n$ need not be flat. It does not assume that taking invariants commutes with an arbitrary quotient. At $n = 1$, the descent datum is precisely the given one on $L[M]$, so its descended torus is T_0 .

More generally, any compatible character data split by these E_n and identified with the same lattice have no additional nilpotent transition parameters. Comparing coefficients in $\Delta(\sum c_m e^m) = (\sum c_m e^m) \otimes (\sum c_m e^m)$ gives $c_m^2 = c_m$, $c_m c_{m'} = 0$ for $m \neq m'$, and $\sum c_m = 1$. Each E_n is local, so exactly one coefficient is 1. A character is therefore one monomial; a torus automorphism is one integral lattice automorphism. A compatible automorphism reducing to a fixed one is that same one. Hence a compatible lattice descent action is the fixed action in (3.3), and (3.5) gives its actual compatible descent. This completes both assertions. $\square$

Choose a finite presentation of the Hopf algebra of $\hat{T}$. Lemma 3.A supplies a finite free, étale, faithfully flat splitting algebra $E/\hat{R}$; choose the resulting splitting isomorphism

$$\mathcal{O}(\hat{T}) \otimes_{\hat{R}} E \simeq E[Z_1^{\pm 1}, \dots, Z_r^{\pm 1}]. \quad (3.1)$$

Lemma 3.B (perfect trace, over an arbitrary base). Let A be a commutative ring and C a commutative A -algebra free of finite rank d . If the determinant of the pairing $(c, z) \mapsto \text{Tr}_{C/A}(cz)$ is a unit, then C/A is finitely presented and formally étale, hence étale. If $d > 0$, it is faithfully flat. These assertions require no Noetherian, reducedness or characteristic hypothesis.

Proof. First, the elementary matrix identities used here hold over every commutative ring. Define a determinant by the signed permutation sum and a cofactor by the corresponding smaller signed determinant. Expanding along a row or column gives the determinant. A matrix with two equal rows or columns has determinant zero: pair each permutation with the permutation interchanging those positions; their contributions have opposite signs, including in characteristic two. Consequently, if M is square, multiplying it on either side by its transposed cofactor matrix gives $(\det M)I$: a diagonal entry is the row or column expansion, and an off-diagonal entry is the expansion of a determinant with two equal rows or columns. A unit determinant therefore gives the inverse $(\det M)^{-1} \text{adj}(M)$. For size zero the determinant is one. To verify the converse implication as well, expand $\det(UV)$ by multilinearity in its columns. The expansion chooses, for each column of UV , one column of U with its coefficient from V . A tuple repeating a column of U contributes zero by the cancellation just proved. Every remaining tuple is a permutation of all the columns of U , whose determinant is its permutation sign times $\det(U)$. Summing its coefficients with these signs is precisely $\det(V)$. Thus $\det(UV) = \det(U) \det(V)$, including for size zero. If $UV = I$, this equation gives $\det(U) \det(V) = 1$, so an invertible matrix has a unit determinant. For square matrices U, V , the finite double sum also gives $\text{tr}(UV) = \text{tr}(VU)$. Thus conjugating a multiplication matrix by an invertible change of basis preserves its trace. Multiplication matrices, their traces and their determinants commute with every scalar extension, entry by entry.

Suppose first that $d > 0$, and choose a basis $u_1, \dots, u_d$. Write $u_i u_j = \sum_k m_{ijk} u_k$ and $1 = \sum_i a_i u_i$. There is an explicit finite algebra presentation

$$C = A[X_1, \dots, X_d] / (X_i X_j - \sum_k m_{ijk} X_k, 1 - \sum_i a_i X_i).$$

Indeed, repeatedly replace a product of two variables by the displayed linear combination, reducing each monomial to degree at most one. Replace its constant term by that constant times $\sum_i a_i X_i$. Every quotient element is therefore represented by $\sum_i b_i X_i$. The map to C is surjective, and if this representative maps to zero, linear independence of the u_i makes every b_i zero. This proves the presentation without an assumption on ideals of A .

The inverse pairing matrix supplies a trace-dual basis $v_1, \dots, v_d$, characterized by $\text{Tr}(u_j v_i) = \delta_{ij}$. In particular $x = \sum_i u_i \text{Tr}(v_i x)$ and $x = \sum_i \text{Tr}(u_i x) v_i$. To check the second equality, pair both sides with each u_j and use invertibility of the pairing; the first is the ordinary basis-coordinate equality.

Put

$$e = \sum_i u_i \otimes v_i \in C \otimes_A C.$$

The A -linear map

$$\Phi : C \otimes_A C \longrightarrow \text{End}_A(C), \quad \Phi(a \otimes b)(x) = a \text{Tr}(bx)$$

is an isomorphism, with inverse $F \mapsto \sum_i F(u_i) \otimes v_i$. The first coordinate equality proves $\Phi\Phi^{-1}(F) = F$; the second proves $\Phi^{-1}\Phi(a \otimes b) = a \otimes b$, and pure tensors generate. Since $\Phi(e)$ is the identity, both $(c \otimes 1)e$ and $(1 \otimes c)e$ have image multiplication by c . Thus

$$(c \otimes 1)e = (1 \otimes c)e \quad (c \in C). \quad (3.6)$$

Let $s = \sum_i u_i v_i$. The diagonal coefficient of multiplication by c in position i is $\text{Tr}(v_i c u_i)$. Summing those coefficients gives $\text{Tr}(c) = \text{Tr}(cs)$ for every c . Invertibility of the pairing now gives

$$\sum_i u_i v_i = 1. \quad (3.7)$$

No division by d , or assertion that d is a unit, occurs.

We prove the lifting property directly. Let $J \subset D$ be a square-zero ideal of a commutative A -algebra, and let $\bar{f} : C \rightarrow D/J$ be an A -algebra map. The free-module lifting calculation in [AG-RG-S01, Lemma 1.0](#) gives an A -linear lift $l : C \rightarrow D$. Its value $l(1)$ is $1 + j$ with $j \in J$, and has inverse $1 - j$. Multiplying the whole lift by that inverse makes $l(1) = 1$ while preserving A -linearity and its reduction. Give J its C -module action via $\bar{f}$: multiply by any lift of $\bar{f}(c)$, which is independent of that lift because $J^2 = 0$.

The defect $b(c, z) = l(cz) - l(c)l(z)$ lies in J , is A -bilinear and has $b(c, 1) = b(1, c) = 0$. Associativity in C and D gives, by expanding the four terms,

$$cb(z, t) - b(cz, t) + b(c, zt) - b(c, z)t = 0. \quad (3.8)$$

Define the A -linear correction $h(c) = \sum_i u_i b(v_i, c)$ in J . Apply the A -linear map $a \otimes z \mapsto ab(z, t)$ to (3.6); it gives $ch(t) = \sum_i u_i b(cv_i, t)$. Using (3.8) with (v_i, c, t) and then (3.7), we obtain

$$ch(t) - h(ct) + h(c)t = \sum_i u_i (b(v_i c, t) - b(v_i, ct) + b(v_i, c)t) = \sum_i u_i v_i b(c, t) = b(c, t).$$

Consequently $f = l + h$ is multiplicative: the defect of this new map is $b(c, t) + h(ct) - ch(t) - h(c)t = 0$, since $h(c)h(t) = 0$. Also $h(1) = 0$. Thus f is a unital A -algebra lift of $\bar{f}$.

For uniqueness, the difference δ of two such lifts takes values in J and satisfies $\delta(ct) = c\delta(t) + \delta(c)t$; expand their products and use $J^2 = 0$. Apply $a \otimes z \mapsto \delta(a)z$ to (3.6). The difference between its two sides is $\delta(c) \sum_i u_i v_i = \delta(c)$, by that identity and (3.7). It is zero, so every lift is unique. If $J^N = 0$, lift successively through $D/J^{r+1} \rightarrow D/J^r$, for $1 \leq r < N$. Its kernel is square-zero because $2r \geq r + 1$. Existence and uniqueness at each step prove formal etaleness for every nilpotent ideal. Together with the explicit finite presentation, this is etaleness by the programme definition.

For $d = 0$, C is the zero algebra with presentation $A/(1)$. If a unital map $C \rightarrow D/J$ exists and J is nilpotent, then $D/J = 0$, so $1 \in J$ and nilpotence forces $D = 0$. There is exactly one lift. Finally, for positive d , tensoring an A -module with C gives d copies of that module; this is exact and detects whether it is zero. This proves faithful flatness. $\square$

This is the finite-free case of the discriminant criterion in the freely accessible [Stacks Project, Lemma 49.3.1, Tag 0BJF](#). The complete lifting proof above supplies the result used here.

Return to the splitting algebra $E/\widehat{R}$ from Lemma 3.A. Its trace determinant is a unit, as we now check rather than adding an unsupported inverse variable. By (3.2), after scalar extension from $\widehat{R}$ to E the algebra becomes $\prod_{\sigma \in \Gamma} E$. Multiplication in this product is diagonal in the coordinate

idempotent basis. Its trace pairing matrix is the identity, hence invertible. The preceding trace and basis-change calculations imply that the original pairing matrix becomes invertible over E , so its determinant Δ becomes a unit there. Then $(\widehat{R}/\Delta\widehat{R}) \otimes_{\widehat{R}} E = 0$. Faithful flatness from Lemma 3.A detects zero modules, giving $\widehat{R}/\Delta\widehat{R} = 0$, which means that Δ is already a unit of $\widehat{R}$.

Fix the finite free basis of E supplied by Lemma 3.A. Its multiplication table, unit and multiplication matrices define finitely many coefficient variables; associativity, commutativity and the unit laws are finite polynomial equations. Add a variable w and the equation $w\Delta = 1$ for the trace determinant. The original completed coefficient tuple satisfies this equation by the preceding argument. After approximation it still satisfies the same equation exactly. Lemma 3.B makes the resulting finite free algebra étale and, since its unchanged rank is positive, faithfully flat.

Add coefficients for the finite presentations, Hopf maps, the isomorphism (3.1), its inverse, and the homomorphism to G . Laurent expressions can be written polynomially by adding inverse variables and relations $Z_i W_i = 1$. The maps need to be specified on finite generating sets. Relations between their images need finitely many witnesses in the finitely generated relation ideals. The two composites of the splitting isomorphism and its inverse are tested on finite generators; equality of these composites with the identity is recorded by their witnesses. The same method records comultiplication, counit, inverse, and the homomorphism identities on a finite list of generators. Coassociativity and every needed descent identity are likewise checked on finite generators, using the finite multiplication tables on the finite overlap algebras. Each expression and witness has finite support.

Retain the closed immersion in this same system. The pullback $\mathcal{O}(G_{\widehat{R}}) \rightarrow \mathcal{O}(\widehat{T})$ is surjective. Choose preimages of a finite list of algebra generators of $\mathcal{O}(\widehat{T})$, and include the equations asserting that their images are those generators. After approximation these equations still prove surjectivity; consequently the resulting homomorphism is a closed immersion.

We have obtained one finite polynomial system over the local model, solved by the original completed coefficient tuple. Apply Theorem 1.1 at order one. The defining equations hold exactly after approximation, while all the chosen coefficients retain their original residues. The preserved splitting maps identify the source Hopf algebra with a Laurent polynomial Hopf algebra on the finite étale faithfully flat cover. Thus it is a torus, with the specified special fibre and closed embedding. If a prescribed higher jet is needed, use that order instead. Lemma 2.1 spreads the data and the neighbourhood back to an open part of the finite-type model; base change returns them to the original base.

The local model is essentially of finite type over $\mathbb{Z}$. The G-ring permanence argument in §5 is therefore a consumed part of this application, including its coefficient-ring inputs. It must be proved before the approximation step is declared complete. An excellence label or an external Stacks link does not replace this work.

Scope of the supporting algebra. The approximation theorem and the desingularization theorem concern Noetherian rings. Every base ring of the finite-type algebra diagrams in their proof is Noetherian. In the foundational complete-intersection arguments below we state and prove the exact Noetherian forms consumed there; arguments explicitly stated over arbitrary rings retain that wider scope. Polynomial presentations with arbitrarily many variables occur only in the two-term cotangent comparison, whose elementary presentation and filtered-colimit proofs are written without Noetherian assumptions. This distinction preserves arbitrary characteristic, imperfect residue fields, and the full Noetherian local G-ring statement.

4. Desingularization proof sequence

The following sequence contains the actual lifting, presentation, local descent, separable, and inseparable arguments. It is adapted from the pinned free source. The source's historical bibliography

is not reproduced. Its statements use the conormal definition of smoothness reproduced among the foundation statements below. The displayed arrow lists specify the maps of the original commutative diagrams.

Singular ideals

Let $R \rightarrow A$ be a ring map. The singular ideal of A over R is the radical ideal in A cutting out the singular locus of the morphism $\text{Spec}(A) \rightarrow \text{Spec}(R)$. Here is a formal definition.

Definition D.1: singular ideal

Adapted from the Stacks project, smoothing.tex, lines 124–132.

Let $R \rightarrow A$ be a ring map. The *singular ideal of A over R* , denoted $H_{A/R}$ is the unique radical ideal $H_{A/R} \subset A$ with

$$V(H_{A/R}) = \{\mathfrak{q} \in \text{Spec}(A) \mid R \rightarrow A \text{ not smooth at } \mathfrak{q}\}$$

This makes sense because the set of primes where $R \rightarrow A$ is smooth is open, see Algebra, Definition F.7. In order to find an explicit set of generators for the singular ideal we first prove the following lemma.

Lemma D.2: find strictly standard

Adapted from the Stacks project, smoothing.tex, lines 141–181.

Let R be a ring. Let $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$. Let $\mathfrak{q} \subset A$ be a prime ideal. Assume $R \rightarrow A$ is smooth at $\mathfrak{q}$. Then there exists an $a \in A$, $a \notin \mathfrak{q}$, an integer c , $0 \leq c \leq \min(n, m)$, subsets $U \subset \{1, \dots, n\}$, $V \subset \{1, \dots, m\}$ of cardinality c such that

$$a = a' \det(\partial f_j / \partial x_i)_{j \in V, i \in U}$$

for some $a' \in A$ and

$$af_\ell \in (f_j, j \in V) + (f_1, \dots, f_m)^2$$

for all $\ell \in \{1, \dots, m\}$.

Proof. Put $P = R[x_1, \dots, x_n]$, $I = (f_1, \dots, f_m)$ and $M = I/I^2$. By F.7, choose $s \notin \mathfrak{q}$ such that A_s is smooth. The presentation comparison F.11, localization F.65, and definition F.6 give the split exact conormal sequence

$$0 \longrightarrow M_s \xrightarrow{d} A_s^n \longrightarrow \Omega_{A_s/R} \longrightarrow 0.$$

Its splitting follows because the last module is projective. Localize at $\mathfrak{q}$ and tensor with $K = \kappa(\mathfrak{q})$. Splitting preserves the injection after this tensor operation. Choose c of the equation classes, indexed by V , giving a basis of $M \otimes_A K$ from its displayed finite spanning list. Their differential columns are linearly independent in K^n , so $c \leq \min(n, m)$.

There are c rows, indexed by U , on which these columns have nonzero determinant. Here is the finite field calculation. For $c > 0$, move a nonzero entry of the first column to the first row, then subtract appropriate multiples of that row from the other rows. Independence makes the remaining $(n-1) \times (c-1)$ block have independent columns. Inductively choose $c-1$ rows with a nonzero minor in that block. On these rows together with the first row, the operations just performed do

not change the original selected determinant, which is the nonzero pivot times that minor. For $c = 0$, use the empty minor 1. Thus

$$\delta = \det(\partial f_j / \partial x_i)_{j \in V, i \in U \in A \setminus \mathfrak{q}}.$$

The module $Q = M / \langle f_j : j \in V \rangle$ is finite, since the m equation classes generate M . Its residue at $\mathfrak{q}$ is zero, so the finite-module Nakayama theorem [F.10](#) gives $Q_{\mathfrak{q}} = 0$. For each member of a finite generating list of Q , the definition of localization gives an element outside $\mathfrak{q}$ killing it. Their product $t \notin \mathfrak{q}$ kills all of Q ; for the zero module take $t = 1$.

Set $a = t\delta$ and $a' = t$. Then $a \notin \mathfrak{q}$ and the claimed determinant equality is exact in A . Also $aQ = 0$, so for every ℓ the conormal class $a[f_\ell]$ is a combination of the selected classes. Lifting the coefficients and a to P gives

$$\tilde{a}f_\ell \in (f_j : j \in V) + I^2.$$

This is precisely the asserted A -module relation; changing the lift of a adds an element of I^2 . The construction uses the original presentation throughout and cancels no possible zero divisor. It includes empty variable, equation and minor lists. $\square$

We will use the notion of a *strictly standard* element in A over R . We also define an *elementary standard* element to be one of the type we found in the lemma above. We compare the different types of elements in [Lemma D.15](#).

Definition D.3: strictly standard

Adapted from the Stacks project, smoothing.tex, lines 191–222.

Let $R \rightarrow A$ be a ring map of finite presentation. We say an element $a \in A$ is *elementary standard in A over R* if there exists a presentation $A = R[x_1, \dots, x_n] / (f_1, \dots, f_m)$ and $0 \leq c \leq \min(n, m)$ such that

$$a = a' \det(\partial f_j / \partial x_i)_{i,j=1,\dots,c} \tag{D.3e1}$$

for some $a' \in A$ and

$$af_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2 \tag{D.3e2}$$

for $j = 1, \dots, m - c$. We say $a \in A$ is *strictly standard in A over R* if there exists a presentation $A = R[x_1, \dots, x_n] / (f_1, \dots, f_m)$ and $0 \leq c \leq \min(n, m)$ such that

$$a = \sum_{I \subset \{1, \dots, n\}, |I|=c} a_I \det(\partial f_j / \partial x_i)_{j=1,\dots,c, i \in I} \tag{D.3s1}$$

for some $a_I \in A$ and

$$af_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2 \tag{D.3s2}$$

for $j = 1, \dots, m - c$.

The following lemma is useful to find implications of [\(D.3s1: strict Jacobian-minor condition\)](#).

Lemma D.4: parse equation strictly standard one

Adapted from the Stacks project, smoothing.tex, lines 228–250.

Let R be a ring. Let $A = R[x_1, \dots, x_n] / (f_1, \dots, f_m)$ and write $I = (f_1, \dots, f_m)$. Let $a \in A$. Then [\(D.3s1: strict Jacobian-minor condition\)](#) implies there exists an A -linear map $\psi : \bigoplus_{i=1,\dots,n} A dx_i \rightarrow A^{\oplus c}$ such that the composition

$$A^{\oplus c} \xrightarrow{(f_1, \dots, f_c)} I/I^2 \xrightarrow{f \mapsto df} \bigoplus_{i=1, \dots, n} \text{Ad} x_i \xrightarrow{\psi} A^{\oplus c}$$

is multiplication by a . Conversely, if such a ψ exists, then a^c satisfies (D.3s1: strict Jacobian-minor condition). The relation condition D.3s2 is a separate hypothesis. This includes $c = 0$: the empty minor is 1 and the module map is the unique map between zero modules.

Proof. Let M be the $n \times c$ differential matrix, with entry $M_{ij} = \partial f_j / \partial x_i$. The $c \times c$ minor on rows indexed by U has the determinant in D.3s1: transposing a square matrix preserves its determinant by reindexing the signed permutation expansion. Thus D.3s1 says that a belongs to the ideal of these minors. Lemma F.70, proved below over arbitrary commutative rings, supplies a $c \times n$ matrix B with $BM = a1_c$. This is the required map ψ . Conversely the displayed composite being $a1_c$ says $BM = a1_c$, and the same lemma puts a^c in the minor ideal, which is exactly D.3s1 for a^c . No relation among the remaining generators is used in this matrix assertion. $\square$

Lemma D.5: elkik

Adapted from the Stacks project, smoothing.tex, lines 252–283.

Let $R \rightarrow A$ be a ring map of finite presentation. The singular ideal $H_{A/R}$ is the radical of the ideal generated by strictly standard elements in A over R and also the radical of the ideal generated by elementary standard elements in A over R .

Proof. Suppose a is strictly standard and put $B = A_a$. Use its presentation $A = P/I$ and the integer c from D.3. The relation condition makes $M = (I/I^2)_a$ generated by the first c equation classes. By D.4, there is a map $\psi : B^n \rightarrow B^c$ whose composite with $B^c \rightarrow M \xrightarrow{d} B^n$ is multiplication by the unit a . Thus $B^c \rightarrow M$ is surjective by generation and injective by this composite. It is an isomorphism. Dividing ψ by the unit a gives a left inverse to the conormal differential. Hence that differential is injective, its cokernel is a finite projective direct summand of B^n , and its first homology is zero. The localization has the finite presentation $B = P[z]/(I, hz - 1)$, where h lifts a , by direct elimination of the inverse variable. The presentation comparison F.11, localization F.65, and definition F.6 now show that B is smooth. If $B = 0$, all modules are zero and the same criterion applies.

Every prime of A not containing a therefore lies in the smooth locus. By D.1, $V(H_{A/R}) \subset V(a)$, whence $a \in H_{A/R}$ because this ideal is radical. Let H_e and H_s be the radicals of the ideals generated by elementary and strictly standard elements. Taking zero coefficients for all other minors makes every elementary element strictly standard. Consequently

$$H_e \subset H_s \subset H_{A/R}.$$

Conversely, if a prime $\mathfrak{q}$ does not contain $H_{A/R}$, the map is smooth at that prime. D.2 supplies an elementary element outside $\mathfrak{q}$, so $\mathfrak{q}$ does not contain H_e . Therefore $V(H_e) \subset V(H_{A/R})$, and the reverse inclusion follows from the displayed ideal inclusion.

For completeness, the radical-to-primes fact used here holds over any ring. If $x \notin \sqrt{J}$, consider ideals containing J and disjoint from $\{1, x, x^2, \dots\}$. A chain union remains disjoint, so a maximal such ideal L exists. If $uv \in L$ with neither factor in L , both $L + (u)$ and $L + (v)$ meet that multiplicative set. Multiplying the two resulting expressions puts a power of x in L , a contradiction. Thus L is prime, contains J , and excludes x . This proves $\sqrt{J}$ is the intersection of its containing primes, including the empty intersection when J is the unit ideal. Equality of the closed sets above therefore gives $H_e = H_{A/R}$, and the intermediate inclusion gives $H_s = H_{A/R}$. $\square$

Example D.6: not quasi compact

Adapted from the Stacks project, smoothing.tex, lines 285–292.

The set of points where a finitely presented ring map is smooth needn't be a quasi-compact open. For example, let k be a field and set

$$R = k[x, y_1, y_2, y_3, \dots]/(xy_i \mid i \geq 1), \quad A = R/(x).$$

The distinguished opens below are in $\text{Spec}(A)$. The smooth locus of $R \rightarrow A$ is

$$U = \bigcup_{i \geq 1} D(y_i) = \text{Spec}(A) \setminus \{\mathfrak{m}\}, \quad \mathfrak{m} = (y_1, y_2, \dots),$$

and this open is not quasi-compact.

Proof. The quotient identifies A with $k[y_1, y_2, \dots]$. The map $R \rightarrow A$ is of finite presentation: it is the quotient by the single generator x , with no algebra variables required. The base ring R is not assumed Noetherian. Evaluation at all $y_i = 0$ gives $A/\mathfrak{m} = k$, so $\mathfrak{m}$ is maximal and is the only prime containing every y_i .

Here is the ideal calculation. Setting all y_i to zero gives a retraction $R \rightarrow k[x]$ of the natural inclusion $k[x] \rightarrow R$, so that inclusion is injective. For $I = (x) \subset R$, the relations $xy_i = 0$ give $xr = xr(x, 0, 0, \dots)$ for every $r \in R$. Therefore

$$I = xk[x], \quad I^2 = x^2k[x], \quad I/I^2 = k\bar{x} \simeq A/\mathfrak{m}.$$

In the last identification every y_i annihilates $\bar{x}$. For the zero-variable polynomial presentation $R \twoheadrightarrow A$, the conormal complex is $[I/I^2 \rightarrow 0]$. The presentation comparison in Lemma F.11 identifies it with the naive cotangent complex used in Definition F.6.

First suppose $y_i \notin \mathfrak{q}$. Inverting y_i in R makes $x = 0$, hence $A_{y_i} = R_{y_i}$. As an R -algebra this is

$$R[t]/(h), \quad h = y_it - 1.$$

The element h is a nonzerodivisor over this arbitrary base ring. Indeed, if $h \sum_{n=0}^N b_n t^n = 0$, its constant coefficient gives $b_0 = 0$, and its successive coefficients $y_i b_{n-1} - b_n = 0$ give every $b_n = 0$. Consequently $(h)/(h^2)$ is free on $\bar{h}$ over $R[t]/(h)$: multiplication by h gives a surjection onto it, and cancellation of the nonzerodivisor h shows that its kernel is exactly (h) . An R -derivation of $R[t]$ is determined by its value on t ; the product rule gives $d(t^n) = nt^{n-1}dt$. Thus $\Omega_{R[t]/R} = R[t]dt$, and this presentation has conormal complex

$$A_{y_i} \bar{h} \xrightarrow{y_i} A_{y_i} dt.$$

Its differential is an isomorphism, since y_i is a unit. Both first homology and the differential module are zero, and the zero module is finite projective. The displayed presentation is finite, so Definition F.6 proves $R \rightarrow A_{y_i}$ smooth. Definition F.7 proves smoothness at every point of $D(y_i)$.

We must also exclude the exceptional point $\mathfrak{m}$ from every smooth neighbourhood. Let $g \in A \setminus \mathfrak{m}$. Its constant coefficient $c = g(0, 0, \dots)$ is nonzero. Regard its polynomial expression as a lift $\tilde{g} \in R$. A presentation of A_g over R is

$$P = R[t] \twoheadrightarrow A_g, \quad J = \ker(P \rightarrow A_g) = (x, \tilde{g}t - 1).$$

The class of x in J/J^2 has relative differential zero, because x belongs to the base ring R . This class is nonzero. To see this, put $C = (R/(x^2))_{\tilde{g}}$. The R -algebra map $P \rightarrow C$ sending t to $\tilde{g}^{-1}$ maps J into the square-zero ideal $(x)C$, and maps J^2 to zero. It therefore maps the class of x in J/J^2 to $x \in C$. The assignments $x \mapsto \epsilon$ and $y_i \mapsto 0$ define

$$R/(x^2) \longrightarrow k[\epsilon]/(\epsilon^2).$$

They respect all the relations, and send $\tilde{g}$ to the unit c . The map extends to C and sends x to the nonzero element ϵ ; its nonzeroness follows directly from the degree bound for a multiple of ϵ^2 . Hence $x \neq 0$ in C , and the class of x is a nonzero kernel element in the conormal complex $[J/J^2 \rightarrow A_g dt]$. By the same presentation comparison, $H_1(\mathrm{NL}_{A_g/R}) \neq 0$. Definition F.6 shows that $R \rightarrow A_g$ is not smooth. This holds for every $g \notin \mathfrak{m}$, so Definition F.7 shows that $R \rightarrow A$ is not smooth at $\mathfrak{m}$.

Every other prime omits some y_i , proving that the smooth locus is exactly the displayed open U . In the notation of Definition D.1, this also gives $H_{A/R} = \mathfrak{m}$.

Finally the open cover $U = \bigcup_{i \geq 1} D(y_i)$ has no finite subcover. Given a finite set F of indices, choose $j \notin F$. Evaluation $y_j \mapsto 1$ and $y_i \mapsto 0$ for $i \neq j$ defines a surjection $A \rightarrow k$ with prime kernel $\mathfrak{q}_j$. This point lies in $D(y_j) \subset U$ but in none of the $D(y_i)$ with $i \in F$. Thus no finite subfamily covers U , and U is not quasi-compact. $\square$

Lemma D.7: strictly standard base change

Adapted from the Stacks project, smoothing.tex, lines 294–309.

Let $R \rightarrow A$ be a ring map of finite presentation. Let $R \rightarrow R'$ be a ring map. If $a \in A$ is elementary, resp. strictly standard in A over R , then $a \otimes 1$ is elementary, resp. strictly standard in $A \otimes_R R'$ over R' .

Proof. If $A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$ is a presentation of A over R , then $A \otimes_R R' = R'[x_1, \dots, x_n]/(f'_1, \dots, f'_m)$ is a presentation of $A \otimes_R R'$ over R' . Here f'_j is the image of f_j in $R'[x_1, \dots, x_n]$. Hence the result follows from the definitions. $\square$

Lemma D.8: final solve

Adapted from the Stacks project, smoothing.tex, lines 311–327.

Let $R \rightarrow A \rightarrow \Lambda$ be ring maps with A of finite presentation over R . Assume that $H_{A/R}\Lambda = \Lambda$. Then there exists a factorization $A \rightarrow B \rightarrow \Lambda$ with B smooth over R .

Proof. If $\Lambda = 0$, take $B = 0$, whose finite presentation $R/(1)$ and zero conormal and differential modules satisfy F.6. Otherwise membership of 1 in the extended ideal supplies finite lists $f_i \in H_{A/R}$ and $\lambda_i \in \Lambda$ with $\sum_i f_i \lambda_i = 1$. Set

$$B = A[t_1, \dots, t_r]/(\sum_i f_i t_i - 1),$$

and map $B \rightarrow \Lambda$ by $t_i \mapsto \lambda_i$. Every relation vanishes, and the composite with $A \rightarrow B$ is the prescribed map.

The algebra B is finitely presented over R : append its finite generator and relation lists to a finite presentation of A . The identity $\sum_i f_i t_i = 1$ makes the $D(f_i)$ cover its spectrum. On such a chart the inverse of f_i solves uniquely for t_i , so

$$B_{f_i} \cong A_{f_i}[t_1, \dots, \hat{t}_i, \dots, t_r].$$

Both substitution maps are inverse: substitute $t_i = f_i^{-1}(1 - \sum_{j \neq i} f_j t_j)$ into the displayed relation, and fix all other variables. Since $f_i \in H_{A/R}$, a prime avoiding f_i cannot contain $H_{A/R}$. Definition D.1 therefore gives smoothness at every point of A_{f_i} , and F.67 gives global smoothness there. A polynomial algebra is standard smooth using its empty equation list, by F.85; smooth composition F.29 makes B_{f_i} smooth over R . The same local criterion F.67 on the finite cover gives smoothness of B/R , proving the factorization. $\square$

Presentations of algebras

Some of the results in this section are due to Elkik. Note that the algebra C in the following lemma is a symmetric algebra over A . Moreover, if R is Noetherian, then C is of finite presentation over R .

Lemma D.9: improve presentation

Adapted from the Stacks project, smoothing.tex, lines 341–439.

Let R be a ring and let A be a finitely presented R -algebra. There exists a finite type R -algebra map $A \rightarrow C$ which has a retraction with the following two properties

1. for each $a \in A$ such that $R \rightarrow A_a$ is a local complete intersection (More on Algebra, Definition F.94) the ring C_a is smooth over A_a and has a presentation $C_a = R[y_1, \dots, y_m]/J$ such that J/J^2 is free over C_a , and
2. for each $a \in A$ such that A_a is smooth over R the module $\Omega_{C_a/R}$ is free over C_a .

Proof. Choose a presentation $A = R[x_1, \dots, x_n]/I$ with $I = (f_1, \dots, f_m)$, and set $M = I/I^2$. Define K by

$$0 \longrightarrow K \longrightarrow A^{\oplus m} \longrightarrow M \longrightarrow 0, \quad e_j \longmapsto f_j \bmod I^2.$$

Put $C = \text{Sym}_A(M)$. Projection onto degree zero is the required retraction. The surjection $R[x_1, \dots, x_n, y_1, \dots, y_m] \rightarrow C$ sends y_j to the class of f_j . Its kernel J is generated by the f_j and the linear forms $\sum h_j y_j$ whose coefficient classes $(h_j \bmod I)$ belong to K . This follows directly from the universal property of the symmetric algebra. In particular C is of finite type over A .

Let $h \in R[x_1, \dots, x_n]$ have image $a \in A$. Use the presentations

$$A_a = R[x_0, x_1, \dots, x_n]/I', \quad C_a = R[x_0, x_1, \dots, x_n, y_1, \dots, y_m]/J',$$

where $I' = (hx_0 - 1, I)$ and $J' = (hx_0 - 1, J)$. The principal-localization conormal calculation F.72 gives

$$I'/(I')^2 = A_a \oplus M_a, \quad J'/(J')^2 = C_a \oplus (J/J^2)_a.$$

Now assume that A_a is a local complete intersection over R . The Koszul-regular local equation lists make M_a locally free. Indeed, if (g_i) is such a list, a conormal relation $\sum b_i g_i \in (g)^2$ can first be corrected by coefficients in (g) to an exact relation. Its coefficients are then Koszul boundaries, by vanishing of first Koszul homology, and hence all belong to (g) . Thus the equation classes form a free conormal basis. Choose a finite principal cover trivializing M_a , using quasi-compactness of its spectrum. On each chart the kernel K_a of the finite free surjection onto M_a is a finite direct summand. Clear denominators of these finitely many chart generators and collect them in K_a ; their cokernel vanishes on the cover, so they generate K_a globally. Consequently M_a is finitely presented. It is flat, because each ideal tensor map is injective on that same free cover and local detection proves its global injectivity. The finite-presentation projectivity theorem AG-CA-07, Theorem 5.3 therefore makes M_a finite projective. The displayed defining sequence of K splits after localization, so K_a is finite projective and $K_a \oplus M_a \cong A_a^{\oplus m}$. Only at this point do we apply the symmetric-algebra conormal identity: the finite-projective quotient hypothesis in F.102 identifies the conormal module of $A_a[y_j] \rightarrow C_a$ with $K_a \otimes_{A_a} C_a$. Also $C_a = \text{Sym}_{A_a}(M_a)$ is smooth over A_a by F.128. On principal neighborhoods trivializing K_a and M_a , the coordinates in F.102 make the kernel of $A_a[y_j] \rightarrow C_a$ an ideal generated by polynomial variables. That is a regular sequence, so its

first Koszul homology vanishes by the exterior-complex cone calculation. Thus the exact conormal injectivity theorem [F.130](#) applies to this particular presentation. The transitivity conormal sequence [F.42](#) for the localized presentations, with left injectivity supplied by [F.130](#), is therefore

$$0 \longrightarrow C_a \oplus (M \otimes_A C_a) \longrightarrow C_a \oplus (J/J^2)_a \longrightarrow K_a \otimes_{A_a} C_a \longrightarrow 0.$$

The right module is projective, so this sequence splits and gives

$$J'/(J')^2 \cong C_a \oplus (M_a \oplus K_a) \otimes_{A_a} C_a \cong C_a^{\oplus(m+1)}.$$

This proves (1), including the asserted smoothness over A_a .

If A_a is smooth over R , its conormal sequence for the original n variables is split exact by the smoothness definition [F.6](#). It makes M_a finite projective and gives $M_a \oplus \Omega_{A_a/R} \cong A_a^{\oplus n}$ independently of (1). Now [F.128](#) and [F.102](#) apply: $\Omega_{C_a/A_a} = M_a \otimes_{A_a} C_a$ and $H_1(L_{C_a/A_a}) = 0$. Thus [F.42](#) gives

$$0 \longrightarrow \Omega_{A_a/R} \otimes_{A_a} C_a \longrightarrow \Omega_{C_a/R} \longrightarrow M_a \otimes_{A_a} C_a \longrightarrow 0.$$

The right module is projective, so this sequence splits. Combining it with the preceding split conormal identity gives

$$\Omega_{C_a/R} \cong (M_a \oplus \Omega_{A_a/R}) \otimes_{A_a} C_a \cong C_a^{\oplus n},$$

which proves (2). $\square$

Proposition D.10: lift smooth

Adapted from the Stacks project, smoothing.tex, lines 447–557.

Let $R \rightarrow R_0$ be a surjective map of Noetherian rings with kernel I . This is the auxiliary lifting scope consumed by the Noetherian desingularization argument below.

1. If $R_0 \rightarrow A_0$ is a syntomic ring map, then there exists a syntomic ring map $R \rightarrow A$ such that $A/IA \cong A_0$.
2. If $R_0 \rightarrow A_0$ is a smooth ring map, then there exists a smooth ring map $R \rightarrow A$ such that $A/IA \cong A_0$.

Proof. Assume $R_0 \rightarrow A_0$ syntomic, in particular a local complete intersection (More on Algebra, Lemma [F.129](#)). Choose a presentation $A_0 = R_0[x_1, \dots, x_n]/J_0$. Set $C_0 = \text{Sym}_{A_0}^*(J_0/J_0^2)$. Note that J_0/J_0^2 is a finite projective A_0 -module (Algebra, Lemma [F.88](#)). By Lemma [D.9](#) the ring map $A_0 \rightarrow C_0$ is smooth and we can find a presentation $C_0 = R_0[y_1, \dots, y_m]/K_0$ with K_0/K_0^2 free over C_0 . By Algebra, Lemma [F.55](#) we can assume $C_0 = R_0[y_1, \dots, y_m]/(\bar{f}_1, \dots, \bar{f}_c)$ where $\bar{f}_1, \dots, \bar{f}_c$ maps to a basis of K_0/K_0^2 over C_0 . Choose $f_1, \dots, f_c \in R[y_1, \dots, y_m]$ lifting $\bar{f}_1, \dots, \bar{f}_c$ and set

$$C = R[y_1, \dots, y_m]/(f_1, \dots, f_c)$$

By construction $C_0 = C/IC$. By Algebra, Lemma [F.66](#) we can after replacing C by C_g assume that C is a relative global complete intersection over R . We conclude that there exists a finite projective A_0 -module P_0 such that $C_0 = \text{Sym}_{A_0}^*(P_0)$ is isomorphic to C/IC for some syntomic R -algebra C .

Choose an integer n and a direct sum decomposition $A_0^{\oplus n} = P_0 \oplus Q_0$. By More on Algebra, Lemma [F.117](#) we can find an étale ring map $C \rightarrow C'$ which induces an isomorphism $C/IC \rightarrow C'/IC'$ and a

finite projective C' -module Q such that Q/IQ is isomorphic to $Q_0 \otimes_{A_0} C/IC$. Then $D = \mathrm{Sym}_{C'}^*(Q)$ is a smooth C' -algebra (see More on Algebra, Lemma F.128). Picture

$$\begin{array}{c}
R \rightarrow R/I \\
R \rightarrow C \\
C \rightarrow C' \\
C \rightarrow C/IC \\
C' \rightarrow D \\
C' \rightarrow C'/IC' \\
D \rightarrow D/ID \\
R/I \rightarrow A_0 \\
A_0 \rightarrow C/IC \\
C/IC \xrightarrow{\cong} C'/IC' \\
C'/IC' \rightarrow D/ID
\end{array}$$

The identities of F.128 give

$$\begin{aligned}
D/ID &= \mathrm{Sym}_{C/IC}^*(Q_0 \otimes_{A_0} C/IC) \\
&= \mathrm{Sym}_{A_0}^*(Q_0) \otimes_{A_0} C/IC \\
&= \mathrm{Sym}_{A_0}^*(Q_0) \otimes_{A_0} \mathrm{Sym}_{A_0}^*(P_0) \\
&= \mathrm{Sym}_{A_0}^*(Q_0 \oplus P_0) \\
&= \mathrm{Sym}_{A_0}^*(A_0^{\oplus n}) \\
&= A_0[x_1, \dots, x_n]
\end{aligned}$$

Choose $f_1, \dots, f_n \in D$ which map to $x_1, \dots, x_n$ in $D/ID = A_0[x_1, \dots, x_n]$. Set $A = D/(f_1, \dots, f_n)$. Note that $A_0 = A/IA$. We claim that $R \rightarrow A$ is syntomic in a neighbourhood of $V(IA)$. If the claim is true, then we can find a $f \in A$ mapping to $1 \in A_0$ such that A_f is syntomic over R and the proof of (1) is finished.

Proof of the claim. Observe that $R \rightarrow D$ is syntomic as a composition of the syntomic ring map $R \rightarrow C$, the étale ring map $C \rightarrow C'$ and the smooth ring map $C' \rightarrow D$ (Algebra, Lemmas F.32 and F.83). The question is local on $\mathrm{Spec}(D)$, hence we may assume that D is a relative global complete intersection (Algebra, Lemma F.87). Say $D = R[y_1, \dots, y_m]/(g_1, \dots, g_s)$. Let $f'_1, \dots, f'_n \in R[y_1, \dots, y_m]$ be lifts of $f_1, \dots, f_n$. Then we can apply Algebra, Lemma F.66 to get the claim.

Proof of (2). Since a smooth ring map is syntomic, we can find a syntomic ring map $R \rightarrow A$ such that $A_0 = A/IA$. By assumption the fibres of $R \rightarrow A$ are smooth over primes in $V(I)$ hence $R \rightarrow A$ is smooth in an open neighbourhood of $V(IA)$ (Algebra, Lemma F.45). Thus we can replace A by a localization to obtain the result we want. $\square$

We know that any syntomic ring map $R \rightarrow A$ is locally a relative global complete intersection, see Algebra, Lemma F.87. The next lemma says that a vector bundle over $\mathrm{Spec}(A)$ is a relative global complete intersection.

Lemma D.11: syntomic complete intersection

Adapted from the Stacks project, smoothing.tex, lines 566–587.

Let $R \rightarrow A$ be a syntomic ring map. Then there exists a smooth R -algebra map $A \rightarrow C$ with a retraction such that C is a global relative complete intersection over R , i.e.,

$$C \cong R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

flat over R and all fibres of dimension $n - c$.

Proof. If $A = 0$, take $C = 0$, with its identity retraction and presentation $R/(1)$. Its zero module is flat, the identity is smooth, and there are no nonempty fibres. Assume $A \neq 0$ and fix a finite presentation

$$A = R[X_1, \dots, X_N]/I, \quad I = (u_1, \dots, u_s).$$

We first supply the arbitrary-base complete-intersection input to D.9.

Fix a prime $\mathfrak{q}$ of A , with polynomial preimage $\mathfrak{Q}$ and contraction $\mathfrak{p}$ in R . Write $k = \kappa(\mathfrak{p})$. In the polynomial fibre local ring $V = k[X]_{\overline{\mathfrak{Q}}}$, the image $\bar{I}$ is a complete-intersection ideal by syntomicity and F.58. By F.57, its conormal is free of some rank e . Select e of the u_i whose fibre conormal classes give a basis modulo the maximal ideal. Nakayama makes the selected map onto the local conormal surjective; its square matrix in a free basis has nonzero residue determinant and is invertible. Thus these selected classes are a conormal basis. Denote the selected polynomials by $v_1, \dots, v_e$. The final assertion of F.57 makes their images a regular sequence generating $\bar{I}$ in V .

Use the local models F.59 for $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{q}}$. They may be indexed by finite-type $\mathbf{Z}$ -subalgebras $R_0 \subset R$ containing every coefficient of the u_i : put $\mathfrak{p}_0 = \mathfrak{p} \cap R_0$, $T_0 = R_0[X]/(u_i)$ and $\mathfrak{q}_0 = \mathfrak{q} \cap T_0$. The local models are

$$R_{0, \mathfrak{p}_0} \longrightarrow (T_0)_{\mathfrak{q}_0}.$$

This indexing is cofinal for $R_{\mathfrak{p}}$: any finite list of fractions in that ring is contained in such a localized subalgebra after adding their numerators and denominators. The target models and their colimit have exactly the localization and finite-presentation descriptions in F.59. Apply F.21 to the module $A_{\mathfrak{q}}$, finitely presented over itself and flat over $R_{\mathfrak{p}}$. At some model $(T_0)_{\mathfrak{q}_0}$ is flat over $R_{0, \mathfrak{p}_0}$.

Its closed fibre is also a complete intersection; here is the required descent calculation. Put $k_0 = \kappa(\mathfrak{p}_0)$ and let $\mathfrak{Q}_0$ be the polynomial preimage of $\mathfrak{q}_0$. The map

$$V_0 = k_0[X]_{\overline{\mathfrak{Q}_0}} \longrightarrow V = k[X]_{\overline{\mathfrak{Q}}}$$

is a local flat map: it is field extension followed by localization, and the source prime is the contraction of the target prime. Flat base change identifies the image ideal and its conormal with those of $\bar{I}$ in V . Consequently the selected conormal map becomes the basis isomorphism just constructed in V . Its kernel and cokernel are finite over the Noetherian local quotient $V_0/\bar{I}_0$, and flatness makes their tensors zero. A finite module with zero tensor along this local map is zero: further tensor with the target residue field, which extends the source residue field, and use Nakayama. Thus the selected classes are a conormal basis in the model as well. The finite module $\bar{I}_0/(v_i)$ likewise has zero tensor, so the v_i generate $\bar{I}_0$. For regularity, flat tensor commutes with the homology of their finite free Koszul complex. Its positive homology tensors to zero, because the final list is regular. Every such homology module is finite over the Noetherian V_0 , so the same local residue-field and Nakayama argument makes it zero. The Noetherian equivalence F.118 makes this list regular in V_0 . This proves the closed-fibre assertion without inferring regularity merely from a free conormal.

Now F.87 applies to the finite presentation $R_0 \rightarrow T_0$ at $\mathfrak{q}_0$: its base is Noetherian, the stated local flatness holds, and its closed fibre is a complete intersection. It gives $g_0 \notin \mathfrak{q}_0$ for which $B_0 = (T_0)_{g_0}$ is a relative global complete intersection over R_0 . Base change to R gives the exact identity

$$R \otimes_{R_0} B_0 \cong A_g, \quad g \notin \mathfrak{q}.$$

It also preserves the relative fibre dimensions by F.14.

We need Koszul regularity on this actual neighbourhood, not merely the dimension assertion. Write the finite polynomial complete-intersection presentation of B_0 as $P_0/(a_1, \dots, a_t)$. Its augmented

Koszul complex is exact by F.126, and B_0 is R_0 -flat by F.76. Every complex term is R_0 -flat, since it is finite free over the polynomial ring P_0 . All successive kernels are R_0 -flat as well. Indeed, in $0 \rightarrow Z \rightarrow F \rightarrow Q \rightarrow 0$ with F, Q flat, quotient flatness makes $Z \otimes L \rightarrow F \otimes L$ injective for every module L ; restricting the injection for F to any inclusion $L' \rightarrow L$ proves that Z is flat. Induct from the flat augmentation quotient B_0 . Tensoring these short exact sequences with R therefore preserves exactness. The base-changed equations have zero positive Koszul homology and present A_g over R .

This construction works at every $\mathfrak{q}$. The arbitrary-ring presentation comparison in F.94 transports these Koszul-regular presentations to the fixed presentation of A ; finite presentation permits clearing the denominators of its finite equation lists. Thus $R \rightarrow A$ is a local complete intersection in precisely the sense required by D.9. No Noetherian condition has been imposed on R .

Apply D.9 with $a = 1$. It constructs

$$M = I/I^2, \quad C = \text{Sym}_A(M),$$

with augmentation $C \rightarrow A$, and proves that M is finite projective, that $A \rightarrow C$ is smooth, and that C has a polynomial presentation with free conormal. It is finitely presented over A by F.128, hence over R by the finite substitution proof F.28. Apply F.55 to obtain

$$C = R[x_1, \dots, x_n]/J, \quad J = (f_1, \dots, f_c),$$

with these equation classes a free conormal basis.

Here the flatness and fibre assertions can be checked directly. On a finite principal cover of $\text{Spec } A$ trivializing M , its symmetric algebra is a polynomial algebra over that chart, by the proved identities in F.128. It is therefore A -flat: tensor an injection with C and detect its kernel on this cover. Combining with the given R -flatness of A makes C flat over R .

For $k = \kappa(\mathfrak{p})$, those same identities give $C_k = \text{Sym}_{A_k}(M \otimes_A A_k)$. The algebra A_k is locally a complete intersection by the original syntomic hypothesis. Trivialize the projective module and restrict further to its complete-intersection charts. On each chart C_k is a polynomial algebra over a field complete intersection. Its regular equation list stays regular under polynomial extension, which is flat, and under localization; the ambient polynomial local rings remain regular. Thus F.58 makes C_k locally a complete intersection. This proves the fibre assertion without invoking smooth/syntomic composition over an arbitrary base.

Finally R -flatness of C makes the tensor of $0 \rightarrow J \rightarrow R[x] \rightarrow C \rightarrow 0$ injective on the left. It identifies $J \otimes_R k$ with the fibre equation ideal $\bar{J}$. Products of the images of elements of J generate $\bar{J}^2$; right exact tensor on $J^2 \rightarrow J \rightarrow J/J^2 \rightarrow 0$ consequently gives

$$\bar{J}/\bar{J}^2 \cong (J/J^2) \otimes_R k \cong C_k^{\oplus c}.$$

For a minimal prime $\mathfrak{r}$ of a nonzero C_k , F.57 identifies this conormal rank with the height of its polynomial preimage, since $\text{height}(\mathfrak{r}) = 0$. Each minimal equation prime therefore has height c . The field polynomial dimension formula AG-CA-09, Theorem 4.3, gives every irreducible component dimension $n - c$. There are finitely many components by Noetherianness of the field algebra, so their maximum is $\dim C_k = n - c$. Empty fibres impose no condition. The augmentation is the required retraction and $A \rightarrow C$ is the already proved smooth map. $\square$

Lemma D.12: smooth standard smooth

Adapted from the Stacks project, smoothing.tex, lines 589–672.

Let $R \rightarrow A$ be a smooth ring map. Then there exists a smooth R -algebra map $A \rightarrow B$ with a retraction such that B is standard smooth over R , i.e.,

$$B \cong R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

and $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$ is invertible in B .

Proof. If $A = 0$, take $B = 0$ and the identity retraction. The presentation $B = R[X]/(1)$ has one equation and one variable; its derivative is zero, which is a unit in the zero ring since $0 = 1$ there. Hence assume $A \neq 0$.

Use the smooth-case construction in the proof of D.9. Here are its precise inputs in this case. For a finite presentation $A = P/I$, smoothness and F.6 give the split conormal sequence, so $M = I/I^2$ is finite projective. A finite free surjection $A^r \rightarrow M$ splits, with finite projective kernel K and $M \oplus K = A^r$. The proof of D.9 constructs $C = \text{Sym}_A(M)$, with its augmentation $C \rightarrow A$. By F.128, $A \rightarrow C$ is smooth and finitely presented. Its displayed conormal calculation applies to these finite projective M and K : on charts trivializing them the extra kernel is generated by polynomial variables, so the injectivity required for that calculation holds. Adding the dummy relation $X_0 - 1$ gives a polynomial presentation of C whose conormal is

$$C \oplus (M \oplus K) \otimes_A C \cong C^{r+1}.$$

These are exactly the arbitrary-ring calculations proved in D.9; no Noetherian syntomic implication is needed here. Apply F.55 to obtain a presentation

$$C = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

whose equation classes form a conormal basis. Smooth composition F.29 makes C smooth over R . It is nonzero because it retracts onto A .

The split conormal criterion in AG-CA-17, Theorem 3.1 and Section 4, now gives a left inverse to its conormal differential. In the chosen bases write it as coefficients $a_{i\ell} \in C$ satisfying

$$\sum_{i=1}^n \frac{\partial f_j}{\partial x_i} a_{i\ell} = \delta_{j\ell} \quad (1 \leq j, \ell \leq c).$$

Reduction at a maximal ideal of the nonzero ring C gives an injection of a c -dimensional vector space into an n -dimensional one, so $c \leq n$. Choose polynomial lifts $\tilde{a}_{i\ell} \in R[x]$ and put $E = C[y_1, \dots, y_c]$. Define

$$h_i = x_i - \sum_{\ell=1}^c \tilde{a}_{i\ell}(x) y_\ell - z_i.$$

Eliminating the z_i gives the exact presentation

$$E \cong R[z_1, \dots, z_n, x_1, \dots, x_n, y_1, \dots, y_c]/(f_1, \dots, f_c, h_1, \dots, h_n).$$

Let $G \in R[x, y]$ be the determinant of the square Jacobian of the ordered equations $(f_1, \dots, f_c, h_1, \dots, h_n)$ with respect to the ordered variables $(x_1, \dots, x_n, y_1, \dots, y_c)$. After setting $y = 0$ and reducing modulo (f) , this matrix is

$$\begin{pmatrix} J & 0 \\ 1_n & -a \end{pmatrix}, \quad Ja = 1_c.$$

Moving the last n rows before the first c rows contributes $(-1)^{nc}$. Subtracting J times the first block row from the second then leaves diagonal blocks 1_n and $Ja = 1_c$. Thus the image g of G in E satisfies

$$g \bmod (y_1, \dots, y_c) = \epsilon = (-1)^{nc}.$$

Set $B = E_g$. The augmentation $E \rightarrow C$, $y \mapsto 0$, sends g to the unit ϵ , so extends to $B \rightarrow C$. Therefore $C \rightarrow B$, and its composite $A \rightarrow B$, have the stated retractions. Polynomial extension

and principal localization are smooth by AG-CA-17, Theorem 4.1 and its composition calculation; combining with $A \rightarrow C$ proves smoothness of $A \rightarrow B$.

Finally write the localization as a polynomial quotient:

$$B \cong R[x_1, \dots, x_n, y_1, \dots, y_c, t, z_1, \dots, z_n] / (f_1, \dots, f_c, h_1, \dots, h_n, tG - 1).$$

There are $n + c + 1$ equations. Select the first $n + c + 1$ variables (x, y, t) . Their square Jacobian has the previous block in its upper left, zero in its upper right, and G in its last diagonal entry. Its determinant is G^2 , a unit because $tG = 1$. This is the required standard smooth presentation over the original ring R . The empty-block conventions give the same calculation when $c = 0$ or $n = 0$. $\square$

Lemma D.13: colimit standard smooth

Adapted from the Stacks project, smoothing.tex, lines 674–691.

Let $R \rightarrow \Lambda$ be a ring map. If Λ is a filtered colimit of smooth R -algebras, then Λ is a filtered colimit of standard smooth R -algebras.

Proof. First take any finitely presented R -algebra A and map $A \rightarrow \Lambda$. The forward implication of F.90 factors it through a smooth R -algebra B . By D.12, there are maps $B \rightarrow C \rightarrow B$ with composite the identity and C standard smooth over R . Thus

$$A \longrightarrow B \longrightarrow C \longrightarrow B \longrightarrow \Lambda$$

is a factorization through a standard smooth algebra.

Here is the full colimit construction for this class. Consider standard smooth finitely presented R -algebras equipped with a map to Λ , and arrows commuting with that map. Choose a set of representatives of finite polynomial presentations, together with their maps to Λ ; this is a small category. It is nonempty: R , with no variables or equations and empty Jacobian determinant 1, is an object. Given objects B_1, B_2 , the tensor product has the finite presentation obtained by adjoining their two disjoint lists of variables and relations. Factor its map to Λ through a standard smooth algebra using the preceding paragraph. The two natural arrows then give a common successor. For parallel arrows $u, v: B \rightrightarrows C$, choose finite algebra generators $b_1, \dots, b_s$ of B . The quotient

$$C / (u(b_1) - v(b_1), \dots, u(b_s) - v(b_s))$$

is finitely presented: append this finite relation list to a presentation of C . Factor its map to Λ through a standard smooth algebra. The resulting arrow out of C equalizes u, v , since equality on the generators implies equality on every polynomial in them. The category is therefore filtered.

Its colimit maps to Λ . For each $\lambda \in \Lambda$, factor $R[T] \rightarrow \Lambda$, $T \mapsto \lambda$, through an object; this proves surjectivity. If $b \in B$ represents an element mapping to zero, the finitely presented quotient $B/(b)$ maps to Λ . A factorization of this map through an object kills b at a successor, so its colimit class is zero. For two representatives with equal images, first move them to a common successor and apply this argument to their difference. This proves injectivity. Hence the colimit is exactly Λ , with every stage standard smooth. $\square$

Lemma D.14: standard smooth include generators

Adapted from the Stacks project, smoothing.tex, lines 693–716.

Let $R \rightarrow A$ be a standard smooth ring map. Let $E \subset A$ be a finite subset of order $|E| = n$. Then there exists a presentation $A = R[x_1, \dots, x_{n+m}] / (f_1, \dots, f_c)$ with $c \geq n$, with $\det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$ invertible in A , and such that E is the set of congruence classes of $x_1, \dots, x_n$.

Proof. Choose a standard smooth presentation

$$A = R[y_1, \dots, y_m] / (g_1, \dots, g_d), \quad d \leq m,$$

whose minor on $y_1, \dots, y_d$ is a unit. Enumerate $E = \{a_1, \dots, a_n\}$ and lift a_i to a polynomial $h_i(y)$. Put

$$Q = R[x_1, \dots, x_n, y_1, \dots, y_m]/(x_1 - h_1, \dots, x_n - h_n, g_1, \dots, g_d).$$

The map $Q \rightarrow A$ sends x_i to a_i and y_j to its original class. Conversely the old presentation maps $A \rightarrow Q$ by those y_j . Their composites are identity: on y_j this is immediate, and on x_i it follows from the relation $x_i = h_i(y)$. Thus this is the asserted presentation of A .

Order its equations as $(x_i - h_i)_{i=1}^n, (g_j)_{j=1}^d$ and select the variables $(x_1, \dots, x_n, y_1, \dots, y_d)$. The matrix with equations as rows and variables as columns is

$$\begin{pmatrix} 1_n & -(\partial h_i / \partial y_j)_{i \leq n, j \leq d} \\ 0 & (\partial g_i / \partial y_j)_{i, j \leq d} \end{pmatrix}.$$

In its signed permutation expansion the first diagonal block forces the determinant to be the determinant of the lower right block. Transposition gives the same determinant for the convention in the statement. It is a unit in A . Set $c = n + d \geq n$, rename the y_j as x_{n+j} , and keep the first n variables in the order just chosen. Their classes are exactly E . Empty blocks have determinant 1, so the calculation also covers $n = 0$ or $d = 0$. $\square$

Lemma D.15: compare standard

Adapted from the Stacks project, smoothing.tex, lines 718–850.

Let $R \rightarrow A$ be a ring map of finite presentation. Let $a \in A$. Consider the following conditions on a :

1. A_a is smooth over R ,
2. A_a is smooth over R and $\Omega_{A_a/R}$ is stably free,
3. A_a is smooth over R and $\Omega_{A_a/R}$ is free,
4. A_a is standard smooth over R ,
5. a is strictly standard in A over R ,
6. a is elementary standard in A over R .

Then we have

1. **(a)** (4) $\Rightarrow$ (3) $\Rightarrow$ (2) $\Rightarrow$ (1),
2. **(b)** (6) $\Rightarrow$ (5),
3. **(c)** (6) $\Rightarrow$ (4),
4. **(d)** (5) $\Rightarrow$ (2),
5. **(e)** (2) $\Rightarrow$ the elements a^e , $e \geq e_0$ are strictly standard in A over R ,
6. **(f)** A_a is smooth over R and $\Omega_{A_a/R}$ has a finite basis $db_1, \dots, db_d$ with $b_i \in A$ if and only if every sufficiently large positive power a^e is elementary standard in A over R . These conditions also follow if any one positive power of a is elementary standard.

The requirement that the differential-basis representatives in (f) belong to A is essential. Standard smoothness of A_a alone does not imply the elementary-power conclusion over arbitrary base rings; the counterexample after the proof makes this distinction explicit.

Proof of (a)–(e). By F.85, a standard smooth algebra is smooth and its differentials are free. A free module is stably free by adding the zero free module; forgetting this condition gives (a). For (b), take the coefficients of the other minors to be zero in D.3s1; transposing the one square minor does not change its determinant. D.3e2 and D.3s2 are the same relation condition.

For (c), use an elementary presentation $A = P/I$, where $P = R[x_1, \dots, x_n]$, $I = (f_1, \dots, f_m)$ and the first c equations have minor Δ . Lift a to $h \in P$. Write

$$B = A_a = P[x_0]/J, \quad J = (x_0h - 1, f_1, \dots, f_m),$$

and $K = (x_0h - 1, f_1, \dots, f_c)$. The localization conormal calculation F.72 and D.3e2 give $J = K + J^2$: after inverting a , the classes of the remaining f_j are combinations of the first c classes. The module J/K is finite, since J has the displayed finite generating list, and satisfies $J/K = J(J/K)$. Choose generators v_i and write $v_i = \sum_j b_{ij}v_j$ with $b_{ij} \in J$. Multiplication by the adjugate of $(\delta_{ij} - b_{ij})$, whose identity is proved in F.70, shows that

$$g = \det(\delta_{ij} - b_{ij}) \in 1 + J, \quad gJ \subset K.$$

For the zero module choose $g = 1$. Thus $J_g = K_g$. Since g maps to 1 in B , adjoining an inverse for it does not change B , and

$$B = P[x_0, z]/(x_0h - 1, f_1, \dots, f_c, gz - 1).$$

For these ordered equations select $(x_0, x_1, \dots, x_c, z)$. The first $c + 1$ rows have zero last column. Their first $c + 1$ columns form the block matrix with upper left entry h , lower left zero, and lower right Jacobian minor Δ . The last diagonal entry of the whole matrix is g . Its determinant is therefore $gh\Delta$. In B , $g = 1$, $h = a$ is a unit, and D.3e1 gives $a = a'\Delta$, so $\Delta^{-1} = a'/a$. This proves standard smoothness, including the empty minor case $c = 0$.

For (d), D.3s2 makes the first c equation classes generate $M = (I/I^2)_a$. D.4 and D.3s1 give a map back from B^n whose composite with the map $B^c \rightarrow M \rightarrow B^n$ is multiplication by the unit a . Thus $B^c \rightarrow M$ is both surjective and injective, and the conormal differential is a split injection. D.5 proves that $B = A_a$ is smooth. The localized conormal sequence, by F.65 and F.72, now gives

$$M \oplus \Omega_{B/R} \cong B^n, \quad M \cong B^c.$$

Consequently $\Omega_{B/R} \oplus B^c \cong B^n$, exactly the required stable freeness.

For (e), start with a finite presentation $A = R[X_1, \dots, X_n]/I$ and $B = A_a$. If $B = 0$, the equality $1/1 = 0$ means $a^L = 0$ for some L . For every $e \geq L$, take $c = 0$ in any finite presentation: the empty minor is 1 and all multiplied relation classes are zero. Such powers are elementary, hence strictly standard. Assume $B \neq 0$. Smoothness, the conormal description of the naive cotangent complex, and F.65/F.72 give the split exact sequence

$$0 \rightarrow M = (I/I^2)_a \rightarrow B^n \rightarrow \Omega_{B/R} \rightarrow 0.$$

If $\Omega_{B/R} \oplus B^s \cong B^t$, then $M \oplus B^t \cong B^{n+s}$. Add t variables Y_i mapping to zero in A . The kernel of this enlarged presentation is $J = (I, Y_1, \dots, Y_t)$, and

$$J/J^2 \cong I/I^2 \oplus A^t.$$

Indeed, evaluation at $Y = 0$ gives the first component, and the coefficients of the terms linear in Y modulo I give the second; their kernel consists precisely of products of two elements of J . The differential sends the new generators to the independent dY_i . Thus $(J/J^2)_a$ is finite free and its differential is a split injection into B^{n+t} .

Choose representatives $f_1, \dots, f_c \in J$ of a basis after localization: a basis element is a fraction of an element of J/J^2 , and multiplying each basis element by its own denominator, a unit in B ,

still gives a basis. Extend them to a finite generating list $f_1, \dots, f_m$ of J . Finiteness follows from F.44, or directly from the old finite list and the new variables. Each remaining class is zero in the localization of

$$Q = (J/J^2)/\langle f_1, \dots, f_c \rangle.$$

For each of its finitely many generators some power of a kills that class. Choose a common exponent L . For a polynomial lift h of a this says

$$h^L f_j \in (f_1, \dots, f_c) + J^2 \quad (j > c).$$

Multiplying by further powers of h gives D.3s2 for every a^e , $e \geq L$.

Let F be the $(n+t) \times c$ matrix of the differentials of the first c equations over A . The split injection after localization has a matrix left inverse P over B . Choose u and a matrix P_0 over A with $(P_0)_a = a^u P$. Every entry of $P_0 F - a^u 1_c$ becomes zero in B , so a common further power a^v kills all those finitely many entries. Hence

$$P_1 F = a^{u+v} 1_c, \quad P_1 = a^v P_0,$$

as an exact equality over A . Increase $u + v$ to a positive integer k by multiplying P_1 by an additional power of a if necessary. No power of a has been cancelled. F.70 puts a^{ck} in the ideal of the $c \times c$ minors of F . Reduction at a prime of the nonzero ring B shows $c \leq n + t$. For every $e \geq \max(L, ck, 1)$, multiply the minor coefficients by a^{e-ck} to obtain D.3s1, together with the already established D.3s2. When $c = 0$, the minor ideal is A and the same conclusion uses the empty determinant 1. This proves (e) over arbitrary rings, including zero divisors. $\square$

Proof of (f). Put $B = A_a$. If $B = 0$, some power a^L is zero. The zero differential module has the empty basis, and, as in (e), every a^e with $e \geq \max(L, 1)$ is elementary using $c = 0$ and the empty minor 1. Assume $B \neq 0$.

First suppose B is smooth and $db_1, \dots, db_d$ is a basis with $b_i \in A$. Choose finite algebra generators $z_1, \dots, z_n$ of A over R and the surjection

$$P = R[X_1, \dots, X_n, Y_1, \dots, Y_d] \longrightarrow A, \quad X_i \longmapsto z_i, \quad Y_j \longmapsto b_j.$$

Its kernel J is finitely generated by F.44. The smoothness definition F.6, the presentation description of the naive cotangent complex F.11, and localization F.65 give the exact sequence

$$0 \longrightarrow M = (J/J^2)_a \xrightarrow{d} B^n \oplus B^d \longrightarrow \Omega_{B/R} \longrightarrow 0.$$

Here the two summands have coordinates dX_i and dY_j . The map from the second summand to $\Omega_{B/R}$ is an isomorphism, because its images are the chosen basis. Consequently projection onto the first summand restricts to an isomorphism $M \rightarrow B^n$: for each $v \in B^n$, there is a unique $w \in B^d$ whose differential image cancels that of v , so (v, w) is in the image of M ; a vector of M with first component zero has second component zero as well. Injectivity of d proves uniqueness in M .

Let $\eta_1, \dots, \eta_n$ be the resulting basis of M whose projected differentials are the standard basis of B^n . Write each η_i as a fraction of a class represented by $f_i \in J$, with denominator a^{k_i} . Thus the classes of f_i are $a^{k_i} \eta_i$, still a basis over B . Their square Jacobian on $X_1, \dots, X_n$ is diagonal after passage to B , with entries a^{k_i} . In particular the image of

$$\delta = \det(\partial f_i / \partial X_j)_{i,j=1}^n \in A$$

is a unit in B . Extend $f_1, \dots, f_n$ to a finite generating list $f_1, \dots, f_m$ of J . The finite module

$$Q = (J/J^2)/\langle f_1, \dots, f_n \rangle$$

has zero localization at a . A common power a^L kills its finite generating list. For a polynomial lift $h \in P$ of a , this means

$$h^L f_j \in (f_1, \dots, f_n) + J^2 \quad (j > n).$$

All higher powers satisfy the same relation condition.

Since δ is a unit in B , choose $u \in A$ and $r \geq 0$ with $\delta(u/a^r) = 1$ in B . The equality in this localization supplies $v \geq 0$ such that

$$a^v(\delta u - a^r) = 0 \quad \text{in } A, \quad a^{r+v} = \delta(a^v u).$$

Increase the exponent to a positive K by multiplying this equality by an additional power of a if necessary. This gives $a^K = \delta u_0$ in A , without cancelling a power of a possible zero divisor. Every $e \geq \max(K, L, 1)$ then satisfies both D.3e1 and D.3e2 for the presentation $A = P/J$, with $c = n$ and the selected variables X_i . This proves the forward direction, including $n = 0$, when δ is the empty determinant 1.

Conversely, suppose a^r is elementary for some $r \geq 1$. Inverting a^r is the same as inverting a , since $a^{-1} = a^{r-1}(a^r)^{-1}$. Part (c), applied to a^r , proves that B is standard smooth, hence smooth. Take its elementary presentation $A = R[x_1, \dots, x_N]/I$ with $I = (g_1, \dots, g_m)$ and selected minor on $x_1, \dots, x_c$. In B , D.3e2 makes every remaining conormal class a combination of the first c . Therefore

$$\Omega_{B/R} = B^N / \langle dg_1, \dots, dg_c \rangle.$$

The selected $c \times c$ Jacobian block is invertible. Subtracting a unique combination of these c differential rows eliminates the first c coordinates of any vector in B^N ; a combination with those coordinates zero has all its coefficients zero. Hence the images of $dx_{c+1}, \dots, dx_N$ form a basis of this quotient. Their representatives are elements of A . The forward direction now proves that every sufficiently large power is elementary. This proves (f). $\square$

Why standard smoothness alone is insufficient. Take

$$R = \mathbf{Z}, \quad A = \mathbf{Z}[t^2, t^3] \subset \mathbf{Z}[t], \quad a = t^2.$$

This algebra has the finite presentation

$$A \cong \mathbf{Z}[X, Y]/(Y^2 - X^3), \quad X \mapsto t^2, \quad Y \mapsto t^3.$$

Indeed division by the monic polynomial $Y^2 - X^3$ writes every class uniquely as $p(X) + Yq(X)$. After substitution, the first summand has even exponents and the second has odd exponents at least 3; their disjoint monomials prove injectivity. The image is precisely the indicated subalgebra.

Inverting a gives

$$B = A_a = \mathbf{Z}[t, t^{-1}], \quad t = Y/X, \quad t^{-1} = Y/X^2.$$

These identities follow from $Y^2 = X^3$ and give both inclusions with the Laurent ring. The presentation $B = \mathbf{Z}[T, U]/(TU - 1)$ is standard smooth: select U for its one equation, whose derivative is the unit T . Every derivation of B is uniquely determined by its value on t , with $d(t^{-1}) = -t^{-2}dt$. Thus $\Omega_{B/\mathbf{Z}} = B dt$.

The units of $\mathbf{Z}[t, t^{-1}]$ are exactly $\pm t^k$. To check this, the lowest and highest nonzero exponents of a product add, since their coefficients are nonzero products in the domain $\mathbf{Z}$. For a product equal to 1, each factor must therefore have width zero and be a monomial; its integer coefficient must be 1 or -1 .

Every $b \in A$ has the form $b_0 + \sum_{k \geq 2} b_k t^k$, with integer coefficients and a finite sum. Consequently

$$db = \left(\sum_{k \geq 2} k b_k t^{k-1} \right) dt.$$

This coefficient cannot be a unit: if it were $\pm t^j$, necessarily $j \geq 1$, and its sole coefficient would require $(j+1)b_{j+1} = \pm 1$, impossible in $\mathbf{Z}$. A basis of the free rank-one module $B dt$ must contain one generator; tensoring with $\mathbf{Q}(t)$ verifies that its length is one. Hence it has no basis consisting of

differentials of elements of A . By the converse in (f), no positive power of a is elementary standard, despite standard smoothness of B .

There is no conflict with (e). In the original cusp presentation the two one-by-one minors are $-3X^2$ and $2Y$, and in A

$$X^3 = (-X)(-3X^2) + (-Y)(2Y).$$

The relation condition is empty because there is only one equation. Thus a^3 , and every a^e with $e \geq 3$ after multiplication by a^{e-3} , is strictly standard. This exact example distinguishes the sum-of-minors condition from the single-minor condition over the same arbitrary base.

The lifting problem

The goal in this section is to prove (Proposition D.18) that the collection of algebras which are filtered colimits of smooth algebras is closed under infinitesimal flat deformations. The proof is elementary and only uses the results on presentations of smooth algebras from Section § presentations.

Lemma D.16: lift once

Adapted from the Stacks project, smoothing.tex, lines 1237–1305.

Let $R \rightarrow \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that

1. $I^2 = 0$, and
2. $\Lambda/I\Lambda$ is a filtered colimit of smooth R/I -algebras.

Let $\varphi : A \rightarrow \Lambda$ be an R -algebra map with A of finite presentation over R . Then there exists a factorization

$$A \rightarrow B/J \rightarrow \Lambda$$

where B is a smooth R -algebra and $J \subset IB$ is a finitely generated ideal.

Proof. Choose a factorization

$$A/IA \rightarrow \bar{B} \rightarrow \Lambda/I\Lambda$$

with $\bar{B}$ standard smooth over R/I ; this is possible by assumption and Lemma D.13. Write

$$\bar{B} = A/IA[t_1, \dots, t_r]/(\bar{g}_1, \dots, \bar{g}_s)$$

and say $\bar{B} \rightarrow \Lambda/I\Lambda$ maps t_i to the class of λ_i modulo $I\Lambda$. Choose $g_1, \dots, g_s \in A[t_1, \dots, t_r]$ lifting $\bar{g}_1, \dots, \bar{g}_s$. Write $\varphi(g_i)(\lambda_1, \dots, \lambda_r) = \sum \epsilon_{ij} \mu_{ij}$ for some $\epsilon_{ij} \in I$ and $\mu_{ij} \in \Lambda$. Define

$$A' = A[t_1, \dots, t_r, \delta_{i,j}]/(g_i - \sum \epsilon_{ij} \delta_{ij})$$

and consider the map

$$A' \longrightarrow \Lambda, \quad a \longmapsto \varphi(a), \quad t_i \longmapsto \lambda_i, \quad \delta_{ij} \longmapsto \mu_{ij}$$

We have

$$A'/IA' = A/IA[t_1, \dots, t_r]/(\bar{g}_1, \dots, \bar{g}_s)[\delta_{ij}] \cong \bar{B}[\delta_{ij}]$$

This is a standard smooth algebra over R/I as $\bar{B}$ is standard smooth. Choose a presentation $A'/IA' = R/I[x_1, \dots, x_n]/(\bar{f}_1, \dots, \bar{f}_c)$ with $\det(\partial \bar{f}_j / \partial x_i)_{i,j=1, \dots, c}$ invertible in A'/IA' . Choose lifts $f_1, \dots, f_c \in R[x_1, \dots, x_n]$ of $\bar{f}_1, \dots, \bar{f}_c$. Then

$$B = R[x_1, \dots, x_n, x_{n+1}]/(f_1, \dots, f_c, x_{n+1} \det(\partial f_j / \partial x_i)_{i,j=1, \dots, c} - 1)$$

is smooth over R . Since smooth ring maps are formally smooth (Algebra, Proposition [AG-CA-17](#), Theorem 3.1 and the split conormal definition of smoothness in this draft) there exists an R -algebra map $B \rightarrow A'$ which is an isomorphism modulo I . Then $B \rightarrow A'$ is surjective by Nakayama's lemma (Algebra, Lemma [F.10](#)). Thus $A' = B/J$ with $J \subset IB$ finitely generated (see Algebra, Lemma [F.44](#)). $\square$

Lemma D.17: lift twice

Adapted from the Stacks project, smoothing.tex, lines 1307–1371.

Let $R \rightarrow \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that

1. $I^2 = 0$,
2. Λ/IA is a filtered colimit of smooth R/I -algebras, and
3. $R \rightarrow \Lambda$ is flat.

Let $\varphi : B \rightarrow \Lambda$ be an R -algebra map with B smooth over R . Let $J \subset IB$ be a finitely generated ideal such that $\varphi(J) = 0$. Then there exists R -algebra maps

$$B \xrightarrow{\alpha} B' \xrightarrow{\beta} \Lambda$$

such that B' is smooth over R , such that $\alpha(J) = 0$ and such that $\beta \circ \alpha = \varphi$.

Proof. If we can prove the lemma in case $J = (h)$, then we can prove the lemma by induction on the number of generators of J . Namely, suppose that J can be generated by n elements $h_1, \dots, h_n$ and the lemma holds for all cases where J is generated by $n - 1$ elements. Then we apply the case $n = 1$ to produce $B \rightarrow B' \rightarrow \Lambda$ where the first map kills h_n . Then we let J' be the ideal of B' generated by the images of $h_1, \dots, h_{n-1}$ and we apply the case for $n - 1$ to produce $B' \rightarrow B'' \rightarrow \Lambda$. It is easy to verify that $B \rightarrow B'' \rightarrow \Lambda$ does the job.

Assume $J = (h)$ and write $h = \sum \epsilon_i b_i$ for some $\epsilon_i \in I$ and $b_i \in B$. Note that $0 = \varphi(h) = \sum \epsilon_i \varphi(b_i)$. As Λ is flat over R , the equational criterion for flatness (Algebra, Lemma [AG-CA-07](#), Theorem 5.1) implies that we can find $\lambda_j \in \Lambda$, $j = 1, \dots, m$ and $a_{ij} \in R$ such that $\varphi(b_i) = \sum_j a_{ij} \lambda_j$ and $\sum_i \epsilon_i a_{ij} = 0$. Set

$$C = B[x_1, \dots, x_m]/(b_i - \sum a_{ij} x_j)$$

with $C \rightarrow \Lambda$ given by φ and $x_j \mapsto \lambda_j$. Choose a factorization

$$C \rightarrow B'/J' \rightarrow \Lambda$$

as in Lemma [D.16](#). Since B is smooth over R we can lift the map $B \rightarrow C \rightarrow B'/J'$ to a map $\alpha : B \rightarrow B'$. Then $\varphi = \beta \circ \alpha$. To finish the proof we check that $\alpha(h) = 0$. Namely, the fact that α lifts $B \rightarrow C \rightarrow B'/J'$ implies that

$$\alpha(b_i) = \sum a_{ij} \xi_j + \theta_i$$

for some $\xi_j \in B'$ and $\theta_i \in J' \subset IB'$. Hence we see that

$$\alpha(h) = \alpha(\sum \epsilon_i b_i) = \sum \epsilon_i a_{ij} \xi_j + \sum \epsilon_i \theta_i = 0$$

because of the relations above and the fact that $I^2 = 0$. $\square$

Proposition D.18: lift

Adapted from the Stacks project, smoothing.tex, lines 1373–1407.

Let $R \rightarrow \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that

1. I is nilpotent,
2. $\Lambda/I\Lambda$ is a filtered colimit of smooth R/I -algebras, and
3. $R \rightarrow \Lambda$ is flat.

Then Λ is a filtered colimit of smooth R -algebras.

Proof. First prove the assertion when $I^2 = 0$. For a finitely presented R -algebra A and an R -algebra map $A \rightarrow \Lambda$, D.16 gives a factorization $A \rightarrow B/J \rightarrow \Lambda$, where B is smooth over R and $J \subset IB$ is finitely generated. The induced map $B \rightarrow \Lambda$ kills J . All hypotheses of D.17 hold: the same ideal has square zero, the quotient target is the assumed smooth colimit, and Λ is flat over R . That lemma gives $B \rightarrow B' \rightarrow \Lambda$ with B' smooth and J killed in B' . Consequently $A \rightarrow B/J \rightarrow B' \rightarrow \Lambda$ is the required smooth factorization. Criterion F.90 proves the square-zero assertion.

Now induct on an integer $n \geq 1$ such that $I^n = 0$. If $n = 1$, then $I = 0$ and the conclusion is the hypothesis. For $n \geq 2$, put $J = I^{n-1}$. We have $J^2 = 0$, since $2n - 2 \geq n$. Set $\bar{R} = R/J$, $\bar{\Lambda} = \Lambda/J\Lambda$ and $\bar{I} = I/J$. Then $\bar{I}^{n-1} = 0$ and

$$\bar{\Lambda}/\bar{I}\bar{\Lambda} \cong \Lambda/I\Lambda.$$

The map $\bar{R} \rightarrow \bar{\Lambda}$ is flat by base change: for a $\bar{R}$ -module M , $M \otimes_{\bar{R}} \bar{\Lambda} \cong M \otimes_R \Lambda$, so its tensor functor preserves exact sequences. The induction hypothesis therefore makes $\bar{\Lambda}$ a filtered colimit of smooth $\bar{R}$ -algebras. Apply the already proved square-zero case to $J \subset R$ and the original flat map $R \rightarrow \Lambda$. It gives the required smooth colimit over R , completing the induction. $\square$

The lifting lemma

The next lemma constructs a finite presentation lifting the smooth locus.

Lemma D.19: lifting

Adapted from the Stacks project, smoothing.tex, lines 1420–1651.

Let R be a Noetherian ring. Let Λ be an R -algebra. Let $\pi \in R$ and assume that $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$ and $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$. Suppose we have R -algebra maps $R/\pi^2 R \rightarrow \bar{C} \rightarrow \Lambda/\pi^2 \Lambda$ with $\bar{C}$ of finite presentation. Then there exists an R -algebra homomorphism $D \rightarrow \Lambda$ and a commutative diagram

$$\begin{aligned}
R/\pi^2 R &\rightarrow \bar{C} \\
R/\pi^2 R &\rightarrow R/\pi R \\
\bar{C} &\rightarrow \Lambda/\pi^2 \Lambda \\
\bar{C} &\rightarrow D/\pi D \\
\Lambda/\pi^2 \Lambda &\rightarrow \Lambda/\pi \Lambda \\
R/\pi R &\rightarrow D/\pi D \\
D/\pi D &\rightarrow \Lambda/\pi \Lambda
\end{aligned}$$

with the following properties

1. **(a)** D is of finite presentation,
2. **(b)** $R \rightarrow D$ is smooth at any prime $\mathfrak{q}$ with $\pi \notin \mathfrak{q}$,
3. **(c)** $R \rightarrow D$ is smooth at any prime $\mathfrak{q}$ with $\pi \in \mathfrak{q}$ lying over a prime of $\bar{C}$ where $R/\pi^2 R \rightarrow \bar{C}$ is smooth, and
4. **(d)** $\bar{C}/\pi \bar{C} \rightarrow D/\pi D$ is smooth at any prime lying over a prime of $\bar{C}$ where $R/\pi^2 R \rightarrow \bar{C}$ is smooth.

Proof. We choose a presentation

$$\bar{C} = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

We also denote $I = (f_1, \dots, f_m)$ and $\bar{I}$ the image of I in $R/\pi^2 R[x_1, \dots, x_n]$. The finite presentation makes $\bar{C}$ Noetherian by AG-CA-03, Theorem 2.1, and its spectrum is Noetherian by Proposition 2.2 of that lesson. The smooth locus is open. Its principal smooth charts have a finite subcover: among their finite unions choose a maximal one using the ascending-chain condition on open sets; adjoining any other chart cannot enlarge it, so that union already covers the entire locus. This also includes the empty smooth locus. Applying Lemma D.2 we may choose a finite list of elements $a_1, \dots, a_r \in R[x_1, \dots, x_n]$ such that

1. the union of the open subspaces $\text{Spec}(\bar{C}_{a_k}) \subset \text{Spec}(\bar{C})$ cover the smooth locus of $R/\pi^2 R \rightarrow \bar{C}$, and
2. for each $k = 1, \dots, r$ there exists a finite subset $E_k \subset \{1, \dots, m\}$ such that $(\bar{I}/\bar{I}^2)_{a_k}$ is freely generated by the classes of f_j , $j \in E_k$.

Set $I_k = (f_j, j \in E_k) \subset I$ and denote $\bar{I}_k$ the image of I_k in $R/\pi^2 R[x_1, \dots, x_n]$. By (2) and Nakayama's lemma we see that $(\bar{I}/\bar{I}_k)_{a_k}$ is annihilated by $1 + b'_k$ for some $b'_k \in \bar{I}_{a_k}$. Suppose b'_k is the image of $b_k/(a_k)^N$ for some $b_k \in I$ and some integer N . After replacing a_k by $a_k((a_k)^N + b_k)$ we get

1. **(3)** $(\bar{I}_k)_{a_k} = (\bar{I})_{a_k}$.

Thus, after possibly replacing a_k by a high power, we may write

1. **(4)** $a_k f_\ell = \sum_{j \in E_k} h_{k,\ell}^j f_j + \pi^2 g_{k,\ell}$

for any $\ell \in \{1, \dots, m\}$ and some $h_{k,\ell}^j, g_{k,\ell} \in R[x_1, \dots, x_n]$. If $\ell \in E_k$ we choose $h_{k,\ell}^j = a_k \delta_{\ell,j}$ (Kronecker delta) and $g_{k,\ell} = 0$. Set

$$D = R[x_1, \dots, x_n, z_1, \dots, z_m] / (f_j - \pi z_j, p_{k,\ell}).$$

Here $j \in \{1, \dots, m\}$, $k \in \{1, \dots, r\}$, $\ell \in \{1, \dots, m\}$, and

$$p_{k,\ell} = a_k z_\ell - \sum_{j \in E_k} h_{k,\ell}^j z_j - \pi g_{k,\ell}.$$

Note that for $\ell \in E_k$ we have $p_{k,\ell} = 0$ by our choices above.

The map $R \rightarrow D$ is the given one. Say $\bar{C} \rightarrow \Lambda/\pi^2\Lambda$ maps x_i to the class of λ_i modulo π^2 . For an element $f \in R[x_1, \dots, x_n]$ we denote $f(\lambda) \in \Lambda$ the result of substituting λ_i for x_i . Then we know that $f_j(\lambda) = \pi^2 \mu_j$ for some $\mu_j \in \Lambda$. Define $D \rightarrow \Lambda$ by the rules $x_i \mapsto \lambda_i$ and $z_j \mapsto \pi \mu_j$. This is well defined because

$$\begin{aligned} p_{k,\ell} &\mapsto a_k(\lambda) \pi \mu_\ell - \sum_{j \in E_k} h_{k,\ell}^j(\lambda) \pi \mu_j - \pi g_{k,\ell}(\lambda) \\ &= \pi \left(a_k(\lambda) \mu_\ell - \sum_{j \in E_k} h_{k,\ell}^j(\lambda) \mu_j - g_{k,\ell}(\lambda) \right) \end{aligned}$$

Substituting $x_i = \lambda_i$ in (4) above we see that the expression inside the brackets is annihilated by π^2 , hence it is annihilated by π as we have assumed $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$. The map $\bar{C} \rightarrow D/\pi D$ is determined by $x_i \mapsto \lambda_i$ (clearly well defined). Thus we are done if we can prove (b), (c), and (d).

Using (4) we obtain the following key equality

$$\begin{aligned} \pi p_{k,\ell} &= \pi a_k z_\ell - \sum_{j \in E_k} \pi h_{k,\ell}^j z_j - \pi^2 g_{k,\ell} \\ &= -a_k(f_\ell - \pi z_\ell) + a_k f_\ell + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j) - \sum_{j \in E_k} h_{k,\ell}^j f_j - \pi^2 g_{k,\ell} \\ &= -a_k(f_\ell - \pi z_\ell) + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j) \end{aligned}$$

The end result is an element of the ideal generated by $f_j - \pi z_j$. In particular, we see that $D[1/\pi]$ is isomorphic to $R[1/\pi][x_1, \dots, x_n, z_1, \dots, z_m] / (f_j - \pi z_j)$ which is isomorphic to $R[1/\pi][x_1, \dots, x_n]$ hence smooth over R . This proves (b).

For fixed $k \in \{1, \dots, r\}$ consider the ring

$$D_k = R[x_1, \dots, x_n, z_1, \dots, z_m] / (f_j - \pi z_j, j \in E_k, p_{k,\ell})$$

The number of equations is $m = |E_k| + (m - |E_k|)$ as $p_{k,\ell}$ is zero if $\ell \in E_k$. Also, note that

$$\begin{aligned} (D_k/\pi D_k)_{a_k} &= R/\pi R[x_1, \dots, x_n, 1/a_k, z_1, \dots, z_m] / (f_j, j \in E_k, p_{k,\ell}) \\ &= (\bar{C}/\pi \bar{C})_{a_k}[z_1, \dots, z_m] / (a_k z_\ell - \sum_{j \in E_k} h_{k,\ell}^j z_j) \\ &\cong (\bar{C}/\pi \bar{C})_{a_k}[z_j, j \in E_k] \end{aligned}$$

In particular $(D_k/\pi D_k)_{a_k}$ is smooth over $(\bar{C}/\pi \bar{C})_{a_k}$. By our choice of a_k we have that $(\bar{C}/\pi \bar{C})_{a_k}$ is smooth over $R/\pi R$ of relative dimension $n - |E_k|$, see (2). Hence for a prime $\mathfrak{q}_k \subset D_k$ containing π and lying over $\text{Spec}(\bar{C}_{a_k})$ the fibre ring of $R \rightarrow D_k$ is smooth at $\mathfrak{q}_k$ of dimension n . Thus $R \rightarrow D_k$ is syntomic at $\mathfrak{q}_k$ by our count of the number of equations above, see Algebra, Lemma F.66. Hence $R \rightarrow D_k$ is smooth at $\mathfrak{q}_k$, see Algebra, Lemma F.45.

To finish the proof, let $\mathfrak{q} \subset D$ be a prime containing π lying over a prime where $R/\pi^2 R \rightarrow \bar{C}$ is smooth. Then $a_k \notin \mathfrak{q}$ for some k by (1). We will show that the surjection $D_k \rightarrow D$ induces an

isomorphism on local rings at $\mathfrak{q}$. Since we know that the ring maps $\bar{C}/\pi\bar{C} \rightarrow D_k/\pi D_k$ and $R \rightarrow D_k$ are smooth at the corresponding prime $\mathfrak{q}_k$ by the preceding paragraph this will prove (c) and (d) and thus finish the proof.

First, note that for any ℓ the equation $\pi p_{k,\ell} = -a_k(f_\ell - \pi z_\ell) + \sum_{j \in E_k} h_{k,\ell}^j(f_j - \pi z_j)$ proved above shows that $f_\ell - \pi z_\ell$ maps to zero in $(D_k)_{a_k}$ and in particular in $(D_k)_{\mathfrak{q}_k}$. Reducing relations (4) modulo π^2 kills their $\pi^2 g_{k,\ell}$ term. They therefore give $a_k \bar{f}_\ell = \sum_{j \in E_k} \bar{h}_{k,\ell}^j \bar{f}_j$ in $\bar{I}/\bar{I}^2$; this assertion is made in the reduced conormal module, not in I/I^2 . Since $(\bar{I}_k/\bar{I}_k^2)_{a_k}$ is free on f_j , $j \in E_k$ we see that

$$a_{k'} h_{k,\ell}^j - \sum_{j' \in E_{k'}} h_{k',\ell}^{j'} h_{k,j'}^j$$

is zero in $\bar{C}_{a_k}$ for every k, k', ℓ and $j \in E_k$. Hence we can find a large integer N such that

$$a_k^N \left(a_{k'} h_{k,\ell}^j - \sum_{j' \in E_{k'}} h_{k',\ell}^{j'} h_{k,j'}^j \right)$$

is in $I_k + \pi^2 R[x_1, \dots, x_n]$. Computing modulo π we have

$$\begin{aligned} & a_k p_{k',\ell} - a_{k'} p_{k,\ell} + \sum h_{k',\ell}^{j'} p_{k,j'} \\ &= -a_k \sum h_{k',\ell}^{j'} z_{j'} + a_{k'} \sum h_{k,\ell}^j z_j + \sum h_{k',\ell}^{j'} a_k z_{j'} - \sum \sum h_{k',\ell}^{j'} h_{k,j'}^j z_j \\ &= \sum \left(a_{k'} h_{k,\ell}^j - \sum h_{k',\ell}^{j'} h_{k,j'}^j \right) z_j \end{aligned}$$

with Einstein summation convention. Combining with the above we see $a_k^{N+1} p_{k',\ell}$ is contained in the ideal generated by I_k and π in $R[x_1, \dots, x_n, z_1, \dots, z_m]$. Thus $p_{k',\ell}$ maps into $\pi(D_k)_{a_k}$. On the other hand, the equation

$$\pi p_{k',\ell} = -a_{k'}(f_\ell - \pi z_\ell) + \sum_{j' \in E_{k'}} h_{k',\ell}^{j'}(f_{j'} - \pi z_{j'})$$

shows that $\pi p_{k',\ell}$ is zero in $(D_k)_{a_k}$. Put $T = (D_k)_{\mathfrak{q}_k}$, already proved smooth and hence flat over R . For $r = 1, 2$, tensor the exact kernel sequence

$$0 \longrightarrow \text{Ann}_R(\pi^r) \longrightarrow R \xrightarrow{\pi^r} R$$

with T . Exactness identifies its left image with $\text{Ann}_T(\pi^r)$. The equality of the two original annihilator submodules therefore gives $\text{Ann}_T(\pi) = \text{Ann}_T(\pi^2)$.

We have proved that the image p of each $p_{k',\ell}$ lies in πT and satisfies $\pi p = 0$. Write $p = \pi u$. Then $\pi^2 u = 0$, so the annihilator equality gives $\pi u = 0$, which is $p = 0$. The relations $f_\ell - \pi z_\ell$ already vanish in T , as proved above. These and the $p_{k',\ell}$ are exactly all additional relations of the surjection $D_k \rightarrow D$. Its localized kernel is consequently zero, giving $D_{\mathfrak{q}} \cong T$. The previously proved smoothness of $R \rightarrow T$ and of the corresponding quotient chart proves (c) and (d). $\square$

The desingularization lemma

The next construction improves the singular ideal.

Lemma D.20: desingularize

Adapted from the Stacks project, smoothing.tex, lines 1664–1809.

Let R be a Noetherian ring. Let Λ be an R -algebra. Let $\pi \in R$ and assume that $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$. Let $A \rightarrow \Lambda$ be an R -algebra map with A of finite presentation. Assume

1. the image of π is strictly standard in A over R , and
2. there exists a section $\rho : A/\pi^4 A \rightarrow R/\pi^4 R$ which is compatible with the map to $\Lambda/\pi^4 \Lambda$.

Then we can find R -algebra maps $A \rightarrow B \rightarrow \Lambda$ with B of finite presentation such that $\mathfrak{a}B \subset H_{B/R}$ where $\mathfrak{a} = \text{Ann}_R(\text{Ann}_R(\pi^2)/\text{Ann}_R(\pi))$.

Proof. Choose a presentation

$$A = R[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

and $0 \leq c \leq \min(n, m)$ such that (D.3s1: strict Jacobian-minor condition) holds for π and such that

$$\pi f_{c+j} \in (f_1, \dots, f_c) + (f_1, \dots, f_m)^2$$

for $j = 1, \dots, m - c$. Say ρ maps x_i to the class of $r_i \in R$. Then we can replace x_i by $x_i - r_i$. Hence we may assume $\rho(x_i) = 0$ in $R/\pi^4 R$. This implies that $f_j(0) \in \pi^4 R$ and that $A \rightarrow \Lambda$ maps x_i to $\pi^4 \lambda_i$ for some $\lambda_i \in \Lambda$. Write

$$f_j = f_j(0) + \sum_{i=1, \dots, n} r_{ji} x_i + \text{h.o.t.}$$

This implies that the constant term of $\partial f_j / \partial x_i$ is r_{ji} . Apply ρ to (D.3s1: strict Jacobian-minor condition) for π and we see that

$$\pi = \sum_{I \subset \{1, \dots, n\}, |I|=c} r_I \det(r_{ji})_{j=1, \dots, c, i \in I} \text{ mod } \pi^4 R$$

for some $r_I \in R$. Thus we have

$$u\pi = \sum_{I \subset \{1, \dots, n\}, |I|=c} r_I \det(r_{ji})_{j=1, \dots, c, i \in I}$$

for some $u \in 1 + \pi^3 R$. By Algebra, Lemma F.70 this implies there exists a $n \times c$ matrix (s_{ik}) such that

$$u\pi \delta_{jk} = \sum_{i=1, \dots, n} r_{ji} s_{ik} \quad \text{for all } j, k = 1, \dots, c$$

(Kronecker delta). We introduce auxiliary variables $v_1, \dots, v_c, w_1, \dots, w_n$ and we set

$$h_i = x_i - \pi^2 \sum_{j=1, \dots, c} s_{ij} v_j - \pi^3 w_i$$

In the following we will use that

$$R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(h_1, \dots, h_n) = R[v_1, \dots, v_c, w_1, \dots, w_n]$$

without further mention. In $R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(h_1, \dots, h_n)$ we have

$$\begin{aligned} f_j &= f_j(x_1 - h_1, \dots, x_n - h_n) \\ &= \pi^2 \sum_{k=1}^c \left(\sum_{i=1}^n r_{ji} s_{ik} \right) v_k + \pi^3 \sum_{i=1}^n r_{ji} w_i \text{ mod } \pi^4 \\ &= \pi^3 v_j + \pi^3 \sum_{i=1}^n r_{ji} w_i \text{ mod } \pi^4 \end{aligned}$$

for $1 \leq j \leq c$. Hence we can choose elements $g_j \in R[v_1, \dots, v_c, w_1, \dots, w_n]$ such that $g_j = v_j + \sum r_{ji} w_i \bmod \pi$ and such that $f_j = \pi^3 g_j$ in $Q/(h_1, \dots, h_n)$, where

$$Q = R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n].$$

We set

$$B = R[x_1, \dots, x_n, v_1, \dots, v_c, w_1, \dots, w_n]/(f_1, \dots, f_m, h_1, \dots, h_n, g_1, \dots, g_c).$$

The map $A \rightarrow B$ is clear. We define $B \rightarrow \Lambda$ by mapping $x_i \rightarrow \pi^4 \lambda_i$, $v_i \mapsto 0$, and $w_i \mapsto \pi \lambda_i$. Then it is clear that the elements f_j and h_i are mapped to zero in Λ . Moreover, it is clear that g_i is mapped to an element t of $\pi\Lambda$ such that $\pi^3 t = 0$ (as $f_i = \pi^3 g_i$ modulo the ideal generated by the h 's). The assumed annihilator equality implies equality with every positive power: if $\pi^r u = 0$ with $r \geq 2$, apply the equality to $\pi^{r-2} u$ to get $\pi^{r-1} u = 0$, and repeat. Here $t = \pi u$ and $\pi^3 t = 0$, so $\pi^4 u = 0$ and the same induction gives $\pi u = 0$. Thus $t = 0$. Thus we are done if we can prove the statement about smoothness.

Note that $B_\pi \cong A_\pi[v_1, \dots, v_c]$ because the equations $g_i = 0$ are implied by $f_i = 0$. Hence B_π is smooth over R as A_π is smooth over R by the assumption that π is strictly standard in A over R , see Lemma D.5.

Set $B' = R[v_1, \dots, v_c, w_1, \dots, w_n]/(g_1, \dots, g_c)$. As $g_i = v_i + \sum r_{ji} w_i \bmod \pi$ we see that $B'/\pi B' = R/\pi R[w_1, \dots, w_n]$. Hence $R \rightarrow B'$ is smooth of relative dimension n at every point of $V(\pi)$ by Algebra, Lemmas F.66 and F.45 (the first lemma shows it is syntomic at those primes, in particular flat, whereupon the second lemma shows it is smooth).

Let $\mathfrak{q} \subset B$ contain π and avoid an element $r \in \mathfrak{a}$. Put $\mathfrak{q}' = B' \cap \mathfrak{q}$ and $T = (B')_{\mathfrak{q}'}$. The preceding paragraph proves that T is smooth and therefore flat over R . The surjection $T \rightarrow B_{\mathfrak{q}}$ has kernel generated by the images F_j of f_{c+j} : the h_i have already eliminated the x_i , and the first c equations $f_i = \pi^3 g_i$ are already zero in B' .

Every F_j lies in $\pi^2 T$, because the substituted x_i are divisible by π^2 and the constant terms of the f_i lie in $\pi^4 R$. The strict relation condition D.3s2, after the first c equations vanish, gives

$$\pi F_j = \sum_{a,b} b_{j,a,b} F_a F_b.$$

For the finite ideal $M = (\pi F_j)_j \subset T$, this implies $M \subset \pi M$, since each first factor F_a is divisible by π^2 . The reverse inclusion is automatic, so $M = \pi M$. The element π is in the maximal ideal of the Noetherian local ring T . Nakayama's proved finite-module form F.10 gives $M = 0$. Thus $\pi F_j = 0$ for every j .

Flatness identifies the two multiplication kernels and their quotient:

$$(\text{Ann}_R(\pi^2)/\text{Ann}_R(\pi)) \otimes_R T \cong \text{Ann}_T(\pi^2)/\text{Ann}_T(\pi).$$

Indeed, tensor the exact sequences $0 \rightarrow \text{Ann}_R(\pi^a) \rightarrow R \xrightarrow{\pi^a} R$ for $a = 1, 2$ and then their submodule quotient sequence. The element r kills the original quotient and is invertible in T , so this quotient is zero. Write $F_j = \pi u_j$. Since $\pi F_j = 0$, we have $\pi^2 u_j = 0$, and the annihilator equality gives $\pi u_j = 0$, or $F_j = 0$. The entire localized kernel vanishes; hence $B_{\mathfrak{q}} \cong T$ is smooth over R .

Primes avoiding π are already smooth by the displayed description of B_π . At every remaining prime avoiding $\mathfrak{a}B$, the preceding argument applies to some $r \in \mathfrak{a}$ outside that prime. Thus every nonsmooth prime contains $\mathfrak{a}B$. Definition D.1 and the radical-prime criterion give $\mathfrak{a}B \subset H_{B/R}$. Explicitly, an element outside a radical ideal has all its powers outside that ideal. An ideal maximal among those containing the ideal and disjoint from those powers is prime: if two elements outside it had product inside it, the product of the two enlarged ideals would contain a power and still lie in it, a contradiction. Consequently that element is avoided by a prime containing the radical ideal. Applying this criterion proves the asserted inclusion. $\square$

Lemma D.21: desingularize strictly standard

Adapted from the Stacks project, smoothing.tex, lines 1811–1859.

Let R be a Noetherian ring. Let Λ be an R -algebra. Let $\pi \in R$ and assume that $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$ and $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$. Let $A \rightarrow \Lambda$ and $D \rightarrow \Lambda$ be R -algebra maps with A and D of finite presentation. Assume

1. π is strictly standard in A over R , and
2. there exists an R -algebra map $A/\pi^4 A \rightarrow D/\pi^4 D$ compatible with the maps to $\Lambda/\pi^4 \Lambda$.

Then we can find an R -algebra map $B \rightarrow \Lambda$ with B of finite presentation and R -algebra maps $A \rightarrow B$ and $D \rightarrow B$ compatible with the maps to Λ such that $H_{D/R}B \subset H_{B/D}$ and $H_{D/R}B \subset H_{B/R}$.

Proof. The ring D is Noetherian: it is a finitely presented algebra over the Noetherian ring R , by the Hilbert-basis and quotient proof in AG-CA-03, Theorem 2.1. Put $C = A \otimes_R D$. It is finitely presented over D , and multiplication of the two given images defines $C \rightarrow \Lambda$. Base change D.7 makes π strictly standard in C over D .

The given map $\phi: A/\pi^4 A \rightarrow D/\pi^4 D$ defines a section

$$\rho: C/\pi^4 C \cong (A/\pi^4 A) \otimes_{R/\pi^4 R} (D/\pi^4 D) \longrightarrow D/\pi^4 D, \quad a \otimes d \longmapsto \phi(a)d.$$

It is the identity on the $D/\pi^4 D$ factor. Compatibility with $\Lambda/\pi^4 \Lambda$ follows from the compatibility assumed for ϕ . Apply D.20 over the base D . This lemma requires the annihilator equality in Λ , which is a hypothesis here; it does not require such an equality in D . We obtain $C \rightarrow B \rightarrow \Lambda$, with B finitely presented over D , and

$$\mathfrak{a}B \subset H_{B/D}, \quad M = \text{Ann}_D(\pi^2)/\text{Ann}_D(\pi), \quad \mathfrak{a} = \text{Ann}_D(M).$$

Finite presentation over R follows by substituting presentations: if $D = R[u]/(g)$ and $B = D[v]/(h)$ with all lists finite, lift the h to polynomials $\tilde{h}$ in $R[u, v]$ to get $B = R[u, v]/(g, \tilde{h})$. The two tensor-factor maps give the required compatible maps $A \rightarrow B$ and $D \rightarrow B$.

Let $\mathfrak{q} \subset D$ be a prime where $R \rightarrow D$ is smooth. Then $D_{\mathfrak{q}}$ is flat over R . Tensoring the two exact multiplication-kernel sequences, as in the final paragraph of D.19, gives equal annihilators of π and π^2 in $D_{\mathfrak{q}}$. Localization is exact, so $M_{\mathfrak{q}} = 0$.

Here is the needed global annihilator calculation. Both annihilator ideals in the Noetherian ring D are finite modules, hence M has finite generators $m_1, \dots, m_v$. Since $m_i/1 = 0$ in $M_{\mathfrak{q}}$, there is $s_i \in D \setminus \mathfrak{q}$ with $s_i m_i = 0$. The product $s = \prod_i s_i$ kills every generator, so $s \in \mathfrak{a} \setminus \mathfrak{q}$. If $M = 0$, take $s = 1$. Thus for every prime $\mathfrak{r} \subset B$ above $\mathfrak{q}$, the inclusion $\mathfrak{a}B \subset H_{B/D}$ gives an element of $H_{B/D}$ outside $\mathfrak{r}$. By definition D.1, $D \rightarrow B$ is smooth there.

We have proved $D(H_{D/R}B) \subset D(H_{B/D})$. Therefore $H_{D/R}B \subset H_{B/D}$, since the latter ideal is radical. Indeed, if an element lies outside a radical ideal, a prime containing that ideal and avoiding the element is obtained by taking an ideal maximal among those disjoint from its powers; the maximal ideal is prime because the product of two enlarged ideals would otherwise meet those powers. Apply this fact to the two complementary closed sets. At the same primes, both successive maps are smooth, so smooth composition F.29 gives smoothness of $R \rightarrow B$. The same radical-ideal argument gives $H_{D/R}B \subset H_{B/R}$. $\square$

Lemma D.22: desingularize lifting apply

Adapted from the Stacks project, smoothing.tex, lines 1861–1894.

Let R be a Noetherian ring. Let Λ be an R -algebra. Let $\pi \in R$ and assume that $\text{Ann}_R(\pi) = \text{Ann}_R(\pi^2)$ and $\text{Ann}_\Lambda(\pi) = \text{Ann}_\Lambda(\pi^2)$. Let $A \rightarrow \Lambda$ be an R -algebra map with A of finite presentation and assume π is strictly standard in A over R . Let

$$A/\pi^8 A \rightarrow \bar{C} \rightarrow \Lambda/\pi^8 \Lambda$$

be a factorization with $\bar{C}$ of finite presentation. Then we can find a factorization $A \rightarrow B \rightarrow \Lambda$ with B of finite presentation such that $R_\pi \rightarrow B_\pi$ is smooth and such that

$$H_{\bar{C}/(R/\pi^8 R)} \cdot \Lambda/\pi^8 \Lambda \subset \sqrt{H_{B/R} \Lambda} \bmod \pi^8 \Lambda.$$

Proof. First record the power calculation. For multiplication by u on a module, $\ker u = \ker u^2$ implies $\ker u^a = \ker u$ for all $a \geq 1$: if $u^{a+1}v = 0$, then $u^{a-1}v \in \ker u^2 = \ker u$, so $u^a v = 0$, and induction finishes. Apply this with $u = \pi$ on R and on Λ . In particular the annihilators of π^4 and π^8 agree in both rings.

Apply D.19 with parameter π^4 and input $\bar{C}$, a finitely presented $R/\pi^8 R$ -algebra mapping to $\Lambda/\pi^8 \Lambda$. It gives a finitely presented R -algebra $D \rightarrow \Lambda$ and a compatible map

$$\bar{C}/\pi^4 \bar{C} \longrightarrow D/\pi^4 D \longrightarrow \Lambda/\pi^4 \Lambda.$$

Its property (b) makes D smooth at primes avoiding π . Property (c) makes it smooth at a prime containing π whenever that prime's contraction through the displayed quotient map is a smooth point of $\bar{C}$ over $R/\pi^8 R$.

The composite $A/\pi^4 A \rightarrow \bar{C}/\pi^4 \bar{C} \rightarrow D/\pi^4 D$ is compatible with the given map to $\Lambda/\pi^4 \Lambda$. Thus every hypothesis of D.21 holds, with the original strictly-standard parameter π in A . It supplies a finitely presented R -algebra B , compatible factorizations $A \rightarrow B \rightarrow \Lambda$ and $D \rightarrow B \rightarrow \Lambda$, and

$$H_{D/R} B \subset H_{B/R}.$$

At a prime of B avoiding π , its contraction to D is smooth, so it avoids $H_{D/R}$. The inclusion shows it also avoids $H_{B/R}$. Therefore every point of $B[1/\pi]$ is smooth, and F.67 proves the required global smoothness over $R[1/\pi]$.

Put $J = H_{B/R} \Lambda$ and let $\mathfrak{l}$ be any prime of Λ containing J . Its contractions $\mathfrak{b} \subset B$ and $\mathfrak{d} \subset D$ contain the respective singular ideals. The preceding smoothness forces $\pi \in \mathfrak{l}$. Consequently $\mathfrak{d}$ defines a prime of $D/\pi^4 D$ and a contraction $\mathfrak{c}$ to $\bar{C}/\pi^4 \bar{C}$. If the corresponding prime of $\bar{C}$ were smooth over $R/\pi^8 R$, D.19(c) would make D smooth at $\mathfrak{d}$, a contradiction. Thus $H_{\bar{C}/(R/\pi^8 R)}$ lies in that contracted prime. Compatibility of all maps makes its image in $\Lambda/\pi^8 \Lambda$ lie in $\mathfrak{l}/\pi^8 \Lambda$.

Every prime containing J contains π , and the prime-intersection proof in D.5 gives $\pi \in \sqrt{J}$. It also gives the asserted inclusion after intersecting the primes $\mathfrak{l}$: the ideal extended from $H_{\bar{C}/(R/\pi^8 R)}$ is contained in the image of $\sqrt{J}$ modulo $\pi^8 \Lambda$. If there are no such primes, $\sqrt{J} = \Lambda$ and this assertion holds as well. This proves both conclusions at the original arbitrary-target scope. $\square$

Warmup: reduction to a base field

In this section we apply the lemmas in the previous sections to prove that it suffices to prove the main result when the base ring is a field, see Lemma D.26.

Situation D.23: global

Adapted from the Stacks project, smoothing.tex, lines 1915–1918.

Here $R \rightarrow \Lambda$ is a regular ring map of Noetherian rings.

Let $R \rightarrow \Lambda$ be as in Situation D.23. We say *PT holds for $R \rightarrow \Lambda$* if Λ is a filtered colimit of smooth R -algebras.

Lemma D.24: product

Adapted from the Stacks project, smoothing.tex, lines 1925–1934.

Let $R_i \rightarrow \Lambda_i$, $i = 1, 2$ be as in Situation D.23. If PT holds for $R_i \rightarrow \Lambda_i$, $i = 1, 2$, then PT holds for $R_1 \times R_2 \rightarrow \Lambda_1 \times \Lambda_2$.

Proof. Write $\Lambda_i = \operatorname{colim}_{a \in I_i} B_{i,a}$ with I_i filtered and every $B_{i,a}$ smooth over R_i . The category $I_1 \times I_2$ is filtered: nonemptiness, a common successor for two objects, and an equalizer for two parallel arrows are constructed in the two coordinates. The $R_1 \times R_2$ -algebra $B_{1,a} \times B_{2,b}$ is smooth, because its spectrum is the disjoint union of the two corresponding smooth spectra. The natural map

$$\operatorname{colim}_{(a,b) \in I_1 \times I_2} (B_{1,a} \times B_{2,b}) \longrightarrow \Lambda_1 \times \Lambda_2$$

is surjective by choosing representatives of the two components, and injective because any equality of pairs becomes an equality at successors in each coordinate. It is a ring isomorphism, proving the assertion. $\square$

Lemma D.25: delocalize base

Adapted from the Stacks project, smoothing.tex, lines 1936–2000.

Let $R \rightarrow A \rightarrow \Lambda$ be ring maps with A of finite presentation over R . Let $S \subset R$ be a multiplicative set. Let $S^{-1}A \rightarrow B' \rightarrow S^{-1}\Lambda$ be a factorization with B' smooth over $S^{-1}R$. Then we can find a factorization $A \rightarrow B \rightarrow \Lambda$ such that some $s \in S$ maps to an elementary standard element (Definition D.3) in B over R .

Proof. If $0 \in S$, take $B = A$ and $s = 0$. The element 0 is elementary standard with $c = 0$: the empty determinant is 1, its coefficient is 0, and all relation conditions are $0 = 0$. Hence assume $0 \notin S$.

By D.12, a standard smooth algebra with a retraction can replace B' while preserving the factorization. Write $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$, with $x_i \mapsto \lambda_i \in \Lambda$. The prescribed-generator construction D.14 gives

$$B' = S^{-1}R[x_1, \dots, x_{n+m}]/(f_1, \dots, f_c),$$

where $c \geq n$, the first n variable classes are the given generator images, and the determinant $\Delta = \det(\partial f_j / \partial x_i)_{i,j=1,\dots,c}$ is invertible in B' . Repeated prescribed images cause no difficulty: the proof of D.14 adds one equation $x_i - h_i$ for each entry of a finite list, and its identity Jacobian block remains invertible even if entries coincide.

For each added variable choose a representation of its image in $S^{-1}\Lambda$ with denominator $d_i \in S$. Replacing that variable by $d_i x_i$ makes its image $\lambda_i/1$ for some $\lambda_i \in \Lambda$. This is an invertible change of variables over $S^{-1}R$. By the chain rule it multiplies each selected Jacobian column by a unit, so the selected determinant stays invertible. Choose $u_j \in S$ clearing the coefficients of each transformed equation and replace f_j by $u_j f_j$. Their ideal over $S^{-1}R$ is unchanged, and the determinant is multiplied by $\prod_j u_j$, another unit. Thus all f_j now lie in $P = R[x_1, \dots, x_{n+m}]$.

The inverse of the determinant in B' has a polynomial representative. Consequently there are $b_0, \dots, b_c \in S^{-1}P$ such that

$$1 = b_0 \Delta + \sum_{j=1}^c b_j f_j \quad \text{in } S^{-1}P.$$

Choose $d \in S$ and polynomial representatives $c_j \in P$ with $c_j/1 = db_j$ in $S^{-1}P$. The difference $d - b_0 \Delta - \sum_j c_j f_j$ maps to zero in $S^{-1}P$, so some $e \in S$ kills it in P . Setting $s_0 = ed$ and

$a_j = ec_j \in P$ gives the exact identity

$$s_0 = a_0\Delta + \sum_{j=1}^c a_j f_j \quad \text{in } P.$$

This extra multiplier e covers possible S -torsion in P .

Each g_i is zero in $S^{-1}(P/(f_1, \dots, f_c))$. The definition of equality in a localization gives $s_i \in S$ with $s_i g_i \in (f_1, \dots, f_c)$ in P . Similarly, $f_j(\lambda)$ is zero in $S^{-1}\Lambda$, so choose $s'_j \in S$ with $s'_j f_j(\lambda) = 0$ in Λ . Put

$$h_j = s'_j f_j, \quad B = P/(h_1, \dots, h_c, g_1, \dots, g_t), \quad s = s_0 \left(\prod_{i=1}^t s_i \right) \left(\prod_{j=1}^c s'_j \right).$$

The finite equations give finite presentation, the g_i give $A \rightarrow B$, and $x_i \mapsto \lambda_i$ defines $B \rightarrow \Lambda$.

We verify both parts of definition D.3. Multiplying $s_i g_i = \sum_j v_{ij} f_j$ by s/s_i expresses sg_i in $(h_1, \dots, h_c)$, since the multiplier contains every s'_j . Put $t_0 = \prod_j s'_j$. Multiplying the exact determinant identity by t_0 gives

$$s_0 t_0 = a_0 \det(\partial h_j / \partial x_i)_{i,j=1,\dots,c} + \sum_{j=1}^c a_j \left(\prod_{\ell \neq j} s'_\ell \right) h_j.$$

In B the relation terms vanish. After multiplication by $\prod_i s_i$, the image of s is the selected determinant times the image of $a_0 \prod_i s_i$. Together with $sg_i \in (h_1, \dots, h_c)$, this is exactly the elementary standard condition. Empty products are 1 and the same calculation includes $c = 0$. $\square$

Lemma D.26: reduce to field

Adapted from the Stacks project, smoothing.tex, lines 2002–2063.

If for every Situation D.23 where R is a field PT holds, then PT holds in general.

Proof. Assume PT for every regular map from a field to a Noetherian ring, and fix the regular Noetherian map $R \rightarrow \Lambda$. Each quotient map $R/I \rightarrow \Lambda/I\Lambda$ is regular by F.123, since R/I is finite type over R . If any quotient fails PT, choose an ideal I maximal among those failures; the ascending-chain condition on ideals guarantees such a choice. Replace the map by that quotient. We then have a map failing PT for which every nonzero further ideal gives a quotient satisfying PT. The zero ring has PT, since its only unital algebra is itself, so the new R is nonzero.

This R is reduced. Indeed, its nilradical J is a finite ideal because R is Noetherian. Choose nilpotent generators a_i with $a_i^{n_i} = 0$. Every monomial of total degree $1 + \sum_i (n_i - 1)$ in these generators has an exponent at least n_i , so that power of J is zero. If $J \neq 0$, the quotient by J has PT by maximality. The original map is flat, so the full nilpotent lifting proof D.18 would then give PT for $R \rightarrow \Lambda$, a contradiction. Thus $J = 0$.

Let $\mathfrak{p}_1, \dots, \mathfrak{p}_t$ be the finite minimal primes of R , as proved in AG-CA-03, Proposition 2.2. Their intersection is zero. To justify this last assertion directly, an element not nilpotent has powers disjoint from the zero ideal; an ideal maximal among those disjoint from its powers is prime, by the product-of-enlarged-ideals argument used in D.21. Thus the intersection of all prime ideals is precisely the nilradical. Every prime contains a minimal prime, so intersecting minimal primes gives the same intersection.

Here is an explicit decomposition of the total fraction ring. Choose $y_i \in \bigcap_{j \neq i} \mathfrak{p}_j \setminus \mathfrak{p}_i$: for each $j \neq i$ choose an element of $\mathfrak{p}_j \setminus \mathfrak{p}_i$ and take their product; if $t = 1$, take $y_1 = 1$. An element outside every minimal prime is a nonzerodivisor, since a product equal to zero can be tested in each domain $R/\mathfrak{p}_i$ and their intersection is zero. Conversely, any $r \in \mathfrak{p}_i$ satisfies $ry_i = 0$ with $y_i \neq 0$, so it is a zerodivisor. Hence the set S of nonzerodivisors is exactly $R \setminus \bigcup_i \mathfrak{p}_i$.

Put $d = \sum_i y_i \in S$, $Q = S^{-1}R$ and $e_i = y_i/d \in Q$. Testing in the domains $R/\mathfrak{p}_i$ shows $e_i e_j = 0$ for $i \neq j$, $e_i^2 = e_i$ and $\sum_i e_i = 1$. The fraction ring embeds in the product of the fraction fields of these domains: a fraction zero in every factor has numerator in $\bigcap_i \mathfrak{p}_i = 0$. Thus multiplication and summation give inverse maps $Q \cong \prod_i e_i Q$. Each nonzero $e_i r/s$ has $r \notin \mathfrak{p}_i$. The element

$$u = e_i r + (1 - e_i) = \frac{y_i r + \sum_{j \neq i} y_j}{d}$$

is invertible in Q , because its numerator avoids every minimal prime. In the component $e_i Q$, the inverse of $e_i r/s$ is $e_i s u^{-1}$. Therefore every $K_i = e_i Q$ is a field, proving $Q \cong \prod_i K_i$ with its full ring maps. This is the total-fraction decomposition needed here; no embedded-prime assertion is left implicit.

The localized map $Q \rightarrow \Lambda_Q = S^{-1}\Lambda$ is regular. Its rings are Noetherian by quotient/localization permanence in AG-CA-03, Theorem 2.1. Flatness follows directly from the tensor identity $M \otimes_Q \Lambda_Q \cong M \otimes_R \Lambda$ for Q -modules. A prime of Q contracts to a prime $\mathfrak{p} \subset R$ disjoint from S , with the same residue field; the fibre is therefore exactly $\Lambda \otimes_R \kappa(\mathfrak{p})$, geometrically regular by the original hypothesis. The finite idempotents split Λ_Q as $\prod_i \Lambda_i$, and each $K_i \rightarrow \Lambda_i$ has these same regular fibres and is flat. By the assumed field case and the finite-product proof D.24, PT holds for $Q \rightarrow \Lambda_Q$.

Fix any finitely presented R -algebra A and a map $A \rightarrow \Lambda$. Its localized presentation is finite, so F.90 gives a factorization $S^{-1}A \rightarrow B' \rightarrow S^{-1}\Lambda$ with B' smooth over Q . Delocalization D.25 gives $A \rightarrow A_1 \rightarrow \Lambda$ with an element $\pi \in S$ elementary standard in A_1/R . Definition D.3 makes it strictly standard too. Replace the algebra to be factored by A_1 .

Since R is nonzero and π is a nonzerodivisor, $\pi^8 \neq 0$. The nonzero ideal $\pi^8 R$ thus gives PT for $R/\pi^8 R \rightarrow \Lambda/\pi^8 \Lambda$, including the possible unit case. Applying F.90 to $A_1/\pi^8 A_1$ gives a smooth factorization through an algebra $\bar{C}$ over $R/\pi^8 R$. Multiplication by every π^r is injective on R . Flatness of Λ preserves the exact sequence $0 \rightarrow R \xrightarrow{\pi^r} R$, so it is injective on Λ as well. In particular both annihilator equalities required by D.19 hold for its parameter π^4 . That lemma gives a finitely presented R -algebra $D \rightarrow \Lambda$ and

$$A_1/\pi^4 A_1 \longrightarrow \bar{C}/\pi^4 \bar{C} \longrightarrow D/\pi^4 D \longrightarrow \Lambda/\pi^4 \Lambda.$$

The map $R \rightarrow D$ is smooth at primes avoiding π , by D.19(b), and at primes containing π , by D.19(c) and smoothness of the entire $\bar{C}$ chart. Finite presentation and these local smooth charts make D smooth over R .

Apply D.21 to A_1 , D and the parameter π . Its annihilator hypotheses have just been proved, and the displayed factorization supplies its compatible quotient map. It gives $A_1 \rightarrow B \rightarrow \Lambda$ with B finitely presented and $H_{D/R}B \subset H_{B/R}$. Since D is smooth, definition D.1 gives $H_{D/R} = D$. Therefore $H_{B/R} = B$, and B is smooth over R . Composing with $A \rightarrow A_1$ supplies the smooth factorization for every A . F.90 gives PT, contradicting the chosen failure. No failing quotient exists, and PT holds for the original map. $\square$

Local tricks

Situation D.27: local

Adapted from the Stacks project, smoothing.tex, lines 2078–2084.

We are given a Noetherian ring R and an R -algebra map $A \rightarrow \Lambda$ and a prime $\mathfrak{q} \subset \Lambda$. We assume A is of finite presentation over R . In this situation we denote $\mathfrak{h}_A = \sqrt{H_{A/R}\Lambda}$.

Let $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27. We say $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ *can be resolved* if there exists a factorization $A \rightarrow B \rightarrow \Lambda$ with B of finite presentation and $\mathfrak{h}_A \subset \mathfrak{h}_B \not\subset \mathfrak{q}$. In this case we will call the factorization $A \rightarrow B \rightarrow \Lambda$ a *resolution* of $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$.

Lemma D.28: lift solution

Adapted from the Stacks project, smoothing.tex, lines 2096–2145.

Let $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27. Let $r \geq 1$ and $\pi_1, \dots, \pi_r \in R$ map to elements of $\mathfrak{q}$. Assume

1. for $i = 1, \dots, r$ we have

$$\mathrm{Ann}_{R/(\pi_1^8, \dots, \pi_{i-1}^8)R}(\pi_i) = \mathrm{Ann}_{R/(\pi_1^8, \dots, \pi_{i-1}^8)R}(\pi_i^2)$$

and

$$\mathrm{Ann}_{\Lambda/(\pi_1^8, \dots, \pi_{i-1}^8)\Lambda}(\pi_i) = \mathrm{Ann}_{\Lambda/(\pi_1^8, \dots, \pi_{i-1}^8)\Lambda}(\pi_i^2)$$

1. for $i = 1, \dots, r$ the element π_i maps to a strictly standard element in A over R .

Then, if

$$R/(\pi_1^8, \dots, \pi_r^8)R \rightarrow A/(\pi_1^8, \dots, \pi_r^8)A \rightarrow \Lambda/(\pi_1^8, \dots, \pi_r^8)\Lambda \supset \mathfrak{q}/(\pi_1^8, \dots, \pi_r^8)\Lambda$$

can be resolved, so can $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$.

Proof. We are going to prove this by induction on r .

The case $r = 1$. Here the assumption is that there exists a factorization $A/\pi_1^8 \rightarrow \bar{C} \rightarrow \Lambda/\pi_1^8$ which resolves the situation modulo π_1^8 . Conditions (1) and (2) are the assumptions needed to apply Lemma D.22. Thus we can “lift” the resolution $\bar{C}$ to a resolution of $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$.

The case $r > 1$. In this case we apply the induction hypothesis for $r - 1$ to the situation $R/\pi_1^8 \rightarrow A/\pi_1^8 \rightarrow \Lambda/\pi_1^8 \supset \mathfrak{q}/\pi_1^8\Lambda$. Note that property (2) is preserved by Lemma D.7. The induction resolves the situation modulo π_1^8 ; the already proved case $r = 1$, applied to π_1 , lifts this resolution to the original situation. $\square$

Lemma D.29: delocalize weak

Adapted from the Stacks project, smoothing.tex, lines 2147–2227.

Let $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27. Let $\mathfrak{p} = R \cap \mathfrak{q}$. Assume that $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$ and that $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}$ can be resolved. Then there exists a factorization $A \rightarrow C \rightarrow \Lambda$ with C of finite presentation such that $H_{C/R}\Lambda \not\subset \mathfrak{q}$.

Proof. In the assumed resolution of the localized problem, the singular radical of the intermediate algebra is outside the maximal ideal of $\Lambda_{\mathfrak{q}}$. Its extended singular ideal is therefore the unit ideal: if that ideal were contained in the maximal prime, its radical would be contained there too. D.8 replaces the intermediate algebra by a smooth one, and D.12 gives a standard smooth factorization, both compatible with the prescribed map from A to $\Lambda_{\mathfrak{q}}$.

Write $A = R[x_1, \dots, x_n]/(g_1, \dots, g_t)$, with $x_i \mapsto \lambda_i \in \Lambda$. The identity-block construction D.14 includes this prescribed generator list in a presentation of the standard smooth intermediate algebra:

$$C = R_{\mathfrak{p}}[x_1, \dots, x_n, u_1, \dots, u_m]/(f_1, \dots, f_c), \quad c \geq n.$$

Its selected c -variable Jacobian determinant δ is a unit. The construction adjoins a separate equation for each member of the list, so repeated generator images cause no difficulty. Multiply

each f_j by a denominator in $R \setminus \mathfrak{p}$ to make its coefficients lie in R . These factors are units over $R_{\mathfrak{p}}$, and multiply δ by their product, so it remains a unit. Put $P = R[x, u]$.

Choose a polynomial representative of δ^{-1} and of its identity with δ modulo the equations over $R_{\mathfrak{p}}$. Clear the finitely many coefficient denominators in that identity; equality of fractions supplies one further element outside $\mathfrak{p}$ annihilating its error. Their product gives $s_0 \notin \mathfrak{p}$ for which δ is a unit in $P_{s_0}/(f_j)$. For each g_i , its class vanishes in $(P/(f_j))_{\mathfrak{p}}$, so a localization witness $s_i \notin \mathfrak{p}$ gives $s_i g_i \in (f_j)$ in P . With $s = s_0 \prod_i s_i$, the algebra

$$C_s = P_s/(f_j)$$

is standard smooth over R_s , all g_i vanish there, and it is smooth over R by principal localization and smooth composition. No assertion that δ fails to be a unit before this localization is needed.

Choose a common denominator $h \in \Lambda \setminus \mathfrak{q}$ and numerators $\nu_\ell \in \Lambda$ representing the target images of the finitely many u_ℓ as ν_ℓ/h . For an empty list take $h = 1$. Homogenize only these variables: for sufficiently large integers $d_j \geq 0$, put

$$F_j(x, U, v) = v^{d_j} f_j(x, U/v) \in R[x, U, v].$$

Every $F_j(\lambda, \nu, h)$ is zero in $\Lambda_{\mathfrak{q}}$. The product of their finite localization annihilator witnesses gives $\lambda_0 \notin \mathfrak{q}$ killing every one of them in Λ .

Set

$$B = R[w, x_1, \dots, x_n, U_1, \dots, U_m, v]/(g_i(x), wF_j(x, U, v)).$$

Map it to Λ by $w \mapsto \lambda_0$, $x_i \mapsto \lambda_i$, $U_\ell \mapsto \nu_\ell$, $v \mapsto h$. Every equation vanishes exactly in Λ , and the g_i relations give the desired factorization through A . Let $b = wvs \in B$. Its image is outside $\mathfrak{q}$, because each of its three factors is outside that prime.

In B_b the factors w, v, s are units. Substitute $u_\ell = U_\ell/v$, with inverse $U_\ell = vu_\ell$. Each equation $wF_j = 0$ becomes $f_j = 0$, since wv^{d_j} is a unit. The equations $g_i = 0$ are already consequences of those equations in P_s . These substitutions give inverse R -algebra maps and the exact isomorphism

$$B_b \cong C_s[w, w^{-1}, v, v^{-1}].$$

This is smooth over R by polynomial extension, principal localization and F.29. Hence $b \in H_{B/R}$, by D.1 and F.67. Since its image is outside $\mathfrak{q}$, $H_{B/R}\Lambda \not\subset \mathfrak{q}$. The displayed finite generator and equation lists make B finitely presented, proving the lemma without adding any Noetherian hypothesis on the target. $\square$

Lemma D.30: delocalize height zero

Adapted from the Stacks project, smoothing.tex, lines 2229–2274.

Let $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27. Let $\mathfrak{p} = R \cap \mathfrak{q}$. Assume

1. $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$,
2. $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}$ can be resolved, and
3. $\dim(\Lambda_{\mathfrak{q}}) = 0$.

Then $R \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. The Noetherian hypothesis is on R ; no Noetherian hypothesis on Λ is needed. By AG-CA-03, Theorem 2.1, the finite type R -algebra A is Noetherian. Choose $H_{A/R} = (a_1, \dots, a_r)$.

A zero-dimensional local ring has only its maximal prime: a strictly smaller prime would give a chain of length one. Thus the nilradical of $\Lambda_{\mathfrak{q}}$ is $\mathfrak{q}\Lambda_{\mathfrak{q}}$, by the prime-intersection argument in D.5.

Since $\mathfrak{q}$ contains $\mathfrak{h}_A$, every a_i has nilpotent image in this localization. Choose one positive exponent N killing their finite list. The definition of localization gives $\lambda_i \notin \mathfrak{q}$ with $\lambda_i a_i^N = 0$ in Λ . Their product $\lambda \notin \mathfrak{q}$ satisfies

$$\lambda a_i^N = 0 \quad (1 \leq i \leq r).$$

For an empty list choose $N = 1$ and $\lambda = 1$. This uses only these finitely many nilpotence witnesses; it does not assert that the local ring is Artinian or that its entire maximal ideal is nilpotent.

Apply D.29 to the assumed localized resolution. It gives $A \rightarrow C \rightarrow \Lambda$, with C finitely presented over R and $\mathfrak{h}_C \not\subset \mathfrak{q}$. There is a finite A -presentation $C = A[x_1, \dots, x_n]/(f_1, \dots, f_m)$: adjoining the generators of a finite R -presentation makes it a finite type A -algebra, and the polynomial ring over the Noetherian A has finite ideal kernels by the same Hilbert theorem. Choose $c \in H_{C/R}$ whose image is outside $\mathfrak{q}$. Such an element exists, since otherwise the radical of $H_{C/R}\Lambda$ would be contained in this prime.

Define

$$B = A[x_1, \dots, x_n, y_1, \dots, y_r, z, t_{ij}] / \left(f_j - \sum_{i=1}^r y_i t_{ij}, z y_i \right),$$

with $1 \leq j \leq m$. This is finitely presented over R . It maps to Λ by the given map on C , by x_i going to the corresponding generator image, by $y_i \mapsto a_i^N$, $z \mapsto \lambda$ and $t_{ij} \mapsto 0$. The displayed annihilator equality makes every equation $z y_i$ vanish, and the other equations reduce to $f_j = 0$.

Inverting z forces every $y_i = 0$, giving the exact elimination

$$B_z \cong C[z, z^{-1}, t_{ij}].$$

Consequently B_{z_c} is smooth over R . Here and below a principal open lying in the smooth locus is globally smooth by F.67. For each ℓ , inverting y_ℓ instead forces $z = 0$ and solves every equation uniquely for $t_{\ell j}$:

$$t_{\ell j} = y_\ell^{-1} \left(f_j - \sum_{i \neq \ell} y_i t_{ij} \right).$$

Hence

$$B_{y_\ell} \cong A[x_j, y_i, y_\ell^{-1}, t_{ij} \ (i \neq \ell)],$$

a polynomial extension and principal localization over A . Since A_{a_ℓ} is smooth over R , $B_{a_\ell y_\ell}$ is smooth over R as well. Thus $z c$ and all $a_\ell y_\ell$ belong to $H_{B/R}$.

The image of $z c$ in Λ is $\lambda c \notin \mathfrak{q}$, so $\mathfrak{h}_B \not\subset \mathfrak{q}$. The image of $a_\ell y_\ell$ is a_ℓ^{N+1} , so each image a_ℓ belongs to the radical $\mathfrak{h}_B$. Taking radicals gives $\mathfrak{h}_A \subset \mathfrak{h}_B$. The finite presentation and the two required radical conditions prove that $A \rightarrow B \rightarrow \Lambda$ is the desired resolution. $\square$

Separable residue fields

In this section we explain how to solve a local problem in the case of a separable residue field extension.

Lemma D.31: ogoma

Adapted from the Stacks project, smoothing.tex, lines 2286–2303.

Let A be a Noetherian ring and let M be a finite A -module. Let $S \subset A$ be a multiplicative set. If $\pi \in A$ and $\ker(\pi : S^{-1}M \rightarrow S^{-1}M) = \ker(\pi^2 : S^{-1}M \rightarrow S^{-1}M)$ then there exists an $s \in S$ such that for any $n > 0$ we have $\ker(s^n \pi : M \rightarrow M) = \ker((s^n \pi)^2 : M \rightarrow M)$.

Proof. Let $K = \ker(\pi : M \rightarrow M)$ and $K' = \{m \in M \mid \pi^2 m = 0 \text{ in } S^{-1}M\}$ and $Q = K'/K$. Note that $S^{-1}Q = 0$ by assumption. Since A is Noetherian we see that Q is a finite A -module. Hence

we can find an $s \in S$ such that s annihilates Q . If $(s^n \pi)^2 m = 0$, then $\pi^2 m = 0$ after localization, so $m \in K'$. The equality $sQ = 0$ gives $sm \in K$, hence $s\pi m = 0$, and therefore $s^n \pi m = 0$ for every $n > 0$. The reverse kernel inclusion is immediate. Thus this single s works for every positive power. $\square$

Lemma D.32: find sequence

Adapted from the Stacks project, smoothing.tex, lines 2305–2349.

Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let $I \subset \mathfrak{q}$ be a prime. Let n, e be positive integers. Assume that $\mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset I \Lambda_{\mathfrak{q}}$ and that $\Lambda_{\mathfrak{q}}$ is a regular local ring of dimension d . Then there exist $\pi_1, \dots, \pi_d \in \Lambda$ such that

1. $(\pi_1, \dots, \pi_d) \Lambda_{\mathfrak{q}} = \mathfrak{q} \Lambda_{\mathfrak{q}}$,
2. $\pi_1^n, \dots, \pi_d^n \in I$, and
3. for $i = 1, \dots, d$ we have

$$\text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i) = \text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_{i-1}^e) \Lambda}(\pi_i^2).$$

Proof. Set $S = \Lambda \setminus \mathfrak{q}$ so that $\Lambda_{\mathfrak{q}} = S^{-1} \Lambda$. First pick $\pi_1, \dots, \pi_d$ with (1) which is possible as $\Lambda_{\mathfrak{q}}$ is regular. By assumption $\pi_i^n \in I \Lambda_{\mathfrak{q}}$. Thus we can find $s_1, \dots, s_d \in S$ such that $s_i \pi_i^n \in I$. Replacing π_i by $s_i \pi_i$ we get (2). Note that (1) and (2) are preserved by further multiplying by elements of S . Suppose that (3) holds for $i = 1, \dots, t$ for some $t \in \{0, \dots, d\}$. Note that $\pi_1, \dots, \pi_d$ is a regular sequence in $S^{-1} \Lambda$, see Algebra, Lemma AG-CA-12, Theorem 6.1. In particular $\pi_1^e, \dots, \pi_t^e, \pi_{t+1}$ is a regular sequence in $S^{-1} \Lambda = \Lambda_{\mathfrak{q}}$ by Algebra, Lemma AG-CA-12, Proposition 1.3. Hence we see that

$$\text{Ann}_{S^{-1} \Lambda/(\pi_1^e, \dots, \pi_{i-1}^e)}(\pi_i) = \text{Ann}_{S^{-1} \Lambda/(\pi_1^e, \dots, \pi_{i-1}^e)}(\pi_i^2).$$

Thus we get (3) for $i = t + 1$ after replacing π_{t+1} by $s \pi_{t+1}$ for some $s \in S$ by Lemma D.31. By induction on t this produces a sequence satisfying (1), (2), and (3). $\square$

Lemma D.33: resolve special

Adapted from the Stacks project, smoothing.tex, lines 2351–2430.

Let $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27 where

1. k is a field,
2. Λ is Noetherian,
3. $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$,
4. $\Lambda_{\mathfrak{q}}$ is a regular local ring, and
5. the field extension $\kappa(\mathfrak{q})/k$ is separable.

Then $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. Put $L = \Lambda_{\mathfrak{q}}$ and $d = \dim L$. If $d = 0$, regularity makes L its residue field: its zero cotangent dimension and finite-module Nakayama give zero maximal ideal. By the separability hypothesis, F.23 writes L as a filtered colimit of smooth k -algebras. Since A is finitely presented, F.90 gives

a smooth factorization of $A \rightarrow L$. Its singular ideal is the unit ideal, so it resolves the localized problem. D.30 gives the required original resolution. Assume $d > 0$.

Set $I = H_{A/k}\Lambda$. Minimality of $\mathfrak{q}$ gives $\sqrt{IL} = \mathfrak{q}L$. The maximal ideal of the Noetherian local ring L has a finite generating list, with a power of each generator in IL . Every monomial of degree greater than the sum of those powers minus one contains the required power of some generator. Thus choose $n > 0$ with $\mathfrak{q}^n L \subset IL$. The algebra A is Noetherian by AG-CA-03, Theorem 2.1, so choose $H_{A/k} = (a_1, \dots, a_r)$.

Put $R = k[x_1, \dots, x_d]$ and

$$B = A[x_1, \dots, x_d, z_{ij}] / \left(x_i^n - \sum_{j=1}^r z_{ij} a_j \right).$$

On inverting a_j , the i th equation solves uniquely for z_{ij} . Therefore B_{a_j} is a polynomial algebra over $A_{a_j}[x_1, \dots, x_d]$, smooth over R by base change and polynomial extension. In B_{x_i} the elements a_j generate the unit ideal, since $1 = x_i^{-n} \sum_j z_{ij} a_j$. Their principal charts cover its spectrum, so F.67 makes B_{x_i} smooth over R .

Use D.9 to construct $B \rightarrow C = \text{Sym}_B(M)$ with retraction $C \rightarrow B$, where M is the conormal of a finite presentation of B . Its smooth-case proof makes M_{x_i} finite projective and C_{x_i} smooth over B_{x_i} , and identifies $\Omega_{C_{x_i}/R}$ as finite free. Thus C_{x_i} is smooth over R with free differentials. The finite type algebra C is finitely presented over the Noetherian R : a finite polynomial surjection has a finite ideal kernel by the Hilbert theorem. Part (e) of D.15 supplies a threshold for each x_i ; choose one common positive c at least all these thresholds. Every x_i^c is strictly standard in C/R . This uses the proved strictly-standard implication, not the disproved original elementary-power implication.

Apply D.32 to $I \subset \Lambda$, with this fixed n and $e = 8c$. It gives $\pi_1, \dots, \pi_d$ generating the maximal ideal of L , satisfying $\pi_i^n \in I$ in Λ and the required annihilator equalities modulo previous e th powers. Write $\pi_i^n = \sum_j a_j \lambda_{ij}$ in Λ . The assignments $x_i \mapsto \pi_i$, $z_{ij} \mapsto \lambda_{ij}$ define $B \rightarrow \Lambda$. Compose with the retraction to get $C \rightarrow \Lambda$; together with $R \rightarrow \Lambda$ these give the required commuting maps from A .

We may pass from the original problem to $R \rightarrow C \rightarrow \Lambda$. Indeed B_{a_j} and C_{a_j} are smooth over R , so the images of a_j belong to $H_{C/R}$. A resolution $C \rightarrow T \rightarrow \Lambda$ therefore has $\mathfrak{h}_A \subset \mathfrak{h}_{T/R}$. Since R is a polynomial algebra over k , smoothness over R implies smoothness over k by composition. Thus $H_{T/R} \subset H_{T/k}$. Also T is finitely presented over k , by appending its finite equations to those for the polynomial base. This gives both radical conditions for an original resolution over k .

Apply D.28 to $x_1^c, \dots, x_d^c$. Its strictly-standard condition was established above. In $R/(x_1^e, \dots, x_{i-1}^e)$ multiplication by the remaining polynomial variable x_i is injective, by comparison of polynomial coefficients. Hence both relevant annihilators are zero. On the target quotient, D.32 gives $\ker(\pi_i) = \ker(\pi_i^2)$. This implies $\ker(\pi_i^r) = \ker(\pi_i)$ for every $r \geq 1$: inductively, $\pi_i^{r+1}v = 0$ puts $\pi_i^{r-1}v$ in $\ker(\pi_i^2) = \ker(\pi_i)$, so $\pi_i^r v = 0$. In particular the kernels for π_i^c and π_i^{2c} agree. These are exactly the target hypotheses of D.28, since $(x_i^c)^8 = x_i^e$ and likewise for π_i .

It remains to resolve the quotient problem over $R_0 = R/(x_1^e, \dots, x_d^e)$. After localization at the target prime its base is

$$S = R_{(x_1, \dots, x_d)} / (x_1^e, \dots, x_d^e),$$

and its target is $L_0 = L/(\pi_1^e, \dots, \pi_d^e)$. The contraction to R is exactly $(x_1, \dots, x_d)$, since a polynomial has residue its constant in the field k . The prequotient base $R_{(x_1, \dots, x_d)}$ is regular local with parameters $x_1, \dots, x_d$. Their images $\pi_1, \dots, \pi_d$ form a regular sequence in the regular local ring L , by AG-CA-12, Proposition 1.3. F.46 gives flatness before quotienting, and base change gives flatness of $S \rightarrow L_0$.

The maximal ideal of S is nilpotent, by the same bounded-degree monomial calculation. Its residue field is k , and L_0 modulo that ideal is $\kappa(\mathfrak{q})$, a separable extension of k . By F.23 this residue field is a smooth colimit; D.18, applied to the nilpotent maximal ideal of S and the flat target, makes L_0 a

filtered colimit of smooth S -algebras. F.90 therefore factors the finitely presented localized algebra $C \otimes_R S$ through a smooth S -algebra.

Finally L_0 is zero dimensional. Its maximal ideal is generated by the π_i , and any sufficiently high-degree monomial contains a π_i^e , so this ideal is nilpotent. If the singular radical of the quotient input is already outside the target prime, the identity factorization resolves that quotient. Otherwise the target prime is minimal over it, since its localization has dimension zero. D.30 delocalizes the smooth factorization just constructed. D.28 then lifts that resolution to $R \rightarrow C \rightarrow \Lambda$, and the preceding change-of-base argument returns it to $k \rightarrow A \rightarrow \Lambda$. $\square$

Inseparable residue fields

In this section we explain how to solve a local problem in the case of an inseparable residue field extension.

Lemma D.34: helper

Adapted from the Stacks project, smoothing.tex, lines 2444–2549.

Let k be a field of characteristic $p > 0$. Let $(\Lambda, \mathfrak{m}, K)$ be an Artinian local k -algebra. Assume that $\dim H_1(L_{K/k}) < \infty$. Then Λ is a filtered colimit of Artinian local k -algebras A with each map $A \rightarrow \Lambda$ flat, with $\mathfrak{m}_A \Lambda = \mathfrak{m}$, and with A essentially of finite type over k .

Proof. Write $i: k \rightarrow \Lambda$ for the specified structure map and identify k with its image in the residue field K . The Artinian-local theorem AG-CA-03, Theorem 4.2, gives a finite generating list $\lambda_1, \dots, \lambda_d$ of $\mathfrak{m}$ and an integer $n \geq 1$ with $\mathfrak{m}^n = 0$.

Every field is formally smooth over its prime field, by the complete arbitrary-field lifting proof AG-CA-19S, Lemma 1.1, recalled in F.91. Square-zero lifting extends to a nilpotent ideal by successively lifting through its powers: the kernel of $E/J^{a+1} \rightarrow E/J^a$ is square zero for $a \geq 1$. Thus there is a coefficient-field section $\sigma: K \rightarrow \Lambda$. This need not agree with i on k . For any such section define $\Psi_\sigma: K[x_1, \dots, x_d] \rightarrow \Lambda$ by $x_j \mapsto \lambda_j$.

We first arrange a section σ and a finitely generated intermediate field $k \subset F_0 \subset K$ such that

$$i(k) \subset \Psi_\sigma(F_0[x_1, \dots, x_d]).$$

Prove this claim by induction on the least nilpotence exponent. For $n = 1$, $\Lambda = K$ and the identity section with $F_0 = k$ works. For $n > 1$ put $I = \mathfrak{m}^{n-1}$ and $\Lambda' = \Lambda/I$. Here $I^2 = 0$ and $\mathfrak{m}I = 0$. The preceding claim supplies a section $\sigma': K \rightarrow \Lambda'$ and a finitely generated field $F' \supset k$ containing the prescribed k -map in the subring generated by $\sigma'(F')$ and the images of the λ_j . Lift σ' to σ by formal smoothness.

The truncated polynomial ring

$$T = F'[x_1, \dots, x_d]/(x_1, \dots, x_d)^n$$

maps to Λ by Ψ_σ and onto that subring of Λ' . The latter surjection has nilpotent kernel: its kernel has zero constant coefficient, since $F' \hookrightarrow K$, and lies in the nilpotent ideal (x_j) . Formal smoothness of $k/\mathbb{F}_p$ lifts the prescribed k -map to a ring map $\tau: k \rightarrow T$. The maps i and $\Psi_\sigma \tau$ agree modulo I . Their difference $\theta: k \rightarrow I$ is an $\mathbb{F}_p$ -derivation, with scalar action through $k \subset K$, by expansion of products and $I^2 = 0$.

The absolute p-basis theorem F.119 gives a basis dy_j of $\Omega_{k/\mathbb{F}_p}$ consisting of differentials of field elements. Formal smoothness and F.17 give $H_1(L_{K/\mathbb{F}_p}) = 0$. The exact field sequence F.131 therefore gives

$$0 \longrightarrow H_1(L_{K/k}) \longrightarrow \Omega_{k/\mathbb{F}_p} \otimes_k K \longrightarrow \Omega_{K/\mathbb{F}_p} \longrightarrow \Omega_{K/k} \longrightarrow 0.$$

Its finite-dimensional first image has a finite basis, each vector of which has finite support in the dy_j . Choose the union J_0 of these supports. The images of dy_j for $j \notin J_0$ are independent: a relation supported outside J_0 would lie in a subspace supported inside J_0 , so it is zero. Prescribe a K -linear map on their span with value $\theta(y_j)$, extend a basis to the whole differential space, and give the added basis vectors value zero in the K -vector space I . The universal derivation gives $D: K \rightarrow I$ with $D(y_j) = \theta(y_j)$ for $j \notin J_0$.

Replace σ by $\sigma + D$. The derivation identity and $I^2 = 0$ show directly that this is a ring map, and it remains a residue section. The change in $\Psi_\sigma \tau(a)$ is exactly $D(a)$: the constant coefficient of $\tau(a)$ has residue a , while a change $D(c)$ in any positive-degree coefficient is killed by its monomial in $\mathfrak{m}$, because $\mathfrak{m}I = 0$. Thus the new error is $\theta - D|_k$. If $da = \sum_j a_j dy_j$, it equals

$$\sum_{j \in J_0} a_j f_j, \quad f_j = (\theta - D|_k)(y_j) \in I.$$

Since $I = \mathfrak{m}^{n-1}$ and $\mathfrak{m}I = 0$, every f_j is a combination of the finitely many degree- $(n-1)$ monomials λ^E , with coefficients from K through the section. Adjoin these finitely many coefficients to F' , giving F_0 . The coefficients a_j above belong to $k \subset F_0$, and $\tau(a)$ has coefficients in F' . Hence both $\Psi_\sigma \tau(a)$ and its correction lie in $\Psi_\sigma(F_0[x])$. This proves the claim for the actual map i , rather than only its residue map.

Fix this section. The map Ψ_σ is surjective. Indeed, subtract the coefficient-field representative of an element's residue, express the remainder using the λ_j , and repeat on the coefficients of that remainder. At degree n the remainder vanishes, giving a finite polynomial expression. Its kernel is generated by finitely many polynomials $g_1, \dots, g_t$ by the Hilbert basis theorem AG-CA-03, Theorem 2.1. Adjoin all their coefficients to F_0 and call the resulting finitely generated field F_* . For every finitely generated intermediate field $F \supset F_*$ put

$$A_F = F[x_1, \dots, x_d] / (g_1, \dots, g_t).$$

Tensoring the displayed presentation gives the exact identity $K \otimes_F A_F = \Lambda$. A basis of K/F containing 1 identifies this tensor product, as an A_F -module, with a direct sum of copies of A_F . The 1-coordinate proves that $A_F \rightarrow \Lambda$ is injective; the same basis proves faithful flatness. The earlier claim makes its image a k -subalgebra for the specified map i .

The ideal $(x_1, \dots, x_d)$ of A_F has n th power zero, by the injection into Λ . Its quotient is F , since every kernel polynomial has zero constant coefficient. Thus A_F is spanned over F by the finitely many monomials of degree below n . It is Artinian local with that maximal ideal: an element with nonzero constant coefficient has an inverse by the finite geometric series in its nilpotent part, and finite vector-space dimension makes every ideal chain stabilize. Its maximal ideal extends to $\mathfrak{m}$, generated by the same λ_j .

The residue map $k \rightarrow F$ is the original subfield inclusion, with F/k finitely generated. It is essentially of finite type, as the fraction field of a finitely generated k -domain. F.40 therefore makes A_F essentially of finite type over k , even though its coefficient-field inclusion need not be the structure map.

Finally these embedded algebras form a directed family: the field generated by two finite field-generator lists is a common successor. Every element of Λ has the finite polynomial expression established above; adjoining its finitely many coefficients to F_* puts it in one A_F . Their union, hence their filtered colimit, is exactly Λ . All transition maps are the actual inclusions, and every member has every asserted property. $\square$

Lemma D.34A: Artinian denominator enlargement

Let k have characteristic $p > 0$. Let $A \rightarrow B$ be a flat local map of Noetherian Artinian local k -algebras, with $\mathfrak{m}_A B = \mathfrak{m}_B$, and suppose $\mathfrak{m}_B^n = 0$. Write F and K for their residue fields. For

finitely many units $b_j \in B$, there is a factorization $A \rightarrow A' \rightarrow B$ by local maps such that $A \rightarrow A'$ is essentially smooth, $A' \rightarrow B$ is faithfully flat, $\mathfrak{m}_{A'}B = \mathfrak{m}_B$, and a p -power of each b_j belongs to A' . If a residue $\bar{b}_j$ is transcendental over the current residue field, the construction adjoins b_j itself. The residue extension F'/F is a finite tower of simple transcendental and finite separable extensions.

Proof. Treat one unit b at a time. If its residue β is transcendental over F , use $A' = A[T]_{\mathfrak{m}_A A[T]}$ with $T \mapsto b$. Its residue field is $F(\beta)$: every nonzero polynomial in $F[T]$ is inverted. Each inverted element maps to a unit of B , so the map is defined and local. This localization is essentially smooth, with $\mathfrak{m}_{A'} = \mathfrak{m}_A A'$. It is Noetherian by the Hilbert basis theorem and localization, and this maximal ideal is nilpotent. Its successive ideal-power quotients are finite-dimensional over $F(\beta)$, because each ideal power is finite in the Noetherian ring. They give a finite composition series; thus the ring is Artinian. This argument also proves Artinity of the finite free algebra in the algebraic case below.

If β is algebraic over F , choose a p -power q_0 for which β^{q_0} is separable over F , by the inseparable exponent calculation already proved. Lift the monic separable minimal polynomial of β^{q_0} from $F[T]$ to $h \in A[T]$ and put $A' = A[T]/(h)$. It is finite free over A . Its quotient modulo $\mathfrak{m}_A$ is the field $F(\beta^{q_0})$, and $\mathfrak{m}_A A'$ is nilpotent, so A' is Artinian local with this maximal ideal. Bézout for the relatively prime residue polynomials $\bar{h}, \bar{h}'$ makes h' a unit modulo $\mathfrak{m}_A A'$; lifting its inverse across a nilpotent ideal makes it a unit in A' . The square Jacobian calculation, AG-CA-17, Theorem 4.1 and Section 6, therefore makes A' étale. The simple-root Hensel proof in AG-CA-19, Theorem 5.1, lifts the specified residue root in the Artinian local ring B and gives $A' \rightarrow B$. This is exactly the finite étale lifting construction in F.54, written here in its consumed form. If $\beta \in F$, use instead $A' = A$, $q_0 = 1$.

In both cases A' is flat over A , and $A'/\mathfrak{m}_A A' = F'$, its residue field. A free A -resolution of F therefore becomes a free A' -resolution of F' after tensoring. Tensor it further with B . Flatness of B over A gives

$$\mathrm{Tor}_1^{A'}(F', B) = \mathrm{Tor}_1^A(F, B) = 0.$$

The Noetherian local criterion, AG-CA-08, Theorem 4.2, applied with finite B -module B , proves that $A' \rightarrow B$ is flat. It is local and hence faithfully flat, so in particular injective. Its maximal ideal still generates $\mathfrak{m}_B$.

In the algebraic case choose $x \in A'$ with residue β^{-q_0} . It is a unit. Then $xb^{q_0} - 1 \in \mathfrak{m}_B$. For a p -power $q_1 \geq n$, Frobenius gives $(xb^{q_0})^{q_1} = 1$, whence $b^{q_0 q_1} = x^{-q_1} \in A'$. In the transcendental case b is already present, so any p -power may be used. Repeating the finite construction proves every assertion.

We also record its differential effect, used immediately below. For a simple transcendental residue extension the polynomial presentation and localization give $\Omega_{F(\beta)/k} = (\Omega_{F/k} \otimes_F F(\beta)) \oplus F(\beta) d\beta$. A finite separable residue extension gives the same old differential space after tensoring: its monic separable presentation has invertible derivative, so its relative conormal complex is acyclic. The field Jacobi–Zariski sequence F.131 likewise gives in both cases

$$H_1(L_{F'/k}) = H_1(L_{F/k}) \otimes_F F'.$$

These are natural identifications, and composition gives them for the finite tower. Thus the image of this first homology in $H_1(L_{K/k})$ is unchanged. $\square$

Lemma D.35: solution modulo

Adapted from the Stacks project, smoothing.tex, lines 2551–2621.

Let k be a field of characteristic $p > 0$. Let Λ be a Noetherian geometrically regular k -algebra. Let

$\mathfrak{q} \subset \Lambda$ be a prime ideal. Let $n \geq 1$ be an integer and let $E \subset \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$ be a finite subset. Then we can find $m \geq 0$ and $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$ with the following properties

1. setting $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$ we have $\mathfrak{q}\Lambda_{\mathfrak{q}} = \mathfrak{p}\Lambda_{\mathfrak{q}}$ and $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$ is flat,
2. there is a factorization by homomorphisms of local Artinian rings

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$$

where the first arrow is essentially smooth and the second is flat,

1. E is contained in D modulo $\mathfrak{q}^n\Lambda_{\mathfrak{q}}$.

Proof. Put $L = \Lambda_{\mathfrak{q}}$, with maximal ideal $\mathfrak{m}$ and residue field K , and put $B = L/\mathfrak{m}^n$. Geometric regularity and F.133 give an injection $H_1(L_{K/k}) \hookrightarrow \mathfrak{m}/\mathfrak{m}^2$, so this first homology is finite-dimensional. By D.34 choose an Artinian local subring $A \subset B$ containing E , essentially of finite type over k , such that $A \rightarrow B$ is faithfully flat and $\mathfrak{m}_A B = \mathfrak{m}_B$. Write F for its residue field.

Here one must ensure global representatives; an arbitrary such A does not supply them. Choose a basis of the finite-dimensional space $\Omega_{F/k}$ consisting of differentials $d\theta_j$ of elements of F , and choose lifts $\tilde{\theta}_j \in A$. Since $L \twoheadrightarrow B$, represent each lift by a fraction a_j/b_j with $a_j \in \Lambda$ and $b_j \in \Lambda \setminus \mathfrak{q}$. By the field left injection F.131, $H_1(L_{F/k}) \otimes_F K$ injects into $H_1(L_{K/k})$. Let V be its image in $\mathfrak{m}/\mathfrak{m}^2$ under the preceding injection. If $n \geq 2$, the equality $\mathfrak{m}_A B = \mathfrak{m}_B$ shows that the images of elements of $\mathfrak{m}_A$ span $\mathfrak{m}/\mathfrak{m}^2$ over K . Choose finitely many $\tilde{\eta}_i \in \mathfrak{m}_A$ whose images form a complement to V . Lift each to a fraction $c_i/d_i \in L$ with $c_i \in \mathfrak{q}$ and $d_i \in \Lambda \setminus \mathfrak{q}$. If $n = 1$, choose instead elements $c_i \in \mathfrak{q}$ whose classes form this complement; their images in $B = K$ are zero, hence already belong to A . Use $d_i = 1$ in that case.

Apply D.34A to this finite list of denominators, as units of B . It enlarges A to an Artinian local subring $A' \subset B$ by an essentially smooth map, retains faithful flatness to B and the maximal-ideal equality, and puts $b_j^{q_j}, d_i^{r_i}$ in A' for suitable p -powers. Consequently the global elements

$$a_j b_j^{q_j-1}, \quad b_j^{q_j}, \quad c_i d_i^{r_i-1}, \quad d_i^{r_i}$$

have images in A' : the first and third are the original fractional lifts multiplied by the indicated powers. Let F' be its residue field. During the enlargement record also the global denominators whose residues were adjoined transcendently; those denominators themselves have images in A' .

The differentials of the residues of the first two displayed global elements, together with those recorded transcendental denominators, span $\Omega_{F'/k}$. Indeed the quotient identity $\theta_j = (a_j b_j^{q_j-1})/b_j^{q_j}$ expresses every old $d\theta_j$ in terms of their differentials. D.34A shows that only the recorded transcendental differentials must be added; a finite separable step adds none. Select a basis from this finite spanning list, and denote its global representatives by $\lambda_1, \dots, \lambda_t \in \Lambda$. All their classes lie in A' . This argument does not assert that $d(b_j^{q_j}) = 0$ inside $\Omega_{F'/k}$: its root need not lie in F' .

The natural first-homology calculation at the end of D.34A gives $H_1(L_{F'/k}) = H_1(L_{F/k}) \otimes_F F'$. Hence its image in $\mathfrak{m}/\mathfrak{m}^2$ is still V . Define $P' = k[y_1, \dots, y_t] \rightarrow \Lambda$ using these λ_j , and let $\mathfrak{p}'$ be the inverse image of $\mathfrak{q}$. Lemma F.109 proves that $P'_{\mathfrak{p}'} \rightarrow L$ is flat and that $L/\mathfrak{p}'L$ is regular. More precisely, the proof of that lemma identifies the image of $(\mathfrak{p}'/\mathfrak{p}'^2) \otimes K$ in $\mathfrak{m}/\mathfrak{m}^2$ with V . To check this identification, write $K_0 = \kappa(\mathfrak{p}')$. The chosen differential basis makes F'/K_0 finitely generated, since F'/k is, and $\Omega_{F'/K_0} = 0$: the chosen $d\lambda_j$ span $\Omega_{F'/k}$, and each λ_j lies in K_0 . Apply F.98 to this exact extension. It gives

$$0 - \dim H_1(L_{F'/K_0}) = \text{trdeg}_{K_0} F'.$$

Both dimensions are nonnegative, so both are zero. A finitely generated algebraic field extension is finite, by multiplying the finitely many generator degrees. The vanishing first homology makes it separable by F.91. Thus F'/K_0 is finite separable. Its acyclic relative conormal complex identifies $H_1(L_{K_0/k}) \otimes_{K_0} K$ with $H_1(L_{F'/k}) \otimes_{F'} K$. The natural conormal square in F.109 then gives exactly the image V .

Set $\lambda_{t+i} = c_i d_i^{r_i-1}$. These are global, have classes in A' , and their cotangent classes are those of the original $\tilde{\eta}_i$ multiplied by the nonzero residues $d_i^{r_i}$. They remain a complement to V , so they map to a regular system of parameters of $L/\mathfrak{p}'L$. Let $P = k[y_1, \dots, y_m]$ append these variables and let $\mathfrak{p}$ be its inverse image of $\mathfrak{q}$. The local ring $P_{\mathfrak{p}}$ is obtained from $P'_{\mathfrak{p}'}[y_{t+1}, \dots, y_m]$ at the prime generated by $\mathfrak{p}'$ and the appended variables, since their residues are zero. Its regular parameters consist of the old parameters and these variables. Their images form a regular parameter list in L : the old list is regular by F.109 and the new list is regular on its regular quotient. The regular-base flatness lemma F.46 gives flatness of $P_{\mathfrak{p}} \rightarrow L$. Their classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$, so Nakayama gives $\mathfrak{p}L = \mathfrak{m}$. This proves property (1).

Every global generator has its class in A' , so the polynomial map to B factors through A' . The elements outside $\mathfrak{p}$ map to units of this local ring. Since $\mathfrak{p}^n B = \mathfrak{m}^n B = 0$ and $A' \hookrightarrow B$, the factorization descends to

$$P_{\mathfrak{p}}/\mathfrak{p}^n \longrightarrow A' \longrightarrow B.$$

Its composite is flat by the flatness just proved and base change. Faithful-flat descent of module flatness, F.48, makes the first map flat. Also $\mathfrak{p}A' = \mathfrak{m}_{A'}$: both ideals become $\mathfrak{m}_B$ in B , and faithful flatness detects equality of ideals. The residue extension $F'/\kappa(\mathfrak{p}) = F'/K_0$ is finite separable as above. The Artinian length-filtration proof F.40 therefore makes A' finite over $P_{\mathfrak{p}}/\mathfrak{p}^n$. The source is Noetherian, so this is a finitely presented algebra. Lemma F.16 now proves that the first map is étale. In particular it is essentially smooth. Taking $D = A'$ proves properties (2) and (3), since $A \subset A'$ retains E . $\square$

Lemma D.36: enlarge solution modulo

Adapted from the Stacks project, smoothing.tex, lines 2623–2669.

Let $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$, n , $\mathfrak{q}$, $\mathfrak{p}$ and

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

be as in Lemma D.35. Then for any $\lambda \in \Lambda \setminus \mathfrak{q}$ there exists an integer $q > 0$ and a factorization

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \rightarrow D \rightarrow D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

such that $D \rightarrow D'$ is an essentially smooth map of local Artinian rings, the last arrow is flat, and λ^q is in D' .

Proof. Put $B = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$. The hypotheses of D.35 make $D \rightarrow B$ flat and local, hence faithfully flat. Its maximal ideal generates that of B : the maximal ideal of the polynomial Artinian source already generates it by property (1), and its image in D is the maximal ideal because the first map in property (2) is local and essentially smooth. Its closed fibre is a localization of a smooth algebra over a field, hence regular by AG-CA-18, Theorem 2.1. It is Artinian local, so its regular dimension is zero; its cotangent space is zero and Nakayama makes its maximal ideal zero. Thus that fibre is a field, which proves the asserted maximal-ideal equality. The equality also holds directly in the étale construction of D.35. Apply D.34A to the unit image of λ . It gives the required essentially smooth Artinian extension $D \rightarrow D'$, flat final map, and a p -power q with $\lambda^q \in D'$. Composing with the original polynomial source supplies the stated factorization. $\square$

Lemma D.37: resolve general

Adapted from the Stacks project, smoothing.tex, lines 2671–3098.

Let $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ be as in Situation D.27 where

1. k is a field of characteristic $p > 0$,
2. Λ is Noetherian and geometrically regular over k ,
3. $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$.

Then $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. The lemma is proven by the following steps in the given order. We will justify each of these steps below.

1. Pick an integer $N > 0$ such that $\mathfrak{q}^N \Lambda_{\mathfrak{q}} \subset H_{A/k} \Lambda_{\mathfrak{q}}$.
2. Pick generators $a_1, \dots, a_r \in A$ of the ideal $H_{A/k}$.
3. Set $d = \dim(\Lambda_{\mathfrak{q}})$.
4. Set $B = A[x_1, \dots, x_d, z_{ij}]/(x_i^{2N} - \sum z_{ij} a_j)$.
5. Consider B as a $k[x_1, \dots, x_d]$ -algebra and let $B \rightarrow C$ be as in Lemma D.9. We also obtain a section $C \rightarrow B$.
6. Choose $c > 0$ such that each x_i^c is strictly standard in C over $k[x_1, \dots, x_d]$.
7. Set $e = 8c$ and $n = \max\{N + dc, d(e - 1) + 1\}$. This choice is made before defining E , D , and all subsequent factorizations.
8. Let $E \subset \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ be the images of generators of A as a k -algebra.
9. Choose an integer m and a k -algebra map $\varphi : k[y_1, \dots, y_m] \rightarrow \Lambda$ and a factorization by local Artinian rings

$$k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

such that the first arrow is essentially smooth, the second is flat, E is contained in D , with $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$ the map $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$ is flat, and $\mathfrak{p} \Lambda_{\mathfrak{q}} = \mathfrak{q} \Lambda_{\mathfrak{q}}$.

1. Choose $\pi_1, \dots, \pi_d \in \mathfrak{p}$ which map to a regular system of parameters of $k[y_1, \dots, y_m]_{\mathfrak{p}}$.
2. Let $R = k[y_1, \dots, y_m, t_1, \dots, t_d]$ and $\gamma_i = \pi_i t_i$.
3. If necessary modify the choice of π_i such that for $i = 1, \dots, d$ we have

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

1. There exist $\delta_1, \dots, \delta_d \in \Lambda$, $\delta_i \notin \mathfrak{q}$ and a factorization $D \rightarrow D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ with D' local Artinian, $D \rightarrow D'$ essentially smooth, the map $D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$ flat such that, with $\pi'_i = \delta_i \pi_i$, we have for $i = 1, \dots, d$
2. $(\pi'_i)^{2N} = \sum a_j \lambda_{ij}$ in Λ where $\lambda_{ij} \bmod \mathfrak{q}^n \Lambda_{\mathfrak{q}}$ is an element of D' ,

3. $\text{Ann}_{\Lambda/(\pi'_1, \dots, \pi'_{i-1})}(\pi'_i) = \text{Ann}_{\Lambda/(\pi'_1, \dots, \pi'_{i-1})}(\pi'^2_i)$,
4. $\delta_i \bmod \mathfrak{q}^n \Lambda_{\mathfrak{q}}$ is an element of D' .
5. Define $B \rightarrow \Lambda$ by sending x_i to π'_i and z_{ij} to λ_{ij} found above. Define $C \rightarrow \Lambda$ by composing the map $B \rightarrow \Lambda$ with the retraction $C \rightarrow B$.
6. Map $R \rightarrow \Lambda$ by φ on $k[y_1, \dots, y_m]$ and by sending t_i to δ_i . Further introduce a map

$$k[x_1, \dots, x_d] \longrightarrow R = k[y_1, \dots, y_m, t_1, \dots, t_d]$$

by sending x_i to $\gamma_i = \pi_i t_i$.

1. It suffices to resolve

$$R \rightarrow C \otimes_{k[x_1, \dots, x_d]} R \rightarrow \Lambda \supset \mathfrak{q}$$

1. Set $I = (\gamma_1^e, \dots, \gamma_d^e) \subset R$.
2. It suffices to resolve

$$R/I \rightarrow C \otimes_{k[x_1, \dots, x_d]} R/I \rightarrow \Lambda/I\Lambda \supset \mathfrak{q}/I\Lambda$$

1. We denote $\mathfrak{r} \subset R = k[y_1, \dots, y_m, t_1, \dots, t_d]$ the inverse image of $\mathfrak{q}$.
2. It suffices to resolve

$$(R/I)_{\mathfrak{r}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/I)_{\mathfrak{r}} \rightarrow \Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}}$$

1. Set $J = (\pi_1^e, \dots, \pi_d^e)$ in $k[y_1, \dots, y_m]$.
2. It suffices to resolve

$$(R/JR)_{\mathfrak{p}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/JR)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}$$

1. It suffices to resolve

$$(R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow C \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

1. It suffices to resolve

$$(R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow B \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}} \supset \mathfrak{q}\Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

1. The ring $D'[t_1, \dots, t_d]$ is given the structure of an $R_{\mathfrak{p}}/\mathfrak{p}^n R_{\mathfrak{p}}$ -algebra by the given map $k[y_1, \dots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow D'$ and by sending t_i to t_i . It suffices to find a factorization

$$B \otimes_{k[x_1, \dots, x_d]} (R/\mathfrak{p}^n R)_{\mathfrak{p}} \rightarrow D'[t_1, \dots, t_d] \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$$

where the second arrow sends t_i to δ_i and induces the given homomorphism $D' \rightarrow \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$.

1. Such a factorization exists by our choice of D' above.

We now give the justification for each of the steps, except that we skip justifying the steps which just introduce notation.

Ad (D.37 / power). This is possible as $\mathfrak{q}$ is minimal over $\mathfrak{h}_A = \sqrt{H_{A/k}\Lambda}$: the finite maximal-ideal monomial argument written in D.33 applies to the Noetherian local ring $\Lambda_{\mathfrak{q}}$.

Ad (D.37 / strictly standard). For each a_j , solve the i th equation for z_{ij} after inverting a_j . This identifies B_{a_j} with a polynomial algebra over $A_{a_j}[x_1, \dots, x_d]$, smooth over $P_0 = k[x_1, \dots, x_d]$. The equations give $1 = x_i^{-2N} \sum_j z_{ij} a_j$ in B_{x_i} , so these charts cover that ring. F.67 makes it smooth over P_0 . The smooth-case proof of D.9 gives smoothness of C_{x_i} over B_{x_i} and finite free $\Omega_{C_{x_i}/P_0}$. Part (e) of D.15 supplies a threshold for each x_i ; choose a common positive c at least all the thresholds. The empty list when $d = 0$ allows $c = 1$. This uses only the proved strictly-standard implication. The finite type algebra C is finitely presented over the Noetherian polynomial ring P_0 , by AG-CA-03, Theorem 2.1: the kernel of a finite polynomial surjection is finitely generated.

Ad (D.37 / NP). This follows by applying Lemma D.35.

Ad (D.37 / choose pii). Since $k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow \Lambda_{\mathfrak{q}}$ is flat and $\mathfrak{p}\Lambda_{\mathfrak{q}} = \mathfrak{q}\Lambda_{\mathfrak{q}}$ by construction we see that $\dim(k[y_1, \dots, y_m]_{\mathfrak{p}}) = d$ by Algebra, Lemma AG-CA-11, Theorem 5.1. Thus we can find $\pi_1, \dots, \pi_d \in \mathfrak{p}$ which map to a regular system of parameters in $k[y_1, \dots, y_m]_{\mathfrak{p}}$.

Ad (D.37 / modify pii). By Algebra, Lemma AG-CA-12, Theorem 6.1 any permutation of the sequence $\pi_1, \dots, \pi_d$ is a regular sequence in $k[y_1, \dots, y_m]_{\mathfrak{p}}$. Hence $\gamma_1 = \pi_1 t_1, \dots, \gamma_d = \pi_d t_d$ is a regular sequence in $R_{\mathfrak{p}} = k[y_1, \dots, y_m]_{\mathfrak{p}}[t_1, \dots, t_d]$, see Algebra, Lemma F.75. Let $S = k[y_1, \dots, y_m] \setminus \mathfrak{p}$ so that $R_{\mathfrak{p}} = S^{-1}R$. Note that $\pi_1, \dots, \pi_d$ and $\gamma_1, \dots, \gamma_d$ remain regular sequences if we multiply our π_i by elements of S . Suppose that

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2)$$

holds for $i = 1, \dots, t$ for some $t \in \{0, \dots, d\}$. Note that $\gamma_1^e, \dots, \gamma_t^e, \gamma_{t+1}$ is a regular sequence in $S^{-1}R$ by Algebra, Lemma AG-CA-12, Proposition 1.3. Hence we see that

$$\text{Ann}_{S^{-1}R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i) = \text{Ann}_{S^{-1}R/(\gamma_1^e, \dots, \gamma_{i-1}^e)R}(\gamma_i^2).$$

Thus we get

$$\text{Ann}_{R/(\gamma_1^e, \dots, \gamma_t^e)R}(\gamma_{t+1}) = \text{Ann}_{R/(\gamma_1^e, \dots, \gamma_t^e)R}(\gamma_{t+1}^2)$$

after replacing π_{t+1} by $s\pi_{t+1}$ for some $s \in S$ by Lemma D.31. By induction on t this produces the desired sequence.

Ad (D.37 / choose deltai). Let $S = \Lambda \setminus \mathfrak{q}$ so that $\Lambda_{\mathfrak{q}} = S^{-1}\Lambda$. Set $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$. Suppose that we have a $t \in \{0, \dots, d\}$ and $\delta_1, \dots, \delta_t \in S$ and a factorization $D \rightarrow D' \rightarrow \Lambda$ as in (D.37 / choose deltai) such that (a), (b), (c) hold for $i = 1, \dots, t$. We have $\pi_{t+1}^N \in H_{A/k}\Lambda_{\mathfrak{q}}$ as $\mathfrak{q}^N \Lambda_{\mathfrak{q}} \subset H_{A/k}\Lambda_{\mathfrak{q}}$ by (D.37 / power). Hence $\pi_{t+1}^N \in H_{A/k}\bar{\Lambda}$. Hence $\pi_{t+1}^N \in H_{A/k}D'$ as $D' \rightarrow \bar{\Lambda}$ is faithfully flat, see Algebra, Lemma AG-CA-08, Theorem 2.2. Recall that $H_{A/k} = (a_1, \dots, a_r)$. Say $\pi_{t+1}^N = \sum a_j d_j$ in D' and choose $c_j \in \Lambda_{\mathfrak{q}}$ lifting $d_j \in D'$. Then $\pi_{t+1}^N = \sum c_j a_j + \epsilon$ with $\epsilon \in \mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset \mathfrak{q}^{n-N} H_{A/k}\Lambda_{\mathfrak{q}}$. Write $\epsilon = \sum a_j c'_j$ for some $c'_j \in \mathfrak{q}^{n-N} \Lambda_{\mathfrak{q}}$. Hence $\pi_{t+1}^{2N} = \sum (\pi_{t+1}^N c_j + \pi_{t+1}^N c'_j) a_j$. Note that $\pi_{t+1}^N c'_j$

maps to zero in $\bar{\Lambda}$; this trivial but key observation will ensure later that (a) holds. Now we choose $s \in S$ such that there exist $\mu_{t+1j} \in \Lambda$ such that on the one hand $\pi_{t+1}^N c_j + \pi_{t+1}^N c'_j = \mu_{t+1j}/s^{2N}$ in $S^{-1}\Lambda$ and on the other $(s\pi_{t+1})^{2N} = \sum \mu_{t+1j} a_j$ in Λ . Here are the exact denominator-clearing details. Let $u_j = \pi_{t+1}^N(c_j + c'_j)$ in $S^{-1}\Lambda$. Choose one $s_0 \in S$ with $s_0^{2N} u_j = \tilde{\mu}_j \in \Lambda$ for every j . The error $r = (s_0 \pi_{t+1})^{2N} - \sum_j a_j \tilde{\mu}_j$ vanishes in $S^{-1}\Lambda$, so some $v \in S$ annihilates r . Set $s = s_0 v$ and $\mu_{t+1j} = v^{2N} \tilde{\mu}_j$. Then the denominator identity still holds and the relation is exact because its error is $v^{2N} r = 0$. Its residue is $s^{2N} \pi_{t+1}^N d_j$, since $\pi_{t+1}^N c'_j$ vanishes modulo $\mathfrak{q}^n$. Apply D.36 to the element $s \notin \mathfrak{q}$. After an essentially smooth Artinian enlargement it puts s^h in the new D' for some $h \geq 1$. Replace s by s^h and each μ_{t+1j} by $s_{\text{old}}^{2N(h-1)} \mu_{t+1j}$. The exact relation and the denominator identity remain true after multiplication by this same factor. Their residues are now $s^{2N} \pi_{t+1}^N d_j$, in the enlarged D' . Every enlargement retains a flat local map to $\bar{\Lambda}$, hence a faithfully flat injection by AG-CA-08, Theorem 2.2; we may identify these source rings with their images. Note that $\pi_1, \dots, \pi_d$ are a regular sequence of parameters in $S^{-1}\Lambda$ by our choice of φ . Hence $\pi_1, \dots, \pi_d$ forms a regular sequence in $\Lambda_{\mathfrak{q}}$ by Algebra, Lemma AG-CA-12, Theorem 6.1. It follows that $\pi_1^e, \dots, \pi_t^e, s\pi_{t+1}$ is a regular sequence in $S^{-1}\Lambda$ by Algebra, Lemma AG-CA-12, Proposition 1.3. Thus we get

$$\text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_t^e)}(s\pi_{t+1}) = \text{Ann}_{S^{-1}\Lambda/(\pi_1^e, \dots, \pi_t^e)}((s\pi_{t+1})^2).$$

Hence we may apply Lemma D.31 to find an $s' \in S$ such that

$$\text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_t^e)}((s')^q s\pi_{t+1}) = \text{Ann}_{\Lambda/(\pi_1^e, \dots, \pi_t^e)}(((s')^q s\pi_{t+1})^2).$$

for any $q > 0$. By Lemma D.36 we can choose q and enlarge D' such that $(s')^q$ maps to an element of D' . Setting $\delta_{t+1} = (s')^q s$ we conclude that (a), (b), (c) hold for $i = 1, \dots, t+1$. For (a) note that $\lambda_{t+1j} = (s')^{2Nq} \mu_{t+1j}$ works. By induction on t we win.

Ad (D.37 / first solve). By construction the radical of $H_{(C \otimes_{k[x_1, \dots, x_d]} R)/R} \Lambda$ contains $\mathfrak{h}_A$. Namely, the elements $a_j \in H_{A/k}$ map to elements of $H_{B/k[x_1, \dots, x_d]}$, hence map to elements of $H_{C/k[x_1, \dots, x_d]}$, hence $a_j \otimes 1$ map to elements of $H_{C \otimes_{k[x_1, \dots, x_d]} R/R}$. Moreover, if we have a solution $C \otimes_{k[x_1, \dots, x_d]} R \rightarrow T \rightarrow \Lambda$ of

$$R \rightarrow C \otimes_{k[x_1, \dots, x_d]} R \rightarrow \Lambda \supset \mathfrak{q}$$

then $H_{T/R} \subset H_{T/k}$ as R is smooth over k . Hence T will also be a solution for the original situation $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$.

Ad (D.37 / second resolve). If $d = 0$, all parameter lists and I are empty, and this reduction is unnecessary. Otherwise apply D.28 to the list $\gamma_1^c, \dots, \gamma_d^c$. D.7 transports the strictly-standard witnesses x_i^c by the base map $x_i \mapsto \gamma_i$. The source and target equalities already arranged above are for γ_i and π_i^e modulo the previous eth powers. For an endomorphism u , $\ker u = \ker u^2$ implies $\ker u^r = \ker u$ for all $r \geq 1$, by the induction written in D.33. Therefore the equalities hold for γ_i^c versus $\gamma_i^{2^c}$, and for $(\pi_i^e)^c$ versus $(\pi_i^e)^{2^c}$. As $e = 8c$, these are precisely all hypotheses of D.28. Thus a resolution after quotienting by $I = (\gamma_i^e)$ lifts to the required resolution before quotienting.

Localization at the target prime and comparison of the two ideals. Put $P = k[y_1, \dots, y_m]$ and $S = P \setminus \mathfrak{p}$. In the displayed coefficient-localization notation, $R_{\mathfrak{p}}$ means $S^{-1}R = P_{\mathfrak{p}}[t_1, \dots, t_d]$; it does not mean localization of R at the extended prime $\mathfrak{p}R$. Localization at the actual prime $\mathfrak{r} \subset R$ is written $R_{\mathfrak{r}}$. Every t_i is outside $\mathfrak{r}$, because its target image is $\delta_i \notin \mathfrak{q}$. Hence, for $J = (\pi_1^e, \dots, \pi_d^e) \subset P$,

$$IR_{\mathfrak{r}} = JR_{\mathfrak{r}}, \quad I\Lambda_{\mathfrak{q}} = J\Lambda_{\mathfrak{q}}.$$

Indeed multiplication of each generator by the corresponding unit t_i^e or δ_i^e changes neither localized ideal. This makes

$$(R/I)_{\mathfrak{r}} \cong (R/JR)_{\mathfrak{r}}, \quad \Lambda_{\mathfrak{q}}/I\Lambda_{\mathfrak{q}} \cong \Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}.$$

A smooth factorization over the coefficient-localized ring $(R/JR)_{\mathfrak{p}}$ can be base changed to $(R/JR)_{\mathfrak{r}}$: every further source denominator has target image a unit, so the target map extends, and smoothness is preserved by this base change.

The latter target quotient is zero dimensional. The images of π_i generate the maximal ideal of the regular local ring $\Lambda_{\mathfrak{q}}$, and their e th powers vanish; every monomial of degree $d(e-1)+1$ contains such a power. When $d=0$, the regular local target is a field and this assertion still holds. In the quotient problem, if its singular radical is outside the target prime, its identity factorization is already a resolution. Otherwise that prime is minimal over the singular radical, because its localization is zero dimensional. D.30 therefore delocalizes the obtained smooth factorization over $(R/I)_{\mathfrak{r}}$ to a quotient resolution over R/I . The preceding D.28 argument then lifts it.

Passage from $\mathfrak{p}^n$ to J . The parameters π_i generate $\mathfrak{p}P_{\mathfrak{p}}$ and their images generate $\mathfrak{q}\Lambda_{\mathfrak{q}}$. Since $n \geq d(e-1)+1$, expansion into degree- n monomials gives

$$\mathfrak{p}^n P_{\mathfrak{p}} \subset JP_{\mathfrak{p}}, \quad \mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset J\Lambda_{\mathfrak{q}}.$$

For $d=0$, these maximal ideals are zero. Set $S_0 = P_{\mathfrak{p}}/\mathfrak{p}^n P_{\mathfrak{p}}$, $S_1 = S_0[t_1, \dots, t_d]$, and $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n \Lambda_{\mathfrak{q}}$. The displayed containments provide quotient maps from S_1 and $\bar{\Lambda}$ to $(R/JR)_{\mathfrak{p}}$ and $\Lambda_{\mathfrak{q}}/J\Lambda_{\mathfrak{q}}$. Thus if the algebra $C \otimes_{P_0} S_1$ factors through a smooth S_1 -algebra T and then $\bar{\Lambda}$, base change by $S_1 \rightarrow (R/JR)_{\mathfrak{p}}$ gives T/JT , smooth over that quotient base, with the required target map. This supplies the coefficient-localized smooth factorization used above.

Retraction and the finite smooth factorization. All enlarged rings D' map flatly and locally to the nonzero local Artinian ring $\bar{\Lambda}$, so the maps are faithfully flat and injective. The chosen images of the finite algebra generators of A belong to D' . Polynomial evaluation in them defines $A \rightarrow D'$: every defining relation maps to zero in $\bar{\Lambda}$, hence in D' by injectivity. The images of π_i , δ_i and λ_{ij} likewise belong to D' . Each δ_i is a unit there, since a local map to $\bar{\Lambda}$ sends it to an element outside the maximal ideal. Injectivity transports the exact target relation to

$$\delta_i^{2N} \pi_i^{2N} = \sum_j a_j \lambda_{ij} \quad \text{in } D'.$$

Define the map to the polynomial ring, using bars for these elements of D' ,

$$B \longrightarrow D'[t_1, \dots, t_d], \quad x_i \longmapsto \bar{\pi}_i t_i, \quad z_{ij} \longmapsto \bar{\lambda}_{ij} \bar{\delta}_i^{-2N} t_i^{2N}.$$

Every defining equation vanishes, since its image is

$$t_i^{2N} \left(\bar{\pi}_i^{2N} - \bar{\delta}_i^{-2N} \sum_j \bar{a}_j \bar{\lambda}_{ij} \right) = 0.$$

The restriction to P_0 agrees with $x_i \mapsto \pi_i t_i$ in S_1 , so this map extends uniquely to $B \otimes_{P_0} S_1$. Under the target map $t_i \mapsto \bar{\delta}_i$, the images become $\pi'_i = \delta_i \pi_i$ and λ_{ij} , respectively. Its composite is exactly the original prescribed map to $\bar{\Lambda}$, not just an unspecified factorization.

The essentially smooth map $S_0 \rightarrow D'$ expresses D' as a localization of a smooth finite-presentation S_0 -algebra. Its finite principal localizations form a filtered system of smooth S_0 -algebras; products give common successors, and fractions give its exact colimit. Polynomial base change therefore expresses $D'[t_i]$ as a filtered colimit of smooth S_1 -algebras. The algebra $B \otimes_{P_0} S_1$ is finitely presented, so F.90 factors its map through one smooth stage. Compose this stage map with the retraction $C \otimes_{P_0} S_1 \rightarrow B \otimes_{P_0} S_1$ to obtain the required smooth factorization for C .

Finally, any resulting R -resolution gives an original k -resolution as in the first-solve argument: the a_j lie in the singular ideal of the changed input, finite presentation composes with the polynomial base, and smooth composition gives $H_{T/R} \subset H_{T/k}$. The quotient, localization and lifting steps just proved return the exact factorization to $k \rightarrow A \rightarrow \Lambda$. This proves the lemma, including $d=0$ and empty parameter and coefficient lists. $\square$

The main theorem

In this section we wrap up the discussion.

Theorem D.38: popescu

Adapted from the Stacks project, smoothing.tex, lines 3112–3135.

Any regular homomorphism of Noetherian rings is a filtered colimit of smooth ring maps.

Proof. By Lemma D.26 it suffices to prove this for $k \rightarrow \Lambda$ where Λ is Noetherian and geometrically regular over k . Let $k \rightarrow A \rightarrow \Lambda$ be a factorization with A a finite type k -algebra. It suffices to construct a factorization $A \rightarrow B \rightarrow \Lambda$ with B of finite type such that $\mathfrak{h}_B = \Lambda$, see Lemma D.8. Hence we may perform Noetherian induction on the ideal $\mathfrak{h}_A$. Pick a prime $\mathfrak{q} \supset \mathfrak{h}_A$ such that $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$. It now suffices to resolve $k \rightarrow A \rightarrow \Lambda \supset \mathfrak{q}$ (as defined in the text following Situation D.27). If the characteristic of k is zero, this follows from Lemma D.33. If the characteristic of k is $p > 0$, this follows from Lemma D.37. $\square$

5. Foundation statements and their free-source proofs

All resolution comparisons, balance of Tor and connecting exact sequences used in the following flatness arguments are proved for arbitrary modules in the earlier AG-RG-S01, Lemma 1.A. This binds their homological inputs to a written programme proof before the approximation argument.

These proofs are included as mathematical text, rather than cited as substitutes for proofs. Their transitive dependencies are checked against the earlier programme locators and the open-obligation register in §6. Inclusion of a proof does not automatically close its unproved inputs.

Cotangent-complex context. For a polynomial presentation $P = R[X_s] \rightarrow S$ with kernel I , write $\mathrm{NL}(P \rightarrow S)$ for the two-term complex $[I/I^2 \rightarrow \bigoplus_s S dX_s]$ in degrees $-1, 0$. The differential takes the class of f to df . The canonical presentation is $R[X_s \mid s \in S] \rightarrow S$, $X_s \mapsto s$. For a commuting square of ring maps $R \rightarrow R'$, $S \rightarrow S'$, a presentation map is a map of these polynomial algebras lifting $S \rightarrow S'$. The full comparison and homotopy calculations below show independence of the polynomial presentation. Localization is likewise the displayed two-term localization proved at the corresponding foundation lemma, not an import of the whole cotangent chapter.

The inseparable exponent calculation. Let α be algebraic over a field F of characteristic p . Its irreducible polynomial is $f(T) = g(T^{p^s})$ for a maximal $s \geq 0$. Such s exists because f has finite positive degree. Maximality gives $g' \neq 0$. Any factorization of g would give a factorization of f , so g is irreducible, and its relatively prime derivative makes it separable. The element α^{p^s} is a root of g , hence is separable over F . This proves the field-theoretic exponent choice used in the enlargement argument.

Dimension notation. Krull dimension is the supremum of the lengths of finite chains of prime ideals; the height of a prime is the dimension of its localization. A system of parameters of a Noetherian local ring of dimension d is a list of d elements generating an ideal whose radical is the maximal ideal. Their existence is AG-CA-11, Theorem 2.1. Minimal primes, irreducible components and the dimension of polynomial local rings are supplied respectively by AG-CA-01, Theorems 4.1–4.3, AG-CA-09, Theorem 4.3, and AG-CA-11, Theorem 6.1. This paragraph fixes notation; it does not assert an additional dimension theorem. The separate dimension-in-a-neighbourhood and codimension arguments consumed below remain separately registered until matched or proved.

Definition F.1: lci

Adapted from the Stacks project, algebra.tex, lines 37874–37881.

A ring map $R \rightarrow S$ is called *syntomic*, or we say S is a *flat local complete intersection over R* if it is flat, of finite presentation, and if all of its fibre rings $S \otimes_R \kappa(\mathfrak{p})$ are local complete intersections, see Definition F.2.

Definition F.2: lci field

Adapted from the Stacks project, algebra.tex, lines 37249–37264.

Let k be a field. Let S be a finite type k -algebra.

1. We say that S is a *global complete intersection over k* if there exists a presentation $S = k[x_1, \dots, x_n]/(f_1, \dots, f_c)$ such that $\dim(S) = n - c$.
2. We say that S is a *local complete intersection over k* if there exists a covering $\operatorname{Spec}(S) = \bigcup D(g_i)$ such that each of the rings S_{g_i} is a global complete intersection over k .

We will also use the convention that the zero ring is a global complete intersection over k .

Definition F.3: lci local ring

Adapted from the Stacks project, algebra.tex, lines 37446–37458.

Let k be a field. Let S be a local k -algebra essentially of finite type over k . We say S is a *complete intersection (over k)* if there exists a local k -algebra R and elements $f_1, \dots, f_c \in \mathfrak{m}_R$ such that

1. R is essentially of finite type over k ,
2. R is a regular local ring,
3. $f_1, \dots, f_c$ form a regular sequence in R , and
4. $S \cong R/(f_1, \dots, f_c)$ as k -algebras.

Definition F.4: relative global complete intersection

Adapted from the Stacks project, algebra.tex, lines 37943–37951.

Let $R \rightarrow S$ be a ring map. We say that $R \rightarrow S$ is a *relative global complete intersection* if there exists a presentation $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$ and every nonempty fibre of $\operatorname{Spec}(S) \rightarrow \operatorname{Spec}(R)$ has dimension $n - c$. We will say “let $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$ be a relative global complete intersection” to indicate this situation.

Definition F.5: separable field extension

Adapted from the Stacks project, algebra.tex, lines 10135–10146.

Let K/k be a field extension.

1. We say K is *separably generated over k* if there exists a transcendence basis $\{x_i; i \in I\}$ of K/k such that the extension $K/k(x_i; i \in I)$ is a separable algebraic extension.
2. We say K is *separable over k* if for every subextension $k \subset K' \subset K$ with K' finitely generated over k , the extension K'/k is separably generated.

Definition F.6: smooth

Adapted from the Stacks project, algebra.tex, lines 38526–38533.

A ring map $R \rightarrow S$ is *smooth* if it is of finite presentation and the naive cotangent complex $\mathrm{NL}_{S/R}$ is quasi-isomorphic to a finite projective S -module placed in degree 0: this means that $H_1(\mathrm{NL}_{S/R}) = 0$ and that $\Omega_{S/R}$ is a finite projective S -module.

Definition F.7: smooth at prime

Adapted from the Stacks project, algebra.tex, lines 38953–38960.

Let $R \rightarrow S$ be a ring map. Let $\mathfrak{q}$ be a prime of S . We say $R \rightarrow S$ is *smooth at $\mathfrak{q}$* if there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is smooth.

Definition F.8: unramified

Adapted from the Stacks project, algebra.tex, lines 42897–42912.

Let $R \rightarrow S$ be a ring map.

1. We say $R \rightarrow S$ is *unramified* if $R \rightarrow S$ is of finite type and $\Omega_{S/R} = 0$.
2. We say $R \rightarrow S$ is *G-unramified* if $R \rightarrow S$ is of finite presentation and $\Omega_{S/R} = 0$.
3. Given a prime $\mathfrak{q}$ of S we say that S is *unramified at $\mathfrak{q}$* if there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is unramified.
4. Given a prime $\mathfrak{q}$ of S we say that S is *G-unramified at $\mathfrak{q}$* if there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is G-unramified.

Lemma F.9: CM over regular flat

Adapted from the Stacks project, algebra.tex, lines 34036–34074.

Let $R \rightarrow S$ be a local homomorphism of Noetherian local rings. Assume

1. R is regular,
2. S is Cohen-Macaulay,
3. $\dim(S) = \dim(R) + \dim(S/\mathfrak{m}_R S)$.

Then $R \rightarrow S$ is flat.

Proof. Induct on $d = \dim R$. If $d = 0$, regularity and Nakayama give $\mathfrak{m}_R = 0$, so R is a field and every R -module is flat. Suppose $d > 0$. Hypothesis (3) gives $\dim S > \dim(S/\mathfrak{m}_R S)$. The CM associated-component theorem AG-CA-12, Theorem 4.1, identifies the associated primes of S with its finitely many minimal primes $\mathfrak{q}_i$, and gives $\dim(S/\mathfrak{q}_i) = \dim S$. Hence $\mathfrak{m}_R S$ cannot be contained in any $\mathfrak{q}_i$: such a containment would make $S/\mathfrak{q}_i$ a quotient of $S/\mathfrak{m}_R S$, contradicting these dimensions.

Thus the contractions $\mathfrak{p}_i = R \cap \mathfrak{q}_i$ are proper subprimes of $\mathfrak{m}_R$. By F.80, choose $x \in \mathfrak{m}_R \setminus \mathfrak{m}_R^2$ outside all $\mathfrak{p}_i$. It avoids every associated prime of S , so multiplication by x on S is injective. The regular-quotient and CM dimension calculations in AG-CA-12, Sections 2 and 4, make S/xS CM of dimension $\dim S - 1$. AG-CA-14, Theorem 1.1 and Corollary 1.2, make x a nonzerodivisor on R and R/xR regular of dimension $d - 1$.

The induced quotient map is local and its closed fibre is still $S/\mathfrak{m}_R S$, since $x \in \mathfrak{m}_R$. Subtracting one from both total dimensions in (3) gives

$$\dim(S/xS) = \dim(R/xR) + \dim(S/\mathfrak{m}_R S).$$

Every induction hypothesis therefore holds, so S/xS is flat over R/xR .

The exact free resolution $0 \rightarrow R \xrightarrow{x} R \rightarrow R/xR \rightarrow 0$ identifies $\mathrm{Tor}_1^R(R/xR, S)$ with the kernel of multiplication by x on S , which is zero. Apply AG-CA-08, Corollary 4.4, to the proper ideal $(x) \subset R$ and the module S , finite over the stipulated Noetherian local overring S . Its quotient is flat and its first Tor vanishes, so S is flat over R . $\square$

Lemma F.10: NAK

Adapted from the Stacks project, algebra.tex, lines 3732–3813.

Let R be a ring with Jacobson radical $\mathrm{rad}(R)$. Let M be an R -module. Let $I \subset R$ be an ideal.

1. If $IM = M$ and M is finite, then there exists an $f \in 1 + I$ such that $fM = 0$.
2. If $IM = M$, M is finite, and $I \subset \mathrm{rad}(R)$, then $M = 0$.
3. If $N, N' \subset M$, $M = N + IN'$, and N' is finite, then there exists an $f \in 1 + I$ such that $fM \subset N$ and $M_f = N_f$.
4. If $N, N' \subset M$, $M = N + IN'$, N' is finite, and $I \subset \mathrm{rad}(R)$, then $M = N$.
5. If $N \rightarrow M$ is a module map, $N/IN \rightarrow M/IM$ is surjective, and M is finite, then there exists an $f \in 1 + I$ such that $N_f \rightarrow M_f$ is surjective.
6. If $N \rightarrow M$ is a module map, $N/IN \rightarrow M/IM$ is surjective, M is finite, and $I \subset \mathrm{rad}(R)$, then $N \rightarrow M$ is surjective.
7. If $x_1, \dots, x_n \in M$ generate M/IM and M is finite, then there exists an $f \in 1 + I$ such that $x_1, \dots, x_n$ generate M_f over R_f .
8. If $x_1, \dots, x_n \in M$ generate M/IM , M is finite, and $I \subset \mathrm{rad}(R)$, then M is generated by $x_1, \dots, x_n$.
9. If $IM = M$, I is nilpotent, then $M = 0$.
10. If $N, N' \subset M$, $M = N + IN'$, and I is nilpotent then $M = N$.
11. If $N \rightarrow M$ is a module map, I is nilpotent, and $N/IN \rightarrow M/IM$ is surjective, then $N \rightarrow M$ is surjective.
12. If $\{x_\alpha\}_{\alpha \in A}$ is a set of elements of M which generate M/IM and I is nilpotent, then M is generated by the x_α .

Proof. Proof of (F.10 / nakayama). Choose generators $y_1, \dots, y_m$ of M over R . For each i we can write $y_i = \sum z_{ij} y_j$ with $z_{ij} \in I$ (since $M = IM$). In other words $\sum_j (\delta_{ij} - z_{ij}) y_j = 0$. Let f be the determinant of the $m \times m$ matrix $A = (\delta_{ij} - z_{ij})$. Note that $f \in 1 + I$ (since the matrix A is entrywise congruent to the $m \times m$ identity matrix modulo I). By Lemma F.70 (1), there exists an $m \times m$ matrix B such that $BA = f1_{m \times m}$. Writing out we see that $\sum_i b_{hi} a_{ij} = f \delta_{hj}$ for all h and j ; hence, $\sum_{i,j} b_{hi} a_{ij} y_j = \sum_j f \delta_{hj} y_j = f y_h$ for every h . In other words, $0 = f y_h$ for every h (since each i satisfies $\sum_j a_{ij} y_j = 0$). This implies that f annihilates M .

By Lemma F.34 an element of $1 + \mathrm{rad}(R)$ is an invertible element of R . Hence we see that (F.10 / nakayama) implies (2). We obtain (3) by applying (1) to M/N which is finite as N'

is finite. We obtain (4) by applying (2) to M/N which is finite as N' is finite. We obtain (5) by applying (3) to M and the submodules $\text{im}(N \rightarrow M)$ and M . We obtain (6) by applying (4) to M and the submodules $\text{im}(N \rightarrow M)$ and M . We obtain (7) by applying (5) to the map $R^{\oplus n} \rightarrow M$, $(a_1, \dots, a_n) \mapsto a_1x_1 + \dots + a_nx_n$. We obtain (8) by applying (6) to the map $R^{\oplus n} \rightarrow M$, $(a_1, \dots, a_n) \mapsto a_1x_1 + \dots + a_nx_n$.

Part (9) holds because if $M = IM$ then $M = I^n M$ for all $n \geq 0$ and I being nilpotent means $I^n = 0$ for some $n \gg 0$. Parts (10), (11), and (12) follow from (9) by the arguments used above. $\square$

Lemma F.11: NL homotopy

Adapted from the Stacks project, algebra.tex, lines 36700–36815.

Let $R \rightarrow R'$ and $S \rightarrow S'$ be a commutative square of ring maps with structure maps $R \rightarrow S$ and $R' \rightarrow S'$. Let $\alpha : P \rightarrow S$ and $\alpha' : P' \rightarrow S'$ be presentations.

1. There exists a morphism of presentations from α to α' .
2. Any two morphisms of presentations induce homotopic morphisms of complexes $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha')$.
3. The construction is compatible with compositions of morphisms of presentations (see proof for exact statement).
4. If $R \rightarrow R'$ and $S \rightarrow S'$ are isomorphisms, then for any map φ of presentations from α to α' the induced map $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha')$ is a homotopy equivalence and a quasi-isomorphism.

In particular, comparing α to the canonical presentation described in the cotangent-complex context above we conclude there is a quasi-isomorphism $\text{NL}(\alpha) \rightarrow \text{NL}_{S/R}$ well defined up to homotopy and compatible with all functorialities (up to homotopy).

Proof. Since P is a polynomial algebra over R we can write $P = R[x_a, a \in A]$ for some set A . As α' is surjective, we can choose for every $a \in A$ an element $f_a \in P'$ such that $\alpha'(f_a) = \phi(\alpha(x_a))$. Let $\varphi : P = R[x_a, a \in A] \rightarrow P'$ be the unique R -algebra map such that $\varphi(x_a) = f_a$. This gives the morphism in (1).

Let φ and φ' be morphisms of presentations from α to α' . Let $I = \ker(\alpha)$ and $I' = \ker(\alpha')$. We have to construct the diagonal map h in the diagram

$$\begin{array}{ccc}
I/I^2 \mathrm{d} & \xrightarrow{\sim} & \Omega_{P/R} \otimes_P S \\
I/I^2 \mathrm{d} & \xrightarrow{\varphi'_1} & I'/(I')^2 \mathrm{d} \\
I/I^2 \mathrm{d} & \xrightarrow{\varphi_1} & I'/(I')^2 \mathrm{d} \\
\Omega_{P/R} \otimes_P S & \xrightarrow{\varphi'_0} & \Omega_{P'/R'} \otimes_{P'} S' \\
\Omega_{P/R} \otimes_P S & \xrightarrow{\varphi_0} & \Omega_{P'/R'} \otimes_{P'} S' \\
\Omega_{P/R} \otimes_P S & \xrightarrow{h} & I'/(I')^2 \mathrm{d} \\
I'/(I')^2 \mathrm{d} & \xrightarrow{\sim} & \Omega_{P'/R'} \otimes_{P'} S'
\end{array}$$

where the vertical maps are induced by φ and φ' , and h must satisfy

$$\varphi_1 - \varphi'_1 = h \circ \mathrm{d} \quad \text{and} \quad \varphi_0 - \varphi'_0 = \mathrm{d} \circ h$$

Consider the map $\varphi - \varphi' : P \rightarrow P'$. Since both φ and φ' are compatible with α and α' we obtain $\varphi - \varphi' : P \rightarrow I'$. This implies that $\varphi, \varphi' : P \rightarrow P'$ induce the same P -module structure on $I'/(I')^2$, since $\varphi(p)i' - \varphi'(p)i' = (\varphi - \varphi')(p)i' \in (I')^2$. Also $\varphi - \varphi'$ is R -linear and

$$(\varphi - \varphi')(fg) = \varphi(f)(\varphi - \varphi')(g) + (\varphi - \varphi')(f)\varphi'(g)$$

Hence the induced map $D : P \rightarrow I'/(I')^2$ is an R -derivation. Thus we obtain a canonical map $h : \Omega_{P/R} \otimes_P S \rightarrow I'/(I')^2$ such that $D = h \circ d$. For $f \in I$, the definition of the universal derivation gives $h(df) = D(f) = \varphi(f) - \varphi'(f) \bmod (I')^2$, so the degree-one difference is $h \circ d$. On a basis element dx_a of $\Omega_{P/R} \otimes_P S$, one has $d(h(dx_a)) = d(\varphi(x_a) - \varphi'(x_a))$, exactly the difference of the degree-zero maps. These two identities verify the homotopy on generators, hence on the whole complexes.

Suppose that we have a commutative diagram

$$\begin{array}{ccc} S & \xrightarrow{\phi} & S' \\ S' & \xrightarrow{\phi'} & S'' \\ R & \rightarrow & R' \\ R & \rightarrow & S \\ R' & \rightarrow & S' \\ R' & \rightarrow & R'' \\ R'' & \rightarrow & S'' \end{array}$$

and that

1. $\alpha : P \rightarrow S$,
2. $\alpha' : P' \rightarrow S'$, and
3. $\alpha'' : P'' \rightarrow S''$

are presentations. Suppose that

1. $\varphi : P \rightarrow P'$ is a morphism of presentations from α to α' and
2. $\varphi' : P' \rightarrow P''$ is a morphism of presentations from α' to α'' .

Then it is immediate that $\varphi' \circ \varphi : P \rightarrow P''$ is a morphism of presentations from α to α'' and that the induced map $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha'')$ of naive cotangent complexes is the composition of the maps $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha')$ and $\text{NL}(\alpha') \rightarrow \text{NL}(\alpha'')$ induced by φ and φ' .

In the simple case of complexes with 2 terms a quasi-isomorphism is just a map that induces an isomorphism on both the cokernel and the kernel of the maps between the terms. Note that homotopic maps of 2 term complexes (as explained above) define the same maps on kernel and cokernel. Hence if φ is a map from a presentation α of S over R to itself, then the induced map $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha)$ is a quasi-isomorphism being homotopic to the identity by part (2). To prove (4) in full generality, consider a morphism φ' from α' to α which exists by (1). The compositions $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha') \rightarrow \text{NL}(\alpha)$ and $\text{NL}(\alpha') \rightarrow \text{NL}(\alpha) \rightarrow \text{NL}(\alpha')$ are homotopic to the identity maps by (2) and (3), hence these maps are homotopy equivalences by definition. It follows formally that both maps $\text{NL}(\alpha) \rightarrow \text{NL}(\alpha')$ and $\text{NL}(\alpha') \rightarrow \text{NL}(\alpha)$ are quasi-isomorphisms. Explicitly, if uv and vu are homotopic to the respective identities, their maps on kernel and cokernel are identities, since a homotopy contributes hd in degree one and dh in degree zero. The maps induced by u and v on both groups are therefore inverse isomorphisms. $\square$

Lemma F.12: Noetherian field extension

Adapted from the Stacks project, algebra.tex, lines 6331–6347.

If A is a Noetherian k -algebra and K/k is a finitely generated field extension, then $A \otimes_k K$ is Noetherian.

Proof. Choose finitely many field generators and let $B \subset K$ be the k -algebra they generate. Its fraction field is K , so $K = (B \setminus \{0\})^{-1}B$. A finite polynomial presentation of B identifies $A \otimes_k B$ with a quotient of a polynomial algebra in finitely many variables over A . Hilbert basis AG-CA-03, Theorem 2.1, makes it Noetherian. Tensor and the fraction universal property identify $A \otimes_k K$ with its localization at the images of the nonzero elements of B . The same Noetherian permanence proof in that earlier theorem makes this localization Noetherian. In the special case of a finite field extension, one can alternatively use its finite field basis to see a finite free A -module, whose ideals are finite submodules. This proves the full finitely generated form consumed in the regularity tests. $\square$

Lemma F.13: another variant local criterion flatness

Adapted from the Stacks project, algebra.tex, lines 24942–24990.

For a local square of Noetherian rings $R \rightarrow S$, $R' \rightarrow S'$ with S' a localization of $S \otimes_R R'$, a finite S -module M , and a proper ideal $I \subset R$, assume M/IM is flat over R/I and the map $\mathrm{Tor}_1^R(M, R/I) \rightarrow \mathrm{Tor}_1^{R'}(M \otimes_S S', R'/IR')$ is zero. Then $M \otimes_S S'$ is R' -flat.

Proof. Write a free presentation $0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0$ over R , allowing an infinite free F . Put $T = \mathrm{Tor}_1^R(M, R/I)$ and let $\bar{K}$ be the image of K/IK in F/IF . The two sequences $0 \rightarrow T \rightarrow K/IK \rightarrow \bar{K} \rightarrow 0$ and $0 \rightarrow \bar{K} \rightarrow F/IF \rightarrow M/IM \rightarrow 0$ are exact. The assumed flatness of the last quotient makes $\bar{K}$ flat by AG-CA-07, Theorem 6.1. Tensor these sequences with any R/I -module N ; their quotient flatness preserves their left injections and identifies

$$\mathrm{Tor}_1^R(M, N) = T \otimes_{R/I} N.$$

This identity follows from the same free-presentation kernel formula for Tor , so does not assume that K is flat.

After base change to R' , let K' be the image of $K \otimes_R R'$ in $F \otimes_R R'$. The map onto K' is surjective. Tensor it with R'/IR' ; every cycle in $\ker(K' \otimes R'/IR' \rightarrow F \otimes R'/IR')$ lifts to an element of $K \otimes_R R'/IR'$, and that lift is automatically a cycle because the two free-module targets are identical. Therefore $\mathrm{Tor}_1^R(M, R'/IR') \rightarrow \mathrm{Tor}_1^{R'}(M \otimes_R R', R'/IR')$ is surjective. By the preceding identity its source is $T \otimes_R R'$. Localizing at the stipulated prime in the algebra commutes with this kernel calculation by exact localization. Thus $T \otimes_R R'$ surjects onto the Tor group of $M' = M \otimes_S S'$. The assumed zero map annihilates this surjection, so that Tor group is zero.

Base change of flatness makes $M'/IR'M'$ flat over R'/IR' . AG-CA-08, Corollary 4.4, is the proved proper-ideal local criterion under these Noetherian local and finite-over- S' hypotheses; it now makes M' flat over R' . This proves the transition criterion rather than invoking an unproved change-of-rings spectral sequence. $\square$

Lemma F.14: base change relative global complete intersection

Adapted from the Stacks project, algebra.tex, lines 38076–38106.

Relative global complete-intersection presentations are preserved by base change, principal localization, and replacing the base by a localization already inverted in the algebra.

Proof. First record the required field-dimension argument. For a finite-type algebra over a field and any extension field, dimension is unchanged. Quotient by its finitely many minimal primes; their

intersection is nilpotent by AG-CA-03, Proposition 2.2. This remains nilpotent after field extension. For each resulting domain, Noether normalization AG-CA-09, Corollary 3.2, gives an injective finite polynomial subalgebra. Field tensor is exact, so the injection and finiteness persist. Dimension is preserved by an injective integral map, AG-CA-05, Theorem 5.1. Thus each component and their maximum retain dimension. This proof applies to an arbitrary, possibly infinite, field extension.

The fibre after any base change is a field extension of the fibre over the contracted base prime. Consequently its dimension stays $n - c$. For localization, each original fibre is equidimensional of dimension $n - c$: the minimal-prime height calculation in F.77, applied just to the polynomial algebra over the fibre field, proves this without any base Noetherianity. A nonempty principal open in an irreducible finite-type field scheme has unchanged dimension, since its coordinate domain has the same fraction field and AG-CA-09, Theorem 4.2, identifies dimension with transcendence degree. Therefore every nonempty localized fibre still has dimension $n - c$. The presentation $S_g = R[X, Z]/(f_1, \dots, f_c, hZ - 1)$ has one additional variable and equation, hence the same expected dimension. Finally if a base element is already inverted in S , tensoring with that base localization gives back S , proving the third assertion. $\square$

Lemma F.15: change base NL

Adapted from the Stacks project, algebra.tex, lines 36963–36981.

For a flat base change $R \rightarrow R'$, the two-term conormal complex of a polynomial presentation $P \twoheadrightarrow S$ base changes to the corresponding complex for $P \otimes_R R' \twoheadrightarrow S \otimes_R R'$.

Proof. Let I be the presentation kernel. Flatness identifies the new kernel with $I \otimes_R R'$, since it preserves the exact sequence $0 \rightarrow I \rightarrow P \rightarrow S \rightarrow 0$. Its square is the image of $I^2 \otimes_R R'$, by writing products of generators. The cokernel description therefore identifies the conormal module with $(I/I^2) \otimes_R R'$. The differential module of the polynomial ring is free on the same dX_s before and after base change. Both differentials send the class of a polynomial f to df , so these are an isomorphism of complexes. Comparison of polynomial presentations and its explicit homotopies, proved above, give the canonical homotopy equivalence. $\square$

Lemma F.16: characterize etale

Adapted from the Stacks project, algebra.tex, lines 40942–40975.

Let $R \rightarrow S$ be a ring map. Let $\mathfrak{q}$ be a prime of S lying over a prime $\mathfrak{p}$ of R . If

1. $R \rightarrow S$ is of finite presentation,
2. $R_{\mathfrak{p}} \rightarrow S_{\mathfrak{q}}$ is flat
3. $\mathfrak{p}S_{\mathfrak{q}}$ is the maximal ideal of the local ring $S_{\mathfrak{q}}$, and
4. the field extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite separable,

then $R \rightarrow S$ is étale at $\mathfrak{q}$.

Proof. Put $k = \kappa(\mathfrak{p})$ and $L = \kappa(\mathfrak{q})$. The local fibre ring is $S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}} = L$ by (3). Since L/k is finite, F.56 supplies $g \notin \mathfrak{q}$ for which $\mathfrak{q}$ is the only point of the fibre of S_g . Replace S by S_g ; finite presentation is preserved by F.35. The fibre $B = S \otimes_R k$ has a unique prime, so is local: every element outside its unique maximal ideal is a unit. Its localization there is $S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}$, hence $B = L$.

Here is the finite separable fibre calculation, including its finite presentation. Choose a finite generating tower $k = E_0 \subset E_1 \subset \dots \subset E_t = L$, with $E_i = E_{i-1}(\alpha_i)$. A finite k -basis provides such a generating list. Let $p_i(T)$ be the monic minimal polynomial of α_i over E_{i-1} . Polynomial division gives $E_i = E_{i-1}[T]/(p_i)$; the quotient is a field by polynomial Bezout. These full field

calculations are proved in [AG-RG-S02, Lemma 4.A](#). Represent the coefficients of p_i by polynomials in the preceding generators, obtaining $P_i(T_1, \dots, T_t)$. Successively taking these monic quotients gives the finite presentation

$$L = k[T_1, \dots, T_t]/(P_1, \dots, P_t).$$

Put $I = (P_1, \dots, P_t) \subset k[T_1, \dots, T_t]$. The minimal polynomial of α_i over E_{i-1} divides its separable minimal polynomial over k . A repeated root of the former would be a repeated root of the latter, so $p'_i(\alpha_i) \neq 0$, as also proved in [Lemma 4.A](#). The Jacobian of this presentation over L is triangular with these nonzero diagonal entries, hence invertible. The map $L^t \rightarrow I/I^2$ given by the equation classes is surjective, and its composite with the conormal differential to L^t is this invertible matrix. Its kernel is therefore zero, and the conormal differential is an isomorphism. Thus $H_1(\mathrm{NL}_{L/k}) = 0$ and $\Omega_{L/k} = 0$, proving that L/k is smooth by [Definition F.6](#). The empty tower gives the same assertion for $L = k$.

The stated local flatness and finite presentation now satisfy [F.45](#), so another principal neighbourhood of $\mathfrak{q}$ is smooth over R . By the arbitrary-base standard-chart assertion of [F.83](#), shrink once more to a standard smooth presentation

$$S = R[X_1, \dots, X_n]/(f_1, \dots, f_c).$$

Every element inverted in these shrinkings has nonzero image in the field L , so the fibre remains L . By [F.85\(2\)](#), $\Omega_{S/R}$ is free on $dX_{c+1}, \dots, dX_n$. The complete base-change calculation in [AG-CA-16](#), Theorem 2.1, identifies its tensor with L with $\Omega_{L/k} = 0$. Hence $L^{n-c} = 0$; a coordinate vector in a nonzero field module shows that this forces $n = c$.

For clarity, the square Jacobian gives étaleness directly. Given an R -map $S \rightarrow D/J$ with $J^2 = 0$, lift its variable images to a tuple $a \in D^n$. Its error vector $f(a)$ lies in J^n . The Jacobian matrix $A(a)$ is invertible: its determinant is a unit modulo J , and a lift of a modular inverse gives $1 + j$, whose inverse is $1 - j$; the cofactor identity of [F.70](#) then gives the matrix inverse. Put $b = a - A(a)^{-1}f(a)$. Polynomial expansion gives $f(b) = f(a) + A(a)(b - a) = 0$, since terms with two corrections vanish in J^2 . Any other lifting root is $b + \epsilon$ with $\epsilon \in J^n$; the same expansion gives $A(b)\epsilon = 0$, so $\epsilon = 0$. Thus every square-zero lifting test has a unique solution. Finite presentation makes this algebra étale by the programme definition, so the original map is étale at $\mathfrak{q}$. Empty variable lists use determinant one and the unique empty tuple. No Noetherian or perfect-field assumption was used. $\square$

Lemma F.17: characterize formally smooth field extension

Adapted from the Stacks project, algebra.tex, lines 45959–45969.

For a field extension K/k , formal smoothness is equivalent to $H_1(\mathrm{NL}_{K/k}) = 0$.

Proof. Use any polynomial presentation, including an infinite one. Its first homology is the kernel of the conormal differential by definition. If it vanishes, the conormal sequence is short exact and its quotient $\Omega_{K/k}$ is a vector space, hence free and projective; choosing and lifting a basis splits it. [AG-CA-17](#), Theorem 3.1, with arbitrary variable sets then proves formal smoothness. Conversely that theorem gives split exactness and hence zero kernel. Polynomial comparison already proved identifies this homology with the presentation-independent $H_1(L_{K/k})$ used in the statement. $\square$

Lemma F.18: characterize smooth kbar

Adapted from the Stacks project, algebra.tex, lines 39962–40064.

Let k be an algebraically closed field. Let S be a finite type k -algebra. Let $\mathfrak{m} \subset S$ be a maximal ideal. The following are equivalent:

1. The ring $S_{\mathfrak{m}}$ is a regular local ring.

2. We have $\dim_{\kappa(\mathfrak{m})} \Omega_{S/k} \otimes_S \kappa(\mathfrak{m}) \leq \dim(S_{\mathfrak{m}})$.
3. We have $\dim_{\kappa(\mathfrak{m})} \Omega_{S/k} \otimes_S \kappa(\mathfrak{m}) = \dim(S_{\mathfrak{m}})$.
4. There exists a $g \in S$, $g \notin \mathfrak{m}$ such that S_g is smooth over k . In other words S/k is smooth at $\mathfrak{m}$.

Proof. By AG-CA-18, Proposition 3.1, the residue field at this closed point is k and

$$\Omega_{S/k} \otimes_S k \cong \mathfrak{m}/\mathfrak{m}^2.$$

AG-CA-14, Section 1, gives $\dim S_{\mathfrak{m}} \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$, with equality precisely for a regular local ring. Consequently (1), (2) and (3) are equivalent. The same complete earlier Proposition 3.1 proves that smoothness at this rational closed point implies regularity, so (4) implies (1).

We write the converse construction explicitly. Put $d = \dim S_{\mathfrak{m}}$ and choose a presentation $S = k[x_1, \dots, x_n]/I$ after subtracting the residue values of the generators. Its polynomial preimage is $\mathfrak{n} = (x_1, \dots, x_n)$. The conormal sequence at the rational point is

$$I/\mathfrak{n}I \longrightarrow \mathfrak{n}/\mathfrak{n}^2 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow 0.$$

Its right space has dimension d by regularity. Choose $c = n - d$ polynomials $f_1, \dots, f_c \in I$ whose linear classes form a basis of the kernel. Let $J = (f_1, \dots, f_c)$ and $T = k[x]/J$, with maximal ideal $\mathfrak{m}' = \mathfrak{n}/J$. The same sequence for J shows that the cotangent space of $T_{\mathfrak{m}'}$ has dimension d . Its surjection onto $S_{\mathfrak{m}}$ implies $\dim T_{\mathfrak{m}'} \geq d$, while the embedding-dimension bound gives the reverse inequality. Thus $T_{\mathfrak{m}'}$ is regular of dimension d .

This local ring is a domain by AG-CA-14, Theorem 1.1. Its surjection to $S_{\mathfrak{m}}$ has zero kernel. Indeed, if a nonzero proper kernel existed, every prime containing it would be nonzero. Any chain of length d in the quotient would lift to such a chain in the domain, and prepending (0) would make a chain of length $d + 1$, contradicting its dimension d . A local Noetherian ring of finite dimension has a chain attaining that integer dimension, so the quotient could not have dimension d .

The Jacobian rows of the f_j at $x = 0$ are their independent linear coefficient vectors. Some $c \times c$ minor Δ is therefore nonzero there; for $c = 0$ take $\Delta = 1$. After reordering variables, T_{Δ} is standard smooth: adjoin an inverse variable u and the equation $u\Delta - 1$. The selected Jacobian block on the original c pivot variables and u has determinant Δ^2 , which is a unit. This is the principal-localization calculation of F.85, so T_{Δ} is smooth over k .

It remains to identify this neighbourhood with one in S . The kernel of $T \rightarrow S$ is finite because T is Noetherian, and its localization at $\mathfrak{m}'$ is zero by the preceding local isomorphism. For each of its finitely many generators choose an annihilating denominator outside $\mathfrak{m}'$. Their product $a \notin \mathfrak{m}'$ kills the entire kernel. Surjectivity and exact localization give $T_a \cong S_a$. Both a and Δ avoid the point, so

$$T_{a\Delta} \cong S_{a\Delta}.$$

Principal localization preserves the standard smooth presentation by F.85(4). Therefore the right side is smooth over k , proving (4) with $g = a\Delta \notin \mathfrak{m}$. This also covers the empty equation list. All assertions retain the stipulated algebraically closed base field. $\square$

Lemma F.19: charpoly

Adapted from the Stacks project, algebra.tex, lines 2856–2879.

If M is an $n \times n$ matrix over a commutative ring, its characteristic polynomial $P(T) = \det(TI - M)$ satisfies $P(M) = 0$.

Proof. The cofactor expansion proves $(TI - M) \operatorname{adj}(TI - M) = P(T)I$ over the polynomial ring. Write the adjugate as $\sum_{j=0}^{n-1} C_j T^j$ and $P(T) = \sum_{j=0}^n p_j T^j$. Comparing coefficients gives $C_{n-1} = I$,

$C_{j-1} - MC_j = p_j I$ for $1 \leq j < n$, and $-MC_0 = p_0 I$. Multiplying these equations respectively by M^j and adding telescopes to $\sum_{j=0}^n p_j M^j = 0$. The case $n = 0$ is the identity on the zero module and causes no exception. $\square$

Lemma F.20: codimension

Adapted from the Stacks project, algebra.tex, lines 29578–29596.

For a surjection of finite-type k -algebras $S' \rightarrow S$ and corresponding primes, $\dim_{x'} \operatorname{Spec} S' - \dim_x \operatorname{Spec} S = \operatorname{height}(\mathfrak{p}') - \operatorname{height}(\mathfrak{p})$.

Proof. AG-CA-09, Theorem 6.1, gives point dimension as local dimension plus the transcendence degree of the residue field. The two residue fields here are equal, since the map is a surjection and the primes correspond. The local dimensions are the heights of those primes, by their chain definition. Subtracting the two identities cancels exactly the equal transcendence degrees, giving the asserted formula. $\square$

Lemma F.21: colimit eventually flat

Adapted from the Stacks project, algebra.tex, lines 34105–34145.

Let $R_\lambda \rightarrow S_\lambda$ be a compatible directed system of local maps of Noetherian local rings, with local transition maps, and put $R = \operatorname{colim} R_\lambda$, $S = \operatorname{colim} S_\lambda$. Assume that for every $\mu \geq \lambda$, S_μ is the stipulated localization of $S_\lambda \otimes_{R_\lambda} R_\mu$. Let M_λ be finite S_λ -modules with the compatible identifications $M_\mu = M_\lambda \otimes_{S_\lambda} S_\mu$, and put $M = \operatorname{colim} M_\lambda$. The constructions F.59 and F.60 supply exactly such models. If M is flat over R , then some model M_μ is flat over R_μ .

Proof. Fix one stage λ and set $T = \operatorname{Tor}_1^{R_\lambda}(M_\lambda, R_\lambda/\mathfrak{m}_\lambda)$. The free-ideal kernel formula identifies it with $\ker(\mathfrak{m}_\lambda \otimes_{R_\lambda} M_\lambda \rightarrow M_\lambda)$. It is finite over S_λ : the ideal and module are finite, and kernels of maps of finite modules over that Noetherian algebra are finite. Choose finitely many generators. In the limit their images in $(\mathfrak{m}_\lambda R) \otimes_R M$ vanish, because multiplication into M is injective by its R -flatness. Tensor and filtered colimits commute by their finite-expression universal properties. Consequently each such vanishing holds at a later stage; a common successor μ kills all generators in $(\mathfrak{m}_\lambda R_\mu) \otimes_{R_\mu} M_\mu$.

Thus the transition map from T to $\operatorname{Tor}_1^{R_\mu}(M_\mu, R_\mu/\mathfrak{m}_\lambda R_\mu)$ is zero. The module $M_\lambda/\mathfrak{m}_\lambda M_\lambda$ is flat over the field $R_\lambda/\mathfrak{m}_\lambda$. The preceding transition criterion applies to the local square $R_\lambda \rightarrow S_\lambda$, $R_\mu \rightarrow S_\mu$, whose localized tensor description was proved in the model construction, with ideal $I = \mathfrak{m}_\lambda$. It makes M_μ flat over R_μ , as required. $\square$

Lemma F.22: colimit formally etale

Adapted from the Stacks project, algebra.tex, lines 42732–42747.

A filtered colimit of formally etale R -algebras is formally etale over R .

Proof. Given a map from the colimit to A/J with $J^2 = 0$, restrict to each stage. Formal etaleness gives a unique lift from that stage to A . Uniqueness makes these maps compatible with every transition arrow, so their colimit supplies a lift. Two such lifts coincide on every stage and hence on the colimit. No finite-presentation assumption is used. $\square$

Lemma F.23: colimit syntomic

Adapted from the Stacks project, algebra.tex, lines 46090–46123.

Every field extension K/k is a filtered union of global complete-intersection k -subalgebras. If it is

separable, it is a filtered union of smooth subalgebras.

Proof. Given a finite subset $E \subset K$, form its finitely generated subfield L/k . Choose a transcendence basis $T_1, \dots, T_d$ and a finite algebraic generator tower $y_1, \dots, y_r$. At the i th step the monic minimal polynomial has coefficients in the preceding field, which is spanned over $k(T)$ by the standard monomials in the preceding generators. Represent those coefficients by polynomials in the preceding y_j with rational-function coefficients in T . Clearing a common denominator $h(T)$ puts all of these monic polynomials P_i in $k[T, 1/h, Y]$. Their successive quotients are free over the preceding coefficient ring on $1, Y_i, \dots, Y_i^{\deg P_i - 1}$, by monic division. In particular each monic polynomial is a nonzerodivisor in the relevant polynomial ring: its highest nonzero coefficient in a putative product cannot disappear. The list is therefore regular.

The resulting algebra C is finite free over $k[T, 1/h]$. After passing to $k(T)$ it is exactly the field L , by the minimal-polynomial tower. Since the coefficient ring is a domain and C is free, $C \rightarrow C \otimes k(T) = L$ is injective. Each member of E is a linear combination of the finite standard monomial basis with coefficients in $k(T)$; enlarging h clears those finitely many denominators and puts E in C . This finite injective extension has dimension d by AG-CA-05, Theorem 5.1, and AG-CA-09, Theorem 4.2. Adjoining the localization variable with $hZ - 1$ gives a polynomial presentation with $d + r + 1$ variables and $r + 1$ equations and dimension d , so it is a global complete intersection. The collection of these subalgebras is directed: to dominate two, apply the construction to their finite generator lists. Every field element is included, proving that the union is K .

In the separable case, Definition F.5 makes the finitely generated subfield containing E separably generated. The forward construction in F.64 gives a smooth finite-type domain inside this subfield with the same fraction field. Express the finitely many elements of E as fractions and invert the product of their nonzero denominators. This principal localization remains smooth and contains E . Applying the same construction to two finite generator lists proves directedness, and every field element is included. This proves the second assertion using the definition directly. $\square$

Lemma F.24: colimits NL

Adapted from the Stacks project, algebra.tex, lines 36983–36999.

The canonical two-term conormal complex commutes with filtered colimits of ring maps.

Proof. Use the canonical polynomial presentation with one variable for each element of the target ring. Every polynomial is a finite sum in finitely many variables and coefficients, so it occurs at one stage of the filtered system. Its being in the kernel is a finite equality in the target, hence holds at a later stage. An element in the square of the kernel is a finite sum of products of such kernel elements, and that equality also holds at a common later stage. These facts prove that the polynomial rings, their kernels, their squared kernels, and their conormal quotients have the indicated filtered colimits. Polynomial differentials are the direct sums on the same variables, and their derivatives are finite expressions, so the differentials commute with these identifications. Here is the exactness calculation for the module colimits being used. For a compatible system of short exact sequences, an element from the left whose image is zero in the middle colimit has zero image at a later stage; injectivity at that stage makes its left representative zero as well. A middle representative mapping to zero in the right colimit maps to zero at a later stage, where exactness lifts it from the left module. Every right representative lifts from the middle module at its own stage by surjectivity. Addition, scalar multiplication and comparison of representatives are checked at a common stage. Thus filtered module colimits preserve the entire short exact sequence even when transition maps are not injective. Applied to the kernels and quotients just described, this gives the asserted equality of two-term complexes. $\square$

Lemma F.25: complete local Noetherian domain finite over regular

Adapted from the Stacks project, algebra.tex, lines 46654–46712.

A complete Noetherian local domain A is finite over a regular complete local subring A_0 with the same residue field. One may take A_0 to be a power-series ring over a field or a Cohen ring.

Proof. The coefficient-ring existence proof is AG-CA-19S, Theorem 5.1. Its characteristic- p lifting argument uses the full p -basis construction proved in Lemma F.119. In particular, that proof divides $T^p - a^p = (T - a)^p$ inside the extension field and uses the coefficient $-da$ of a putative proper factor, with $1 \leq d < p$, to prove the required degree- p assertion. This supplies the field step for arbitrary residue fields. Since A is a domain its coefficient image is a field in equal characteristic, and is an embedded Cohen DVR in mixed characteristic. Let $D = \dim A$. In the field case choose D parameters x_i , by AG-CA-11, Theorem 2.1, and set $A_0 = k[[X_1, \dots, X_D]]$. In the mixed case p is a nonzerodivisor; AG-CA-11, Theorem 3.2, gives $\dim A/pA = D - 1$. Choose parameters $x_1, \dots, x_{D-1}$ of that quotient and set $A_0 = C[[X_1, \dots, X_{D-1}]]$. Here C is its coefficient Cohen ring. In either case the image I of the maximal ideal of A_0 is primary to the maximal ideal of A . A power of the latter lies in I by AG-CA-03, Proposition 2.2, making the two filtrations cofinal. Thus A is I -adically complete and separated. Evaluation of a power series at the x_i is well defined, by the same convergence and multiplication proof as AG-CA-19S, Theorem 6.1.

Choose lifts $a_1, \dots, a_s$ of a basis of the finite-dimensional residue quotient A/I over k . Every $a \in A$ can be expressed modulo I as a coefficient combination of them. Express the error in I as a sum of its chosen generators times elements of A , and repeat the same residue decomposition on those elements. After n steps this constructs coefficients $c_{jn} \in A_0$, compatible modulo its n -th maximal-ideal power, such that $a - \sum_j c_{jn} a_j \in I^n$. Completeness of A_0 gives limits c_j and separatedness gives $a = \sum_j c_j a_j$. Hence A is finite as an A_0 -module. The quotient A/I is finite-dimensional because its maximal ideal is nilpotent and its finitely many successive maximal-ideal layers are finite-dimensional, by AG-CA-03, Theorem 4.2.

The power-series ring A_0 is the maximal-ideal completion of the regular polynomial local ring over its coefficient field or DVR, as is checked directly on the finite quotients. Regularity of the polynomial local ring is AG-CA-14, Proposition 3.3, with the DVR base regular of dimension one; its dimension is the number of displayed parameters, including p in mixed characteristic, by AG-CA-11, Theorem 5.1. Completion preserves dimension and regularity by AG-CA-19, Theorem 4.1. Thus A_0 is a regular local domain of dimension D . If the evaluation kernel contained $0 \neq f$, the quotient $A_0/(f)$ would have dimension $D - 1$ by AG-CA-11, Theorem 3.2. The injective finite integral map $A_0/\ker \rightarrow A$ preserves dimension by AG-CA-05, Theorem 5.1, but $\dim(A_0/\ker) \leq D - 1$ would contradict $\dim A = D$. The evaluation is therefore injective. Its residue map is the coefficient-field identification, completing the proof. For $D = 0$ the domain is a field and the assertion takes $A_0 = A$. $\square$

Lemma F.26: completion at quasi finite prime

Adapted from the Stacks project, algebra.tex, lines 32412–32464.

Let R be Noetherian, $R \rightarrow S$ finite type, and q a quasi-finite prime over p . Then $\widehat{R}_p \otimes_R S \cong \widehat{S}_q \times D$, with its first projection the natural completed local map.

Proof. The earlier full affine conductor theorem Zariski Main, Theorem C4.3 gives an integral subalgebra $C \subset S$ and $g \in C \setminus q$ with $C_g = S_g$. Choose finitely many numerators from C that, together with g^{-1} , generate S_g over R . Let $S' \subset C$ be generated by them and g . Its generators are integral, hence it is a finite R -module algebra, and $S'_g = S_g$. Put $q' = q \cap S'$. Localizing this equality again gives $S'_{q'} = S_q$, so their maximal-adic completions agree.

The finite-extension completion argument of F.27 gives a product

$$\widehat{R}_p \otimes_R S' = A \times D', \quad A = \widehat{S'_{q'}} = \widehat{S}_q.$$

Let e be the central idempotent of this selected factor. Its image splits $T = \widehat{R_p} \otimes_R S$ as $eT \times (1-e)T$. The element g maps to a unit in A , because it is outside q' , so it is a unit in eT through the map $A \rightarrow eT$. Consequently $eT = (eT)_g$. But localizing the tensor map at g is an isomorphism, since $S'_g = S_g$. Its selected e -factor therefore identifies $(eT)_g$ with $A_g = A$. This proves $eT = A$ and the desired product decomposition, including its adic factor.

To check its first projection is the natural map, it agrees on $\widehat{R_p}$ and on S' by the finite-extension completion construction. For any $s \in S$, equality after localizing at g supplies an exponent a with $g^a s \in S'$: write its fraction using S'_g , then multiply by one further power to kill the localization error in S . Since g is a unit in $\widehat{S_g}$, agreement on $g^a s$ and on g forces agreement on s . Pure tensors generate T , so its projection is precisely the natural completed-local map. This is an explicit adic comparison, not an identification of finite scheme completion with ring completion. $\square$

Lemma F.27: completion finite extension

Adapted from the Stacks project, algebra.tex, lines 24144–24184.

For a finite map $R \rightarrow S$ of Noetherian rings and a prime $\mathfrak{p}$ of R , with primes $\mathfrak{q}_i$ above it, one has $\widehat{R_p} \otimes_R S = \prod_i \widehat{S_{\mathfrak{q}_i}}$.

Proof. Localize the base at $\mathfrak{p}$. All maximal ideals of the finite algebra lie above this maximal ideal, by AG-CA-05, Theorem 3.3, and there are finitely many by Theorem 3.4. For each n , the finite algebra $S/\mathfrak{p}^n S$ is a finite module over the Artinian ring $R/\mathfrak{p}^n$, hence Artinian by AG-CA-03, Theorem 3.3 and its finite maximal-ideal-layer proof. The product theorem 4.2 of that lesson gives $S/\mathfrak{p}^n S = \prod_i S_{\mathfrak{q}_i}/\mathfrak{p}^n S_{\mathfrak{q}_i}$. On each factor $\mathfrak{p} S_{\mathfrak{q}_i}$ has radical $\mathfrak{q}_i S_{\mathfrak{q}_i}$. A power of the latter ideal lies in the former, by AG-CA-03, Proposition 2.2; their adic filtrations are cofinal, so their inverse limits are the same. Taking the finite product of the inverse limits now gives the asserted product of local completions. Finally the completion-tensor equality for the finite R_p -module S_p is AG-CA-19, Theorem 3.1. $\square$

Lemma F.28: compose finite type

Adapted from the Stacks project, algebra.tex, lines 547–571.

Finite-type and finitely presented ring maps compose. If $R \rightarrow S' \rightarrow S$ and S is finite type over R , it is finite type over S' . If additionally S/R is finitely presented and S'/R finite type, then S/S' is finitely presented.

Proof. Generators of S'/R , followed by generators of S/S' , generate the composite. For presentations $S' = R[Y]/(a)$ and $S = S'[X]/(b)$, lift the finitely many coefficients of b to $R[Y]$ to obtain the finite composite presentation $R[Y, X]/(a, \tilde{b})$. The same generators of S/R generate it over S' . Finally write $S = R[X]/(f_1, \dots, f_m)$ and $S' = R[Y]/I$, with finitely many Y_i mapping to polynomials $h_i(X)$ in S . Then $S = S'[X]/(f_j, Y_i - h_i(X))$. To check this equality, substitute $Y_i = h_i(X)$; every relation in I vanishes in S and so belongs to the ideal of the f_j . This proves both inverse maps directly without assuming I finitely generated. $\square$

Lemma F.29: compose smooth

Adapted from the Stacks project, algebra.tex, lines 39039–39053.

A composition of smooth ring maps is smooth.

Proof. Let $R \rightarrow S \rightarrow T$ be smooth maps. Lemma F.28 gives finite presentation of T/R . The Jacobi–Zariski sequence F.42 places $H_1(\mathrm{NL}_{T/R})$ between $H_1(\mathrm{NL}_{S/R} \otimes_S T)$ and $H_1(\mathrm{NL}_{T/S})$. The latter is zero by smoothness. For the former, the last assertion of F.42 identifies it with $H_1(\mathrm{NL}_{S/R}) \otimes_S T = 0$:

its two Tor hypotheses hold because $\Omega_{S/R}$ is projective. Hence $H_1(\mathrm{NL}_{T/R}) = 0$. The same sequence gives

$$0 \longrightarrow \Omega_{S/R} \otimes_S T \longrightarrow \Omega_{T/R} \longrightarrow \Omega_{T/S} \longrightarrow 0.$$

The last module is projective, so this sequence splits. The first is finite projective because a finite direct-summand presentation for $\Omega_{S/R}$ remains such a presentation after tensoring with T ; the last is finite projective by smoothness of T/S . Their direct sum is therefore finite projective. Definition F.6 proves smoothness of T/R . $\square$

Lemma F.30: compose standard smooth

Adapted from the Stacks project, algebra.tex, lines 38815–38878.

A composite of standard smooth ring maps is standard smooth.

Proof. Write $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$ and $T = S[Y_1, \dots, Y_m]/(g_1, \dots, g_d)$, with their selected Jacobian minors invertible. Lift the coefficients of the g_i to $R[X]$. The resulting R -presentation of T has equations f_i, g_j and pivot variables the c selected X variables followed by the d selected Y variables. Its Jacobian matrix is block lower triangular, since $\partial f_i / \partial Y_j = 0$. Its determinant is the product of the two given unit determinants. This is the required standard smooth presentation. Any principal localization is encoded by one more variable z and equation $hz - 1$, whose additional diagonal Jacobian entry h is a unit. Thus the assertion covers the localized presentations as well. $\square$

Lemma F.31: composition essentially of finite type

Adapted from the Stacks project, algebra.tex, lines 13129–13138.

Essentially finite type, and essentially finite presentation, are preserved by composition.

Proof. Write the first algebra as $U^{-1}R[Y]/(a)$, and the second before its final localization as a polynomial algebra over it modulo finitely many chosen relations in the finite-presentation case. Only finitely many denominators from U occur in those relations; absorb them in one element $u \in U$. This finite algebra then descends to $(R[Y]/(a))_u$, and its further localization gives the original composite. The composite is therefore a localization of a finite-type R -algebra, or of a finitely presented one in the second case. The remaining elements of U and the second localization set are simply included in the final localization set. The finite-type argument only needs the finite generator list and uses the same construction with the possibly infinite relation ideal. $\square$

Lemma F.32: composition syntomic

Adapted from the Stacks project, algebra.tex, lines 38404–38453.

Let $R \rightarrow S, S \rightarrow S'$ be ring maps.

1. If R is Noetherian and $R \rightarrow S$ and $S \rightarrow S'$ are syntomic, then $R \rightarrow S'$ is syntomic.
2. If $R \rightarrow S$ and $S \rightarrow S'$ are relative global complete intersections, then $R \rightarrow S'$ is a relative global complete intersection.

Proof. Proof of (2). Say $R \rightarrow S$ and $S \rightarrow S'$ are relative global complete intersections and we have presentations $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$ and $S' = S[y_1, \dots, y_m]/(h_1, \dots, h_d)$ as in Definition F.4. Then

$$S' \cong R[x_1, \dots, x_n, y_1, \dots, y_m]/(f_1, \dots, f_c, h'_1, \dots, h'_d)$$

for some lifts $h'_j \in R[x_1, \dots, x_n, y_1, \dots, y_m]$ of the h_j . Hence it suffices to bound the dimensions of the fibre rings. Thus we may assume $R = k$ is a field. In this case we see that we have a ring, namely S , which is of finite type over k and equidimensional of dimension $n - c$, and a finite type ring map $S \rightarrow S'$ all of whose nonempty fibre rings are equidimensional of dimension $m - d$. Then, by Lemma AG-CA-11, Theorem 5.1 for example applied to localizations at maximal ideals of S' , we see that $\dim(S') \leq n - c + m - d$ as desired.

We will reduce part (1) to part (2). Assume R Noetherian and $R \rightarrow S$ and $S \rightarrow S'$ are syntomic. Their finite presentation makes S and S' Noetherian by the earlier Hilbert basis theorem, so both applications of F.87 have their stated base hypothesis. Let $\mathfrak{q}' \subset S'$ be a prime ideal lying over $\mathfrak{q} \subset S$. By Lemma F.87 there exists a $g' \in S'$, $g' \notin \mathfrak{q}'$ such that $S \rightarrow S'_{g'}$ is a relative global complete intersection. Similarly, we find $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is a relative global complete intersection. By Lemma F.14 the ring map $S_g \rightarrow S'_{gg'}$ is a relative global complete intersection. By part (2) we see that $R \rightarrow S'_{gg'}$ is a relative global complete intersection and $gg' \notin \mathfrak{q}'$. Since $\mathfrak{q}'$ was arbitrary combining Lemmas F.87 and F.62 we see that $R \rightarrow S'$ is syntomic (this also uses that the spectrum of S' is quasi-compact, see Lemma AG-CA-01, Proposition 2.3). $\square$

Lemma F.33: conormal module localize

Adapted from the Stacks project, algebra.tex, lines 37203–37225.

For finite polynomial presentations $R[X_1, \dots, X_n] \twoheadrightarrow S$ and $R[Y_1, \dots, Y_m] \twoheadrightarrow S_g$ with kernels I, J , there is an isomorphism $(I/I^2)_g \oplus S_g^m \cong J/J^2 \oplus S_g^n$.

Proof. Put $C = S_g$, $P = R[X_1, \dots, X_n]$, and $Q = R[Y_1, \dots, Y_m]$. Choose $f \in P$ representing g . Then $P_f \twoheadrightarrow C$ has kernel I_f , by exact localization. Choose $u_j \in P_f$ lifting the images of the Y_j in C , and choose $v_i \in Q$ lifting the images of the X_i . The latter lifts exist because $Q \twoheadrightarrow C$ is surjective.

Map $W = P[Y_1, \dots, Y_m]$ to C using both specified presentations. Its kernel is $L = (J, X_i - v_i(Y))$: quotienting by this ideal eliminates the X_i and gives exactly $Q/J = C$. Translation makes $W = Q[s_1, \dots, s_n]$, with $s_i = X_i - v_i(Y)$, and $L = (J, s)$. Expanding a polynomial by its s degree proves

$$L/L^2 \cong J/J^2 \oplus C^m.$$

Its constant part is read modulo J^2 , its linear coefficients modulo J , and all terms of degree at least two belong to L^2 . These rules define inverse maps, and justify the formula over every base ring.

The image of f in C is the unit g , so localization leaves this conormal module unchanged. Ideal products and quotients localize by their fraction formulas, giving $L_f/L_f^2 = L/L^2$. On the other hand $W_f = P_f[Y]$, and its kernel for the same map to C is $(I_f, Y_j - u_j)$. Eliminating the Y_j gives $P_f/I_f = C$, which proves this kernel equality. Translating these variables and applying the identical constant/linear calculation gives

$$L_f/L_f^2 \cong (I_f/I_f^2) \oplus C^m \cong (I/I^2)_g \oplus C^m.$$

Combine the two displayed isomorphisms. No free summand is cancelled and no flatness of C is assumed. The formulas also hold for empty variable lists and $C = 0$. $\square$

Lemma F.34: contained in radical

Adapted from the Stacks project, algebra.tex, lines 3648–3678.

An ideal I is contained in the Jacobson radical exactly when every element of $1 + I$ is a unit. In that case units modulo I lift to units.

Proof. If I lies in every maximal ideal, $1 + i$ lies in none and therefore is a unit: a nonunit generates a proper ideal, which lies in a maximal ideal by Zorn's lemma. Conversely, if $I \not\subset \mathfrak{m}$, write $1 = i + x$ with $i \in I$, $x \in \mathfrak{m}$; then $x = 1 - i$ is a nonunit, contradiction. If $ab = 1 + i$ after lifting a modular inverse, its unit status gives an inverse $b(ab)^{-1}$ for a . $\square$

Lemma F.35: cover upstairs

Adapted from the Stacks project, algebra.tex, lines 4346–4405.

Let $g_1, \dots, g_r \in S$ generate the unit ideal. Finite type and finite presentation of S/R can be checked on the finitely many S_{g_i} .

Proof. First record the module detection calculation on a finite principal cover. Let $t_1, \dots, t_r$ generate one in a ring T , say $\sum c_i t_i = 1$, and suppose a T -module M has $M_{t_i} = 0$ for each i . For any $m \in M$, choose positive integers e_i with $t_i^{e_i} m = 0$, using the equality criterion for localization. Expand

$$1 = \left(\sum c_i t_i \right)^N, \quad N = 1 + \sum_i (e_i - 1).$$

Every monomial contains some $t_i^{e_i}$, so multiplying by m gives $m = 0$. Thus $M = 0$, with no finiteness hypothesis on M . An empty cover forces $T = 0$, when this assertion also holds.

Choose $h_i \in S$ with $\sum h_i g_i = 1$. For finite type, let $S_0 \subset S$ be the R -subalgebra generated by all h_i, g_i and the numerators of finite R -algebra generator lists for the S_{g_i} . The maps $(S_0)_{g_i} \rightarrow S_{g_i}$ are injective by the localization equality criterion and surjective by these chosen generators. The g_i generate one in S_0 as well. The cokernel S/S_0 , viewed as an S_0 -module, vanishes on this principal cover, so the preceding calculation gives $S = S_0$.

For finite presentation first use finite type to write $S = P/J$, $P = R[X_1, \dots, X_n]$. Lift g_i, h_i to $u_i, v_i \in P$. The surjection $P[Z] \rightarrow S_{g_i}$ has kernel $JP[Z] + (u_i Z - 1)$, since its quotient is $S[Z]/(g_i Z - 1) = S_{g_i}$ by the fraction universal property. This kernel is finitely generated because S_{g_i} is finitely presented. Here is the precise presentation-independence calculation: given $C = R[Y_1, \dots, Y_m]/(a_1, \dots, a_s)$ and a finite polynomial surjection $R[W_1, \dots, W_t] \rightarrow C$, choose polynomials $U_j(W)$ representing the images of Y_j and $V_l(W)$ representing the images of W_l . The ideal

$$(a_j(U(W)), W_l - V_l(U(W)))_{j,l}$$

is its kernel. Indeed substitution $Y_j \mapsto U_j(W)$ defines a map from C to the displayed quotient, inverse to its map to C : the first list makes substitution well-defined, the second makes the composite on the W_l the identity, and the chosen images make the other composite on the Y_j the identity.

Expand each finite kernel generator for $P[Z] \rightarrow S_{g_i}$ as a finite combination of elements of J and $u_i Z - 1$. Collect the finitely many elements $a_{ij} \in J$ that occur. These elements together with $u_i Z - 1$ generate exactly that kernel: they belong to it and generate each of its chosen generators. Set $J_0 = (\sum v_i u_i - 1, a_{ij}) \subset J$. The classes of the u_i generate one in P/J_0 . The equality of the preceding kernels shows that every map $(P/J_0)_{u_i} \rightarrow (P/J)_{u_i}$ is an isomorphism. Its global kernel J/J_0 therefore vanishes by the module detection calculation, so $J = J_0$ is finitely generated.

Conversely, if S is of finite type, then S_{g_i} is generated by its finite algebra generators and g_i^{-1} . If $S = P/(a_1, \dots, a_s)$ is finitely presented and $u_i \in P$ lifts g_i , then $S_{g_i} = P[Z]/(a_1, \dots, a_s, u_i Z - 1)$ is finitely presented. This proves both directions, including zero rings and empty generator lists. $\square$

Lemma F.36: criterion flatness fibre

Adapted from the Stacks project, algebra.tex, lines 34374–34496.

Let R, S, S' be local rings and let $R \rightarrow S \rightarrow S'$ be local ring homomorphisms. Let M be an

S' -module. Let $\mathfrak{m} \subset R$ be the maximal ideal. Assume

1. The ring maps $R \rightarrow S$ and $R \rightarrow S'$ are essentially of finite presentation.
2. The module M is of finite presentation over S' .
3. The module M is not zero.
4. The module $M/\mathfrak{m}M$ is a flat $S/\mathfrak{m}S$ -module.
5. The module M is a flat R -module.

Then S is flat over R and M is a flat S -module.

Proof. Use the local Noetherian models of F.59: $R = \operatorname{colim} R_\lambda$, with local transition maps, maximal ideals $\mathfrak{p}_\lambda$, and each R_λ essentially of finite type over $\mathbf{Z}$. The finite presentations of S and S' and the finitely many coefficients giving $S \rightarrow S'$ can be put into a common model. Explicitly, present S by a finite variable and relation list over R before its localization. Present S' by another such finite list. For every generator of S choose in S' a polynomial representative with its finitely many localization denominators. Put those denominators in the localization set, and add the finite equations identifying those representatives with the chosen S -generators. Eliminating the latter recovers the presentation of S' over R . Thus we obtain compatible local models

$$S_\lambda = (R_\lambda[x]/(f_\lambda))_{\mathfrak{q}_\lambda}, \quad S'_\lambda = (R_\lambda[x, y]/(f_\lambda, g_\lambda))_{\mathfrak{q}'_\lambda}$$

with both relation lists included, and with the primes contracted from the actual local rings. The finite equality witnesses required for the maps hold after one common successor, by the finite-expression colimit calculation in F.59. All these rings are Noetherian.

Choose a finite presentation $(S')^s \xrightarrow{A} (S')^t \rightarrow M \rightarrow 0$. The model construction F.60 puts the finitely many matrix entries into one stage. Set $M_\lambda = \operatorname{coker}((S'_\lambda)^s \xrightarrow{A_\lambda} (S'_\lambda)^t)$. For $\mu \geq \lambda$, the maps

$$S_\lambda \otimes_{R_\lambda} R_\mu \rightarrow S_\mu, \quad S'_\lambda \otimes_{S_\lambda} S_\mu \rightarrow S'_\mu$$

are localizations, and $M_\lambda \otimes_{S'_\lambda} S'_\mu \cong M_\mu$. The same descriptions hold with the final rings. In particular $M_\lambda = 0$ would imply $M = 0$, so every such model is nonzero. Each M_λ is finite over S'_λ , as required for the Noetherian local criterion.

Apply F.21 to the R -flat module M . It gives a stage where M_λ is flat over R_λ . That property persists at every later stage: the displayed module and ring identities express the later module as a base change followed by localization, both of which preserve flatness by their exact tensor functors. The local transitions give $\mathfrak{m} = \operatorname{colim} \mathfrak{p}_\lambda$, and right exact tensor products give the quotient colimits

$$S/\mathfrak{m}S = \operatorname{colim} S_\lambda/\mathfrak{p}_\lambda S_\lambda, \quad S'/\mathfrak{m}S' = \operatorname{colim} S'_\lambda/\mathfrak{p}_\lambda S'_\lambda, \quad M/\mathfrak{m}M = \operatorname{colim} M_\lambda/\mathfrak{p}_\lambda M_\lambda.$$

Reducing the same polynomial presentations and matrices gives, for $\mu \geq \lambda$, the localization

$$\frac{S'_\lambda}{\mathfrak{p}_\lambda S'_\lambda} \otimes_{S_\lambda/\mathfrak{p}_\lambda S_\lambda} \frac{S_\mu}{\mathfrak{p}_\mu S_\mu} \longrightarrow \frac{S'_\mu}{\mathfrak{p}_\mu S'_\mu}$$

and the corresponding exact module base-change isomorphism. These are precisely the quotient local finite-presentation models in F.60. Apply F.21 to the assumed flat module $M/\mathfrak{m}M$ over $S/\mathfrak{m}S$. After taking a common successor, both model flatness assertions hold: M_λ is R_λ -flat and $M_\lambda/\mathfrak{p}_\lambda M_\lambda$ is flat over $S_\lambda/\mathfrak{p}_\lambda S_\lambda$.

Now F.37 applies to the local Noetherian maps $R_\lambda \rightarrow S_\lambda \rightarrow S'_\lambda$ and the nonzero finite module M_λ . It gives R_λ -flatness of S_λ and S_λ -flatness of M_λ . The final ring S is a localization of $S_\lambda \otimes_{R_\lambda} R$, and the final module M is the corresponding localization of $M_\lambda \otimes_{S_\lambda} S$. Tensor base change and localization give the two asserted flatness conclusions. No Noetherian hypothesis has been imposed on the original rings. $\square$

Lemma F.37: criterion flatness fibre Noetherian

Adapted from the Stacks project, algebra.tex, lines 24999–25045.

Let $R \rightarrow S \rightarrow T$ be local maps of Noetherian local rings, and $M \neq 0$ finite over T . If M is R -flat and $M/\mathfrak{m}_R M$ is flat over $S/\mathfrak{m}_R S$, then M is faithfully flat over S and S is flat over R .

Proof. Put $I = \mathfrak{m}_R S$. The map $\mathfrak{m}_R \otimes_R M \rightarrow I \otimes_S M$ is surjective, because the images of $\mathfrak{m}_R$ generate I . Its composite to M is injective by R -flatness, so both maps are injective. Thus $\mathrm{Tor}_1^S(S/I, M) = 0$. The ideal form of the local flatness criterion, AG-CA-08, Corollary 4.4, applied to the proper ideal I gives S -flatness. Nakayama gives $M/\mathfrak{m}_T M \neq 0$, hence $M/\mathfrak{m}_S M \neq 0$; the faithful-flatness criterion for modules, AG-CA-08, Theorem 1.1, gives faithfulness over S .

Tensor the exact sequence $0 \rightarrow \mathrm{Tor}_1^R(R/\mathfrak{m}_R, S) \rightarrow \mathfrak{m}_R \otimes_R S \rightarrow I \rightarrow 0$ with this flat S -module. Its first term maps to the kernel of the already injective map $\mathfrak{m}_R \otimes_R M \rightarrow I \otimes_S M$, hence is zero. Faithfulness kills the original Tor module. AG-CA-08, Theorem 4.2, then gives R -flatness of S . All modules to which the local criterion is applied are finite over the stipulated Noetherian local overring. $\square$

Lemma F.38: descent regular

Adapted from the Stacks project, algebra.tex, lines 48353–48371.

Regularity descends along faithfully flat maps of Noetherian rings.

Proof. Fix a source prime and choose a target prime above it, which exists by faithful flatness. The localized map $A \rightarrow B$ is flat local, hence faithfully flat by AG-CA-08, Theorem 2.1. The target is regular local. Take a free resolution of the source residue field k over A . After tensoring with B it remains a free resolution of $k \otimes_A B$, by flatness. Consequently

$$\mathrm{Tor}_i^A(k, k) \otimes_A B = \mathrm{Tor}_i^B(k \otimes_A B, k \otimes_A B).$$

AG-CA-14, Theorem 2.2, gives finite global dimension of B , so these groups vanish for all sufficiently large i . Faithfulness gives vanishing of the original source Tor groups. The minimal-free-resolution calculation AG-CA-13, Theorem 2.2, makes the source residue field have finite projective dimension; AG-CA-14, Theorem 2.2, then makes A regular. Doing this at every prime proves the assertion. $\square$

Lemma F.39: dimension fibres bounded open upstairs

Adapted from the Stacks project, algebra.tex, lines 32674–32690.

For a finite-type map of Noetherian rings $R \rightarrow S$, if the fibre dimension at a point q is n , some principal neighbourhood of q has fibre point-dimension at most n everywhere. This is the exact form used here.

Proof. Let $p = q \cap R$. Choose a principal neighbourhood whose $\kappa(p)$ -fibre has dimension exactly n , using the point-dimension definition and the finite-type field component formula. Apply Noether normalization to that fibre. The coordinate changes in AG-CA-09, Lemma 2.1 and Corollary 3.2, use polynomials with integer coefficients in the algebra generators: at each step replace a generator by it minus a sufficiently high power of the last generator, so one selected relation has a constant nonzero leading coefficient; dividing that coefficient makes the last generator integral. Inducting yields n such parameters and finiteness over their polynomial algebra. Thus these parameters lift to elements $t_i \in S$, and $R[T_1, \dots, T_n] \rightarrow S$, $T_i \mapsto t_i$, has a finite fibre over p . At the chosen point it is quasi-finite: its further fibre over the contracted polynomial prime is a finite algebra over that prime's residue field.

The earlier affine conductor proof [Zariski Main, Theorem C5.1](#) proves the open quasi-finite locus from its full Theorem C4.3. Shrink once more around q so this polynomial map is quasi-finite

everywhere. In a fibre over any base prime p' , its algebra U is quasi-finite over $\kappa(p')[T_1, \dots, T_n]$. At a minimal prime η of U , the residue field is finite over that of its contracted polynomial prime, by the isolated-fibre criterion proved above. Its transcendence degree over $\kappa(p')$ is therefore at most n . AG-CA-09, Theorem 4.2, gives $\dim(U/\eta)$ equal to that transcendence degree. Taking the maximum over the finitely many minimal primes gives $\dim U \leq n$. Every fibre point-dimension is at most that fibre dimension. This proves the claimed neighbourhood bound. Its earlier conductor input includes the polynomial-normality proof in [Zariski Main over an arbitrary base](#), [Corollary B1.2](#), so no old external proof tag is being substituted. $\square$

Lemma F.40: essentially of finite type into artinian local

Adapted from the Stacks project, algebra.tex, lines 13151–13207.

Let $R \rightarrow S$ be a ring map. Assume S is an Artinian local ring with maximal ideal $\mathfrak{m}$. Then

1. $R \rightarrow S$ is finite if and only if $R \rightarrow S/\mathfrak{m}$ is finite,
2. $R \rightarrow S$ is of finite type if and only if $R \rightarrow S/\mathfrak{m}$ is of finite type.
3. $R \rightarrow S$ is essentially of finite type if and only if the composition $R \rightarrow S/\mathfrak{m}$ is essentially of finite type.

Proof. If $R \rightarrow S$ is finite, then $R \rightarrow S/\mathfrak{m}$ is finite by Lemma [AG-CA-05, Theorem 1.2](#). Conversely, assume $R \rightarrow S/\mathfrak{m}$ is finite. As S has finite length over itself (Lemma [AG-CA-03, Theorem 4.2](#)) we can choose a filtration

$$0 \subset I_1 \subset \dots \subset I_n = S$$

by ideals such that $I_i/I_{i-1} \cong S/\mathfrak{m}$ as S -modules. Thus S has a filtration by R -submodules I_i such that each successive quotient is a finite R -module. Thus S is a finite R -module by Lemma [F.43](#).

If $R \rightarrow S$ is of finite type, then $R \rightarrow S/\mathfrak{m}$ is of finite type by Lemma [F.28](#). Conversely, assume that $R \rightarrow S/\mathfrak{m}$ is of finite type. Choose $f_1, \dots, f_n \in S$ which map to generators of $S/\mathfrak{m}$. Then $A = R[x_1, \dots, x_n] \rightarrow S$, $x_i \mapsto f_i$ is a ring map such that $A \rightarrow S/\mathfrak{m}$ is surjective (in particular finite). Hence $A \rightarrow S$ is finite by part (1) and we see that $R \rightarrow S$ is of finite type by Lemma [F.28](#).

If $R \rightarrow S$ is essentially of finite type, then $R \rightarrow S/\mathfrak{m}$ is essentially of finite type by Lemma [F.31](#). Conversely, assume that $R \rightarrow S/\mathfrak{m}$ is essentially of finite type. Suppose $S/\mathfrak{m}$ is the localization of $R[x_1, \dots, x_n]/I$. Choose $f_1, \dots, f_n \in S$ whose congruence classes modulo $\mathfrak{m}$ correspond to the congruence classes of $x_1, \dots, x_n$ modulo I . Consider the map $R[x_1, \dots, x_n] \rightarrow S$, $x_i \mapsto f_i$ with kernel J . Set $A = R[x_1, \dots, x_n]/J \subset S$ and $\mathfrak{p} = A \cap \mathfrak{m}$. Note that $A/\mathfrak{p} \subset S/\mathfrak{m}$ is equal to the image of $R[x_1, \dots, x_n]/I$ in $S/\mathfrak{m}$. Hence $\kappa(\mathfrak{p}) = S/\mathfrak{m}$. Thus $A_{\mathfrak{p}} \rightarrow S$ is finite by part (1). We conclude that S is essentially of finite type by Lemma [F.31](#). $\square$

Lemma F.41: étale standard smooth

Adapted from the Stacks project, algebra.tex, lines 40737–40760.

Any étale ring map is standard smooth. More precisely, if $R \rightarrow S$ is étale, then there exists a presentation $S = R[x_1, \dots, x_n]/(f_1, \dots, f_n)$ such that the image of $\det(\partial f_j / \partial x_i)$ is invertible in S .

Proof. If $S = 0$, use $S = R[X]/(1)$. Its one-by-one Jacobian is zero, a unit in the zero ring, giving the required presentation. Suppose $S \neq 0$.

Choose a finite polynomial presentation $S = P/I$, $P = R[x_1, \dots, x_n]$. The complete formal/conormal comparison in AG-CA-17, Theorem 3.1 and Section 4, makes its conormal

differential

$$I/I^2 \longrightarrow S^{\oplus n}, \quad \bar{f} \longmapsto \sum_i \frac{\partial f}{\partial x_i} dx_i$$

an isomorphism: étaleness means smoothness and zero differentials, and the split conormal sequence therefore has zero cokernel and zero kernel. In particular I/I^2 is finite free.

Apply F.55. It supplies a possibly different finite presentation $S = R[y_1, \dots, y_m]/J$, with $J = (g_1, \dots, g_c)$ and these equation classes a conormal basis. Apply the same étale conormal criterion to this new presentation as well. It identifies $S^c \cong J/J^2$ with S^m . Reducing this isomorphism at any maximal ideal of the nonzero ring S gives an isomorphism of finite-dimensional vector spaces, hence $c = m$.

In these bases the conormal isomorphism is the square Jacobian matrix H of the actual new equations. A matrix inverse exists over S , so multiplicativity of determinants gives $\det(H) \det(H^{-1}) = 1$. Thus its determinant is a unit. Relabeling the m variables and equations gives exactly the asserted standard smooth presentation, with equal numbers of variables and equations. $\square$

Lemma F.42: exact sequence NL

Adapted from the Stacks project, algebra.tex, lines 36838–36913.

Let $A \rightarrow B \rightarrow C$ be ring maps. Choose a presentation $\alpha : A[x_s, s \in S] \rightarrow B$ with kernel I . Choose a presentation $\beta : B[y_t, t \in T] \rightarrow C$ with kernel J . Let $\gamma : A[x_s, y_t] \rightarrow C$ be the induced presentation of C with kernel K . Then we get a canonical commutative diagram

$$\begin{array}{c} 0 \rightarrow \Omega_{A[x_s]/A} \otimes C \\ \Omega_{A[x_s]/A} \otimes C \rightarrow \Omega_{A[x_s, y_t]/A} \otimes C \\ \Omega_{A[x_s, y_t]/A} \otimes C \rightarrow \Omega_{B[y_t]/B} \otimes C \\ \Omega_{B[y_t]/B} \otimes C \rightarrow 0 \\ I/I^2 \otimes C \rightarrow K/K^2 \\ I/I^2 \otimes C \rightarrow \Omega_{A[x_s]/A} \otimes C \quad , \\ K/K^2 \rightarrow J/J^2 \\ K/K^2 \rightarrow \Omega_{A[x_s, y_t]/A} \otimes C \\ J/J^2 \rightarrow 0 \\ J/J^2 \rightarrow \Omega_{B[y_t]/B} \otimes C \end{array}$$

with exact rows. We get the following exact sequence of homology groups

$$H_1(\text{NL}_{B/A} \otimes_B C) \rightarrow H_1(L_{C/A}) \rightarrow H_1(L_{C/B}) \rightarrow C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$$

of C -modules extending the sequence of Lemma AG-CA-16, Theorem 3.1. If $\text{Tor}_1^B(\Omega_{B/A}, C) = 0$ and $\text{Tor}_2^B(\Omega_{B/A}, C) = 0$, then $H_1(\text{NL}_{B/A} \otimes_B C) = H_1(L_{B/A}) \otimes_B C$.

Proof. On differentials, the first map sends dx_s to dx_s , the second sends every dx_s to zero and dy_t to dy_t . On conormal modules the first map sends $(f \bmod I^2) \otimes c$ to $cf \bmod K^2$, using any lift of c to $A[x_s, y_t]$; the ambiguity changes the answer by K^2 . The second sends $g \bmod K^2$ to its image in J/J^2 . Derivatives of these representatives verify every commutative square. The connecting homology map is obtained by lifting a cycle from J/J^2 to K/K^2 , taking its derivative (which lies in the image of the left differential module), and reducing modulo the image of $I/I^2 \otimes C$. Changing the lift changes this by a boundary. The same lifting calculation proves exactness at each homology term. The exactness of the top row follows as the dx_s, dy_t form a basis for the middle module. The map γ factors

$$A[x_s, y_t] \rightarrow B[y_t] \rightarrow C$$

with surjective first arrow and second arrow equal to β . Thus we see that $K \rightarrow J$ is surjective. Moreover, the kernel of the first displayed arrow is $IA[x_s, y_t]$. Hence $I/I^2 \otimes C$ surjects onto the kernel of $K/K^2 \rightarrow J/J^2$. Finally, we can use Lemma F.11 to identify the terms as homology groups of the naive cotangent complexes.

The final assertion is a statement in homological algebra. Recall that $NL_{B/A} = (N^{-1} \rightarrow N^0)$ is a two term complex of B -modules with N^0 free and cohomology modules $H^0 = \Omega_{B/A}$ and $H^{-1} = H_1(L_{B/A})$. Write $M \subset N^0$ for the image of the differential. If $\text{Tor}_1^B(H^0, C) = 0$, then we have an exact sequence

$$0 \rightarrow M \otimes_B C \rightarrow N^0 \otimes_B C \rightarrow H^0 \otimes_B C \rightarrow 0$$

Since N^0 is free, we also see that $\text{Tor}_2^B(H^0, C) = \text{Tor}_1^B(M, C)$. Hence if $\text{Tor}_2^B(H^0, C) = 0$ then we also have an exact sequence

$$0 \rightarrow H^{-1} \otimes_B C \rightarrow N^{-1} \otimes_B C \rightarrow M \otimes_B C \rightarrow 0$$

Putting everything together we see that if $\text{Tor}_1^B(H^0, C) = 0$ and $\text{Tor}_2^B(H^0, C) = 0$, then $H^{-1} \otimes_B C$ is the kernel of $N^{-1} \otimes_B C \rightarrow N^0 \otimes_B C$ as desired. $\square$

Lemma F.43: extension

Adapted from the Stacks project, algebra.tex, lines 398–483.

For $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$, finite generation is preserved under extensions and quotients; finite presentation is preserved under extensions; a quotient of a finitely presented module by a finite submodule is finitely presented; and if M_3 is finitely presented and M_2 finite, then M_1 is finite.

Proof. Generators of M_1 and lifts of generators of M_3 generate M_2 ; images of generators generate any quotient. For the last assertion take a finite free surjection $F \rightarrow M_2$. Its composite $F \rightarrow M_3$ has finite kernel K : to see presentation independence, compare it with a fixed finite presentation of M_3 by lifting each of the two free bases to the other free module; the finite defining relations and the finite substitution differences generate the kernel. The image of K in M_2 is precisely M_1 , which is consequently finite.

If $M_2 = F/K$ has finite generators and relations and M_1 is finite, lift its finite generator list to F ; adjoining those lifts to the finite generator list of K presents M_3 . Finally for an extension of two finitely presented modules take finite free surjections $F_i \rightarrow M_i$ for $i = 1, 3$ and lift the second basis to M_2 . Then $F_1 \oplus F_3 \rightarrow M_2$ is surjective. Its kernel fits into a short exact sequence of the two presentation kernels: projection to F_3 gives the map onto the third kernel, and subtracting a suitable F_1 lift proves surjectivity; its kernel is the first presentation kernel. Both outer kernels are finite, so the generator argument at the start makes the middle kernel finite. This proves finite presentation of M_2 . $\square$

Lemma F.44: finite presentation independent

Adapted from the Stacks project, algebra.tex, lines 573–589.

Let $R \rightarrow S$ be a ring map of finite presentation. For any surjection $\alpha : R[x_1, \dots, x_n] \rightarrow S$ the kernel of α is a finitely generated ideal in $R[x_1, \dots, x_n]$.

Proof. Write $S = R[y_1, \dots, y_m]/(f_1, \dots, f_k)$. Choose $g_i \in R[y_1, \dots, y_m]$ which are lifts of $\alpha(x_i)$. Then we see that $S = R[x_i, y_j]/(f_l, x_i - g_i)$. Choose $h_j \in R[x_1, \dots, x_n]$ such that $\alpha(h_j)$ corresponds

to $y_j \bmod (f_1, \dots, f_k)$. Consider the map $\psi : R[x_i, y_j] \rightarrow R[x_i]$, $x_i \mapsto x_i$, $y_j \mapsto h_j$. Then the kernel of α is the image of $(f_i, x_i - g_i)$ under ψ and we win. $\square$

Lemma F.45: flat fibre smooth

Adapted from the Stacks project, algebra.tex, lines 39115–39149.

Let $R \rightarrow S$ be a ring map. Let $\mathfrak{q} \subset S$ be a prime lying over the prime $\mathfrak{p}$ of R . Assume

1. there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is of finite presentation,
2. the local ring homomorphism $R_{\mathfrak{p}} \rightarrow S_{\mathfrak{q}}$ is flat,
3. the fibre $S \otimes_R \kappa(\mathfrak{p})$ is smooth over $\kappa(\mathfrak{p})$ at the prime corresponding to $\mathfrak{q}$.

Then $R \rightarrow S$ is smooth at $\mathfrak{q}$.

Proof. We prove the stated arbitrary-ring assertion. First replace S by the given S_s , with $s \notin \mathfrak{q}$, and keep the corresponding prime denoted by $\mathfrak{q}$. Choose a finite presentation

$$P = R[X_1, \dots, X_n] \longrightarrow S = P/I, \quad I = (a_1, \dots, a_m).$$

Let $\mathfrak{r} \subset P$ be the inverse image of $\mathfrak{q}$, put $A = R_{\mathfrak{p}}$, $Q = P_{\mathfrak{r}}$, and $B = S_{\mathfrak{q}}$. The exact sequence $0 \rightarrow I_{\mathfrak{r}} \rightarrow Q \rightarrow B \rightarrow 0$ is a sequence of A -modules, and its quotient B is A -flat by hypothesis. We will use only finite generation of I , not a chain condition on any of these rings.

Here is the tensor-injection calculation needed for that sequence. More generally, suppose $0 \rightarrow K \rightarrow M \rightarrow F \rightarrow 0$ is exact over a ring A and F is flat; M need not be flat or free. For an arbitrary A -module N , choose a free presentation $F_1 \xrightarrow{d} F_0 \rightarrow N \rightarrow 0$, and put $T = \ker d$ and $L = \operatorname{im} d$. The free presentations, direct-sum exactness and right-exact tensor calculations are written in [AG-RG-S01, Lemma 1.0](#). If $z \in K \otimes_A N$ maps to zero in $M \otimes_A N$, lift it to $x \in K \otimes_A F_0$. Right exactness gives $y \in M \otimes_A F_1$ whose image in $M \otimes_A F_0$ is the image of x . The image of y in $F \otimes_A F_1$ is killed by $1 \otimes d$. Tensoring $0 \rightarrow T \rightarrow F_1 \rightarrow L \rightarrow 0$ with the flat F , and using the injection $L \rightarrow F_0$, shows that this kernel is the image of $F \otimes_A T$. Lift such a preimage to $M \otimes_A T$, using the surjection $M \rightarrow F$ and right exactness, and subtract its image from y . The resulting y' still maps to the image of x , and its image in $F \otimes_A F_1$ is zero. Since F_1 is free, coordinatewise exactness lifts y' from $K \otimes_A F_1$. Its image in $K \otimes_A F_0$ is x : this follows from the injection $K \otimes_A F_0 \rightarrow M \otimes_A F_0$, again coordinatewise because F_0 is free. Thus x comes from $K \otimes_A F_1$, so $z = 0$. We have proved the tensor injection with no Tor assertion.

Set $k = \kappa(\mathfrak{p}) = A/\mathfrak{p}A$. Apply that calculation to $N = k$. Right exactness and the quotient-tensor formula identify

$$J := \ker(Q/\mathfrak{p}Q \rightarrow B/\mathfrak{p}B) \simeq I_{\mathfrak{r}}/\mathfrak{p}I_{\mathfrak{r}}.$$

Write $\overline{Q} = Q/\mathfrak{p}Q$ and $\overline{B} = B/\mathfrak{p}B$. Then

$$\overline{Q} = k[X_1, \dots, X_n]_{\overline{\mathfrak{r}}}, \quad \overline{B} = \overline{Q}/J.$$

Indeed localization commutes with quotient and coefficient change; every polynomial over k lifts coefficientwise to A , and being outside $\overline{\mathfrak{r}}$ is exactly being outside the prime before reduction. The ring $\overline{Q}$ is local, J is contained in its maximal ideal, and $\overline{B}$ is the local ring of the fibre at the specified point. In particular J is finite, generated by the images of the a_j .

The smooth-fibre hypothesis makes $\overline{B}$ formally smooth over k . Here the equivalence of the definitions is used at its proved scope: Definition F.6 and the presentation comparison in Lemma F.11 give, on a smooth fibre neighbourhood, an injective conormal map with projective differential cokernel. Projectivity splits that quotient, and the full conormal criterion [AG-CA-17, Theorem 3.1](#) gives

formal smoothness. Its Theorem 1.2 proves that arbitrary further localizations preserve formal smoothness: a lift on the unlocalized algebra sends the inverted elements to units, since units lift across a square-zero ideal, and hence extends to the localization. Thus this step does not assume that a local ring is itself finitely presented.

We obtain the needed conormal splitting in this same localized polynomial presentation, without changing presentations. Formal smoothness lifts the identity of $\bar{B}$ to a k -algebra section

$$\sigma : \bar{B} \longrightarrow \bar{Q}/J^2$$

of its square-zero quotient. The map

$$D(q) = q \bmod J^2 - \sigma(q \bmod J) \quad : \quad \bar{Q} \longrightarrow J/J^2$$

is a k -derivation: writing $q = \sigma(\bar{q}) + D(q)$, and discarding products of two elements of J/J^2 , gives $D(qq') = \bar{q}D(q') + \bar{q}'D(q)$. For $j \in J$, it has $D(j) = j \bmod J^2$. The universal differential construction therefore supplies a left inverse to

$$J/J^2 \longrightarrow \Omega_{\bar{Q}/k} \otimes_{\bar{Q}} \bar{B} = \bar{B}^n, \quad [j] \longmapsto \left(\frac{\partial j}{\partial X_\nu} \right)_\nu.$$

The free differential basis and its localization here are precisely [AG-CA-16, Theorems 5.1 and 2.2](#); localization extends a derivation by $D(q/t) = t^{-1}D(q) - qt^{-2}D(t)$. Thus J/J^2 is a finite direct summand of $\bar{B}^n$.

We spell out the finite local-module argument, including the required Nakayama uses. Over any local ring $(D, \mathfrak{n})$, a finite module M with $M = HM$ for $H \subset \mathfrak{n}$ is zero: choose generators and write $m_i = \sum_j h_{ij}m_j$ with $h_{ij} \in H$. The cofactor identity of Lemma 3.B applied to $I - (h_{ij})$ shows that its determinant annihilates every generator. That determinant is congruent to one modulo $\mathfrak{n}$, hence is a unit. An element outside the unique maximal ideal is a unit, because a nonunit generates a proper ideal, which lies in a maximal ideal by the proper-ideal Zorn argument. Therefore $M = 0$. It follows also that lifts of a basis of $M/\mathfrak{n}M$ generate a finite M , by applying this assertion to their quotient.

Choose from the finite list $a_1, \dots, a_m$ elements $f_1, \dots, f_c$ whose conormal classes give a basis of $(J/J^2) \otimes_{\bar{B}} \kappa(\mathfrak{q})$; a finite spanning list over a field contains a basis by successively adjoining independent members. Nakayama over $\bar{B}$ makes these classes generate J/J^2 . They are in fact a free basis. To see this, the surjection $\bar{B}^c \rightarrow J/J^2$ splits: lift the projection $\bar{B}^n \rightarrow J/J^2$ through it by choosing lifts of the n coordinate images, and restrict that lift to the direct summand J/J^2 . Its kernel is then a finite direct summand of $\bar{B}^c$, generated by the projections of its coordinate vectors. Modulo the maximal ideal the surjection is an isomorphism by our basis choice, so this kernel has zero residue quotient. The just-proved Nakayama argument kills it.

Let H be the c by n equation-by-variable matrix $(\partial f_i / \partial X_\nu)$, with coefficients reduced to $\bar{B}$. The split conormal injection has matrix $H^\top$ and a left inverse V , so $VH^\top = I_c$. We write the minor argument here. Expanding $\det(VH^\top)$ by multilinearity in its columns chooses c columns of V with their coefficients from $H^\top$. Repeated choices have determinant zero by the signed-permutation cancellation. Grouping the remaining choices by their distinct index set U , in increasing order, gives

$$1 = \sum_{|U|=c} \det(V^U) \det((H^\top)_U).$$

The signed-permutation formula also gives $\det(W^\top) = \det W$, by replacing a permutation with its inverse. If $c > n$ every choice repeats, contradicting $1 \neq 0$ in the local ring $\bar{B}$. Thus $c \leq n$, and at least one c by c minor of H is a unit of $\bar{B}$: if all belonged to its maximal ideal the displayed sum could not be one. Choose its variable indices U . The corresponding polynomial

$$\Delta = \det \left(\frac{\partial f_i}{\partial X_\nu} \right)_{1 \leq i \leq c, \nu \in U} \in P$$

is outside $\mathfrak{r}$, since its image in $\kappa(\mathfrak{q})$ is nonzero. For $c = 0$ this argument uses the empty determinant $\Delta = 1$ and the empty generator list.

The chosen f_i generate the whole fibre ideal J in $\overline{Q}$, not merely its conormal quotient. Indeed $J = (\overline{f}_1, \dots, \overline{f}_c) + J^2$. Its finite quotient by the displayed ideal is equal to J times that quotient. Since J is contained in the maximal ideal of the local ring $\overline{Q}$, Nakayama makes the quotient zero. Now put $M = I_{\mathfrak{r}}/(f_1, \dots, f_c)Q$, a finite Q -module. The tensor-injection identification above gives

$$M/\mathfrak{p}M = J/(\overline{f}_1, \dots, \overline{f}_c) = 0.$$

The ideal $\mathfrak{p}Q$ lies in the maximal ideal of Q , so another application of finite Nakayama gives $M = 0$. This is the step where flatness of the local quotient B is essential.

For each of the finitely many original generators a_j of I , this equality expresses $a_j/1$ as a combination of the $f_i/1$ in $P_{\mathfrak{r}}$. Clear the finitely many coefficient denominators and also any multiplier outside $\mathfrak{r}$ witnessing each equality in that localization. Their finite product $v \notin \mathfrak{r}$ gives $I_v = (f_1, \dots, f_c)P_v$. This keeps the killed-cross-difference condition and does not cancel a possible zero divisor. Consequently a neighbourhood of $\mathfrak{q}$ is

$$S_{v\Delta} = R[X_1, \dots, X_n]_{v\Delta}/(f_1, \dots, f_c).$$

It is finitely presented over R : add one variable Y with the relation $(v\Delta)Y - 1$, together with the c equations. These further principal localizations of the earlier S_s give a principal neighbourhood in the original $\text{Spec } S$: write their elements with numerators in the original S and denominators powers of s , and invert s and those finitely many numerators.

Finally we prove formal smoothness of this neighbourhood by the square-zero calculation. Given an R -algebra D , an ideal K with $K^2 = 0$, and a map from the displayed algebra to D/K , lift the variable images to $b_\nu \in D$. The errors $r_i = f_i(b)$ lie in K . The selected square Jacobian block $H_U(b)$ has unit determinant, because its determinant is $\Delta(b)$ and its image modulo K is a unit. The cofactor calculation in Lemma 3.B gives its inverse. Change only the selected coordinates by

$$(b'_\nu)_{\nu \in U} = (b_\nu)_{\nu \in U} - H_U(b)^{-1}(r_i)_{1 \leq i \leq c},$$

leaving the others fixed. All corrections lie in K . Expanding each polynomial monomial shows $f_i(b') = f_i(b) + \sum_\nu (\partial f_i / \partial X_\nu)(b)(b'_\nu - b_\nu)$, because every term with two corrections lies in K^2 . Thus all $f_i(b')$ are zero. The image of $v\Delta(b')$ modulo K is unchanged and is a unit; it is a unit in D as well. Explicitly, if a lift of its inverse gives a product $1 + j$ with $j \in K$, multiply that lift by $1 - j$. Evaluation therefore extends through the quotient and localization to an R -algebra lift. For $c = 0$ there are no errors to correct; choosing arbitrary variable lifts and using the same unit argument suffices. We have proved formal smoothness for every square-zero test. Together with the finite presentation it is smooth, by the written conormal criterion and its agreement with Definition F.6. The neighbourhood contains $\mathfrak{q}$, proving the conclusion. No Noetherian hypothesis has been used. $\square$

Lemma F.46: flat over regular

Adapted from the Stacks project, algebra.tex, lines 34076–34098.

Let $R \rightarrow S$ be a homomorphism of Noetherian local rings. Assume that R is a regular local ring and that a regular system of parameters maps to a regular sequence in S . Then $R \rightarrow S$ is flat.

Proof. Suppose that $x_1, \dots, x_d$ are a system of parameters of R which map to a regular sequence in S . Note that $S/(x_1, \dots, x_d)S$ is flat over $R/(x_1, \dots, x_d)$ as the latter is a field. Then x_d is a nonzerodivisor in $S/(x_1, \dots, x_{d-1})S$ hence $S/(x_1, \dots, x_{d-1})S$ is flat over $R/(x_1, \dots, x_{d-1})$ by the local criterion of flatness (see Lemma AG-CA-08, Corollary 4.4 and equation (6) and remarks following). Then x_{d-1} is a nonzerodivisor in $S/(x_1, \dots, x_{d-2})S$ hence $S/(x_1, \dots, x_{d-2})S$ is flat over $R/(x_1, \dots, x_{d-2})$ by the local criterion of flatness (see Lemma AG-CA-08, Corollary 4.4 and equation (6) and remarks following). Continue till one reaches the conclusion that S is flat over R . $\square$

Lemma F.47: flat over regular with regular fibre

Adapted from the Stacks project, algebra.tex, lines 28729–28751.

A flat local map of Noetherian local rings with regular source and regular closed fibre has regular target.

Proof. Let the source dimension be d and fibre dimension e . Choose d source parameters and lift e fibre parameters. Together they generate the target maximal ideal: the fibre generators generate it modulo the source maximal ideal, which the source parameters already generate. The flat local dimension formula AG-CA-11, Theorem 5.1, gives target dimension $d + e$. Its embedding dimension is at most the number $d + e$ of these generators, and at least its dimension by the height theorem AG-CA-11, Theorem 3.1. Equality is the definition of a regular local ring. $\square$

Lemma F.48: flatness descends more general

Adapted from the Stacks project, algebra.tex, lines 9135–9161.

Let R be a ring. Let $S \rightarrow S'$ be a flat map of R -algebras. Let M be a module over S , and set $M' = S' \otimes_S M$.

1. If M is flat over R , then M' is flat over R .
2. If $S \rightarrow S'$ is faithfully flat, then M is flat over R if and only if M' is flat over R .

Proof. Let $N \rightarrow N'$ be an injection of R -modules. By the flatness of $S \rightarrow S'$ we have

$$\ker(N \otimes_R M \rightarrow N' \otimes_R M) \otimes_S S' = \ker(N \otimes_R M' \rightarrow N' \otimes_R M')$$

If M is flat over R , then the left hand side is zero and we find that M' is flat over R by the second characterization of flatness in Lemma AG-CA-07, Theorem 2.1. If M' is flat over R then we have the vanishing of the right hand side and if in addition $S \rightarrow S'$ is faithfully flat, this implies that $\ker(N \otimes_R M \rightarrow N' \otimes_R M)$ is zero which in turn shows that M is flat over R . $\square$

Lemma F.49: formally smooth extensions easy

Adapted from the Stacks project, algebra.tex, lines 45971–46012.

Let K/k be an extension of fields.

1. If K is purely transcendental over k , then K is formally smooth over k .
2. If K is separable algebraic over k , then K is formally smooth over k .
3. If K is separable over k , then K is formally smooth over k .

Proof. For (1) write $K = k(x_j; j \in J)$. Suppose that A is a k -algebra, and $I \subset A$ is an ideal of square zero. Let $\varphi : K \rightarrow A/I$ be a k -algebra map. Let $a_j \in A$ be an element such that $a_j \bmod I = \varphi(x_j)$. Then it is easy to see that there is a unique k -algebra map $K \rightarrow A$ which maps x_j to a_j and which reduces to $\varphi \bmod I$. Hence $k \subset K$ is formally smooth.

In case (2), each finite subset of K generates a finite separable extension of k : adjoining its finitely many algebraic generators gives only finitely many finite degrees. These subextensions form a filtered family, by adjoining two finite generating lists together, and their union is K . AG-CA-17, Proposition 6.3 proves that every such finite extension is étale; its separable-generator tower proof

explicitly gives the unique square-zero lifts, hence formal étaleness. Hence this case follows from Lemma F.22 and the trivial observation that a formally étale ring map is formally smooth.

In case (3), write $K = \operatorname{colim} K_i$ as the filtered colimit of its finitely generated k -subextensions. By Definition F.5 each K_i is separable algebraic over a purely transcendental extension of k . Hence K_i/k is formally smooth by cases (1) and (2) and Lemma AG-CA-17, Theorem 1.2. Thus $H_1(L_{K_i/k}) = 0$ by Lemma F.17. Hence $H_1(L_{K/k}) = 0$ by Lemma F.24. Hence K/k is formally smooth by Lemma F.17 again. $\square$

Lemma F.50: geometrically regular

Adapted from the Stacks project, algebra.tex, lines 48753–48798.

For a Noetherian k -algebra, regularity after every finitely generated field extension is equivalent to regularity after every finite purely inseparable field extension.

Proof. One implication is immediate. For the other, given a finitely generated K/k , choose k'/k and K'/K by the explicit separabilization proof F.69. The assumed $A \otimes_k k'$ is regular and Noetherian.

Since K/k is finitely generated and K'/K is finite, K'/k' is finitely generated too. Choose field generators $a_1, \dots, a_n$ with $K' = k'(a_1, \dots, a_n)$ and put $B = k'[a_1, \dots, a_n] \subset K'$. This is a finite-type domain with fraction field K' . Its generic local ring $B_{(0)} = K'$ is regular, and its residue field is separably generated over k' by Lemma F.69. The proved separable-residue criterion AG-CA-18, Theorem 3.2 therefore makes B smooth at its generic point. By the definition of smoothness at a point, some $0 \neq g \in B$ has B_g smooth over k' . This remains a finite-type domain, and

$$K' = (B_g \setminus \{0\})^{-1} B_g.$$

This constructs the smooth field model used here, with the stated finitely generated and separably generated hypotheses.

Base change gives the smooth $(A \otimes_k k')$ -algebra $C = (A \otimes_k k') \otimes_{k'} B_g$: finite presentation base changes by the polynomial quotient presentation, and formal smoothness base changes by AG-CA-17, Theorem 1.2. The ring C is Noetherian because it is finite type over the Noetherian ring $A \otimes_k k'$. Smooth maps are flat by AG-CA-18, Theorem 6.1, and their residue-field fibres are regular by AG-CA-18, Theorem 2.1; regularity ascent proved above makes this base-changed algebra regular. Localization then makes $A \otimes_k K'$ regular. Choose a K -basis of the finite extension K'/K containing 1. It identifies $A \otimes_k K'$ with a positive finite number of copies of $A \otimes_k K$ as a module. Tensoring with it takes the same positive number of copies of every module, so preserves exact sequences and detects the zero module. Thus $A \otimes_k K \rightarrow A \otimes_k K'$ is faithfully flat. F.38 now gives regularity of $A \otimes_k K$. These rings are Noetherian because finitely generated field tensor is a localization of a finite-type polynomial extension, the Noetherian field-tensor argument already written in this lesson. $\square$

Lemma F.51: geometrically regular descent

Adapted from the Stacks project, algebra.tex, lines 48815–48834.

Geometric regularity descends through faithfully flat maps of Noetherian k -algebras.

Proof. Let $A \rightarrow B$ be the given faithfully flat map and let k'/k be finite purely inseparable. Put $A' = A \otimes_k k'$ and $B' = B \otimes_k k'$. Both are Noetherian by F.12. Tensor associativity gives $B' = A' \otimes_A B$ and, for every A' -module M , a natural identification

$$M \otimes_{A'} B' = M \otimes_A B.$$

Restriction from A' -modules to A -modules preserves exact sequences and the zero module. Since tensoring with B over A is exact and detects the zero module, the displayed identification proves

that $A' \rightarrow B'$ is faithfully flat. The target B' is regular by geometric regularity of B . Regularity descent [F.38](#) therefore makes A' regular. Applying the finite purely inseparable criterion [F.50](#) to every such k' proves that A is geometrically regular over k . $\square$

Lemma F.52: geometrically regular over subfields

Adapted from the Stacks project, algebra.tex, lines 48851–48866.

If A is Noetherian and geometrically regular over k , then $A \otimes_k E$ is regular for any field extension E/k for which this tensor product is Noetherian. If $A \otimes_k E$ is Noetherian, it is geometrically regular over E . Also if k is a directed union of subfields k_i and A is geometrically regular over every k_i , then it is geometrically regular over k .

Proof. Write E as the directed union of the intermediate fields E_i/k generated by finite subsets of E : the union of two finite generating lists gives a common successor. Put $C_i = A \otimes_k E_i$ and $C = A \otimes_k E$. The natural inclusion $C_i \rightarrow C$ is injective. Indeed, an E_i -basis of E containing 1 identifies $C = C_i \otimes_{E_i} E$ with a direct sum of copies of C_i , with this inclusion as the summand corresponding to 1. The same basis shows that C is a free, hence flat, C_i -module. Each element of C is a finite tensor sum, so all its field coefficients lie in one E_i .

Assume C is Noetherian and fix a prime $\mathfrak{q} \subset C$. Its local ring $T = C_{\mathfrak{q}}$ is Noetherian; choose a finite list $x_1, \dots, x_r$ generating its maximal ideal $\mathfrak{m}$. Write $x_j = a_j/s_j$ with $a_j \in \mathfrak{q}$ and $s_j \notin \mathfrak{q}$. Including the finite field coefficients of all these numerators and denominators in one E_i puts every a_j, s_j in C_i . At the contracted prime $\mathfrak{q}_i = \mathfrak{q} \cap C_i$, put $B_i = (C_i)_{\mathfrak{q}_i}$. Every s_j is invertible in B_i , and each a_j lies in $\mathfrak{q}_i$. The map $B_i \rightarrow T$ is local and flat: localizing the free C_i -algebra C first away from $\mathfrak{q}_i$ and then at $\mathfrak{q}$ preserves flatness. Therefore $\mathfrak{q}_i T$ lies in $\mathfrak{m}$ and contains each of its chosen generators, giving

$$\mathfrak{q}_i T = \mathfrak{m}, \quad T \otimes_{B_i} \kappa(\mathfrak{q}_i) = T/\mathfrak{m}.$$

The closed fibre is the residue field of T , hence a regular ring. The source B_i is Noetherian by [F.12](#) and localization, and regular by geometric regularity of A over k , since E_i/k is finitely generated. Flat local regularity ascent [F.47](#) makes T regular. The empty generating list when $\mathfrak{m} = 0$ satisfies the same argument. This proves regularity at every prime of C .

For any finite purely inseparable extension F/E , [F.12](#) makes

$$(A \otimes_k E) \otimes_E F = A \otimes_k F$$

Noetherian. Apply the assertion just proved to the field extension F/k ; its tensor ring is regular. The criterion [F.50](#) now proves geometric regularity of $A \otimes_k E$ over E .

Finally suppose k is the directed union of the k_i and A is geometrically regular over every k_i . Let k'/k be finite purely inseparable. If $k' = k$, take $k'_i = k_i$ at any index. Otherwise the characteristic is $p > 0$. Choose a k -basis $b_1 = 1, b_2, \dots, b_d$ of k' . There is a common exponent e such that every $b_j^{p^e}$ belongs to k . Write the finitely many multiplication identities

$$b_j b_\ell = \sum_{h=1}^d c_{j\ell h} b_h, \quad c_{j\ell h} \in k.$$

Choose one k_i containing all these coefficients and all the $b_j^{p^e}$. The subspace

$$V = \bigoplus_{j=1}^d k_i b_j \subset k'$$

contains 1 and is closed under multiplication by the displayed identities. It is a finite-dimensional domain over k_i . For $0 \neq v \in V$, multiplication by v is an injective endomorphism of this finite-dimensional vector space, hence surjective; a preimage of 1 is its inverse. Thus V is a field, which

we call k'_i . For every $v = \sum_j a_j b_j \in k'_i$ the Frobenius identity gives

$$v^{p^e} = \sum_j a_j^{p^e} b_j^{p^e} \in k_i.$$

This proves that k'_i/k_i is finite purely inseparable, including every element of the constructed field. The multiplication map

$$k \otimes_{k_i} k'_i \longrightarrow k', \quad a \otimes v \longmapsto av,$$

is an isomorphism: the tensors $1 \otimes b_j$ form a k -basis of the source and map to the chosen k -basis of the target. Tensor associativity consequently gives

$$A \otimes_k k' \cong A \otimes_{k_i} k'_i.$$

The right side is regular by geometric regularity over k_i . Criterion F.50 proves geometric regularity over k . This also covers characteristic zero, in which a finite purely inseparable extension is trivial. $\square$

Lemma F.53: grothendieck regular sequence

Adapted from the Stacks project, algebra.tex, lines 24513–24526.

For a flat local map of Noetherian local rings $R \rightarrow P$, a list whose reduction is regular in the closed fibre is regular in P , and every successive quotient remains flat over R .

Proof. For the first element apply AG-CA-08, Lemma 5.1, to multiplication $P \xrightarrow{f_1} P$. The target is R -flat, both modules are finite over the Noetherian local ring P , and the map on the closed fibre is injective. That exact proved lemma gives injectivity and flatness of $P/(f_1)$. Repeat on the successive local quotient, whose closed fibre is the corresponding regular quotient. Local Nakayama ensures that every quotient remains nonzero, since the elements belong to the maximal ideal. This proves every assertion by induction. $\square$

Lemma F.54: henselian cat finite etale

Adapted from the Stacks project, algebra.tex, lines 43976–44032.

For a henselian local ring $(R, \mathfrak{m}, k)$, reduction modulo $\mathfrak{m}$ gives an equivalence between finite etale R -algebras and finite products of finite separable extensions of k .

Proof. A finite algebra over R is a product of local rings by AG-CA-20, Theorem 2.2. Each factor of a finite etale algebra is etale, by the lifting property restricted by its idempotent, and its residue is a finite separable field by AG-CA-17, Theorem 7.2. For full faithfulness it suffices to treat local factors S_1, S_2 : a field map between their residues lifts uniquely to $S_1 \rightarrow S_2$ by AG-CA-20, Theorem 1.2, since a finite local algebra over henselian R is henselian by Proposition 3.1 there. Ring maps to a local factor select one source idempotent; its residue selects the same idempotent. Thus the local-factor statement proves full faithfulness for products.

For essential surjectivity, choose a finite separable tower $k = k_0 \subset k_1 \subset \cdots \subset k_t = L$ with each step generated by one element. Starting with $S_0 = R$, lift the monic minimal polynomial to $S_{i-1}[T]$ and put $S_i = S_{i-1}[T]/(f_i)$. This ring is finite free over S_{i-1} and has residue field k_i . Every maximal ideal contracts to the maximal ideal of S_{i-1} by the integral maximal-ideal theorem, so S_i is local with maximal ideal $\mathfrak{m}S_i$. The derivative f'_i has nonzero residue in this field and hence is a unit. The one-equation Jacobian lift proves S_i/S_{i-1} etale. Its finite etale composite over R lifts L . Finite products lift the general object. This proof does not consume a decomposition theorem for arbitrary finite-type algebras over a henselian ring. $\square$

Lemma F.55: huber

Adapted from the Stacks project, algebra.tex, lines 37957–37984.

Let S be a finitely presented R -algebra which has a presentation $S = R[x_1, \dots, x_n]/I$ such that I/I^2 is free over S . Then S has a presentation $S = R[y_1, \dots, y_m]/(f_1, \dots, f_c)$ such that $(f_1, \dots, f_c)/(f_1, \dots, f_c)^2$ is free with basis given by the classes of $f_1, \dots, f_c$.

Proof. Note that I is a finitely generated ideal by Lemma F.44. Let $f_1, \dots, f_c \in I$ be elements which map to a basis of I/I^2 . By Nakayama's lemma (Lemma F.10) there exists a $g \in 1 + I$ such that

$$g \cdot I \subset (f_1, \dots, f_c)$$

and $I_g \cong (f_1, \dots, f_c)_g$. Hence we see that

$$S \cong R[x_1, \dots, x_n]/(f_1, \dots, f_c)[1/g] \cong R[x_1, \dots, x_n, x_{n+1}]/(f_1, \dots, f_c, gx_{n+1} - 1)$$

as desired. It follows that $f_1, \dots, f_c, gx_{n+1} - 1$ form a basis for $(f_1, \dots, f_c, gx_{n+1} - 1)/(f_1, \dots, f_c, gx_{n+1} - 1)^2$ for example by applying Lemma F.72. $\square$

Lemma F.56: isolated point fibre

Adapted from the Stacks project, algebra.tex, lines 31543–31578.

For a finite-type map $R \rightarrow S$ and a fibre point q , the following are equivalent: it is isolated in its fibre; its local fibre algebra is finite over the residue field; it is a closed fibre point with zero-dimensional local fibre ring; and one principal neighbourhood contains no other point of that fibre. These are also equivalent to zero dimension at that fibre point and to finite residue extension together with zero-dimensional local fibre ring.

Proof. Work in the finite-type field algebra $T = S \otimes_R \kappa(\mathfrak{p})$. Its dimension-at-a-point formula AG-CA-09, Theorem 6.1, is $\dim_x \operatorname{Spec} T = \dim T_q + \operatorname{trdeg} \kappa(q)$. Both summands are nonnegative. Thus zero point-dimension is precisely local dimension zero and finite residue extension; the latter finiteness is equivalent to closedness by AG-CA-06, Theorem 2.1. A zero-dimensional neighbourhood exists by the definition of point-dimension and principal opens forming a basis. Its finite-type field algebra is Noetherian of dimension zero, hence Artinian by AG-CA-03, Theorem 4.2. Its finite product decomposition and finite residue fields imply it is finite-dimensional over the field: its finitely many maximal-ideal-power quotients have finite-dimensional successive layers because each ideal is finite. Its points are finitely many isolated points, and selecting the factor of q gives a principal singleton neighbourhood.

Conversely an isolated point admits such a principal neighbourhood. Its only prime is q , so it is a zero-dimensional local finite-type algebra and the same Artinian-layer proof makes it finite-dimensional; localizing it again leaves it unchanged. If the local fibre ring is finite-dimensional, it is Artinian and its residue field is finite, giving the preceding zero-point-dimension criterion. Finally principal opens of T lift to principal opens of S , after clearing the one coefficient denominator from the base. This identifies the last neighbourhood statement with the original fibre statement and proves every equivalence. $\square$

Lemma F.57: lci

Adapted from the Stacks project, algebra.tex, lines 37341–37441.

Let k be a field. Let S be a finite type k -algebra. Let $\mathfrak{q}$ be a prime of S . Choose any presentation $S = k[x_1, \dots, x_n]/I$. Let $\mathfrak{q}'$ be the prime of $k[x_1, \dots, x_n]$ corresponding to $\mathfrak{q}$. Set $c = \operatorname{height}(\mathfrak{q}') -$

$\text{height}(\mathfrak{q})$, in other words $\dim_{\mathfrak{q}}(S) = n - c$ (see Lemma F.20). The following are equivalent

1. There exists a $g \in S$, $g \notin \mathfrak{q}$ such that S_g is a global complete intersection over k .
2. The ideal $I_{\mathfrak{q}'} \subset k[x_1, \dots, x_n]_{\mathfrak{q}'}$ can be generated by c elements.
3. The conormal module $(I/I^2)_{\mathfrak{q}}$ can be generated by c elements over $S_{\mathfrak{q}}$.
4. The conormal module $(I/I^2)_{\mathfrak{q}}$ is a free $S_{\mathfrak{q}}$ -module of rank c .
5. The ideal $I_{\mathfrak{q}'}$ can be generated by a regular sequence in the regular local ring $k[x_1, \dots, x_n]_{\mathfrak{q}'}$.

In this case any c elements of $I_{\mathfrak{q}'}$ which generate $I_{\mathfrak{q}'}/\mathfrak{q}'I_{\mathfrak{q}'}$ form a regular sequence in the local ring $k[x_1, \dots, x_n]_{\mathfrak{q}'}$.

Proof. Set $R = k[x_1, \dots, x_n]_{\mathfrak{q}'}$. Iterating AG-CA-14, Proposition 3.3, starting with k , makes this a regular local ring. The full proof of AG-CA-12, Theorem 6.1, makes it Cohen–Macaulay. Its dimension is $\text{height}(\mathfrak{q}')$ by prime-localization correspondence. Moreover, $\bar{R} = R/IR = R/I_{\mathfrak{q}'} = S_{\mathfrak{q}}$ is a quotient of dimension $\text{height}(\mathfrak{q})$. Let $f_1, \dots, f_c \in I_{\mathfrak{q}'}$ be elements which generate $(I/I^2)_{\mathfrak{q}}$. By Lemma F.10 we see that $f_1, \dots, f_c$ generate $I_{\mathfrak{q}'}$. Since the dimensions work out, we conclude by Proposition AG-CA-12, Theorem 4.2 and Corollary 4.3 that $f_1, \dots, f_c$ is a regular sequence in R . By Lemma F.74 we see that $(I/I^2)_{\mathfrak{q}}$ is free. These arguments show that (2), (3), (4) are equivalent and that they imply the last statement of the lemma, and therefore they imply (5).

If (5) holds, say $I_{\mathfrak{q}'}$ is generated by a regular sequence of length e , then $\text{height}(\mathfrak{q}) = \dim(S_{\mathfrak{q}}) = \dim(k[x_1, \dots, x_n]_{\mathfrak{q}'} - e = \text{height}(\mathfrak{q}') - e$ by dimension theory, see Section the dimension notation. We conclude that $e = c$. Thus (5) implies (2).

We continue with the notation introduced in the first paragraph. For each f_i we may find $d_i \in k[x_1, \dots, x_n]$, $d_i \notin \mathfrak{q}'$ such that $f'_i = d_i f_i \in k[x_1, \dots, x_n]$. Then it is still true that $I_{\mathfrak{q}'} = (f'_1, \dots, f'_c)R$. Hence there exists a $g' \in k[x_1, \dots, x_n]$, $g' \notin \mathfrak{q}'$ such that $I_{g'} = (f'_1, \dots, f'_c)$. Moreover, pick $g'' \in k[x_1, \dots, x_n]$, $g'' \notin \mathfrak{q}'$ such that $\dim(S_{g''}) = \dim_{\mathfrak{q}} \text{Spec}(S)$. By Lemma F.20 this dimension is equal to $n - c$. Finally, set g equal to the image of $g'g''$ in S . Then we see that

$$S_g \cong k[x_1, \dots, x_n, x_{n+1}]/(f'_1, \dots, f'_c, x_{n+1}g'g'' - 1)$$

and by our choice of g'' this ring has dimension $n - c$. Therefore it is a global complete intersection. Thus each of (2), (3), and (4) implies (1).

Assume (1). Let $S_g \cong k[y_1, \dots, y_m]/J$, with $J = (f_1, \dots, f_t)$ and $\dim S_g = m - t$, be a global complete-intersection presentation. Let $\mathfrak{q}''$ be the polynomial prime corresponding to $\mathfrak{q}_{S_g}$. First justify the dimension at this particular point. Every minimal prime $\mathfrak{a}$ of J has height at most t by AG-CA-11, Theorem 3.1. Its component has dimension $m - \text{height}(\mathfrak{a})$ by AG-CA-09, Theorem 4.3. The maximum of these finitely many component dimensions is $\dim S_g = m - t$. Thus each such height is at least t as well, and every component has dimension exactly $m - t$.

The complete point-dimension calculation in AG-CA-09, Theorem 6.1, now gives $\dim_{\mathfrak{q}_{S_g}} \text{Spec } S_g = m - t$: every component through this point has that dimension. The local rings $(S_g)_{\mathfrak{q}_{S_g}}$ and $S_{\mathfrak{q}}$ are identical, so their heights agree. Applying F.20 to the polynomial presentation therefore gives

$$t = m - \dim_{\mathfrak{q}_{S_g}} \text{Spec } S_g = \text{height}(\mathfrak{q}'') - \text{height}(\mathfrak{q}).$$

The ambient polynomial local ring is regular by iterating AG-CA-14, Proposition 3.3, starting with the field k . It is Cohen–Macaulay by AG-CA-12, Theorem 6.1. Its quotient has dimension $\text{height}(\mathfrak{q})$, so the displayed equality and the expected-dimension criterion AG-CA-12, Theorem 4.2, prove that the actual chosen $f_1, \dots, f_t$ form a regular sequence there. This includes the empty-list case. By Lemma F.74 we see that $(J/J^2)_{\mathfrak{q}}$ is free of rank t . By Lemma F.33 we have

$$J/J^2 \oplus S_g^n \cong (I/I^2)_g \oplus S_g^m$$

Localize this isomorphism at $\mathfrak{q}$. Since its left side is finite free, $(I/I^2)_{\mathfrak{q}}$ is a finite direct summand of a finite free module. It is flat by restricting tensor injections and finitely presented because its complementary summand is finite. The complete local calculation in [AG-RG-S01, Lemma 1.2](#) makes it finite free. Reduction to the residue field in this same isomorphism computes its rank. Thus $(I/I^2)_{\mathfrak{q}}$ is free of rank $t + n - m = m - \dim(S_g) + n - m = n - \dim(S_g) = \text{height}(\mathfrak{q}') - \text{height}(\mathfrak{q}) = c$. Thus we obtain (4). $\square$

Lemma F.58: lci at prime

Adapted from the Stacks project, algebra.tex, lines 37592–37613.

For a finite-type algebra S over a field and a prime $\mathfrak{q}$, being a complete-intersection local ring is equivalent to a global complete-intersection principal neighbourhood and to its polynomial presentation kernel being generated locally by a regular sequence.

Proof. A polynomial local presentation $P_{\mathfrak{q}'} \twoheadrightarrow S_{\mathfrak{q}}$ has regular local ambient ring by [AG-CA-14, Proposition 3.3](#) and [Theorem 3.2](#). If its kernel is a regular list of length c , the conormal is free of rank c by [F.74](#). Conversely, if the conormal can be generated by $c = \dim P_{\mathfrak{q}'} - \dim S_{\mathfrak{q}}$ elements, Nakayama lifts them to generators of the ideal, and the CM expected-dimension criterion [AG-CA-12, Theorem 4.2](#), makes them regular. Stable comparison of conormal modules under a change of polynomial presentation, proved above, preserves this assertion: the two ambient local dimensions differ by the difference of their variable counts, since [AG-CA-09, Theorem 4.3](#), gives $\dim k[X]_{\mathfrak{q}'} = n - \text{trdeg}_k \kappa(\mathfrak{q})$. A stably free finite module over the local quotient is free, with the resulting prescribed rank.

For the definition allowing an arbitrary regular local ambient Q essentially finite type over k , present Q as a quotient of a polynomial local ring. Both ambient and quotient are regular. [AG-CA-14, Proposition 1.4](#), proves that its kernel is generated by part of a regular parameter system. Lift the regular equations defining $S_{\mathfrak{q}}$ from Q and append them to that system. By the definition of successive injectivity this concatenated list is regular. Hence the polynomial criterion applies; the reverse implication uses that polynomial ambient itself as Q .

To obtain a principal neighbourhood, clear the finitely many denominators in the local regular generator list. Kill the finite cokernel of its map onto the presentation ideal, then kill its finitely many positive Koszul homology modules by another element outside $\mathfrak{q}'$. At every prime on this neighbourhood containing the ideal the list is regular by the Noetherian H_1 criterion. Every minimal prime of the resulting equation ideal therefore has height exactly c : its quotient local ring has dimension zero, while each regular equation lowers dimension by one by [AG-CA-12, Corollary 6.2](#). Each component has dimension $n - c$ by [AG-CA-09, Theorem 4.3](#). Adjoining the localization variable gives a global presentation with $n + 1$ variables and $c + 1$ equations and this same dimension. This is a global complete intersection. A global complete-intersection presentation in turn has regular local equation lists by the minimal-prime height and CM argument proved above, completing the equivalence. $\square$

Lemma F.59: limit essentially finite presentation

Adapted from the Stacks project, algebra.tex, lines 33722–33770.

A local essentially finitely presented map $R \rightarrow S$ is a directed colimit of local maps $R_{\lambda} \rightarrow S_{\lambda}$ essentially of finite type over $\mathbb{Z}$, with transition maps obtained by tensoring the base and localizing at the contracted prime.

Proof. Write $S = (R[X_1, \dots, X_n]/(f_1, \dots, f_c))_{\mathfrak{q}}$. For each finite subset of R containing every equation coefficient, let $A_{\lambda} \subset R$ be the subring it generates over the image of $\mathbb{Z}$, and set $R_{\lambda} =$

$(A_\lambda)_{\mathfrak{m}_R \cap A_\lambda}$. These are local subrings of R with local inclusions. Their directed union is R , since each element is included in some finite subset. Form $C_\lambda = R_\lambda[X]/(f_i)$ and localize it at the inverse image $\mathfrak{q}_\lambda$ of the chosen target prime, giving S_λ . The maps are local and essentially finite type, and every stage is Noetherian by the Hilbert basis theorem.

Tensoring C_λ with R_μ gives C_μ . After initially inverting the complement of $\mathfrak{q}_\lambda$, localizing at the contracted prime inverts exactly the additional complement of $\mathfrak{q}_\mu$; the fraction universal property identifies this with S_μ . Every fraction in S has a numerator and denominator involving finitely many coefficients and hence occurs at one stage. If two fractions become equal, cross multiplication gives an equality in the polynomial quotient after one denominator outside $\mathfrak{q}$ is multiplied in. Its finite ideal-membership expression uses only finitely many coefficients, so the equality holds at a later stage. This proves both surjectivity and injectivity of $\text{colim } S_\lambda \rightarrow S$, completing every claimed colimit identification. $\square$

Lemma F.60: limit module essentially finite presentation

Adapted from the Stacks project, algebra.tex, lines 33804–33859.

A finitely presented module M over the local algebra in the preceding approximation has finitely presented models M_λ at sufficiently large stages, compatible by tensoring with S_μ , with colimit M .

Proof. Choose a finite matrix presentation $S^a \xrightarrow{H} S^b \rightarrow M \rightarrow 0$. Its finitely many entries are fractions and occur in one S_{λ_0} by the preceding representative proof. Use their images to define $M_\lambda = \text{coker}(S_\lambda^a \xrightarrow{H_\lambda} S_\lambda^b)$ at all later stages. Tensor is right exact, so $M_\lambda \otimes_{S_\lambda} S_\mu = M_\mu$. Taking the colimit of the finite free presentations gives the original cokernel: a vector represents zero precisely when it equals the image of a finite vector, and this finite equality holds at a common later stage. Thus the colimit is M , each model is finitely presented, and all ring-transition assertions remain those proved above. $\square$

Lemma F.61: local dimension zero henselian

Adapted from the Stacks project, algebra.tex, lines 44097–44121.

Let $(R, \mathfrak{m})$ be a local ring of dimension 0. Then R is henselian.

Proof. Let $R \rightarrow S$ be a finite ring map. By Lemma AG-CA-20, Theorem 1.2 and Theorem 2.2 it suffices to show that S is a product of local rings. By Lemma AG-CA-05, Theorem 3.4 S has finitely many primes $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ which all lie over $\mathfrak{m}$. There are no inclusions among these primes, see Lemma AG-CA-05, Theorem 3.3, hence they are all maximal. Every element of $\mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_r$ is nilpotent by Lemma AG-CA-01, Theorem 1.2 and Proposition 2.1. It follows S is the product of the localizations of S at the primes $\mathfrak{m}_i$ by Lemma F.73. $\square$

Lemma F.62: local syntomic

Adapted from the Stacks project, algebra.tex, lines 37927–37941.

If finitely many principal opens cover $\text{Spec } S$ and $R \rightarrow S_{g_i}$ is syntomic on each, then $R \rightarrow S$ is syntomic.

Proof. Finite presentation follows from F.35, including its explicit finite substitution calculation. For flatness, tensor any injection of R -modules with S and localize its kernel at each g_i . Each kernel is zero by the assumed chart flatness; local detection on this principal cover makes the original kernel zero. In every residue-field fibre these same opens cover, and their local complete-intersection covers combine to give a cover by global complete-intersection neighbourhoods. This is precisely the fibre definition of syntomicity. No Noetherian hypothesis is required for this gluing step. $\square$

Lemma F.63: localization colimit

Adapted from the Stacks project, algebra.tex, lines 1260–1278.

Localization of a module is the filtered colimit of copies of that module with transition maps given by multiplication by denominators. In particular a localization of a ring is the filtered colimit of its principal localizations.

Proof. For the first assertion, index the copies by finite products of elements of the multiplicative set, ordered with multiplication maps (or use the filtered divisibility category). Send an element m in the copy indexed by s to m/s . Every fraction has such a representative. Equality $m/s = n/t$ means $u(tm - sn) = 0$ for some denominator u , precisely the equality at the common successor ust . Thus the map is bijective and respects addition and scalars. For rings use the principal subalgebras R_s : two such stages map to R_{st} ; every element and every equality in the full localization is represented or becomes equal at a principal successor by the same fraction criterion. This proves the ring colimit assertion. $\square$

Lemma F.64: localization smooth separable

Adapted from the Stacks project, algebra.tex, lines 46075–46088.

A finitely generated separably generated extension K/k is the fraction field of a smooth finite-type k -domain. Conversely a fraction field of a smooth finite-type k -domain is separably generated.

Proof. For the forward implication choose a finite separating transcendence basis T and a finite tower of separable algebraic generators $y_1, \dots, y_r$. At each step, express the coefficients of its monic minimal polynomial in the standard monomial basis of the preceding field over $k(T)$. Clear the finitely many rational-function denominators by one $0 \neq h(T) \in k[T]$. This gives monic equations $F_i \in k[T, 1/h, Y_1, \dots, Y_i]$ and

$$C = k[T, 1/h, Y_1, \dots, Y_r]/(F_1, \dots, F_r).$$

Successive monic division gives the standard monomials as a free basis over $k[T, 1/h]$. Over $k(T)$ the quotient is exactly the chosen field tower K . Since the coefficient ring is a domain and C is free, $C \rightarrow K$ is injective. Its fraction field is K , because it contains T and all the y_i .

The evaluated derivatives $\Delta_i = (\partial F_i / \partial Y_i)(y)$ are nonzero, since each tower step is separable. Put $\Delta = \prod_i \Delta_i$. The algebra C_Δ is a succession of standard étale extensions over $k[T, 1/h]$: at step i invert Δ_i and retain the preceding inverses. AG-CA-17, Proposition 6.1, proves the unique lifts for each such equation. Polynomial algebras and localization are formally smooth by its Theorem 1.2, as are their composites with these maps. All variables, equations and inverted elements are finite in number, so C_Δ is finitely presented over k . It is therefore smooth. It remains a domain inside K with fraction field K . Empty towers use $\Delta = 1$, and the same construction applies.

For the converse take a nonempty principal standard smooth chart D in the given domain, using AG-CA-17, Theorem 5.1. Its fraction field is still K . Reorder its variables so that the selected square Jacobian uses the pivot variables, and let $U_1, \dots, U_t$ be the nonpivot variables. Viewed over $R_0 = k[U_1, \dots, U_t]$, this presentation has only the pivot variables and the same square invertible Jacobian. The explicit lifting calculation AG-CA-17, Theorem 4.1, gives formal smoothness and zero relative differentials; Theorem 2.2 gives formal étaleness. Its finite presentation makes the map étale, hence flat by AG-CA-18, Corollary 6.2.

The map $R_0 \rightarrow D$ is injective. Indeed, multiplication by a nonzero element of the polynomial domain is injective, and flatness preserves this injection after tensoring with D . An element of the structural kernel would act as zero on the nonzero ring D , contradicting that injection. Thus the U_i are algebraically independent in K . AG-CA-18, Theorem 2.1, at the generic point gives $t = \text{trdeg}_k K$. They are consequently a transcendence basis. The extension $K/k(U_1, \dots, U_t)$ is

finite: it is algebraic and has finitely many field generators, whose finite degrees multiply along a finite tower.

Formal étaleness survives base change and localization by AG-CA-17, Theorem 1.2. First invert the nonzero elements of R_0 , and then those of D ; this gives the formally étale field map $k(U_1, \dots, U_t) \rightarrow K$. Its differentials vanish by Theorem 2.2. The finite-field converse AG-CA-17, Lemma 7.1, therefore makes it separable. This constructs the required separating transcendence basis directly, without using Proposition F.91. $\square$

Lemma F.65: localize NL

Adapted from the Stacks project, algebra.tex, lines 37118–37133.

Let $A \rightarrow B$ be a ring map. Let $S \subset B$ be a multiplicative subset. The canonical map $\mathrm{NL}_{B/A} \otimes_B S^{-1}B \rightarrow \mathrm{NL}_{S^{-1}B/A}$ is a quasi-isomorphism.

Proof. We have $S^{-1}B = \mathrm{colim}_{g \in S} B_g$ where we think of S as a directed set (ordering by divisibility), see Lemma F.63. By Lemma F.72 each of the maps $\mathrm{NL}_{B/A} \otimes_B B_g \rightarrow \mathrm{NL}_{B_g/A}$ is a quasi-isomorphism. The lemma follows from Lemma F.24. $\square$

Lemma F.66: localize relative complete intersection

Adapted from the Stacks project, algebra.tex, lines 38108–38158.

Let R be Noetherian. Let $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$. We will find $h \in R[x_1, \dots, x_n]$ which maps to $g \in S$ such that

$$S_g = R[x_1, \dots, x_n, x_{n+1}]/(f_1, \dots, f_c, hx_{n+1} - 1)$$

is a relative global complete intersection with a presentation as in Definition F.4 in each of the following cases:

1. Let $I \subset R$ be an ideal. If the fibres of $\mathrm{Spec}(S/IS) \rightarrow \mathrm{Spec}(R/I)$ have dimension $n - c$, then we can find (h, g) as above such that g maps to $1 \in S/IS$.
2. Let $\mathfrak{p} \subset R$ be a prime. If $\dim(S \otimes_R \kappa(\mathfrak{p})) = n - c$, then we can find (h, g) as above such that g maps to a unit of $S \otimes_R \kappa(\mathfrak{p})$.
3. Let $\mathfrak{q} \subset S$ be a prime lying over $\mathfrak{p} \subset R$. If $\dim_{\mathfrak{q}}(S/R) = n - c$, then we can find (h, g) as above such that $g \notin \mathfrak{q}$.

Proof. Ad (1). By Lemma F.39 there exists an open subset $W \subset \mathrm{Spec}(S)$ containing $V(IS)$ such that all fibres of $W \rightarrow \mathrm{Spec}(R)$ have dimension $\leq n - c$. Say $W = \mathrm{Spec}(S) \setminus V(J)$. Then $V(J) \cap V(IS) = \emptyset$ hence we can find a $g \in J$ which maps to $1 \in S/IS$. Let $h \in R[x_1, \dots, x_n]$ be any preimage of g .

Ad (2). By Lemma F.39 there exists an open subset $W \subset \mathrm{Spec}(S)$ containing $\mathrm{Spec}(S \otimes_R \kappa(\mathfrak{p}))$ such that all fibres of $W \rightarrow \mathrm{Spec}(R)$ have dimension $\leq n - c$. Say $W = \mathrm{Spec}(S) \setminus V(J)$. Then $V(J \cdot S \otimes_R \kappa(\mathfrak{p})) = \emptyset$. Hence we can find a $g \in J$ which maps to a unit in $S \otimes_R \kappa(\mathfrak{p})$. Indeed, the image of J generates the unit ideal in this fibre. Write a finite relation $1 = \sum_j \bar{a}_j \bar{g}_j$ with $g_j \in J$. Choose a common denominator $s \in R \setminus \mathfrak{p}$ for the $\bar{a}_j$. Multiplication gives $g = \sum_j a_j g_j \in J$ whose fibre image is the nonzero scalar s , hence is a unit. Let $h \in R[x_1, \dots, x_n]$ be any preimage of g .

Ad (3). By Lemma F.39 there exists a $g \in S$, $g \notin \mathfrak{q}$ such that all nonempty fibres of $R \rightarrow S_g$ have dimension $\leq n - c$. Let $h \in R[x_1, \dots, x_n]$ be any element that maps to g .

In every case, a nonempty fibre of this principal localization has dimension at least $n - c$: its ambient polynomial algebra has $n + 1$ variables and the displayed ideal has $c + 1$ generators, so the height bound AG-CA-11, Theorem 3.1 and the finite-type field dimension formula give that lower bound. The upper bound already proved therefore gives dimension exactly $n - c$. This is precisely the relative global complete-intersection definition for the displayed presentation. $\square$

Lemma F.67: locally smooth

Adapted from the Stacks project, algebra.tex, lines 39014–39037.

A ring map is smooth if and only if it is smooth on a principal neighbourhood of every target prime.

Proof. For the forward direction, take a finite polynomial presentation of the smooth algebra. Its conormal sequence is split and injective on the left by AG-CA-17, Theorem 3.1. Localization preserves this splitting and injection. By F.72 and the presentation comparison F.11, the conormal complex of a principal localization differs from the localized complex by a contractible summand. Thus that localization again has an injective conormal map and a projective differential module, giving formal smoothness by the same split-conormal criterion. Adding one variable and the relation $gZ - 1$ preserves finite presentation by F.35, so the principal localization is smooth.

Conversely, choose for each target prime a principal neighbourhood on which the map is smooth. The elements of these principal opens generate one: otherwise the ideal they generate lies in a maximal ideal, which is an uncovered prime. A finite expression for one gives a finite subcover $D(g_1), \dots, D(g_r)$. By F.35, S is finitely presented over R . Choose $S = P/I$, where $P = R[X_1, \dots, X_n]$ and I is finitely generated. The conormal presentation

$$I/I^2 \longrightarrow S^n \longrightarrow M = \Omega_{S/R} \longrightarrow 0$$

makes M finitely presented: the differentials of a finite generating list of I generate the kernel of $S^n \rightarrow M$. This is the conormal and polynomial-differential calculation of AG-CA-16, Section 3.

On each $D(g_i)$, F.72 and F.11 identify the first homology with that of the smooth chart, which is zero by AG-CA-17, Theorem 3.1. Exact localization commutes with this kernel, and the module detection calculation in F.35 makes the global first homology zero. On the same cover, the split-conormal criterion makes M_{g_i} a direct summand of a finite free module. It is therefore flat: tensor injections remain injective after restricting to that summand. To prove M flat over S , tensor any injection of S -modules with M . The kernel localizes to zero at every g_i , since the M_{g_i} are flat; F.35 again makes the kernel zero.

The finitely presented flat module M is finite projective by AG-CA-07, Theorem 5.3. Hence its conormal surjection splits, and the already proved vanishing of first homology makes the left conormal map injective. AG-CA-17, Theorem 3.1, gives formal smoothness. Together with finite presentation this is smoothness. If the target ring is zero, the empty presentation with the relation $1 = 0$ gives finite presentation, all these modules vanish, and a square-zero extension of the zero ring is zero because its unit is square-zero. Hence every lifting problem has the unique required map. Thus the empty-spectrum case is included. $\square$

Lemma F.68: make etale map prescribed residue field

Adapted from the Stacks project, algebra.tex, lines 41287–41319.

For a prime $\mathfrak{p}$ of R and a finite separable extension $L/\kappa(\mathfrak{p})$, some finitely presented etale R -algebra has a point over $\mathfrak{p}$ with this prescribed residue field.

Proof. Choose finitely many elements generating L as a field, and adjoin them one at a time. At a given step their monic minimal polynomial over the preceding residue field has nonzero derivative at the chosen element, since it is separable. Lift its finitely many coefficients to the local ring of

the already constructed algebra at the chosen prime. Clear finitely many denominators to obtain that polynomial over one principal neighbourhood. Adjoin a root of this monic polynomial and invert its derivative. The square one-equation Jacobian calculation of AG-CA-17, Proposition 6.1, makes this algebra étale; its chosen residue-field evaluation maps the root to the required element. The kernel of this evaluation at the chosen point has exactly the enlarged residue field, since the localized base residue field and the root generate it. The polynomial coefficients and inverses involved at each of the finitely many steps descend from the local ring by finitely many denominator clearings. Composing the resulting finitely presented étale maps gives the asserted R -algebra. This construction uses a finite separable tower and does not assume a primitive element theorem. $\square$

Lemma F.69: make separably generated

Adapted from the Stacks project, algebra.tex, lines 10182–10253.

For a finitely generated field extension K/k there are finite purely inseparable extensions k'/k and K'/K with a commutative field square and K'/k' separably generated.

Proof. Choose a finite field-generating list for K/k and a maximal algebraically independent sublist $T = (T_1, \dots, T_d)$. Every remaining generator is algebraic over $F = k(T)$: a polynomial dependence involving it has a nonzero coefficient after evaluating the variables in T , because those variables are independent. Adjoining the remaining algebraic generators gives a finite tower. At a simple step, division by the monic minimal polynomial gives a unique remainder of smaller degree. The resulting finite-dimensional domain is a field: multiplication by a nonzero element is injective and hence surjective, giving its inverse. It therefore equals the simple field extension, with the powers below the minimal-polynomial degree as a basis. Products of these bases form a basis of the whole tower: they span by successive expansion, and grouping a putative relation by the last generator's powers proves independence inductively. This proves finiteness and multiplication of all degrees used below. In characteristic zero every irreducible polynomial has nonzero derivative, hence is separable; the finite-tower separability proof in AG-CA-17, Lemma 7.1, gives the assertion with $k' = k$ and $K' = K$.

Now assume characteristic $p > 0$. When a finite list of p th roots is needed, adjoin them successively: choose an irreducible factor of each $X^p - a$ and take its field quotient. Such a factor exists by descent on the positive degree, and the Euclidean algorithm makes the quotient a field. Each polynomial has at most one root in a field extension, because $u^p = v^p$ gives $(u - v)^p = 0$, hence $u = v$. Retain the existing fields as subfields at every adjunction; all roots and composita below use these compatible inclusions. We first construct the separable subfield of any finite extension E/F . If $x \in E$ has monic irreducible polynomial f , write

$$f(X) = g(X^{p^e})$$

with e as large as possible. The polynomial g is irreducible, since a factorization would substitute to one of f , and $g' \neq 0$. Euclidean division gives $\gcd(g, g') = 1$: irreducibility and the smaller degree of g' exclude a nonconstant common factor. Thus no root is repeated, and x^{p^e} is separable over F .

If x and y are separable over F , the minimal polynomial of y over $F(x)$ divides its separable polynomial over F , so the two-step tower $F \subset F(x) \subset F(x, y)$ has separable steps. The complete finite-tower argument AG-CA-17, Lemma 7.1, makes every element of $F(x, y)$ separable over F . Consequently the separable elements in E are closed under sums, products and nonzero inverses. They form a field E_s , finite-dimensional over F as a subspace of E , and every one of its elements is separable over F .

For each of a finite generating list of E/F , the preceding irreducible-polynomial calculation puts some p -power in E_s . Choose a common exponent e . Frobenius applied to a rational expression in these generators then puts every p^e th power from E in E_s . Hence E/E_s is purely inseparable. Its

degree is a power of p : for each generator, adjoin successively its powers in reverse order, beginning with the power already in E_s . Every nontrivial step has the form $u^p \in H$, $u \notin H$, and has degree p by the proper-factor proof in F.119. Multiplication of the finite tower degrees proves the assertion.

Apply this construction to K/F , writing its separable subfield as K_s . We use strong induction on $m = [K : K_s]$ for all finitely generated extensions together with the selected transcendence basis. If $m = 1$, that basis already separates K/k . If $m > 1$, choose $x \in K \setminus K_s$ and let $e \geq 1$ be the least exponent with $x^{p^e} \in K_s$. Put $\beta = x^{p^{e-1}}$ and $\alpha = \beta^p$. Then $\beta \notin K_s$ and $\alpha \in K_s$. The same degree- p proof gives

$$[K_s(\beta) : K_s] = p, \quad [K : K_s(\beta)] = m/p.$$

Write the monic separable minimal polynomial of α over F as

$$P(X) = X^n + \sum_{i < n} a_i X^i.$$

Each a_i is a rational function in T involving finitely many coefficients of k . Using the finite-root construction over K , adjoin the p th roots of all those coefficients. Let k' be the subfield they generate over k . It is finite and purely inseparable over k , with $(k')^p \subset k$. Next adjoin $U_j = T_j^{1/p}$ by the same construction. A polynomial relation among the U_j over k' would, after taking its p th power, be a nonzero polynomial relation among the T_j over k : every coefficient of k' has its p th power in k , and Frobenius is injective on this field. Thus the U_j are algebraically independent over k' .

Set $F' = k'(U_1, \dots, U_d)$ and $L = KF'$. This extension of K is finite and purely inseparable: it adjoins only the finitely many coefficient roots and the U_j , all with p th powers in K ; the p th power of any rational expression in them lies in K . For each a_i , take the coefficientwise p th roots of its numerator and denominator and substitute the U_j . The denominator is nonzero because its p th power is the original nonzero denominator. We obtain $b_i \in F'$ with $b_i^p = a_i$. Put

$$Q(X) = X^n + \sum_{i < n} b_i X^i, \quad Q(X)^p = P(X^p).$$

Evaluation at β gives $Q(\beta)^p = P(\alpha) = 0$, hence $Q(\beta) = 0$. Also

$$Q'(\beta)^p = P'(\alpha) \neq 0.$$

If h is the irreducible minimal polynomial of β over F' , write $Q = hq$. Evaluating $Q' = h'q + hq'$ at β makes $h'(\beta)q(\beta) \neq 0$. Thus $h' \neq 0$, so β is separable over F' .

For each separable irreducible minimal polynomial, the gcd calculation above gives a Euclidean identity with its derivative equal to one. That identity remains equal to one after mapping the coefficients to a field extension, so the polynomial still has no repeated root. Each factor, in particular each new minimal polynomial, consequently has distinct roots. Choose finite generators of K_s/F ; their separable polynomials therefore make the successive generator tower for $K_s F'/F'$ separable. Adjoining the now separable β gives a further separable step. The finite-tower proof AG-CA-17, Lemma 7.1, gives

$$H = K_s(\beta)F' \subset L_s,$$

where L_s is the separable subfield of L/F' constructed above.

Here is the needed degree bound after base change. Choose a basis of K over $K_s(\beta)$ of size m/p . Its H -linear span inside L is a subalgebra: products of basis elements expand with coefficients in $K_s(\beta) \subset H$. It is a finite-dimensional domain over H , hence a field, since multiplication by a nonzero element is an injective linear map and therefore surjective. It contains both K and F' , so it is exactly their compositum L . Consequently

$$[L : L_s] \leq [L : H] \leq m/p < m.$$

The extension L/k' is finitely generated, the U_j are its transcendence basis, and L/F' is finite. Strong induction applied to this data gives finite purely inseparable k''/k' and L'/L with L'/k'' separably generated.

Finally these purely inseparable extensions compose. If $z \in L'$, some p -power lies in L and a further p -power lies in K , so L'/K is purely inseparable; the same argument applies to k''/k . Their degrees are finite by the tower basis calculation. Taking the final $k' = k''$ and $K' = L'$ gives the required commutative field square. This also covers the empty transcendence basis and trivial root extensions. $\square$

Lemma F.70: matrix left inverse

Adapted from the Stacks project, algebra.tex, lines 2709–2745.

Let R be a ring. Let $n \geq m$. Let A be an $n \times m$ matrix with coefficients in R . Let $J \subset R$ be the ideal generated by the $m \times m$ minors of A .

1. For any $f \in J$ there exists a $m \times n$ matrix B such that $BA = f1_{m \times m}$.
2. If $f \in R$ and $BA = f1_{m \times m}$ for some $m \times n$ matrix B , then $f^m \in J$.

Proof. For $m = 0$, the only minor is the empty determinant 1, so $J = R$; both assertions hold for the unique matrix between the zero modules. Assume $m > 0$. For $I \subset \{1, \dots, n\}$ with $|I| = m$, let $E_I : R^n \rightarrow R^m$ select those coordinates in increasing order, and write $A_I = E_I A$.

For a square matrix C , its adjugate is the transpose of its signed cofactor matrix. The (i, j) entry of $C \operatorname{adj}(C)$ is the cofactor expansion along row j , replacing that row by row i . It is $\det C$ when $i = j$ and zero when $i \neq j$, since then two rows coincide. The analogous column expansion gives $\operatorname{adj}(C)C = (\det C)1_m$ as well. The row and column expansions are obtained by grouping the signed permutation formula according to the chosen entry. Thus both identities hold over every commutative ring.

If $f = \sum_I c_I \det A_I$, set $B = \sum_I c_I \operatorname{adj}(A_I) E_I$. The identity just proved gives $BA = f1_m$, proving (1).

For (2), expand $\det(BA)$ by multilinearity in its m columns, each a linear combination of columns of B . Terms selecting a repeated column vanish by alternation. Group the remaining terms by their m distinct selected column indices I , in increasing order. The signed sum of the coefficients for each group is $\det A_I$, while its column determinant is $\det B^I$, where B^I consists of columns indexed by I . Hence

$$\det(BA) = \sum_{|I|=m} \det(B^I) \det(A_I).$$

If $BA = f1_m$, its determinant is f^m , so the displayed identity gives $f^m \in J$. This proves (2) with no field, domain or reducedness hypothesis. $\square$

Lemma F.71: presentation sym exterior

Adapted from the Stacks project, algebra.tex, lines 2282–2310.

For $M_2 \rightarrow M_1 \rightarrow M \rightarrow 0$, the kernel of $\operatorname{Sym}^n M_1 \rightarrow \operatorname{Sym}^n M$ is the image of $M_2 \otimes \operatorname{Sym}^{n-1} M_1$; the analogous statement holds for exterior powers.

Proof. Put $N = \operatorname{im}(M_2 \rightarrow M_1)$. The symmetric algebra is the tensor algebra modulo the pairwise commutation relations. Quotienting it further by the degree-one elements of N gives the symmetric algebra of $M_1/N = M$, because maps from either quotient to a commutative algebra are exactly linear maps from M_1 annihilating N . In degree n the ideal generated by N consists precisely of finite

sums $n_i u_i$ with u_i of degree $n - 1$; lifting each n_i from M_2 gives the asserted image. Surjectivity follows from lifting every factor of a pure symmetric tensor. The exterior algebra is the tensor algebra modulo $x \otimes x$ for every degree-one element x ; the same quotient and degree argument gives the exterior assertion, with product denoted by wedge. This includes $n = 1$; for $n = 0$ the kernel is zero. $\square$

Lemma F.72: principal localization NL

Adapted from the Stacks project, algebra.tex, lines 37031–37116.

Let $A \rightarrow B$ be a ring map. Let $g \in B$. Suppose $\alpha : P \rightarrow B$ is a presentation with kernel I . Then a presentation of B_g over A is the map

$$\beta : P[x] \longrightarrow B_g$$

extending α and sending x to $1/g$. The kernel J of β is generated by I and the element $fx - 1$ where $f \in P$ is an element mapped to $g \in B$ by α . In this situation we have

1. $J/J^2 = (I/I^2)_g \oplus B_g(fx - 1)$,
2. $\Omega_{P[x]/A} \otimes_{P[x]} B_g = \Omega_{P/A} \otimes_P B_g \oplus B_g dx$,
3. $\text{NL}(\beta) \cong \text{NL}(\alpha) \otimes_B B_g \oplus (B_g \xrightarrow{g} B_g)$.

Hence the canonical map $\text{NL}_{B/A} \otimes_B B_g \rightarrow \text{NL}_{B_g/A}$ is a homotopy equivalence.

Proof. Since $P[x]/(I, fx - 1) = B[x]/(gx - 1) = B_g$ we get the statement about I and $fx - 1$ generating J . Consider the commutative diagram

$$\begin{array}{c} 0 \rightarrow \Omega_{P/A} \otimes_P B_g \\ \Omega_{P/A} \otimes_P B_g \rightarrow \Omega_{P[x]/A} \otimes_{P[x]} B_g \\ \Omega_{P[x]/A} \otimes_{P[x]} B_g \rightarrow \Omega_{B[x]/B} \otimes_{B[x]} B_g \\ \Omega_{B[x]/B} \otimes_{B[x]} B_g \rightarrow 0 \\ (I/I^2)_g \rightarrow J/J^2 \\ (I/I^2)_g \rightarrow \Omega_{P/A} \otimes_P B_g \\ J/J^2 \rightarrow (gx - 1)/(gx - 1)^2 \\ J/J^2 \rightarrow \Omega_{P[x]/A} \otimes_{P[x]} B_g \\ (gx - 1)/(gx - 1)^2 \rightarrow 0 \\ (gx - 1)/(gx - 1)^2 \rightarrow \Omega_{B[x]/B} \otimes_{B[x]} B_g \end{array}$$

with exact rows of Lemma F.42. The B_g -module $\Omega_{B[x]/B} \otimes_{B[x]} B_g$ is free of rank 1 on dx . The element dx in the B_g -module $\Omega_{P[x]/A} \otimes_{P[x]} B_g$ provides a splitting for the top row. The element $gx - 1 \in (gx - 1)/(gx - 1)^2$ is mapped to gdx in $\Omega_{B[x]/B} \otimes_{B[x]} B_g$ and hence $(gx - 1)/(gx - 1)^2$ is free of rank 1 over B_g . Here is a direct conormal calculation that also includes $B_g = 0$. Put $q = gx - 1$. Multiplication by q on $B[x]$ is injective: for a nonzero polynomial, the coefficient of its lowest nonzero degree in its product with q is the negative of that coefficient. The map

$$B[x]/(q) \longrightarrow (q)/(q^2), \quad [a] \longmapsto [aq],$$

is surjective. If $aq = bq^2$, injectivity of multiplication by q gives $a = bq$, so it is injective as well. Thus $(q)/(q^2)$ is the free rank-one B_g -module with generator q . When g is nilpotent, $B_g = 0$

and both modules are zero, so the same calculation applies without a proper regular-sequence hypothesis.

Let us prove $(I/I^2)_g \rightarrow J/J^2$ is injective. Consider the P -algebra map

$$\pi : P[x] \rightarrow (P/I^2)_f = P_f/I_f^2$$

sending x to $1/f$. Since J is generated by I and $fx - 1$ we see that $\pi(J) \subset (I/I^2)_f = (I/I^2)_g$. Since this is an ideal of square zero we see that $\pi(J^2) = 0$. If $a \in I$ maps to an element of J^2 in J , then $\pi(a) = 0$, which implies that a maps to zero in I_f/I_f^2 . This proves the desired injectivity.

Thus we have a short exact sequence of two term complexes

$$0 \rightarrow \mathrm{NL}(\alpha) \otimes_B B_g \rightarrow \mathrm{NL}(\beta) \rightarrow (B_g \xrightarrow{g} B_g) \rightarrow 0$$

For this sequence the following splittings commute with the differentials:

$$J/J^2 = (I/I^2)_g \oplus B_g(fx - 1) \quad \text{and} \quad \Omega_{P[x]/A} \otimes B_g = \Omega_{P/A} \otimes B_g \oplus B_g(g^{-2}df + dx).$$

This works because $d(fx - 1) = xdf + fdx = g(g^{-2}df + dx)$ in $\Omega_{P[x]/A} \otimes B_g$. The second summand is contractible: multiplication by g^{-1} from degree zero to degree one is a homotopy, since its two composites with the differential g are identity. Consequently inclusion and projection on the first summand are homotopy inverses. This proves the claimed homotopy equivalence. $\square$

Lemma F.73: product local

Adapted from the Stacks project, algebra.tex, lines 13035–13062.

A ring with finitely many maximal ideals and locally nilpotent Jacobson radical is the product of its localizations at those ideals; every prime is maximal.

Proof. Write $I = \bigcap_i \mathfrak{m}_i$. Every element of I is nilpotent, so every prime contains I , hence the product of the finitely many $\mathfrak{m}_i$, and therefore one maximal ideal. Thus these are all the primes. The Chinese remainder calculation gives $R/I = \prod_i R/\mathfrak{m}_i$: for two comaximal ideals choose $a + b = 1$ with a in the first and b in the second and use $ub + va$ to lift a prescribed pair; induction gives the finite version. Idempotents lift uniquely across a nil ideal by AG-CA-01, Lemma 5.1. Lift the coordinate idempotents to e_i . For $i \neq j$, the product $e_i e_j$ is an idempotent in the nil ideal, hence zero: an idempotent $e = e^2$ satisfies $e = e^N$ for every positive N , so nilpotence makes it zero. The lifted idempotents are therefore pairwise orthogonal, and their sum is idempotent. Hence $1 - \sum_i e_i$ is also an idempotent; its residue is zero, so it lies in the nil ideal and is zero by the same argument. Consequently $R \rightarrow \prod_i e_i R$, $r \mapsto (e_i r)$, has inverse $(r_i) \mapsto \sum r_i$. The factor $e_i R$ has just the prime corresponding to $\mathfrak{m}_i$, so is local and is the localization $R_{\mathfrak{m}_i}$ by the fraction universal property. This proves the claimed product. $\square$

Lemma F.74: regular quasi regular

Adapted from the Stacks project, algebra.tex, lines 17680–17766.

If $f_1, \dots, f_c$ is a regular sequence in A , the conormal module of $I = (f_1, \dots, f_c)$ is free over A/I on the classes of the f_i . Only this consequence of quasi-regularity is used here.

Proof. First prove that every relation $\sum a_i f_i = 0$ is generated by the pairwise Koszul relations. For $c = 1$, injectivity of f_1 gives $a_1 = 0$. For $c > 1$, reduction modulo $(f_1, \dots, f_{c-1})$ and injectivity of f_c give $a_c = \sum_{i < c} b_i f_i$. Subtract from the relation the sum of the relations $b_i(f_i e_c - f_c e_i)$, making its last coefficient zero. Apply induction to the remaining coefficients. In particular all coefficients of every relation lie in I . Now if $\sum a_i f_i \in I^2$, subtract a displayed finite sum of products $f_i f_j$,

changing each a_i by an element of I , to obtain a relation equal to zero. The preceding calculation puts its coefficients in I , hence puts the original a_i in I . This proves injectivity of $(A/I)^c \rightarrow I/I^2$; generation proves surjectivity. $\square$

Lemma F.75: regular sequence in polynomial ring

Adapted from the Stacks project, algebra.tex, lines 17576–17624.

Let R be a ring. Let $f_1, \dots, f_r \in R$ which do not generate the unit ideal. The following are equivalent:

1. any permutation of $f_1, \dots, f_r$ is a regular sequence,
2. any subsequence of $f_1, \dots, f_r$ (in the given order) is a regular sequence, and
3. $f_1x_1, \dots, f_rx_r$ is a regular sequence in the polynomial ring $R[x_1, \dots, x_r]$.

Proof. It is clear that (1) implies (2). We prove (2) implies (1) by induction on r . The case $r = 1$ is trivial. The case $r = 2$ says that if $a, b \in R$ are a regular sequence and b is a nonzerodivisor, then b, a is a regular sequence. The kernel isomorphism has the following explicit form. If $ax = by$, send the class of x modulo (b) to the class of y modulo (a) . The element y is unique because b is a nonzerodivisor. Replacing x by $x + bz$ replaces y by $y + az$, so the map is well-defined; its image is in the kernel of multiplication by b modulo (a) . Conversely, from $by = ax$ recover the unique x because a is a nonzerodivisor. These constructions are inverse and linear. Thus vanishing of the latter kernel gives vanishing of the former. The case $r > 2$. Assume (2) holds and say we want to prove $f_{\sigma(1)}, \dots, f_{\sigma(r)}$ is a regular sequence for some permutation σ . We already know that $f_{\sigma(1)}, \dots, f_{\sigma(r-1)}$ is a regular sequence by induction. Hence it suffices to show that f_s where $s = \sigma(r)$ is a nonzerodivisor modulo $f_1, \dots, \hat{f}_s, \dots, f_r$. If $s = r$ we are done. If $s < r$, then note that f_s and f_r are both nonzerodivisors in the ring $R/(f_1, \dots, \hat{f}_s, \dots, f_{r-1})$ (by induction hypothesis again). Since we know f_s, f_r is a regular sequence in that ring we conclude by the case of sequence of length 2 that f_r, f_s is too.

Note that $R[x_1, \dots, x_r]/(f_1x_1, \dots, f_rx_r)$ as an R -module is a direct sum of the modules

$$R/I_E \cdot x_1^{e_1} \dots x_r^{e_r}$$

indexed by multi-indices $E = (e_1, \dots, e_r)$ where I_E is the ideal generated by f_j for $1 \leq j \leq i$ with $e_j > 0$. Hence $f_{i+1}x_{i+1}$ is a nonzerodivisor on this if and only if f_{i+1} is a nonzerodivisor on R/I_E for all E . For any subset $U \subset \{1, \dots, i\}$, take E with $e_j = 1$ for $j \in U$ and $e_j = 0$ otherwise. Then $I_E = (f_j \mid j \in U)$, and (3) makes f_{i+1} a nonzerodivisor on this quotient. Apply this successively to the earlier members of any chosen subsequence; it proves every required multiplication injective. Its ideal is proper because it is contained in the original proper ideal. Thus (3) implies (2). Conversely, if (2) holds, then any subsequence of $f_1, \dots, f_i, f_{i+1}$ is a regular sequence in particular f_{i+1} is a nonzerodivisor on all R/I_E . In this way we see that (2) implies (3). $\square$

Lemma F.76: relative global complete intersection

Adapted from the Stacks project, algebra.tex, lines 38280–38292.

A relative global complete intersection over a Noetherian base is syntomic.

Proof. It is finitely presented by its displayed finite equation list. Its fibres are global complete intersections by the dimension definition. Its flatness is the quotient-flatness conclusion of F.77, proved immediately below using F.53. These are exactly the three syntomic conditions. $\square$

Lemma F.77: relative global complete intersection conormal

Adapted from the Stacks project, algebra.tex, lines 38204–38278.

Let R be Noetherian and let $S = R[X_1, \dots, X_n]/(f_1, \dots, f_c)$ have all its nonempty fibres of dimension $n - c$. At every prime above the equation ideal, the f_i form a regular sequence in the corresponding polynomial local ring, its successive quotients are R -flat, and $(f)/(f)^2$ is free over S on their classes. This is the Noetherian form consumed in desingularization.

Proof. Fix a base prime $\mathfrak{p}$ and put $k = \kappa(\mathfrak{p})$. In $k[X]$ every minimal prime of the equation ideal has height at most c , by AG-CA-11, Theorem 3.1. Its component has dimension n minus that height, by AG-CA-09, Theorem 4.3. The asserted fibre dimension $n - c$ forces every such height to equal c . At a prime $\mathfrak{r}$ containing the equations, $A = k[X]_{\mathfrak{r}}$ is regular by iterating AG-CA-14, Proposition 3.3 from the field, hence CM by AG-CA-12, Theorem 6.1. For a minimal equation prime $\mathfrak{a} \subset \mathfrak{r}$, prime-localization correspondence preserves all primes below $\mathfrak{a}$, so $\dim A_{\mathfrak{a}A} = \text{height}(\mathfrak{a}) = c$. The CM dimension formula AG-CA-12, Lemma 5.4, applied in A at $\mathfrak{a}A$ therefore gives $\dim A/\mathfrak{a}A = \dim A - c$ for this component of the quotient. Taking the maximum gives that same quotient dimension. The expected-dimension criterion AG-CA-12, Theorem 4.2, makes the c equations a regular sequence in this fibre local ring.

The polynomial local ring over $R_{\mathfrak{p}}$ is flat, since polynomial extension is free and localization is flat. Apply F.53, whose full Noetherian lifting proof uses AG-CA-08, Lemma 5.1, to obtain regularity before reduction and flatness of every successive quotient. The elementary regular-sequence relation calculation proved at F.74 makes its conormal module free on the classes of the f_i at every prime of S . The global map $S^c \rightarrow (f)/(f)^2$ is surjective and has zero kernel after every prime localization, so AG-CA-02, Theorem 5.1, makes it an isomorphism. Flatness of the quotient is likewise detected locally by the ideal injection criterion and local detection. $\square$

Lemma F.78: relative global complete intersection smooth

Adapted from the Stacks project, algebra.tex, lines 39071–39113.

For the preceding Noetherian presentation, smoothness at a prime is equivalent to one $c \times c$ Jacobian minor being outside that prime.

Proof. The preceding conormal calculation identifies the two-term complex with $S^c \xrightarrow{J} S^n$. If a chosen maximal minor is invertible, the inverse of that square block gives a left inverse to J ; elimination identifies its cokernel with a free module of rank $n - c$. The split conormal criterion AG-CA-17, Theorem 3.1, gives smoothness on that localization. Conversely smoothness supplies a left inverse after localization by that same criterion. Reducing at the prime preserves the left inverse. Thus the matrix over the residue field has rank c , and some c -minor is nonzero. Principal localization of the conormal complex is the explicit contractible-summand computation proved at F.72. $\square$

Lemma F.79: section smooth

Adapted from the Stacks project, algebra.tex, lines 39859–39914.

For a smooth ring map $R \rightarrow S$ with section $\sigma : S \rightarrow R$, put $I = \ker \sigma$. Then I/I^2 is finite projective over R . If it is free of rank d , the I -adic completion is $R[[T_1, \dots, T_d]]$.

Proof. The map $s \mapsto s - \sigma(s) \bmod I^2$ is an R -derivation to I/I^2 and sends each $i \in I$ to its class. Universal differentials give the inverse to the natural map $I/I^2 \rightarrow \Omega_{S/R} \otimes_S R$: both composites agree on the algebra generators and on the ideal classes. Smoothness makes $\Omega_{S/R}$ finite projective by the split conormal criterion AG-CA-17, Theorem 3.1, so its base change is finite projective. The ideal I is finitely generated: for finite algebra generators x_j of S/R , subtracting their section values shows that $x_j - \sigma(x_j)$ generate the section kernel.

Choose $t_i \in I$ representing a free conormal basis. Define a map $b : R[[T]] \rightarrow \widehat{S}$ by convergent evaluation $T_i \mapsto t_i$. It is surjective: start with the residue coefficient in R , then successively correct the error in I^n/I^{n+1} by a homogeneous degree- n polynomial in the t_i . These monomials generate that quotient because the t_i generate I/I^2 ; expanding products proves this in every degree. The corrections converge in the completion.

The chosen conormal identification defines a first-order R -map $S \rightarrow R[[T]]/(T)^2$ sending t_i to T_i and reducing to σ . Formal smoothness of S/R , proved equivalent to the split conormal property in AG-CA-17, Theorem 3.1, lifts this map successively through the square-zero kernels $(T)^n/(T)^{n+1}$. Its compatible inverse limit sends I into (T) , hence extends to a continuous map $a : \widehat{S} \rightarrow R[[T]]$. The composite ab sends T_i to T_i plus terms of degree at least two. Such a substitution is an automorphism: recursively correct each inverse variable one homogeneous degree at a time; the linear term is identity, so the correction at each degree is uniquely its negative error, and the series converge. The same recursion proves that both composites with the resulting inverse are identity in every finite quotient and hence in the separated limit. Replace a by $(ab)^{-1}a$. Then $ab = 1$. Since b is already surjective it is also injective and gives the required isomorphism. $\square$

Lemma F.80: silly

Adapted from the Stacks project, algebra.tex, lines 2605–2636.

Let $I_1, \dots, I_r, J$ be ideals of a ring with $J \not\subset I_i$ for every i , and at most two of the I_i not prime. Then J contains an element outside all I_i .

Proof. For two ideals, choose $x \in J \setminus I_1$ and $y \in J \setminus I_2$. If neither already works, then $x \in I_2$, $y \in I_1$, and $x + y$ works. Remove any I_i contained in another listed ideal, since avoidance of the larger ideal implies avoidance of the smaller. Induct on the number of remaining ideals. For $r \geq 3$ renumber so I_r is prime. Choose x avoiding the first $r - 1$ by induction. If $x \in I_r$, choose $y \in JI_1 \cdots I_{r-1} \setminus I_r$: such y exists because none of these factors is contained in the prime I_r , and products of separately chosen elements outside a prime stay outside it. Then $y \in I_i$ for $i < r$, so $x + y$ still avoids them, and it avoids I_r because $x \in I_r$ and $y \notin I_r$. $\square$

Lemma F.81: size extension pth roots

Adapted from the Stacks project, algebra.tex, lines 45787–45812.

If $a_1, \dots, a_n$ in a characteristic- p field have independent absolute differentials, adjoining their p th roots has degree p^n .

Proof. First prove the absolute differential calculation used here. A maximal p -independent set is a p -basis, by the Zorn and degree- p argument in AG-CA-19S, Lemma 1.1. In its monomial expansion define the partial derivation of one basis element by differentiating the corresponding monomial and killing all p th-power coefficients. The relations $T_b^p - b^p$ have derivative zero, so these are well-defined derivations. Their values on the basis elements show that their differentials are linearly independent; differentiating every expansion shows that they span. An expansion has all partial derivatives zero precisely when every exponent is zero, hence precisely when it lies in the subfield of p th powers. Thus $da = 0$ if and only if $a \in k^p$.

For $n = 1$ this proves a_1 is not a p th power. The monic polynomial $T^p - a_1$ is irreducible: any proper factor over k is $(T - \alpha)^e$ in an algebraic closure, $1 \leq e < p$, and its coefficient $-e\alpha$ would put $\alpha \in k$. Suppose by induction the first $n - 1$ roots give degree p^{n-1} , with their standard monomial basis. If a_n became a p th power there, write a root in that basis and raise its expansion to p . Then $a_n = \sum_E \lambda_E^p \prod_{i < n} a_i^{e_i}$. Its differential belongs to the span of the preceding da_i , a contradiction. The same degree- p irreducibility argument proves the last extension degree p , and multiplication of finite vector-space degrees gives p^n . $\square$

Lemma F.82: smooth over field

Adapted from the Stacks project, algebra.tex, lines 38622–38676.

A smooth finite-type algebra over a field is locally a complete intersection.

Proof. At a given prime choose a standard smooth neighbourhood by AG-CA-17, Theorem 5.1. In its polynomial ambient local ring, the classes of its chosen equations in the maximal ideal modulo its square are linearly independent. Indeed, differentiating an element of the square gives zero after reduction to the residue field; a linear relation between those equation classes would therefore give the same relation between their differential rows, contradicting the invertible Jacobian minor. Extend the independent classes to a basis of the maximal ideal modulo its square. The ambient polynomial local ring is regular by iterating AG-CA-14, Proposition 3.3, starting with the field; therefore AG-CA-12, Theorem 6.1, makes this full minimal generator list a regular sequence; its initial equation list is regular. The complete-intersection neighbourhood criterion just proved gives a global complete-intersection chart around the prime. Such charts cover the spectrum, proving the local statement without any algebraic-closure reduction or omitted diagram calculation. $\square$

Lemma F.83: smooth syntomic

Adapted from the Stacks project, algebra.tex, lines 38880–38951.

Let $R \rightarrow S$ be a smooth ring map. There exists an open covering of $\text{Spec}(S)$ by standard opens $D(g)$ such that each S_g is standard smooth over R . If R is Noetherian, then $R \rightarrow S$ is syntomic.

Proof. If $S = 0$, take the empty standard-open cover. The map $R \rightarrow 0$ is finitely presented by $R/(1)$, its zero module is flat, and its fibres are empty, so the syntomic consequence holds as well. Assume $S \neq 0$.

Choose a presentation $\alpha : R[x_1, \dots, x_n] \rightarrow S$ with kernel $I = (f_1, \dots, f_m)$. For every subset $E \subset \{1, \dots, m\}$ consider the open subset U_E where the classes $f_e, e \in E$ freely generate the finite projective S -module I/I^2 . Here is the full basis-open calculation over an arbitrary base. Write $M = I/I^2$. The split conormal sequence of AG-CA-17, Theorem 3.1, makes M a finite direct summand of S^n . Its complementary summand is finite, generated by the projections of the coordinate vectors, so this splitting is a finite presentation of M . Tensoring an injection with M is injective by restricting the injection obtained with S^n . Thus M is finitely presented and flat, and the complete proof of AG-RG-S01, Lemma 1.2 gives finite freeness over each local ring.

At a point $\mathfrak{q}$, choose a basis of $M \otimes_S \kappa(\mathfrak{q})$ from the spanning classes of the finitely many f_i , and let E be its indices, of cardinality r . Over $S_{\mathfrak{q}}$, Nakayama makes the resulting map $\varphi : S^r \rightarrow M$ surjective; its square coordinate matrix in a free local basis is invertible modulo the maximal ideal, so the cofactor identity makes $\varphi_{\mathfrak{q}}$ an isomorphism. The finite cokernel of φ vanishes there. Clearing the finitely many denominators annihilating its generators gives $a \notin \mathfrak{q}$ for which φ_a is surjective. Projectivity of M_a splits this surjection, so its kernel is a finite direct summand of S_a^r . Its localization at $\mathfrak{q}$ is zero. Clear the finitely many annihilating denominators of its generators as well, taking their numerators in S , to obtain $b \notin \mathfrak{q}$ with φ_{ab} an isomorphism. Hence $D(ab) \subset U_E$.

Apply the same argument at any point where a fixed selected list is a basis. It gives a principal neighbourhood contained in its basis locus, proving that every U_E is open. Since a spanning list over a field contains a basis, these opens cover $\text{Spec } S$. Empty products are one, and the argument also covers rank zero. Choose for each point a principal neighbourhood $D(g)$ contained in its selected U_E , using the basis-open calculation just proved. These opens cover $\text{Spec } S$, and every chosen S_g is nonzero because its open contains that point. For such a g we look at the presentation

$$\beta : R[x_1, \dots, x_n, x_{n+1}] \longrightarrow S_g$$

mapping x_{n+1} to $1/g$. Setting $J = \ker(\beta)$ we use Lemma F.72 to see that $J/J^2 \cong (I/I^2)_g \oplus S_g$

is free. We may and do replace S by S_g . Then using Lemma F.55 we may assume we have a presentation $\alpha : R[x_1, \dots, x_n] \rightarrow S$ with kernel $I = (f_1, \dots, f_c)$ such that I/I^2 is free on the classes of $f_1, \dots, f_c$.

Using the presentation α obtained at the end of the previous paragraph, we more or less repeat this argument with the basis elements $dx_1, \dots, dx_n$ of $\Omega_{R[x_1, \dots, x_n]/R}$. Namely, for any subset $E \subset \{1, \dots, n\}$ of cardinality c we may consider the open subset U_E of $\text{Spec}(S)$ where the differential of $\text{NL}(\alpha)$ composed with the projection

$$S^{\oplus c} \cong I/I^2 \longrightarrow \Omega_{R[x_1, \dots, x_n]/R} \otimes_{R[x_1, \dots, x_n]} S \longrightarrow \bigoplus_{i \in E} S dx_i$$

is an isomorphism. Here is the full minor-cover calculation. Let A be the $n \times c$ matrix of the conormal differential in these bases. The split conormal sequence of AG-CA-17, Theorem 3.1, gives a $c \times n$ matrix B with $BA = 1_c$. Reduction at any maximal ideal gives an injection of a c -dimensional vector space into an n -dimensional one, hence $c \leq n$. By F.70(2), the $c \times c$ minors Δ_E of A generate the unit ideal. For a fixed E , the cofactor identity makes the selected square map invertible exactly where its determinant is a unit, so $U_E = D(\Delta_E)$. Thus these finitely many principal opens cover $\text{Spec } S$, and we may use them as the required $D(g)$. For $c = 0$, the empty determinant is one and the unique empty selected list covers the whole spectrum. By renumbering, we may assume $E = \{1, \dots, c\}$. For a g with $D(g) \subset U_E$ we look at the presentation

$$\beta : R[x_1, \dots, x_n, x_{n+1}] \rightarrow S_g$$

mapping x_{n+1} to $1/g$. Setting $J = \ker(\beta)$ we conclude from Lemma F.72 that $J = (f_1, \dots, f_c, f x_{n+1} - 1)$ where $\alpha(f) = g$ and that the composition

$$J/J^2 \longrightarrow \Omega_{R[x_1, \dots, x_{n+1}]/R} \otimes_{R[x_1, \dots, x_{n+1}]} S_g \longrightarrow \bigoplus_{i=1}^c S_g dx_i \oplus S_g dx_{n+1}$$

is an isomorphism: in the equation-class bases, its matrix is block triangular, with the selected square Jacobian block first and g in the last diagonal slot. Its determinant is $g\Delta_E$, a unit in S_g , so the cofactor identity gives its inverse. Reordering the coordinates as $x_1, \dots, x_c, x_{n+1}, x_{c+1}, \dots, x_n$ we conclude that S_g is standard smooth over R as desired.

For the final consequence, suppose R is Noetherian. Standard smooth algebras are syntomic (Lemmas F.85 and F.76) and being syntomic over R is local on S (Lemma F.62). $\square$

Lemma F.84: snake

Adapted from the Stacks project, algebra.tex, lines 311–343.

Given a commutative diagram

$$\begin{array}{ccc} X & \rightarrow & Y \\ X & \xrightarrow{\alpha} & U \\ Y & \rightarrow & Z \\ Y & \xrightarrow{\beta} & V \\ Z & \rightarrow & 0 \\ Z & \xrightarrow{\gamma} & W \\ 0 & \rightarrow & U \\ U & \rightarrow & V \\ V & \rightarrow & W \end{array}$$

of abelian groups with exact rows, there is a canonical exact sequence

$$\ker(\alpha) \rightarrow \ker(\beta) \rightarrow \ker(\gamma) \rightarrow \operatorname{coker}(\alpha) \rightarrow \operatorname{coker}(\beta) \rightarrow \operatorname{coker}(\gamma)$$

Moreover: if $X \rightarrow Y$ is injective, then the first map is injective; if $V \rightarrow W$ is surjective, then the last map is surjective.

Proof. The map $\partial : \ker(\gamma) \rightarrow \operatorname{coker}(\alpha)$ is defined as follows. Take $z \in \ker(\gamma)$. Choose $y \in Y$ mapping to z . Then $\beta(y) \in V$ maps to zero in W . Hence $\beta(y)$ is the image of some $u \in U$. Set $\partial z = \bar{u}$, the class of u in the cokernel of α . Changing the lift y by the image of $x \in X$ changes u by $\alpha(x)$, so the connecting map is well defined and additive. Exactness at $\ker \beta$ follows because a lift in X of an element of Y with zero image in Z has $\alpha(x) = 0$, using injectivity $U \rightarrow V$. Exactness at $\ker \gamma$ follows because $\partial z = 0$ means $u = \alpha(x)$ for some x ; then $y - x$ has zero image under β and still maps to z . Conversely any such lift has connecting class zero. At $\operatorname{coker} \alpha$, if the image of u in V equals $\beta(y)$, its image z in Z is in $\ker \gamma$ and has $\partial z = [u]$; conversely this is the definition of ∂ . At $\operatorname{coker} \beta$, if v maps to $\gamma(z)$ in W , lift z to $y \in Y$ and replace v by $v - \beta(y)$. Exactness of the bottom row puts it in the image of U , so its class comes from $\operatorname{coker} \alpha$. The converse follows from the row being a complex. If $X \rightarrow Y$ is injective, so is its restriction to $\ker \alpha$. If $V \rightarrow W$ is surjective, every class modulo $\gamma(Z)$ has a representative lifted from V , proving the final surjectivity. $\square$

Lemma F.85: standard smooth

Adapted from the Stacks project, algebra.tex, lines 38716–38780.

Let $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c) = R[x_1, \dots, x_n]/I$ be a standard smooth algebra. Then

1. the ring map $R \rightarrow S$ is smooth,
2. the S -module $\Omega_{S/R}$ is free on $dx_{c+1}, \dots, dx_n$,
3. the S -module I/I^2 is free on the classes of $f_1, \dots, f_c$,
4. for any $g \in S$ the ring map $R \rightarrow S_g$ is standard smooth,
5. for any ring map $R \rightarrow R'$ the base change $R' \rightarrow R' \otimes_R S$ is standard smooth,
6. if $f \in R$ maps to an invertible element in S , then $R_f \rightarrow S$ is standard smooth, and
7. the ring S is a relative global complete intersection over R .

Proof. Consider the naive cotangent complex of the given presentation

$$(f_1, \dots, f_c)/(f_1, \dots, f_c)^2 \longrightarrow \bigoplus_{i=1}^n S dx_i.$$

Let us compose this map with the projection onto the first c direct summands of the direct sum. According to the definition of a standard smooth algebra the classes $f_i \bmod (f_1, \dots, f_c)^2$ map to a basis of $\bigoplus_{i=1}^c S dx_i$. We conclude that $(f_1, \dots, f_c)/(f_1, \dots, f_c)^2$ is free of rank c with a basis given by the elements $f_i \bmod (f_1, \dots, f_c)^2$, and that the homology in degree 0, i.e., $\Omega_{S/R}$, of the naive cotangent complex is a free S -module with basis the images of dx_{c+j} , $j = 1, \dots, n - c$. In particular, this proves $R \rightarrow S$ is smooth.

For (4), represent g by $h(x)$ and add a variable z and the equation $hz - 1$. The Jacobian block on $x_1, \dots, x_c, z$ is block triangular, with determinant $h \det(\partial f_i / \partial x_j)$. Both factors are units in the localization, proving it standard smooth. For (6), use the same presentation over R_f and the same Jacobian minor: it remains a unit, while $R_f[x]/(f_i) \cong S$ because f was already a unit in S .

Let $\varphi : R \rightarrow R'$ be any ring map. Set $S' = R'[x_1, \dots, x_n]/(f_1^\varphi, \dots, f_c^\varphi)$ where f^φ is the polynomial obtained from $f \in R[x_1, \dots, x_n]$ by applying φ to all the coefficients. Then $S' \cong R' \otimes_R S$. Moreover, the determinant of Definition AG-CA-17, Section 4, equations (6) and (7) for S'/R' is equal to g^φ . Its image in S' is therefore the image of g via $R[x_1, \dots, x_n] \rightarrow S \rightarrow S'$ and hence invertible. This proves (5).

To prove (7) it suffices to show that every nonzero fibre $S \otimes_R \kappa(\mathfrak{p})$ has dimension $n - c$ for every prime $\mathfrak{p} \subset R$. By (5) it suffices to prove that any standard smooth algebra $k[x_1, \dots, x_n]/(f_1, \dots, f_c)$ over a field k , if nonzero, has dimension $n - c$. Put $B = k[x_1, \dots, x_n]/I \neq 0$. It is locally a complete intersection by F.82. For a prime $\mathfrak{q}$ of B , with polynomial preimage $\mathfrak{q}'$, the complete equivalence in F.57 therefore identifies the free rank of $(I/I^2)_{\mathfrak{q}}$ with $\text{height}(\mathfrak{q}') - \text{height}(\mathfrak{q})$. The standard-smooth conormal calculation above makes that rank exactly c ; equality of free ranks is checked by reducing to the residue field.

In particular, if $\mathfrak{q}$ is a minimal prime of B , its height is zero, and its polynomial preimage is a minimal prime over I of height c . By AG-CA-09, Theorem 4.3, this irreducible component has dimension $n - c$. There are finitely many such components, and their union is $\text{Spec } B$, by AG-CA-03, Proposition 2.2. The finite-component dimension formula in AG-CA-09, Section 1, gives $\dim B = n - c$. This proves the assertion for every nonzero fibre, including empty equation lists and imperfect fields. Zero fibres impose no dimension condition. $\square$

Lemma F.86: sum two terms

Adapted from the Stacks project, algebra.tex, lines 37135–37186.

Two homotopy-equivalent two-term complexes $A_1 \xrightarrow{d_A} A_0$ and $B_1 \xrightarrow{d_B} B_0$ satisfy $A_1 \oplus B_0 \cong B_1 \oplus A_0$.

Proof. Let φ, ψ be inverse comparison maps up to homotopies $h : A_0 \rightarrow A_1$, $h' : B_0 \rightarrow B_1$, chosen with $1 - \psi\varphi = hd_A$ in degree one and $d_A h$ in degree zero, and the analogous equations on B . Form the block maps

$$U = \begin{pmatrix} \psi_1 & h \\ -d_B & \varphi_0 \end{pmatrix}, \quad V = \begin{pmatrix} \varphi_1 & -h' \\ d_A & \psi_0 \end{pmatrix}.$$

Their product UV has blocks $\psi_1\varphi_1 + hd_A = 1$, $-\psi_1h' + h\psi_0$, $-d_B\varphi_1 + \varphi_0d_A = 0$, and $d_Bh' + \varphi_0\psi_0 = 1$. Reversing the order gives blocks 1, $\varphi_1h - h'\varphi_0$, $d_A\psi_1 - \psi_0d_B = 0$, and 1. Both products are upper triangular with identity diagonal, hence invertible by negating their off-diagonal block. A map with an invertible product on each side has a left and a right inverse, which coincide: $(UV)^{-1}U$ and $U(VU)^{-1}$ are the inverses of V . Thus U and V are isomorphisms. $\square$

Lemma F.87: syntomic

Adapted from the Stacks project, algebra.tex, lines 38315–38381.

Let $R \rightarrow S$ be a ring map with R Noetherian. Let $\mathfrak{q} \subset S$ be a prime lying over the prime $\mathfrak{p}$ of R . The following are equivalent:

1. There exists an element $g \in S$, $g \notin \mathfrak{q}$ such that $R \rightarrow S_g$ is syntomic.
2. There exists an element $g \in S$, $g \notin \mathfrak{q}$ such that S_g is a relative global complete intersection over R .
3. There exists an element $g \in S$, $g \notin \mathfrak{q}$, such that $R \rightarrow S_g$ is of finite presentation, the local ring map $R_{\mathfrak{p}} \rightarrow S_{\mathfrak{q}}$ is flat, and the local ring $S_{\mathfrak{q}}/\mathfrak{p}S_{\mathfrak{q}}$ is a complete intersection ring over $\kappa(\mathfrak{p})$ (see Definition F.3).

Proof. The implication (1) $\Rightarrow$ (3) is Lemma F.58. The implication (2) $\Rightarrow$ (1) is Lemma F.76. It remains to show that (3) implies (2).

Assume (3). After replacing S by S_g for some $g \in S$, $g \notin \mathfrak{q}$, we may assume S is finitely presented over R . Choose a presentation $S = R[x_1, \dots, x_n]/I$. Let $\mathfrak{q}' \subset R[x_1, \dots, x_n]$ be the prime corresponding to $\mathfrak{q}$. Write $\kappa(\mathfrak{p}) = k$. Note that $S \otimes_R k = k[x_1, \dots, x_n]/\bar{I}$ where $\bar{I} \subset k[x_1, \dots, x_n]$ is the ideal generated by the image of I . Let $\bar{\mathfrak{q}}' \subset k[x_1, \dots, x_n]$ be the prime ideal generated by the image of $\mathfrak{q}'$. By Lemma F.58 the equivalent conditions of Lemma F.57 hold for $\bar{I}$ and $\bar{\mathfrak{q}}'$. Say the dimension of $\bar{I}_{\bar{\mathfrak{q}}'}/\bar{\mathfrak{q}}'\bar{I}_{\bar{\mathfrak{q}}'}$ over $\kappa(\bar{\mathfrak{q}}')$ is c . Pick $f_1, \dots, f_c \in I$ mapping to a basis of this vector space. The images $\bar{f}_j \in \bar{I}$ generate $\bar{I}_{\bar{\mathfrak{q}}'}$ (by Lemma F.57). Set $S' = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$. Let J be the kernel of the surjection $S' \rightarrow S$. Since S is of finite presentation J is a finitely generated ideal (Lemma F.28). Consider the short exact sequence

$$0 \rightarrow J \rightarrow S' \rightarrow S \rightarrow 0.$$

As $S_{\mathfrak{q}}$ is flat over R we see that $J_{\mathfrak{q}'} \otimes_R k \rightarrow S'_{\mathfrak{q}'} \otimes_R k$ is injective (Lemma AG-CA-07, Theorem 2.1). However, by construction $S'_{\mathfrak{q}'} \otimes_R k$ maps isomorphically to $S_{\mathfrak{q}} \otimes_R k$. Hence we conclude that $J_{\mathfrak{q}'} \otimes_R k = J_{\mathfrak{q}'}/\mathfrak{p}J_{\mathfrak{q}'} = 0$. By Nakayama's lemma (Lemma F.10) we conclude that there exists a $g \in R[x_1, \dots, x_n]$, $g \notin \mathfrak{q}'$ such that $J_g = 0$. In other words $S'_g \cong S_g$. After further localizing we see that S' (and hence S) becomes a relative global complete intersection by Lemma F.66 as desired. $\square$

Lemma F.88: syntomic presentation ideal mod squares

Adapted from the Stacks project, algebra.tex, lines 38383–38402.

Let R be Noetherian. Let $S = R[x_1, \dots, x_n]/I$ for some finitely generated ideal I . If $g \in S$ is such that S_g is syntomic over R , then $(I/I^2)_g$ is a finite projective S_g -module.

Proof. By Lemma F.87 there exist finitely many elements $g_1, \dots, g_m \in S$ which generate the unit ideal in S_g such that each S_{gg_j} is a relative global complete intersection over R . Since it suffices to prove that $(I/I^2)_{gg_j}$ is finite projective, see Lemma AG-CA-07, Theorem 5.3, we may assume that S_g is a relative global complete intersection. In this case the result follows from Lemmas F.33 and F.77. $\square$

Lemma F.89: total ring fractions no embedded points

Adapted from the Stacks project, algebra.tex, lines 4748–4775.

Let R be a ring. Assume that R has finitely many minimal primes $\mathfrak{q}_1, \dots, \mathfrak{q}_t$, and that $\mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_t$ is the set of zerodivisors of R . Then the total ring of fractions $Q(R)$ is equal to $R_{\mathfrak{q}_1} \times \dots \times R_{\mathfrak{q}_t}$.

Proof. There are natural maps $Q(R) \rightarrow R_{\mathfrak{q}_i}$ since any nonzerodivisor lies in $R \setminus \mathfrak{q}_i$. Hence a natural map $Q(R) \rightarrow R_{\mathfrak{q}_1} \times \dots \times R_{\mathfrak{q}_t}$. For any nonminimal prime $\mathfrak{p} \subset R$ we see that $\mathfrak{p} \not\subset \mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_t$ by Lemma F.80. Hence $\text{Spec}(Q(R)) = \{\mathfrak{q}_1, \dots, \mathfrak{q}_t\}$ (as subsets of $\text{Spec}(R)$, see Lemma AG-CA-02, Theorem 3.1). Therefore $\text{Spec}(Q(R))$ is a finite discrete set and it follows that $Q(R) = A_1 \times \dots \times A_t$ with $\text{Spec}(A_i) = \{\mathfrak{q}_i\}$, see Lemma AG-CA-01, Theorem 5.2 and Corollary 5.3. Moreover A_i is a local ring, which is a localization of R . Hence $A_i \cong R_{\mathfrak{q}_i}$. $\square$

Lemma F.90: when colimit

Adapted from the Stacks project, algebra.tex, lines 33328–33357.

Let $R \rightarrow \Lambda$ be a ring map. Then Λ is a filtered colimit of smooth R -algebras if and only if every map $A \rightarrow \Lambda$, with A finitely presented over R , factors through a smooth R -algebra.

Proof. For the forward direction, choose finitely many generators and relations for A . Representatives of all generator images lie at one stage of the filtered diagram. The finitely many relations become zero at a common later stage, giving the factorization.

Conversely, consider the category of smooth finitely presented R -algebras equipped with a map to Λ , with morphisms commuting with that map. It is nonempty because it contains R . Given two objects B_1, B_2 , their tensor product is finitely presented; factoring its map to Λ through a smooth algebra gives a common successor. Given parallel arrows $u, v: B \rightrightarrows C$, choose finite algebra generators b_i of B and factor the finitely presented algebra $C/(u(b_i) - v(b_i))$ through a smooth algebra. The resulting arrow out of C equalizes u, v . Thus the category is filtered. One may choose a set of representatives of finite presentations, so size causes no difficulty.

Its colimit maps onto Λ : for $\lambda \in \Lambda$, factor $R[T] \rightarrow \Lambda$, $T \mapsto \lambda$, through a smooth algebra. For injectivity, if an element represented by $b \in B$ maps to zero, factor $B/(b) \rightarrow \Lambda$ through a smooth algebra; then b is already zero at a successor. Apply this to the difference of any two representatives after moving them to a common stage. This proves the ring map from the colimit to Λ is an isomorphism. $\square$

Proposition F.91: characterize separable field extensions

Adapted from the Stacks project, algebra.tex, lines 46037–46069.

For any field extension K/k , separability, geometric reducedness, algebraic formal smoothness, vanishing of $H_1(L_{K/k})$, and injectivity of $K \otimes_k \Omega_{k/P} \rightarrow \Omega_{K/P}$ are equivalent; here P is the prime field. In characteristic p , separability means that a p -basis of k remains p -independent over K^p . In characteristic zero all these conditions hold. For finitely generated extensions this agrees with having a separating transcendence basis.

Proof. Every field is formally smooth over its prime field by the full arbitrary-field proof AG-CA-19S, Lemma 1.1. Suppose the displayed differential map is injective. In a square-zero lifting test for K/k , first obtain a prime-field lift $u: K \rightarrow E$ by that lemma. Its restriction differs from the given map of k by a derivation $D: k \rightarrow J$, where J is a K -vector space through the map modulo J . The corresponding K -linear map $K \otimes_k \Omega_{k/P} \rightarrow J$ extends to $\Omega_{K/P}$ by choosing a basis complement. Subtract the resulting derivation from u . The square-zero product identity shows that this is still a ring map, and its restriction to k is the prescribed one. Thus K/k is formally smooth.

Conversely for any P -derivation $D: k \rightarrow V$, with V a K -vector space, use the test algebra $K \oplus V$ with structural map $a \mapsto (a, D(a))$ and target quotient K . A lift of the identity gives a P -derivation of K extending D . Hence every linear functional on $K \otimes_k \Omega_{k/P}$ extends to $\Omega_{K/P}$. If the displayed map had a nonzero kernel vector, a linear functional nonzero on that vector could not extend. It is therefore injective. The conormal/vector-space argument in the preceding formal-field lemma equates formal smoothness with the H_1 condition.

In characteristic zero choose a transcendence basis of k/P and extend it to one of K/P . Algebraic extensions in characteristic zero are separable. The polynomial and separable algebraic differential formulas AG-CA-16, Theorem 6.2, identify their absolute differentials with the free spaces on those bases, proving the indicated injection. The same formula for arbitrary algebraic unions follows because each finite polynomial expression and relation occurs in a finite subextension. The lifting proof in the first paragraph consequently applies.

In characteristic p , the absolute instance of F.119 says that an absolute p -basis B of k has differentials forming a basis of $\Omega_{k/P}$. The same proved equivalence for K/P says that their images in $\Omega_{K/P}$ are independent exactly when the monomials in B , with exponents below p , are independent over K^p . This proves equivalence of the differential injection with the p -basis condition. Its equivalence with the definition in F.5 will be proved below, rather than assumed here.

That condition is equivalent to $K \otimes_k k^{1/p}$ being reduced. For a finite subset of B the tensor product

is $K[T_b]/(T_b^p - b)$, free on the standard monomials. It is a field exactly when those roots have degree $p^{\#B_0}$, which is exactly the indicated monomial independence. If the degree is smaller, its map to the field generated by the roots has a nonzero kernel; every element of that kernel has zero p th power, so it is not reduced. The whole tensor product is the directed union of these rings; these inclusions are injective by their monomial bases. Reducedness is therefore equivalent to independence of all finite subsets.

The independence persists at every root height. Indeed, a relation $\sum_a c_a^{p^r} B^a = 0$ with $0 \leq a_b < p^r$ can be grouped by $a = pu + v$. Independence of the monomials B^v , $0 \leq v_b < p$, over K^p makes each grouped coefficient, a p th power, zero. Taking its unique p th root and induction on r makes every coefficient zero. Iterating the p -basis expansion proves that these are the bases for adjoining all p^r th roots of k . Consequently tensoring K with any finite purely inseparable extension of k gives a field: it injects into the corresponding full-root tensor field and is finite-dimensional over K .

For any finitely generated extension E/k , F.69 gives finite purely inseparable k'/k and E'/E with E'/k' separably generated. Put $F = K \otimes_k k'$, a field by the preceding root-height argument. Choose a finite separating basis T for E'/k' and its finite separable algebraic generator tower over $k'(T)$. The ring $F \otimes_{k'} k'(T)$ is the localization of the polynomial ring $F[T]$ at the nonzero polynomials from $k'[T]$, so it is Noetherian and regular by AG-CA-14, Proposition 3.3 and Theorem 3.2. Tensor the separable tower with F . At every step its monic polynomial and invertible derivative give a finite étale algebra, by AG-CA-17, Proposition 6.3 and base change. The residue-field fibre algebras are finite products of finite separable fields by its Theorem 7.2. Thus flatness and regularity ascent F.47 make every resulting ring regular; the rings are Noetherian because each step is finite over the preceding one. Regular local rings are domains, hence these rings are reduced by local detection. Consequently $K \otimes_k E' = F \otimes_{k'} E'$ is reduced.

Tensoring the inclusion $E \subset E'$ with the k -vector space K preserves injectivity: extend a basis of E to a basis of E' to see the split vector-space inclusion. Hence $K \otimes_k E$ is reduced. Every field extension is the directed union of its finitely generated subfields; their tensor maps are injective by the same argument, and every nilpotent element occurs at one stage. This proves geometric reducedness for every field extension. The converse follows by testing $k^{1/p}$.

Here is the finitely generated bridge to a separating transcendence basis. Suppose first that L/k is finitely generated and formally smooth. Choose field generators and a domain $B = k[a_1, \dots, a_n] \subset L$ with fraction field L . Write $B = P/I$, where $P = k[X_1, \dots, X_n]$. The ideal I is finitely generated by the Hilbert basis theorem AG-CA-03, Theorem 2.1. Formal smoothness of L/k splits the conormal sequence at its generic point by AG-CA-17, Theorem 3.1. Choose $f_1, \dots, f_c$ among the generators of I whose conormal classes are a basis over L . Their differentials are independent in L^n , so some c -column Jacobian determinant Δ lies outside I .

The polynomial local ring P_I has maximal ideal IP_I and residue field L . Nakayama's lemma F.10 makes the f_i generate IP_I . The finite module $I/(f_1, \dots, f_c)$ therefore vanishes there. For each of its finite generators choose an annihilating denominator outside I , and multiply those denominators to obtain $h_0 \notin I$ with $I_{h_0} = (f_1, \dots, f_c)_{h_0}$. Put $h = h_0\Delta$ and let g be its nonzero image in B . Then

$$B_g = k[X_1, \dots, X_n, Z]/(f_1, \dots, f_c, hZ - 1).$$

The minor using the selected variable columns and Z has block determinant $h\Delta$, a unit. This is a standard smooth domain by AG-CA-17, Theorem 4.1, with fraction field L . The independent converse construction in F.64 gives a separating transcendence basis. Conversely a finitely generated separably generated field is formally smooth by the transcendental and separable algebraic cases of F.49 and formal composition. This proves the finitely generated comparison without identifying separability with the p -basis condition in advance.

Now use the exact definition F.5. If K/k is separable by that definition, F.49(3) proves formal smoothness, hence all the conditions already compared above. Conversely, in characteristic p , suppose the p -basis condition holds for K/k . For every finitely generated intermediate field L , it

also holds for L/k : a monomial relation with coefficients in $L^p \subset K^p$ would already be a relation in K^p . Thus the differential map for L/k is injective, its square-zero lifting proof above gives formal smoothness, and the finitely generated construction makes L/k separably generated. Definition F.5 therefore makes the original, possibly infinitely generated, K/k separable. In characteristic zero, every finitely generated intermediate field has a finite algebraic tower over a transcendence basis, and all its irreducible minimal polynomials are separable; the same definition applies. This proves the definition bridge and finishes every comparison in the full stated field scope. $\square$

Definition F.92: G ring

Adapted from the Stacks project, more-algebra.tex, lines 12788–12793.

A ring R is called a *G-ring* if R is Noetherian and for every prime $\mathfrak{p}$ of R the ring map $R_{\mathfrak{p}} \rightarrow (R_{\mathfrak{p}})^{\wedge}$ is regular.

Definition F.93: excellent

Adapted from the Stacks project, more-algebra.tex, lines 13961–13970.

Let R be a ring.

1. We say R is *quasi-excellent* if R is Noetherian, a G-ring, and J-2.
2. We say R is *excellent* if R is quasi-excellent and universally catenary.

Definition F.94: local complete intersection

Adapted from the Stacks project, more-algebra.tex, lines 8947–8952.

A ring map $A \rightarrow B$ is called a *local complete intersection* if it is of finite type and for some (equivalently any) presentation $B = A[x_1, \dots, x_n]/I$ the ideal I is Koszul-regular.

Lemma F.94.A: arbitrary-ring presentation independence

Adapted from the freely accessible Stacks project, Tag 07CZ, with the finite-generation, change-of-basis and complex arguments written out below. This argument retains the lesson's GNU Free Documentation License terms.

Let A be any ring and let C have two finite polynomial presentations

$$P = A[T_1, \dots, T_n] \twoheadrightarrow C = P/I, \quad Q = A[X_1, \dots, X_m] \twoheadrightarrow C = Q/J.$$

If I is a Koszul-regular ideal in the sense of Definition F.95, then so is J . In particular the parenthetical assertion in Definition F.94 holds without a Noetherian hypothesis, and such a finite-type algebra is finitely presented.

Proof. First prove the required finite generation. At every prime containing I , a principal neighbourhood makes I generated by a finite Koszul-regular list. The corresponding principal opens cover $V(I)$. If their elements are a_λ , no prime contains $I + (a_\lambda)$, so that ideal is the unit ideal: a proper ideal lies in a prime by the full prime-separation proof in AG-RG-S04, Lemma P0.1. Thus finitely many $a_1, \dots, a_s$ suffice, and

$$1 = \sum_{i=1}^s b_i a_i + d, \quad d \in I.$$

Choose in I numerators of the finite generators of each I_{a_i} and let N be the ideal generated by those numerators and d . Then $N_{a_i} = I_{a_i}$. At a prime outside all $D(a_i)$, the displayed relation makes d a unit, so N and I are both the unit ideal there. Every prime is in one of these cases. Local detection applied to I/N gives $I = N$. Hence I is finite and C is finitely presented. Lemma F.44, proved for arbitrary rings, now makes J finite too. If $C = 0$, both kernels are unit ideals and their Koszul-regularity condition is vacuous; this also gives a finite presentation. We may henceforth fix a prime $\mathfrak{q}$ of the nonzero C .

Choose $p_j(T) \in P$ representing the images of X_j and $q_i(X) \in Q$ representing the images of T_i . In $W = A[T, X]$ the common kernel is

$$L = (I, X_j - p_j(T)) = (J, T_i - q_i(X)).$$

Quotienting either displayed ideal eliminates its added variables and gives precisely the specified map to C , proving both equalities. Let $\mathfrak{r}$ be the inverse image of $\mathfrak{q}$ in W and put $s_i = T_i - q_i(X)$. Translation identifies W with $Q[s_1, \dots, s_n]$ and L with $(J, s_1, \dots, s_n)$. Expanding polynomials by their degree in the s_i gives

$$L/L^2 \simeq (J/J^2) \oplus C^{\oplus n}.$$

Indeed its constant part is J/J^2 , its linear coefficients are read modulo J , and all terms with two s factors lie in L^2 . These descriptions define mutually inverse maps, so require no finiteness or flatness assumption.

The Koszul complex on $f_1, \dots, f_c$ has free degree- k basis indexed by the k -element subsets; its differential deletes one index with the alternating sign and multiplies by the corresponding f_i . The terms of its square cancel in pairs. If its first homology is zero, the classes of the f_i form a free basis of $(f)/(f)^2$. To prove independence, correct a relation $\sum b_i f_i \in (f)^2$ by coefficients in (f) so that its scalar product is exactly zero. First-homology vanishing expresses that relation as a sum of the skew Koszul boundaries, whose coefficients all belong to (f) . The original b_i therefore belong to (f) as well. Surjectivity follows because the f_i generate the ideal.

By the hypothesis on I , choose $a \in P$ outside the contracted prime and a Koszul-regular list f generating I_a . Polynomial extension and localization preserve its positive-homology vanishing: they are flat, and the exact flat-tensor homology calculation is proved in AG-RG-S01, Lemma 1.0. Append $u_j = X_j - p_j(T)$. Modulo I these are successive polynomial variables translated by constants, and eliminating them leaves P/I . Multiplication by each such variable is injective, as comparison of polynomial coefficients shows, also after localization. Appending one element to a Koszul complex is the cone of its multiplication map; the connecting-homomorphism construction in AG-CA-06S, Section 2 then identifies its positive homology with the kernel and higher homology of that map. The higher groups are zero and the degree-zero multiplication is injective. Induction therefore makes the whole list (f, u) Koszul-regular in W_a . Its conormal classes give a free basis at $\mathfrak{r}$ by the preceding relation calculation.

Consequently $(J/J^2)_{\mathfrak{q}}$ is a finite direct summand of that free conormal module. Such a summand is flat by restricting the free tensor injection, and is finitely presented because the complementary summand is generated by the projections of a finite basis. AG-RG-S01, Lemma 1.2 makes it finite free over the local ring $C_{\mathfrak{q}}$. Choose from a finite generating list of J elements $g_1, \dots, g_r$ whose conormal residue classes form a basis. Nakayama makes those classes generate the local conormal module; in a free basis their square coordinate matrix has unit determinant, so they form a basis there. They also generate J in the local ring $Q_{\mathfrak{q}_Q}$, where $\mathfrak{q}_Q$ is the inverse image of $\mathfrak{q}$. In fact $J = (g) + J^2$ there; the finite module $J/(g)$ equals J times itself, and J is contained in that local maximal ideal. The same proved Nakayama calculation makes this quotient zero.

Thus (g, s) and (f, u) are two generator lists for $L_{\mathfrak{r}}$ whose conormal classes are bases. They have the same length, since reduction gives two bases of $L_{\mathfrak{r}}/\mathfrak{r}L_{\mathfrak{r}}$; here $L^2 \subset \mathfrak{r}L$. Expressing one list in the other gives a square matrix invertible modulo the maximal ideal, hence invertible by the cofactor identity. Its exterior powers give an isomorphism of their Koszul complexes: the alternating

deletion formula above checks commutation with the differential on every basis wedge, and the exterior powers of the inverse give the inverse maps. Permuting the list is the same argument with a permutation matrix. In particular the Koszul complex on (s, g) has zero positive homology at this stage.

We need a principal neighbourhood, not only this stalk. Clear the finitely many denominators in both generator comparisons and in the invertible matrix, and invert its determinant. For equalities in a localization, also include an element outside $\mathfrak{r}$ killing each cross-difference; no zero divisor is cancelled. Together with a and the finite generator comparison for J , their product $h \in W \setminus \mathfrak{r}$ makes the list (s, g) generate L_h and its Koszul complex isomorphic to the Koszul-regular complex on (f, u) over W_h . Thus its positive homology vanishes over the actual neighbourhood W_h .

Finally set $v = h(q_i(X), X) \in Q$. It is outside $\mathfrak{q}Q$, because evaluation modulo (s) takes $\mathfrak{r}$ to $\mathfrak{q}Q$. We have

$$W_h/(s_1, \dots, s_n) = Q_v.$$

The Koszul complex on s is a free resolution of this quotient: before localization it is the successive polynomial-variable complex, whose acyclicity follows from the same cone calculation; flat localization preserves it. Tensor this resolution with the finite free terms of the Koszul complex on g . The augmentation of the resulting total complex to the Koszul complex on g over Q_v is a homology isomorphism. Here is the finite elimination argument: include the quotient as augmented degree -1 in each column. Those columns are exact. In a total cycle choose its component with largest g degree, lift its vertical cycle in that exact column, and subtract the total boundary. The largest degree disappears and only smaller degrees are changed. Repetition terminates and makes every cycle a boundary. This proves the augmentation assertion without a spectral-sequence or Noetherian assumption. The total complex is precisely the Koszul complex on (s, g) , so g has zero positive Koszul homology over Q_v . The generator equality also descends to $J_v = (g)$. We have therefore produced the neighbourhood required by Definition F.95 at every prime. Empty lists obey the same formulas. This proves every assertion. $\square$

Definition F.95: regular ideal

Adapted from the Stacks project, more-algebra.tex, lines 8710–8731.

Let R be a ring and let $I \subset R$ be an ideal.

The sequence terms used here have the following meanings. The Koszul complex on a finite list $f_1, \dots, f_r$ is the alternating exterior-power complex described in Lemma F.94A. The list is *Koszul-regular* if its positive homology vanishes, and *H_1 -regular* if its first homology vanishes. With $J = (f_1, \dots, f_r)$, it is *quasi-regular* if the graded map

$$(R/J)[T_1, \dots, T_r] \longrightarrow \bigoplus_{n \geq 0} J^n/J^{n+1}, \quad T_1^{e_1} \dots T_r^{e_r} \longmapsto f_1^{e_1} \dots f_r^{e_r} \bmod J^{1+\sum e_i}$$

is an isomorphism. In degree one this identifies $(R/J)^r$ with J/J^2 on the displayed generator classes. This is the freely accessible [Stacks project definition, Tag 061P](#), written here to specify the exact condition being used.

1. We say I is a *regular ideal* if for every $\mathfrak{p} \in V(I)$ there exists a $g \in R$, $g \notin \mathfrak{p}$ and a regular sequence $f_1, \dots, f_r \in R_g$ such that I_g is generated by $f_1, \dots, f_r$.
2. We say I is a *Koszul-regular ideal* if for every $\mathfrak{p} \in V(I)$ there exists a $g \in R$, $g \notin \mathfrak{p}$ and a Koszul-regular sequence $f_1, \dots, f_r \in R_g$ such that I_g is generated by $f_1, \dots, f_r$.
3. We say I is a *H_1 -regular ideal* if for every $\mathfrak{p} \in V(I)$ there exists a $g \in R$, $g \notin \mathfrak{p}$ and an H_1 -regular sequence $f_1, \dots, f_r \in R_g$ such that I_g is generated by $f_1, \dots, f_r$.

4. We say I is a *quasi-regular ideal* if for every $\mathfrak{p} \in V(I)$ there exists a $g \in R$, $g \notin \mathfrak{p}$ and a quasi-regular sequence $f_1, \dots, f_r \in R_g$ such that I_g is generated by $f_1, \dots, f_r$.

Lemma F.96: G ring goes up quasi finite

Adapted from the Stacks project, more-algebra.tex, lines 12823–12870.

Let $R \rightarrow R'$ be a finite type map of Noetherian rings. Let $\mathfrak{q}' \subset \mathfrak{p}'$ be primes of R' with contractions $\mathfrak{q} \subset \mathfrak{p}$ in R , and assume the map is quasi-finite at $\mathfrak{p}'$.

1. If $\widehat{R_{\mathfrak{p}}} \otimes_R \kappa(\mathfrak{q})$ is geometrically regular over $\kappa(\mathfrak{q})$, then $\widehat{R'_{\mathfrak{p}'}} \otimes_{R'} \kappa(\mathfrak{q}')$ is geometrically regular over $\kappa(\mathfrak{q}')$.
2. If every formal fibre of $R_{\mathfrak{p}}$ is geometrically regular, then every formal fibre of $R'_{\mathfrak{p}'}$ is geometrically regular.
3. If $R \rightarrow R'$ is quasi-finite and R is a G-ring, then R' is a G-ring.

Proof. Put $\widehat{R} = \widehat{R_{\mathfrak{p}}}$ and $\widehat{S} = \widehat{R'_{\mathfrak{p}'}}$. Both completions are Noetherian by AG-CA-19, Theorem 3.3. The finite-type algebra $T = \widehat{R} \otimes_R R'$ is Noetherian by AG-CA-03, Theorem 2.1. The explicit quasi-finite completion-factor proof F.26 gives

$$T \cong \widehat{S} \times D,$$

with first projection the natural completed-local map.

Write $k = \kappa(\mathfrak{q})$ and $E = \kappa(\mathfrak{q}')$. The injective map $R/\mathfrak{q} \rightarrow R'/\mathfrak{q}'$ extends to $k \subset E$: every nonzero element of the source domain has nonzero image and becomes invertible in the target fraction field. Put $F = \widehat{R} \otimes_R k$, geometrically regular by the hypothesis of (1). Tensor associativity identifies

$$H = T \otimes_{R'} E \cong F \otimes_k E.$$

Here H is Noetherian: if $S = R' \setminus \mathfrak{q}'$, then $E = S^{-1}(R'/\mathfrak{q}')$, so $H = S^{-1}(T/\mathfrak{q}'T)$. Quotients and localizations preserve Noetherianness by the same earlier Hilbert-basis and permanence proof. The full field-extension result F.52 therefore makes H geometrically regular over E .

Tensoring the displayed product decomposition gives

$$H \cong (\widehat{S} \otimes_{R'} E) \times (D \otimes_{R'} E).$$

To check geometric regularity of its first factor H_1 , let E'/E be finite purely inseparable. The ring $H \otimes_E E'$ is regular, and it is the product of the two corresponding factor tensors. The first factor is its localization at the idempotent $(1, 0)$, hence regular: its local rings are local rings of that regular product. It is Noetherian as a quotient of the product. The first factor H_1 is itself Noetherian as a quotient of H . Criterion F.50 now proves that $H_1 = \widehat{S} \otimes_{R'} E$ is geometrically regular over E . Zero factors cause no exception, since they have no prime local rings. This proves (1).

For (2), let $\mathfrak{q}'$ run over all primes contained in $\mathfrak{p}'$; their contractions lie in $\mathfrak{p}$, so (1) applies to every formal fibre. Completion is flat by AG-CA-19, Theorem 3.2. Thus these geometrically regular fibres give a regular completion map. Finally, if R is a G-ring and the map is quasi-finite at every prime, apply this argument at every $\mathfrak{p}'$. Definition F.92 says exactly that the resulting regular completion maps make R' a G-ring. $\square$

Lemma F.97: another helper G ring

Adapted from the Stacks project, more-algebra.tex, lines 13103–13159.

Let R be any Noetherian complete local domain with fraction field K . First, for a prime q of $R[X]$, $\widehat{R[X]_q} \otimes_R K$ is regular. If q lies over the maximal ideal and $0 \neq r \subset q$ lies over $(0) \subset R$, then $\widehat{R[X]_q} \otimes_{R[X]} \kappa(r)$ is geometrically regular over $\kappa(r)$.

Proof. Choose the already proved finite regular complete local subring $A_0 \subset R$, and put $F = \text{Frac}(A_0)$. Let q_0 contract q to $A_0[X]$. The finite-extension completion decomposition gives

$$\widehat{A_0[X]_{q_0}} \otimes_{A_0[X]} R[X] = \prod_i \widehat{R[X]_{q_i}},$$

with the finitely many primes over q_0 , one of which is q . Tensor with K over R . The left side becomes $\widehat{A_0[X]_{q_0}} \otimes_{A_0} K$: the finite domain R over A_0 has $R \otimes_{A_0} F = K$, since a finite domain over a field is a field. The strong first assertion of the preceding helper makes $\widehat{A_0[X]_{q_0}} \otimes_{A_0} F$ geometrically regular over F . Its base change to the finite extension K/F is regular by the field-extension proof. Each factor, including $\widehat{R[X]_q} \otimes_R K$, is regular. This proves the first assertion without any use of a general formal-smoothness/regularity equivalence.

For the second, $\kappa(r)/K$ is finite, because the nonzero prime $rK[X]$ is generated by an irreducible polynomial. Given a finite purely inseparable $L/\kappa(r)$, choose a finite integral R -subalgebra $B \subset L$ with fraction field L , by scaling finitely many algebraic generators into monic equations. As a finite R -module, B is Noetherian. Apply Lemma F.27 to the finite map $R \rightarrow B$ at the maximal ideal of R . Since R is complete, its left side is B , and it gives the canonical isomorphism $B \cong \prod_i \widehat{B_{\mathfrak{n}_i}}$, with the finitely many maximal ideals $\mathfrak{n}_i$ of B . Each factor has its nonzero residue-field quotient. A domain has only the idempotents zero and one, so this product has exactly one factor. Thus B has one maximal ideal $\mathfrak{n}$, $B = B_{\mathfrak{n}}$, and the displayed map identifies B with its maximal-ideal completion. This proves that B is a complete local domain directly from the already written finite-completion decomposition. Let r' be the kernel of $B[X] \rightarrow L$ sending X to its specified image from $\kappa(r)$; then $\kappa(r') = L = \text{Frac}(B)$.

Finite-extension completion and tensor associativity identify the tested fibre with a product of

$$\widehat{B[X]_{q_i}} \otimes_{B[X]} \kappa(r'),$$

over primes q_i above q ; factors not meeting this fibre are zero. In each remaining factor, $r'L[X] = (X - a)$ for some $a \in L$, so this is

$$(\widehat{B[X]_{q_i}} \otimes_B L)/(X - a).$$

The parenthesized ring is regular by the first assertion, applied to the complete local domain B . Lemma F.104 extends $\partial/\partial X$ from $B[X]$ through its prime localization and completion, and then through localization at $B \setminus \{0\}$. It kills B , so the fraction formula in that proof makes it zero on $L = \text{Frac}(B)$ as well. Thus it sends $X - a$ to one. Lemma F.122 applies to this regular parenthesized ring and makes each quotient factor regular. Thus every finite purely inseparable test extension L gives a regular fibre, and the geometric-regularity criterion gives the assertion. This is the exact completion argument consumed by polynomial permanence of G-rings. $\square$

Lemma F.98: cartier equality

Adapted from the Stacks project, more-algebra.tex, lines 9182–9206.

For a finitely generated field extension K/k , both $\Omega_{K/k}$ and $H_1(L_{K/k})$ are finite-dimensional and their dimension difference is $\text{trdeg}_k K$.

Proof. Use the explicit finite global complete-intersection domain $C \subset K$ constructed in the monic-tower proof, with fraction field K . Write its polynomial presentation with n variables and c regular equations; the localization variable has already been counted, so $\dim C = n - c$. Its conormal module is free on the equation classes by the regular-relation calculation, and polynomial differentials are free on the variables. Localization and homotopy comparison give the two-term complex $K^c \rightarrow K^n$ for K/k . If the matrix rank is r , its homology dimensions are $c - r$ and $n - r$, respectively. Their difference is $n - c = \dim C = \text{trdeg}_k K$, by AG-CA-09, Theorem 4.2. This also proves their finiteness. $\square$

Lemma F.99: check G ring easy

Adapted from the Stacks project, more-algebra.tex, lines 12803–12821.

A Noetherian ring R is a G-ring if and only if for every $\mathfrak{q} \subset \mathfrak{p}$ the ring $(R/\mathfrak{q})_{\mathfrak{p}}^{\wedge} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$ is geometrically regular over $\kappa(\mathfrak{q})$.

Proof. The completion map of each Noetherian local ring is flat by AG-CA-19, Theorem 3.2. Its fibre at a contracted prime $\mathfrak{q}$ is $\widehat{R}_{\mathfrak{p}} \otimes_R \kappa(\mathfrak{q})$. Exactness of finite-module completion, AG-CA-19, Theorem 3.1, gives $\widehat{R}_{\mathfrak{p}}/\mathfrak{q}\widehat{R}_{\mathfrak{p}} = \widehat{(R/\mathfrak{q})}_{\mathfrak{p}}$. Tensoring this equality with the fraction field of $R/\mathfrak{q}$ identifies the displayed fibre with the ring in the assertion. A flat map is regular precisely when all its residue-field fibres are geometrically regular, by the stated definition. Applying this to every local completion gives both directions. $\square$

Lemma F.99A: completion of a flat local map

Let $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ be a flat local map of Noetherian local rings. Its induced map on maximal-ideal completions $\widehat{A} \rightarrow \widehat{B}$ is faithfully flat. The residue fields need not be equal.

Proof. A local map sends $\mathfrak{m}^r$ into $\mathfrak{n}^r$; passing to compatible finite quotients therefore defines the map of completions. It remains local because their maximal ideals are $\mathfrak{m}\widehat{A}$ and $\mathfrak{n}\widehat{B}$, and their residue fields are $A/\mathfrak{m}$ and $B/\mathfrak{n}$, respectively. These quotient and Noetherian assertions are AG-CA-19, Proposition 2.2 and Theorem 3.3.

Put $\kappa = A/\mathfrak{m}$. Choose a free A -resolution $F_{\bullet} \rightarrow \kappa$, as constructed in AG-CA-06S, Section 1. Completion is flat by AG-CA-19, Theorem 3.2, so $F_{\bullet} \otimes_A \widehat{A}$ is a free $\widehat{A}$ -resolution of $\kappa \otimes_A \widehat{A} = \kappa$. The ring $\widehat{B}$ is flat over B by the same theorem and hence flat over A by composition. Consequently

$$\mathrm{Tor}_1^{\widehat{A}}(\kappa, \widehat{B}) = H_1(F_{\bullet} \otimes_A \widehat{B}) = \mathrm{Tor}_1^A(\kappa, \widehat{B}) = 0.$$

Apply the proved local criterion for flatness, AG-CA-08, Theorem 4.2, to the Noetherian local map $\widehat{A} \rightarrow \widehat{B}$ and the finite $\widehat{B}$ -module $\widehat{B}$. It proves flatness. Locality gives $\mathfrak{m}\widehat{B} \subseteq \mathfrak{n}\widehat{B}$, so the closed fibre $\widehat{B}/\mathfrak{m}\widehat{B}$ surjects onto $\widehat{B}/\mathfrak{n}\widehat{B} = B/\mathfrak{n}$. This residue field is nonzero, hence the closed fibre is nonzero. The local faithful-flatness criterion AG-CA-08, Theorem 2.1 now makes the map faithfully flat. This proves the assertion. $\square$

Lemma F.100: check G ring maximal ideals

Adapted from the Stacks project, more-algebra.tex, lines 13001–13037.

Let R be a Noetherian ring. Then R is a G-ring if and only if $R_{\mathfrak{m}}$ has geometrically regular formal fibres for every maximal ideal $\mathfrak{m}$ of R .

Proof. Assume $R_{\mathfrak{m}} \rightarrow R_{\mathfrak{m}}^{\wedge}$ is regular for every maximal ideal $\mathfrak{m}$ of R . Let $\mathfrak{p}$ be a prime of R and choose a maximal ideal $\mathfrak{p} \subset \mathfrak{m}$. Since $R_{\mathfrak{m}} \rightarrow R_{\mathfrak{m}}^{\wedge}$ is faithfully flat we can choose a prime $\mathfrak{p}'$ in $R_{\mathfrak{m}}^{\wedge}$ lying over $\mathfrak{p}R_{\mathfrak{m}}$. Consider the commutative diagram

$$\begin{array}{ccc} R_{\mathfrak{m}}^{\wedge} & \rightarrow & (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'} \\ (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'} & \rightarrow & (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}^{\wedge} \\ R_{\mathfrak{m}} & \rightarrow & R_{\mathfrak{m}}^{\wedge} \\ R_{\mathfrak{m}} & \rightarrow & R_{\mathfrak{p}} \\ R_{\mathfrak{p}} & \rightarrow & (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'} \\ R_{\mathfrak{p}} & \rightarrow & R_{\mathfrak{p}}^{\wedge} \\ R_{\mathfrak{p}}^{\wedge} & \rightarrow & (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}^{\wedge} \end{array}$$

By assumption the ring map $R_{\mathfrak{m}} \rightarrow R_{\mathfrak{m}}^{\wedge}$ is regular. By Proposition F.132 $(R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}^{\wedge}$ is regular. The localization $R_{\mathfrak{m}}^{\wedge} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}$ is regular. Hence $R_{\mathfrak{m}} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}$ is regular by Lemma F.124. Since it factors through the localization $R_{\mathfrak{p}}$, also the ring map $R_{\mathfrak{p}} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}^{\wedge}$ is regular. The local map $R_{\mathfrak{p}} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}$ is flat, being a localization of the flat completion map. Lemma F.99A makes the induced map $R_{\mathfrak{p}}^{\wedge} \rightarrow (R_{\mathfrak{m}}^{\wedge})_{\mathfrak{p}'}^{\wedge}$ faithfully flat. Thus Lemma F.125 applies to this actual factorization and proves that $R_{\mathfrak{p}} \rightarrow R_{\mathfrak{p}}^{\wedge}$ is regular. $\square$

Lemma F.101: conormal sequence H1 regular ideal

Adapted from the Stacks project, more-algebra.tex, lines 8844–8854.

If $I \subset J$ and J/I is an H_1 -regular ideal in A/I , then $I \cap J^2 = IJ$.

Proof. This equality of modules is local. Away from $V(J)$ it reads $I = I$, so localize at a prime containing J . Choose lifts $g_i \in J$ of local generators whose Koszul first homology over A/I vanishes. For $x \in I \cap J^2$, write $x = \sum_i g_i b_i + u$ with $b_i \in J$, $u \in IJ$; this follows by expanding $J = I + (g_i)$ and grouping products. Modulo I , the vector (b_i) is a relation on the (g_i) . Vanishing of H_1 says that it is a finite sum of the skew Koszul relations. Lift their coefficients to A and subtract those relations from (b_i) . Their scalar products with the g_i are zero identically, while the corrected coefficients all lie in I . Hence $\sum_i g_i b_i \in IJ$ and $x \in IJ$. The opposite inclusion is immediate. Local detection proves the global identity. $\square$

Lemma F.102: cotangent complex symmetric algebra

Adapted from the Stacks project, more-algebra.tex, lines 2066–2109.

Let A be a ring and let $0 \rightarrow K \rightarrow A^{\oplus m} \rightarrow M \rightarrow 0$ be exact, with M a finite projective A -module. Put $C = \text{Sym}_A(M)$ and let $\alpha : A[y_1, \dots, y_m] \rightarrow C$ be its induced presentation, with kernel J . Then the natural map

$$K \otimes_A C \longrightarrow J/J^2, \quad (k_1, \dots, k_m) \otimes c \longmapsto c \sum_i k_i y_i \bmod J^2,$$

is an isomorphism. Under it the two-term conormal complex is

$$\text{NL}(\alpha) = [K \otimes_A C \longrightarrow C^{\oplus m}],$$

with differential the tensor of $K \hookrightarrow A^{\oplus m}$. It follows that $\Omega_{C/A} = M \otimes_A C$.

Proof. Projectivity of M splits the defining exact sequence and makes K a finite projective module too. At each prime of A , localize further on a principal neighborhood on which both summands are free; their two bases combine to a basis of $A^{\oplus m}$. Its invertible linear change of variables changes the polynomial presentation into

$$A[U_1, \dots, U_r, V_1, \dots, V_s] \longrightarrow A[V_1, \dots, V_s], \quad U_i \longmapsto 0, \quad V_j \longmapsto V_j.$$

Here the coefficients denote the localized ring. Its kernel is $(U_1, \dots, U_r)$. In the polynomial monomial basis its square consists of monomials containing at least two U factors, so its conormal module is free over $A[V]$ on the classes of the U_i . The displayed natural map is exactly the identity on these bases. It is therefore an isomorphism at every prime; local detection of the kernel and cokernel proves the global assertion. The derivative of $\sum k_i y_i$ relative to A is $\sum k_i dy_i$, proving the claimed differential and its cokernel. The finite-projective quotient hypothesis is essential for this conormal identity. $\square$

Lemma F.103: degree p extension regular

Adapted from the Stacks project, more-algebra.tex, lines 12453–12467.

If R is Noetherian regular and $D(f)$ is a unit for some derivation, then $R[Z]/(P(Z) - f)$ is regular for every $P \in \mathbb{Z}[Z]$.

Proof. Polynomial extension of a Noetherian regular ring is regular by AG-CA-14, Proposition 3.3. Extend the derivation coefficientwise and set $D(Z) = 0$. Every integer and hence $P(Z)$ is annihilated. The derivative of $P(Z) - f$ is $-D(f)$, still a unit in the quotient, so the preceding hypersurface proof applies at every prime. The special case $P(Z) = Z^p$ is the one used in the inseparable induction. $\square$

Lemma F.104: derivation extends

Adapted from the Stacks project, more-algebra.tex, lines 12363–12405.

A derivation $D : R \rightarrow R$ extends to every localization and canonically to every ideal-adic completion. If $R \subset R'$ is finite type and $R_g = R'_g$ with g a nonzerodivisor in R' , some $g^N D$ preserves R' .

Proof. In a localization the forced formula is $D(r/s) = D(r)/s - rD(s)/s^2$. If $r/s = r'/s'$, a multiplier kills $rs' - r's$; differentiating this equality and using the original equality after localization proves equality of the two displayed values. Direct expansion verifies addition and the product rule; differentiating $s(s^{-1}) = 1$ proves uniqueness.

For completion, Leibniz gives $D(I^{n+1}) \subset I^n$. For a compatible family $a_n \in R/I^n$, choose a representative of a_{n+1} and define the n th coordinate of $\widehat{D}(a)$ by its derivative modulo I^n . It is independent of that representative and compatible as n varies. Represent two families modulo I^{n+1} ; the product rule there reduces modulo I^n to the product rule for the new coordinate. Addition is checked in the same way. Every coordinate therefore satisfies the derivation identities, so their inverse limit does too. This fills the completion verification.

For finite type choose generators x_i of R'/R and n with $g^n x_i \in R$. Inside R'_g ,

$$g^{n+1} D(x_i) = -ng^n x_i D(g) + gD(g^n x_i) \in R'.$$

Because R' injects into R'_g , the extended derivation $g^{n+1} D$ can be restricted to R' : it preserves the coefficient subring and the generators, and the product rule shows that it preserves every polynomial in them. All relations continue to hold in this subring. This proves the assertion. $\square$

Lemma F.105: find D

Adapted from the Stacks project, more-algebra.tex, lines 12469–12520.

If a characteristic- p domain B is finite type over a Noetherian complete local ring and $f \in B$ is not a p th power in its fraction field, there exists a derivation $D : B \rightarrow B$ with $D(f) \neq 0$.

Proof. Replace the complete base by its image in B . It is its quotient by a prime, hence again a complete Noetherian local domain by exact completion AG-CA-19, Theorem 3.1. The finite-over-regular-subring construction already proved above reduces it to a finite extension of $R = k[[X_1, \dots, X_n]]$. Thus B is finite type over R .

Apply Noether normalization AG-CA-09, Corollary 3.2, to $B \otimes_R \text{Frac}(R)$, a finite-type domain over that field. Clearing finitely many denominators gives algebraically independent $y_j \in B$ and a nonzero $g \in R$ such that B_g is finite over $R_g[y]$. Here scaling normalized parameters clears their denominators without losing algebraic independence. For each finite generator b_i of $B/R[y]$, take its monic integral equation over $R_g[y]$. For one sufficiently large N , the generator $g^N b_i$ satisfies a monic equation over $R[y]$, because multiplying the coefficient of its degree- j term by $g^{N(d-j)}$ clears

that coefficient's denominator. The subalgebra B' generated by these scaled generators is finite over $R[y]$, by reducing monomials with those monic equations, and $B'_g = B_g$.

For a sufficiently large r , $f' = g^{pr}f$ belongs to B' . It is still not a p th power in its fraction field, since the multiplier is a p th power. Set $L = \text{Frac}(B')$ and $K = \text{Frac}(R[y])$. By the power-series-subfield lemma the finite coefficient subrings A_J have fraction fields K_J intersecting in K^p . By the finite-extension intersection lemma, $\bigcap_J L^p K_J = L^p$. Therefore some J has $f' \notin L^p K_J$. The relative p -basis construction extends f' to a p -basis of L/K_J , and its partial derivation gives a K_J -derivation $\delta : L \rightarrow L$ with $\delta(f') = 1$.

Since B' is finite over $R[y]$ and that ring is finite over A_J , it has finitely many A_J -module generators $b_1, \dots, b_t$. Write $\delta(b_i)$ as fractions of B' and multiply δ by their nonzero denominator product h . Then $h\delta(B') \subset B'$: differentiate each finite module expression and use that δ kills A_J . Its value on f' is $h \neq 0$. The finite-type derivation-extension proof above now supplies a power g^s for which $D = g^s h \delta$ preserves B . The fraction formula gives $D(f) = g^{s-pr} h \delta(f') = g^{s-pr} h \neq 0$ in its fraction field, hence in B . This proves the required derivation without assuming k perfect or using characteristic zero differentiation. $\square$

Lemma F.106: formally smooth

Adapted from the Stacks project, more-algebra.tex, lines 9774–9820.

Algebraic formal smoothness implies formal smoothness for any continuous map with a pre-adic target topology. For an adic target, changing a continuous adic topology on the base to the discrete topology does not change formal smoothness.

Proof. In a lifting test with square-zero kernel J , continuity of the target map modulo J says some ideal power $\mathfrak{n}^r$ maps to zero. Every algebraic lift sends $\mathfrak{n}^r$ into J , hence sends $\mathfrak{n}^{2r}$ to zero; it is continuous. Conversely in a test with discrete base, continuity of the original structural map supplies $\varphi(\mathfrak{m}^s) \subset \mathfrak{n}$. Since $\mathfrak{n}^r$ is killed modulo J , the given base map sends $\mathfrak{m}^{sr}$ into J and consequently kills $\mathfrak{m}^{2sr}$. It was therefore continuous for the original adic base topology. Applying that version of the lifting property gives exactly the desired lift. These two arguments prove both implications without any finiteness or Noetherian assumption. $\square$

Lemma F.107: formally smooth completion

Adapted from the Stacks project, more-algebra.tex, lines 9833–9860.

For continuous maps between Noetherian rings with finitely generated adic ideals, formal smoothness in the target adic topology is unchanged by completing the target or both rings.

Proof. AG-CA-19, Proposition 2.2, identifies each finite quotient of a completion with the original quotient: $\widehat{R}/I^n \widehat{R} = R/I^n$, and likewise for the target. A continuous map from either ring to a discrete test algebra kills an ideal power, hence factors through one of these common finite quotients. Thus maps from R and $\widehat{R}$ correspond uniquely, as do maps from the target and its completion. The same is true for each map to the quotient test algebra. The square-zero lifting diagrams and their continuous dotted lifts are consequently in bijection. Existence of a lift in one version is existence in the other. This is the Noetherian version actually used below. $\square$

Lemma F.108: gamma commutative diagram

Adapted from the Stacks project, more-algebra.tex, lines 9230–9281.

For a commutative field square $k \rightarrow k', k \rightarrow K, k' \rightarrow K', K \rightarrow K'$ with k'/k and K'/K finitely generated, let $\alpha : \Omega_{K/k} \otimes_K K' \rightarrow \Omega_{K'/k'}$ and $\beta : H_1(L_{K/k}) \otimes_K K' \rightarrow H_1(L_{K'/k'})$. Their kernels and cokernels are finite-dimensional and $\dim \ker \alpha - \dim \text{coker } \alpha - \dim \ker \beta + \dim \text{coker } \beta =$

$\text{trdeg}_k k' - \text{trdeg}_K K'$.

Proof. Write the two Jacobi–Zariski sequences, now proved with their left zeros, for $k \subset k' \subset K'$ and $k \subset K \subset K'$. Denote their common first middle space by $V = H_1(L_{K'/k})$ and common last middle space by $U = \Omega_{K'/k}$. Their respective first injected spaces are $P_1 = H_1(L_{k'/k}) \otimes K'$ and $P_2 = H_1(L_{K/k}) \otimes K'$. Their next spaces are $W_1 = H_1(L_{K'/k'})$ and $W_2 = H_1(L_{K'/K})$. The next spaces are $Q_1 = \Omega_{k'/k} \otimes K'$ and $Q_2 = \Omega_{K/k} \otimes K'$, and the last spaces are $Z_1 = \Omega_{K'/k'}$ and $Z_2 = \Omega_{K'/K}$. Thus both rows have the form $0 \rightarrow P_i \rightarrow V \rightarrow W_i \rightarrow Q_i \rightarrow U \rightarrow Z_i \rightarrow 0$. The maps in the question are $P_2 \rightarrow V \rightarrow W_1$ and $Q_2 \rightarrow U \rightarrow Z_1$, by their polynomial functorial construction.

Here is the finite-dimensional bookkeeping even when V, U, P_2, Q_2 are infinite. Put $I = P_1 \cap P_2$, $Y = V/(P_1 + P_2)$, $A_i = \ker(Q_i \rightarrow U)$, and $B_i = \text{im}(Q_i \rightarrow U)$. Set $i = \dim I$, $r = \dim(P_1/I)$, $y = \dim Y$, $a_i = \dim A_i$, $j = \dim(B_1 \cap B_2)$, $s = \dim(B_1/(B_1 \cap B_2))$, and $z = \dim(U/(B_1 + B_2))$. These numbers are finite: P_1, Q_1, W_2, Z_2 are finite by Cartier equality, Y is a quotient of $V/P_2 \subset W_2$, A_2 a quotient of W_2 , and $U/(B_1 + B_2)$ a quotient of Z_2 .

The kernel of β is I . Its cokernel has exact sequence $0 \rightarrow Y \rightarrow \text{coker } \beta \rightarrow A_1 \rightarrow 0$, giving dimension $y + a_1$. The kernel of α has exact sequence $0 \rightarrow A_2 \rightarrow \ker \alpha \rightarrow B_1 \cap B_2 \rightarrow 0$, giving dimension $a_2 + j$; its cokernel is $U/(B_1 + B_2)$, of dimension z . The rows also give $\dim P_1 = i + r$, $\dim W_2 = r + y + a_2$, $\dim Q_1 = a_1 + j + s$, and $\dim Z_2 = s + z$. Consequently the required expression is $a_2 + j - z - i + y + a_1 = \dim Q_1 - \dim P_1 - \dim Z_2 + \dim W_2$. Applying Cartier equality to k'/k and K'/K gives exactly the asserted transcendence-degree difference. No alternating subtraction of infinite dimensions has been used. $\square$

Lemma F.109: geometrically regular over field

Adapted from the Stacks project, more-algebra.tex, lines 9460–9501.

Let k be a field of characteristic $p > 0$. Let $(A, \mathfrak{m}, K)$ be a Noetherian local k -algebra. Assume A is geometrically regular over k . Let $K/F/k$ be a finitely generated subextension. Let $\varphi : k[y_1, \dots, y_m] \rightarrow A$ be a k -algebra map such that y_i maps to an element of F in K and such that $dy_1, \dots, dy_m$ map to a basis of $\Omega_{F/k}$. Set $\mathfrak{p} = \varphi^{-1}(\mathfrak{m})$. Then

$$k[y_1, \dots, y_m]_{\mathfrak{p}} \rightarrow A$$

is flat and $A/\mathfrak{p}A$ is regular.

Proof. Set $A_0 = k[y_1, \dots, y_m]_{\mathfrak{p}}$, with maximal ideal $\mathfrak{m}_0$. Its residue field K_0 is the fraction field of the image $k[\bar{y}_1, \dots, \bar{y}_m] \subset F$, so $K_0 = k(\bar{y}_1, \dots, \bar{y}_m)$. Polynomial differentials and localization make $\Omega_{A_0/k}$ free on the dy_i . These differentials span $\Omega_{K_0/k}$ by the quotient and fraction rules. They are independent there: any K_0 -linear relation maps to a relation in $\Omega_{F/k}$, where their images are a basis by hypothesis. Thus $\Omega_{A_0/k} \otimes K_0 \rightarrow \Omega_{K_0/k}$ is an isomorphism.

The ring A_0 is regular by the polynomial and localization theorems AG-CA-14, Proposition 3.3 and Theorem 3.2. After every finite purely inseparable k'/k , its tensor product is again a localization of the polynomial ring $k'[y_1, \dots, y_m]$, hence regular by those same theorems. The proved field test F.50 therefore makes A_0 geometrically regular. Proposition F.133 gives injectivity of $H_1(L_{K_0/k}) \rightarrow \mathfrak{m}_0/\mathfrak{m}_0^2$. The quotient sequence F.42 makes it surjective, since the following differential map is the isomorphism just proved. Hence it is an isomorphism.

Now consider

$$\begin{aligned} H_1(L_{K_0/k}) \otimes K &\rightarrow H_1(L_{K/k}) \\ H_1(L_{K_0/k}) \otimes K &\rightarrow \mathfrak{m}_0/\mathfrak{m}_0^2 \otimes K \\ \mathfrak{m}_0/\mathfrak{m}_0^2 \otimes K &\rightarrow \mathfrak{m}/\mathfrak{m}^2 \\ H_1(L_{K/k}) &\rightarrow \mathfrak{m}/\mathfrak{m}^2 \end{aligned}$$

Since the left vertical arrow is injective by Lemma F.131 and the lower horizontal by Proposition F.133 we conclude that the right vertical one is too. Hence a regular system of parameters in A_0 maps to part of a regular system of parameters in A . The images of the parameters can be completed to a regular system of parameters of A , since their classes are independent in $\mathfrak{m}/\mathfrak{m}^2$; lift a completing residue-field basis and use Nakayama's lemma F.10. AG-CA-12, Theorem 6.1, makes this parameter list regular, so F.46 proves flatness of $A_0 \rightarrow A$. For the asserted quotient regularity, apply the proved parameter-quotient theorem AG-CA-14, Corollary 1.2 successively to the images of the parameters of A_0 . They generate $\mathfrak{p}A = \mathfrak{m}_0A$, so the resulting regular quotient is exactly $A/\mathfrak{p}A$. $\square$

Lemma F.110: helper G ring

Adapted from the Stacks project, more-algebra.tex, lines 12915–12963.

Let A_0 be a regular complete Noetherian local domain and $P = A_0[Y_1, \dots, Y_m]$. For every prime q of P , both $\widehat{P}_q \otimes_{A_0} \text{Frac}(A_0)$ and $\widehat{P}_q \otimes_P \text{Frac}(P)$ are geometrically regular over their indicated fields. The second assertion includes the original power-series helper; the first is the extra consumed form that removes a general formal-smoothness theorem from the completion proof.

Proof. The polynomial ring P is regular by AG-CA-14, Proposition 3.3, and its localization is regular. Completion preserves the cotangent dimension and Krull dimension by AG-CA-19, Theorem 4.1, so $\widehat{P}_q$ is regular. Both displayed rings are localizations of it and therefore regular. In characteristic zero the finite purely inseparable test is trivial, so they are geometrically regular by that criterion.

In positive characteristic Cohen structure AG-CA-19S, Theorem 6.1, identifies A_0 with a power-series ring over its residue field: its minimal parameter list gives a surjection from that power-series ring, equality of dimensions and the nonzero-prime dimension drop AG-CA-11, Theorem 3.2, make its kernel zero. This is also the regular complete specialization already proved above. Treat either indicated fraction field F , and induct on a finite purely inseparable degree $[L : F]$. At a step $F \subset M \subset L$, $L = M(z)$ with $z^p = f \notin M^p$. Let Q denote A_0 for the first assertion and P for the second. Choose a finite Q -subalgebra $B \subset M$ with fraction field M and $f \in B$. To justify this choice, scale each finite algebraic field generator so it satisfies a monic equation over Q , and scale z by an element of Q to put its p th power f in that algebra; this does not alter its generated field. Here is the denominator step: write $f = b/d$ with nonzero $d \in B$. A monic integral equation for d may be shortened by cancelling powers of d until its constant coefficient $a \in Q$ is nonzero, since B is a domain. That equation gives $a/d \in B$. Replacing z by az changes its p th power to $a^p f = b a^{p-1}(a/d) \in B$, as required. The finitely generated integral algebra is finite. Put $C = B[Z]/(Z^p - f) \subset L$; it is a finite domain with fraction field L .

For the first assertion use the finite extension $P \subset B[Y]$; for the second use $P \subset B$. The base Q is normal: it is regular Noetherian, and AG-CA-15, Corollary 4.5, proves its normality. If $b \in B$, some b^{p^r} lies in $\text{Frac}(Q)$ by pure inseparability and is integral over Q because b is. Normality puts that power in Q . The same applies to C . These extensions therefore have every element's sufficiently high p th power in the smaller ring. Each prime has exactly one prime above it. Indeed membership of an element in a prime is equivalent to membership of that power in its contracted prime; lying over supplies existence. The finite-extension completion proof already written identifies the tensor of $\widehat{P}_q$ with the finite overring with the completion of its unique local factor. The same holds for C (with the polynomial Y variables in the first assertion). Hence the ring at the M stage is a localization of this completed B -ring, and the ring at the L stage is exactly its quotient extension $E[Z]/(Z^p - f)$. Finite-module completion commutes with this monic finite algebra by AG-CA-19, Theorem 3.1; this is the needed equality rather than an assumed equality of unrelated completions. The induction hypothesis says E is regular. The derivation lemma above applies to B , which is finite type over the complete local ring A_0 in either case, and gives $D : B \rightarrow B$ with $D(f) \neq 0$. Extend it

coefficientwise over the added Y variables, then through the prime localization, completion and the localization to M , using the fully checked derivation extension formula. Its value on f remains the nonzero element $D(f)$ of the field M , hence a unit of E . The hypersurface derivation proof makes $E[Z]/(Z^p - f)$ regular. This completes the induction for every finite purely inseparable extension, and the criterion gives both geometric-regularity assertions. $\square$

Lemma F.111: helper integral

Adapted from the Stacks project, more-algebra.tex, lines 1959–1975.

For an integral map $A \rightarrow B$, if $y \in B$ becomes idempotent modulo IB , there is a monic annihilating polynomial $f(Y)$ with $f \bmod I = Y^d(Y - 1)^d$.

Proof. Write $z = y^2 - y = \sum a_i b_i$ with $a_i \in I$. The algebra generated by y and these finitely many b_i is a finite A -module C , since all are integral. Choose a finite module generator list including one. Multiplication by z sends C into IC , so on these generators it is represented by a matrix with entries in I . The adjugate identity for $z\mathbf{1} - M$ shows that the monic determinant polynomial $g(Z)$ annihilates each generator, including one, so $g(z) = 0$. Its reduction is Z^d because every matrix entry belongs to I . Set $f(Y) = g(Y^2 - Y)$. It is monic, annihilates y , and reduces to $(Y^2 - Y)^d = Y^d(Y - 1)^d$. $\square$

Lemma F.112: intersection subfields

Adapted from the Stacks project, more-algebra.tex, lines 12067–12128.

Let K have characteristic p and downward directed subfields F_λ containing K^p with intersection K^p . Then the maps $\Omega_{K/\mathbb{F}_p} \rightarrow \Omega_{K/F_\lambda}$ have zero common kernel. For every finite L/K , the fields $L^p F_\lambda$ have intersection L^p .

Proof. A nonzero absolute differential is a finite linear combination of differentials of distinct absolute p -basis elements. If it vanishes over every F_λ , those finitely many elements have dependent relative differentials, hence their monomials have a nonzero relation over every F_λ by the preceding p -basis proof. Apply the finite-subspace intersection lemma to the space of all monomial coefficient relations inside K^{p^n} . It gives a nonzero relation over K^p , contradicting the absolute p -basis property. This proves the first assertion.

For the second, finite extensions decompose into towers of finite separable simple extensions and degree- p purely inseparable simple extensions by the repeated-minimal-polynomial and separable-subfield construction given above. It suffices to treat one step and iterate: if the assertion holds for an intermediate M , then the fields $M^p F_\lambda$ satisfy the same three assumptions inside M .

For a separable simple extension $L = K(\theta)$ choose its generator in L^p . This is possible because $K(\theta^p) = K(\theta)$: the intervening extension is purely inseparable and also separable, so trivial. Frobenius identifies $[L^p : K^p] = [L : K] = d$. Thus both L^p and each $L^p F_\lambda$ have the common basis $1, \theta, \dots, \theta^{d-1}$ over K^p and F_λ , respectively. Independence over F_λ follows from independence over the larger field K . Uniqueness of coefficient expansions makes the intersection precisely the vectors with all coefficients in K^p , namely L^p .

For a degree- p step $L = K(\theta)$, $\theta^p = t \notin K^p$, one has $L^p = K^p(t)$. Choose an index with $t \notin F_{\lambda_0}$; it exists by the intersection assumption. On the cofinal family of subfields refined inside F_{λ_0} , the elements $1, t, \dots, t^{p-1}$ are independent by the same degree- p argument. Thus $L^p F_\lambda = F_\lambda(t)$ has this basis. The unique expansion coefficients of an element in the intersection lie in the intersection of that cofinal family, which is K^p , proving it lies in $K^p(t)$. The opposite inclusion is automatic. This cofinal restriction is necessary: at other indices t may already belong to F_λ , so a degree- p basis cannot be assumed there. $\square$

Lemma F.113: intersection subfields subspace

Adapted from the Stacks project, more-algebra.tex, lines 12027–12065.

If subfields $F_\lambda \subset K$ are downward directed and have intersection F , then a K -linear subspace $V \subset K^n$ meets every F_λ^n nontrivially if and only if it meets F^n nontrivially.

Proof. The reverse implication is immediate. Induct on n for the other. If $V \cap (0 \oplus F^{n-1}) \neq 0$, the conclusion already holds. Otherwise induction supplies some λ_0 for which $V \cap (0 \oplus F_{\lambda_0}^{n-1}) = 0$. The nonzero intersection with $F_{\lambda_0}^n$ contains a vector with nonzero first coordinate, which can be normalized to $v = (1, v_2, \dots, v_n)$. Subtracting the first coordinate times v shows that this intersection is exactly the line $F_{\lambda_0} v$. For every refined subfield inside F_{λ_0} , the nonzero intersection similarly has a normalized vector; its uniqueness in the larger line makes it the same v . Thus all its coordinates belong to every refined subfield. Their intersection is F , because downward directedness refines λ_0 with each arbitrary index. Hence $v \in F^n$. The case $n = 1$ is immediate, completing the induction. $\square$

Lemma F.114: Ici local

Adapted from the Stacks project, more-algebra.tex, lines 8957–8993.

For a finitely presented algebra over a Noetherian ring, a polynomial presentation kernel is locally generated by a regular sequence if and only if this holds on a principal cover of the target.

Proof. Here is the presentation comparison, including the step that stable freeness of the conormal module alone would not prove. Suppose $P = R[X] \twoheadrightarrow S$ and $Q = R[Y] \twoheadrightarrow S$ are two finite polynomial presentations. Choose polynomials $p_j(X)$ representing the images of Y_j and $q_i(Y)$ representing the images of X_i . In the common polynomial ring $W = R[X, Y]$ its kernel K is both

$$K = (I, Y_j - p_j(X)) = (J, X_i - q_i(Y)).$$

Indeed quotienting either displayed ideal eliminates the added variables and gives S , with exactly its stipulated generator images. At the prime of W above a chosen prime of S , a regular generator list for J remains regular after polynomial extension and localization, by flatness; appending the $X_i - q_i(Y)$ remains regular, since each successive quotient eliminates one polynomial variable. Thus K has a regular generator list.

The conormal comparison proved above makes I/I^2 stably free over the local quotient, hence free by AG-CA-07, Theorems 5.2–5.3. Lift a basis to elements f_i of I in the polynomial local ring. Nakayama makes them generate I there. In the common local ring the classes of the $f_i, Y_j - p_j(X)$ form a conormal basis: translation in the Y variables identifies this conormal module with $(I/I^2) \oplus S^{\#Y}$. Consequently this is a minimal generator list of K . Any two minimal lists of generators of a finite ideal over a local ring have equal lengths and an invertible change matrix: reduce to the vector space $K/\mathfrak{m}_W K$, then lift the invertible residue determinant. Exterior powers of this matrix identify their Koszul complexes. The list $f_i, Y_j - p_j(X)$ therefore has zero first Koszul homology. Permute the list to put the added-variable equations first. The Noetherian first-homology criterion proved above makes this reordered list regular. Quotienting its initial added-variable equations gives the original polynomial local ring, so the remaining f_i are regular there.

For a principal chart S_g , first present it by $P[Z]/(I, uZ - 1)$, where u lifts g . At the prime over the chosen point the same argument compares this presentation with the chart's regular presentation. Its kernel has regular minimal generators $uZ - 1, f_i$, because its conormal module is $(I/I^2)_g \oplus S_g$ and the preceding invertible-matrix argument applies. Putting $uZ - 1$ first and eliminating Z recovers the required regular list for I in P_q . Finally clear denominators so the finite list generates I on one principal neighbourhood. Its finitely many positive Koszul homology modules are finite, so one further element outside the prime annihilates them. The same local first-homology criterion

proves regularity throughout this neighbourhood. Conversely, restricting such neighbourhoods proves the chart condition. $\square$

Lemma F.115: lift factorization monic

Adapted from the Stacks project, more-algebra.tex, lines 1843–1897.

A coprime monic factorization of a monic polynomial modulo an ideal I lifts after a finitely presented étale base change $A \rightarrow A'$ inducing $A/I \cong A'/IA'$.

Proof. Write the prospective monic factors G, H with respectively d, e indeterminate lower coefficients and impose the $d + e$ coefficient equations $GH = f$. Their Jacobian is the linear map $(u, v) \mapsto Hu + Gv$ on polynomial spaces of degrees below d, e . At the stipulated residue factorization this is an isomorphism. Indeed modulo G , coprimality makes H a unit, so u is uniquely the remainder of $H^{-1}w$ modulo G ; then $v = (w - Hu)/G$ is polynomial of degree below e . This gives the inverse over A/I by monic division and a Bezout identity. Invert the determinant of that Jacobian in the universal coefficient algebra C . AG-CA-17, Theorem 4.1 and Theorem 2.2, make C/A étale. The given coefficients define a section $C/IC \rightarrow A/I$.

Here is the component selection needed to make the closed fibre exactly A/I . For any finitely presented étale D -algebra E with a section $\sigma : E \rightarrow D$, its kernel J is finite, generated by $x_i - \sigma(x_i)$ for finite algebra generators. The section conormal calculation gives $J/J^2 = \Omega_{E/D} \otimes_E D = 0$, hence $J = J^2$. Choose generators j_i and write $j_i = \sum m_{ij} j_j$ with $m_{ij} \in J$. The adjugate identity makes $a = \det(1 - M)$ annihilate every j_i , and $a \equiv 1 \pmod{J}$. Put $e = 1 - a \in J$. Then $ej = j$ for $j \in J$, and in particular $e^2 = e$. Thus $J = eE$ and $E_{1-e} = E/J = D$. Apply this to $E = C/IC$, lift $1 - e$ to $h \in C$, and set $A' = C_h$. Its closed fibre is A/I , it is still finitely presented étale by principal localization, and its universal G, H give the required lifted factorization. $\square$

Lemma F.116: lift idempotent upstairs

Adapted from the Stacks project, more-algebra.tex, lines 1977–2017.

For an integral map $A \rightarrow B$ and an idempotent $\bar{e} \in B/IB$, there is a finitely presented étale $A \rightarrow A'$ inducing $A/I \cong A'/IA'$ and an idempotent in $B \otimes_A A'$ lifting $\bar{e}$.

Proof. Choose a lift y . The preceding integral determinant construction gives $f(y) = 0$ with $f \bmod I = Y^d(Y - 1)^d$. These two monic factors are coprime: expand $(Y + (1 - Y))^{2d-1} = 1$, and each term is divisible by one of $Y^d, (1 - Y)^d$. The monic factorization lifting proof above gives an étale base with this factorization $f = GH$, with $G \bmod I = Y^d$ and $H \bmod I = (Y - 1)^d$. Replace the base by it, and put $b_1 = G(y)$, $b_2 = H(y)$; then $b_1 b_2 = 0$ and their residues are $\bar{e}$ and $(-1)^d(1 - \bar{e})$.

The closed set $V(b_1, b_2)$ is disjoint from $V(IB)$. Its image in $\text{Spec } A$ is closed by the integral closed-image theorem AG-CA-05, Theorem 3.4: it is $V((b_1, b_2) \cap A)$ by lying over for the quotient. The disjointness says $I + ((b_1, b_2) \cap A) = A$. Choose $a \in (b_1, b_2) \cap A$ with $a \equiv 1 \pmod{I}$, and localize at a . This is another finitely presented étale base change with unchanged quotient modulo I . Now $ub_1 + vb_2 = 1$ in B for some u, v . The element $e = ub_1$ is idempotent because $e(1 - e) = uvb_1 b_2 = 0$. Multiplying the residue of that relation by $\bar{e}$ gives $\bar{u}\bar{e} = \bar{e}$, so e reduces to $\bar{e}$. The tensor base change preserves integrality, so every use of the closed-image theorem remains justified. Composing the two étale base changes proves the assertion. $\square$

Lemma F.117: lift projective module

Adapted from the Stacks project, more-algebra.tex, lines 2019–2064.

Let A be a ring, let $I \subset A$ be an ideal. Let $\bar{P}$ be a finite projective A/I -module. Then there exists

an étale ring map $A \rightarrow A'$ which induces an isomorphism $A/I \rightarrow A'/IA'$ and a finite projective A' -module P' lifting $\bar{P}$.

Proof. We can choose an integer n and a direct sum decomposition $(A/I)^{\oplus n} = \bar{P} \oplus \bar{K}$ for some A/I -module $\bar{K}$. Choose a lift $\varphi : A^{\oplus n} \rightarrow A^{\oplus n}$ of the projector $\bar{p}$ associated to the direct summand $\bar{P}$. Let $f \in A[x]$ be the characteristic polynomial of φ . Set $B = A[x]/(f)$. By Cayley-Hamilton (Algebra, Lemma F.19) there is a map $B \rightarrow \text{End}_A(A^{\oplus n})$ mapping x to φ . For every prime $\mathfrak{p} \supset I$ the image of f in $\kappa(\mathfrak{p})$ is $(x-1)^r x^{n-r}$ where r is the dimension of $\bar{P} \otimes_{A/I} \kappa(\mathfrak{p})$. Hence $(x-1)^n x^n$ maps to zero in $B \otimes_A \kappa(\mathfrak{p})$ for all $\mathfrak{p} \supset I$. Thus $x(1-x)$ is contained in every prime ideal of B/IB . Hence $x^N(1-x)^N$ is contained in IB for some $N \geq 1$. It follows that $x^N + (1-x)^N$ is a unit in B/IB and that

$$\bar{e} = \text{image of } \frac{x^N}{x^N + (1-x)^N} \text{ in } B/IB$$

is an idempotent as both assertions hold in $\mathbf{Z}[x]/(x^N(x-1)^N)$. The image of $\bar{e}$ in $\text{End}_{A/I}((A/I)^{\oplus n})$ is

$$\frac{\bar{p}^N}{\bar{p}^N + (1-\bar{p})^N} = \bar{p}$$

as $\bar{p}$ is an idempotent. After replacing A by an étale extension A' as in the lemma, we may assume there exists an idempotent $e \in B$ which maps to $\bar{e}$ in B/IB , see Lemma F.116. Then the image of e under the map

$$B = A[x]/(f) \longrightarrow \text{End}_A(A^{\oplus n}).$$

is an idempotent element p which lifts $\bar{p}$. Set $P = \text{im}(p)$. The decomposition $A^{\oplus n} = \text{im}(p) \oplus \text{im}(1-p)$ follows from $v = pv + (1-p)v$; applying p makes their intersection zero. The images of the finite coordinate basis generate both summands, so P is finite projective. Reducing this split decomposition modulo I identifies P/IP with the image of $\bar{p}$, namely $\bar{P}$. These are exactly the required properties. For an empty original basis, $\bar{P} = 0$ and one may directly take $A' = A$ and $P' = 0$. $\square$

Lemma F.118: noetherian finite all equivalent

Reading source: [Stacks, Tag 09CC](#), together with the regular-sequence and Koszul arguments cited there. The proof below is independently written for finite modules.

For a nonzero finite module over a Noetherian local ring and a list in its maximal ideal, regularity, vanishing of all positive Koszul homology, vanishing of first Koszul homology, and the polynomial associated-graded condition are equivalent.

Proof. Write $(A, \mathfrak{m})$ for the ring, M for the module, and $J = (f_1, \dots, f_r)$. The fourth assertion means that the natural graded map

$$\theta : (M/JM)[T_1, \dots, T_r] \longrightarrow \bigoplus_{n \geq 0} J^n M / J^{n+1} M, \quad \bar{m}T^a \longmapsto [mf^a],$$

is an isomorphism. It is always surjective. If $r = 0$, the Koszul complex has only M in degree zero, $J = 0$, and θ is the identity in degree zero. The empty list is regular because $M \neq 0$. Thus all four assertions hold in that case; assume $r > 0$.

Use the exterior complex on the symbols $e_1, \dots, e_r$, with coefficients in M , differential $d(e_i) = f_i$, and the signed product rule. Separating wedges which contain e_r identifies it with the cone of multiplication by f_r on the complex for the first $r-1$ entries. Its homology sequence contains

$$H_1(K_{r-1}) \xrightarrow{f_r} H_1(K_{r-1}) \longrightarrow H_1(K_r) \longrightarrow H_0(K_{r-1}) \xrightarrow{f_r} H_0(K_{r-1}).$$

For the two consequences needed here, exactness can also be checked directly. A degree-one cycle of K_{r-1} becomes a boundary in K_r when $H_1(K_r) = 0$. Writing that boundary as the sum of a wedge without e_r and a wedge containing e_r shows that its class in $H_1(K_{r-1})$ is f_r times a class represented by a cycle. This gives surjectivity of the first arrow. If $f_r m \in (f_1, \dots, f_{r-1})M$, choose coefficients for this equality to obtain a degree-one cycle in K_r with last coefficient m . A degree-two boundary has last coefficient in $(f_1, \dots, f_{r-1})M$, so $[m] = 0$ in $H_0(K_{r-1})$. This gives injectivity of the last arrow.

If $H_1(K_r) = 0$, the preceding H_1 is a finite A -module, since its complex consists of finite modules over a Noetherian ring. Nakayama, with $f_r \in \mathfrak{m}$, makes it zero by the surjectivity just proved. Induction therefore gives all successive multiplication injections, including the final one displayed above. Their last quotient is nonzero by Nakayama applied to M/JM . This proves regularity. Conversely, assume the list regular and the preceding complex acyclic in positive degrees. A cycle in K_r of degree k has a component v multiplying e_r , and v is a cycle of degree $k-1$ in K_{r-1} . For $k \geq 2$, preceding acyclicity makes v a boundary. For $k = 1$, the cycle identity says $f_r[v] = 0$ in $H_0(K_{r-1})$; regularity again makes v a boundary. Subtract the differential of a suitable wedge containing e_r to remove v . What remains is a degree- k cycle in K_{r-1} and is a boundary by preceding acyclicity. This proves acyclicity in every positive degree by induction. In particular it implies vanishing of H_1 .

We next prove that a regular list makes θ injective. In degree zero this is the defining quotient M/JM . In degree one, suppose $\sum_i f_i m_i \in J^2 M$. Express this element as $\sum_i f_i u_i$ with $u_i \in JM$, and subtract. The vector $(m_i - u_i)$ is a degree-one Koszul cycle. By the preceding acyclicity it is a boundary, whose coefficients all belong to JM , since the degree-two differential has coefficients f_i . Consequently every $m_i \in JM$. This proves directly the module version of the conormal calculation in F.74.

Here is a coefficient calculation for any degree $n \geq 1$. In a relation

$$\sum_{|a|=n} m_a f^a \in J^{n+1} M,$$

collect every term involving one of $f_1, \dots, f_{r-1}$ into their generated submodule. The only remaining term on the left is $m_{(0, \dots, 0, n)} f_r^n$; the only remaining type on the right is $u f_r^{n+1}$. Thus

$$(m_{(0, \dots, 0, n)} - u f_r) f_r^n + \sum_{i < r} f_i v_i = 0$$

for suitable $u, v_i \in M$. The list $f_1, \dots, f_{r-1}, f_r^n$ is regular by AG-CA-12, Proposition 1.3, whose power argument applies to modules. Its first Koszul homology is zero by the cone argument. Hence the last coefficient of this relation belongs to $(f_1, \dots, f_{r-1})M$, so $m_{(0, \dots, 0, n)} \in JM$.

To recover the coefficients of all monomials, put $B = A[U_1, \dots, U_r, U_r^{-1}]$, $N = M \otimes_A B$, and

$$g_i = f_i - (U_i/U_r) f_r \quad (i < r), \quad g_r = f_r/U_r.$$

The ring B is free and faithfully flat over A , with its Laurent monomials as a basis. Tensoring preserves Koszul acyclicity. The displayed change of generators is invertible, so its exterior powers identify the two Koszul complexes and give $H_1(K(g; N)) = 0$.

Fix a prime $\mathfrak{p}$ of B . If $N_{\mathfrak{p}} = 0$, every term of every Koszul complex with these coefficients is zero. If $N_{\mathfrak{p}} \neq 0$ and $JB \subset \mathfrak{p}$, the local criterion already proved makes $g_1, \dots, g_r$ regular on the nonzero finite $B_{\mathfrak{p}}$ -module $N_{\mathfrak{p}}$. Its power list $g_1, \dots, g_{r-1}, g_r^n$ is regular and has zero H_1 . Finally, if $JB \not\subset \mathfrak{p}$, at least one g_i is a unit in $B_{\mathfrak{p}}$. The power list still contains a unit, so choose a_i with $\sum_{i < r} a_i g_i + a_r g_r^n = 1$. Wedge multiplication by $\sum_i a_i e_i$ satisfies $dh + hd = 1$ by the signed product rule, and contracts its Koszul complex. These three cases show that the power list has zero first homology at every prime. Localization commutes with the complex and its homology; a module whose localizations at all primes vanish is zero. The global H_1 therefore vanishes as well.

The preceding pure-last-coefficient calculation now applies to g on N . Substituting $f_i = g_i + U_i g_r$ for $i < r$ and $f_r = U_r g_r$ makes the coefficient of g_r^n equal to $\sum_{|a|=n} m_a U^a$. Thus this element belongs to JN . Since $N/JN = (M/JM) \otimes_A B$ is the direct sum of its Laurent monomial copies of M/JM , each coefficient m_a belongs to JM . This proves injectivity of θ in every degree.

For the converse, suppose θ is an isomorphism. Krull intersection, AG-CA-03, Theorem 6.1, gives $\bigcap_{n \geq 0} J^n M = 0$. Every nonzero $m \in M$ therefore has a finite order v , characterized by $m \in J^v M \setminus J^{v+1} M$. Its initial class is nonzero; multiplication by T_1 is injective on $(M/JM)[T_1, \dots, T_r]$, so $f_1 m$ is nonzero and has order $v + 1$. In particular f_1 acts injectively on M . The same calculation gives

$$f_1 M \cap J^n M = f_1 J^{n-1} M \quad (n \geq 1) :$$

if $f_1 m \in J^n M$ and $m \neq 0$, its order forces $m \in J^{n-1} M$; the reverse inclusion follows from $f_1 \in J$.

Set $M' = M/f_1 M$ and let J' be the ideal induced by J . In degree $n \geq 1$, the last equality identifies the kernel of $J^n M/J^{n+1} M \rightarrow J'^n M'/J'^{n+1} M'$ with multiplication by T_1 from degree $n - 1$. Indeed an element in that kernel is $y + f_1 m$, with $y \in J^{n+1} M$; then $f_1 m \in J^n M$, so $m \in J^{n-1} M$. Degree zero is unchanged. It follows that

$$\mathrm{gr}_{J'} M' \simeq (M/JM)[T_2, \dots, T_r].$$

Repeat the order and quotient argument for the remaining entries. Nakayama ensures that each finite intermediate module and the final quotient are nonzero. This proves regularity and hence all four equivalences. $\square$

Lemma F.119: p basis

Adapted from the Stacks project, more-algebra.tex, lines 11972–12025.

For a characteristic- p extension L/F , a subset is p -independent over F if its monomials of exponents below p are independent over FL^p . This is equivalent to independence of its relative differentials. Every such subset extends to a p -basis, whose differentials are a basis of $\Omega_{L/F}$.

Proof. Put $E = FL^p$. A chain of p -independent subsets has independent union, because every monomial relation uses only finitely many elements and coefficients. Zorn's lemma therefore extends any such subset to a maximal one B . The subalgebra $E[B]$ is a field: every element belongs to a subalgebra on finitely many basis elements, which is a finite-dimensional domain over E and hence a field. Here the last assertion follows directly: multiplication by a nonzero element is an injective linear map of a finite-dimensional space, hence surjective, so that element has an inverse.

If $x \in L$ lies outside $E[B]$, then $x^p \in E$. The polynomial $T^p - x^p$ is irreducible over $E[B]$. Inside $L[T]$ it is $(T - x)^p$. In a factorization of this polynomial, evaluation at $T = x$ makes at least one factor vanish; division of that factor by $T - x$ and induction on the exponent show that every monic factor is a power of $T - x$. A monic proper factor over $E[B]$ would therefore be $(T - x)^r$ with $1 \leq r < p$; its coefficient of T^{r-1} is $-rx$, so x would belong to $E[B]$, since r is invertible in characteristic p . Thus adjoining x has degree p and preserves monomial independence, contradicting maximality. Consequently $E[B] = L$. Reducing exponents by the monic relations, and then using monomial independence, proves the presentation

$$L = E[T_b : b \in B]/(T_b^p - b^p : b \in B).$$

All expressions use finitely many variables, so this argument also covers infinite B .

The product and inverse rules make every F -derivation, including the universal one, vanish on E . In the displayed presentation, monomial differentiation defines a partial F -derivation for each $b \in B$: it kills the coefficient field E and all the defining relations. These derivations send the basis elements to their respective coordinate values, proving that the db are independent. Differentiating

the finite monomial expansion of each element proves that they span $\Omega_{L/F}$. In particular, every p -independent subset has independent differentials after extending it to B .

For the converse, suppose a finite list $db_1, \dots, db_r$ is independent. Here is the required extension of its coordinate functionals, including when $\Omega_{L/F}$ is infinite-dimensional. Prescribe a coordinate functional on their span. Order pairs consisting of a larger subspace and a linear functional extending this prescription by extension. A chain has an upper bound obtained by taking the union of its subspaces and compatible functionals. Zorn's lemma gives a maximal pair. If its subspace omits a vector v , extend to its direct sum with Lv by assigning value zero to v ; independence of v makes this extension well-defined, contradicting maximality. The maximal subspace is therefore all of $\Omega_{L/F}$. By the universal differential property, these extended coordinate functionals give F -derivations $D_i : L \rightarrow L$ with $D_i(b_j) = \delta_{ij}$, and they kill E .

If the b_j have a nonzero monomial relation over E with all exponents below p , choose one of least total degree. Its degree is positive, since a nonzero constant cannot vanish in a field. Some partial derivative of its polynomial is nonzero: choose a variable having positive exponent in a nonconstant term; that exponent is between one and $p - 1$, and different monomials give different derivative monomials. This derivative has smaller total degree. Applying the corresponding D_i to the original relation makes its evaluation zero, contradicting minimality. Thus the finite list is p -independent. For an arbitrary subset, every relation would involve a finite list, so the same conclusion holds. This proves the converse, the equivalence, and the basis assertion in the full stated field scope. $\square$

Lemma F.120: power series ring subfields

Adapted from the Stacks project, more-algebra.tex, lines 12130–12171.

For $A = k[[X_1, \dots, X_n]][Y_1, \dots, Y_m]$ in characteristic p , choose a p -basis B of k . For each finite subset $J \subset B$, let $k_J = k^p(B \setminus J)$ and $A_J = k_J[[X_1^p, \dots, X_n^p]][Y_1^p, \dots, Y_m^p]$. Then A is finite free over A_J and the fraction fields K_J are downward directed, contain K^p , and intersect in K^p .

Proof. The monomials in the finite set J , in the X_i , and in the Y_j , all with exponents below p , form a basis of A over A_J . Group power-series terms by the X exponents modulo p , polynomial terms by the Y exponents modulo p , and each coefficient by the finite basis of k/k_J ; uniqueness of each expansion proves both spanning and independence. This proves finite freeness directly, including imperfect k .

Every p th power lies in each A_J , hence $K^p \subset K_J$, and adjoining more excluded basis elements shrinks K_J , proving directedness. To compute the intersection, write $u = f/g^p$ with $f, g \in A$, $g \neq 0$; every fraction admits this form by multiplying its numerator by g^{p-1} . If $u \in K_J$, write it as a/b^p with $a \in A_J$ and nonzero $b \in A$: the inverse of any denominator $h \in A_J$ is h^{p-1}/h^p , so this form is available. Then $b^p f = ag^p \in A_J$. Any A_J -derivation of A therefore kills f , since differentiating gives $b^p D(f) = 0$ in the domain. The power-series and polynomial partial derivatives imply every nonzero exponent of f in the X, Y variables is divisible by p . The partial derivations on the excluded coefficient p -basis elements, extended coefficientwise to the power series, imply its coefficients lie in k_J : their monomial basis expansion shows that their common kernel is k_J . For every finite J this holds, so each coefficient lies in $\bigcap_J k_J = k^p$, again by its finite p -basis monomial expansion. Thus $f \in A^p$ (take coefficient roots and divide all exponents by p), and $u \in K^p$. This proves the intersection assertion. $\square$

Lemma F.121: quasi regular ideal finite projective

Adapted from the Stacks project, more-algebra.tex, lines 8769–8778.

Let $I \subset R$ be a quasi-regular ideal of a ring. Then I/I^2 is a finite projective R/I -module.

Proof. Put $S = R/I$ and $M = I/I^2$. If $S = 0$, then $I = R$ and $M = 0$, which is finite projective. Otherwise Definition F.95 gives, around each prime of S , a principal open $D(\bar{g})$ where $I_{\bar{g}}$ is generated

by a quasi-regular list. The degree-one isomorphism in that definition and exact localization identify $M_{\bar{g}}$ with a finite free $S_{\bar{g}}$ -module. The elements $\bar{g}$ generate the unit ideal: a proper ideal would lie in a prime missing every one of these opens, contrary to their cover. Hence finitely many $g_1, \dots, g_s \in S$ suffice, with $(g_1, \dots, g_s) = S$.

We use the following finite-cover detection calculation for arbitrary modules. If $N_{g_i} = 0$ for every i and $x \in N$, then $g_i^{e_i} x = 0$ for some positive e_i . The powers $g_i^{e_i}$ still generate the unit ideal, since a prime containing all of them would contain every g_i . Writing $1 = \sum b_i g_i^{e_i}$ gives $x = 0$, so $N = 0$.

For each local basis of M_{g_i} , clear its finitely many denominators to choose elements of M whose localized images are a basis; multiplying basis elements by powers of the unit g_i preserves that property. Let $M_0 \subset M$ be generated by all these finitely many elements. The quotient M/M_0 vanishes on every open, so the detection calculation gives $M = M_0$. Choose a finite free surjection $F = S^n \rightarrow M$ and let K be its kernel. On each open the quotient M_{g_i} is free, so the localized surjection splits. Thus K_{g_i} is a direct summand of $S_{g_i}^n$, generated by the projections of its finite basis. Clear denominators in those kernel generators to obtain finitely many elements of K ; the submodule they generate has the same localization as K on every open. Apply the same detection calculation to their quotient to conclude that K is finitely generated. Therefore M is finitely presented.

At every prime one of the g_i is invertible, and M localizes there to a free module. The proved local-flatness criterion [AG-CA-07, Theorem 3.3](#) makes M flat. Its finite presentation now supplies the full hypothesis of [AG-CA-07, Theorem 5.3](#), giving finite projectivity. No Noetherian assumption, or finite presentation of the initial relation module, was used. $\square$

Lemma F.122: quotient regular

Adapted from the Stacks project, more-algebra.tex, lines 12407–12424.

If R is Noetherian regular and a derivation $D : R \rightarrow R$ sends f to a unit modulo (f) , then $R/(f)$ is regular.

Proof. At a prime containing f , extend D to the local ring by the fraction calculation above. If $f \in \mathfrak{m}^2$, Leibniz on a finite expression $f = \sum a_i b_i$, $a_i, b_i \in \mathfrak{m}$, gives $D(f) \in \mathfrak{m}$, contradicting that its class is a unit. Hence $f \notin \mathfrak{m}^2$. In the regular local ring its class extends to a regular parameter system, and [AG-CA-14, Corollary 1.2](#), gives regularity of the parameter quotient. These quotients are exactly all local rings of $R/(f)$. $\square$

Lemma F.123: regular base change

Adapted from the Stacks project, more-algebra.tex, lines 11022–11047.

Let $R \rightarrow \Lambda$ be a regular ring map. For any finite type ring map $R \rightarrow R'$ the base change $R' \rightarrow \Lambda \otimes_R R'$ is regular too.

Proof. Flatness survives base change by [AG-CA-07, Proposition 3.2](#). Fix $\mathfrak{p}' \subset R'$ with contraction $\mathfrak{p} \subset R$. Set $k = \kappa(\mathfrak{p})$ and $E = \kappa(\mathfrak{p}')$. The field extension E/k is finitely generated, as follows directly from the finite-type hypothesis. Choose algebra generators $u_1, \dots, u_n$ of R'/R . The injection $R/\mathfrak{p} \rightarrow R'/\mathfrak{p}'$ makes

$$(R \setminus \mathfrak{p})^{-1}(R'/\mathfrak{p}') = k[\bar{u}_1, \dots, \bar{u}_n]$$

a domain with fraction field E . Thus $E = k(\bar{u}_1, \dots, \bar{u}_n)$.

The original fibre $F = \Lambda \otimes_R k$ is Noetherian and geometrically regular. Tensor associativity identifies the new fibre with

$$(\Lambda \otimes_R R') \otimes_{R'} E \cong F \otimes_k E.$$

It is Noetherian by [F.12](#), using the field generation just proved. Its geometric regularity over E is

the full base-field conclusion of F.52. Flatness and these fibre assertions are precisely the regularity conditions. No assumption that E/k is algebraic or separable has been added. $\square$

Lemma F.124: regular composition

Adapted from the Stacks project, more-algebra.tex, lines 11049–11065.

A composition of regular maps of Noetherian rings is regular.

Proof. Let $A \rightarrow B \rightarrow C$ be the maps. Their composite is flat by AG-CA-07, Proposition 3.2. For $\mathfrak{a} \subset A$, put $k_0 = \kappa(\mathfrak{a})$ and $S = A \setminus \mathfrak{a}$. Both original fibres are Noetherian: for example

$$C \otimes_A k_0 = S^{-1}(C/\mathfrak{a}C).$$

The right side is a localization of a quotient of the Noetherian ring C . Thus the Noetherian composite-fibre condition is automatic here.

Let K/k_0 be finite purely inseparable, and set $D = B \otimes_A K$ and $E = C \otimes_A K$. Lemma F.12 makes them Noetherian, since each is the finite field tensor of its original Noetherian fibre. Regularity of $A \rightarrow B$ makes D regular. The identification $E = C \otimes_B D$ and the same flat base-change proposition make $D \rightarrow E$ flat.

Fix $\mathfrak{r} \subset E$ and let $\mathfrak{s} \subset D$ and $\mathfrak{b} \subset B$ be its successive contractions. The contraction of $\mathfrak{b}$ to A is $\mathfrak{a}$: elements of $\mathfrak{a}$ vanish in D , whereas every element outside $\mathfrak{a}$ becomes invertible through K . There is a field inclusion $k_0 \subset \kappa(\mathfrak{b})$. Tensoring D with this residue field gives

$$Q = D \otimes_B \kappa(\mathfrak{b}) \cong \kappa(\mathfrak{b}) \otimes_{k_0} K.$$

A finite k_0 -basis of K makes Q a finite-dimensional $\kappa(\mathfrak{b})$ -algebra. The prime $\mathfrak{s}$ descends to a prime $\bar{\mathfrak{s}}$ of Q : first quotient by $\mathfrak{b}D$ and then invert $B \setminus \mathfrak{b}$. Its quotient $Q/\bar{\mathfrak{s}}$ is a finite-dimensional domain over $\kappa(\mathfrak{b})$, hence a field. Indeed, multiplication by a nonzero element is injective and therefore surjective, giving its inverse. Localizing this quotient at its nonzero elements changes nothing. The quotient-localization construction therefore identifies it with $\kappa(\mathfrak{s})$, proving

$$[\kappa(\mathfrak{s}) : \kappa(\mathfrak{b})] < \infty.$$

The fibre of $D \rightarrow E$ at $\mathfrak{s}$ is

$$E \otimes_D \kappa(\mathfrak{s}) \cong (C \otimes_B \kappa(\mathfrak{b})) \otimes_{\kappa(\mathfrak{b})} \kappa(\mathfrak{s}).$$

The original B -fibre is Noetherian and geometrically regular. Its displayed finite field tensor is Noetherian by F.12 and regular by F.52. The closed fibre of the flat local map $D_{\mathfrak{s}} \rightarrow E_{\mathfrak{r}}$ is a localization of this regular ring, hence regular. The source is a regular local ring and both local rings are Noetherian. F.47 makes $E_{\mathfrak{r}}$ regular. This holds for every target prime, so E is regular. Criterion F.50, applied to every finite purely inseparable K/k_0 , proves geometric regularity of the composite fibre. Doing this at every $\mathfrak{a}$ proves the composition assertion. Empty fibres have no target primes and satisfy the same criterion. $\square$

Lemma F.125: regular permanence

Adapted from the Stacks project, more-algebra.tex, lines 11116–11129.

If $A \rightarrow B \rightarrow C$ are maps of Noetherian rings, $A \rightarrow C$ is regular and $B \rightarrow C$ faithfully flat, then $A \rightarrow B$ is regular.

Proof. For an injection $N_1 \rightarrow N_2$ of A -modules, put $K = \ker(N_1 \otimes_A B \rightarrow N_2 \otimes_A B)$. Flatness of C over B preserves the exact sequence

$$0 \longrightarrow K \longrightarrow N_1 \otimes_A B \longrightarrow N_2 \otimes_A B.$$

Tensor associativity identifies the last map after tensoring with C with $N_1 \otimes_A C \rightarrow N_2 \otimes_A C$. It is injective by A -flatness of C . Consequently $K \otimes_B C = 0$, and faithful flatness makes $K = 0$. This proves that B is flat over A .

For $\mathfrak{a} \subset A$, set $k = \kappa(\mathfrak{a})$, $F = B \otimes_A k$ and $G = C \otimes_A k$. These rings are Noetherian: they are respectively localizations of $B/\mathfrak{a}B$ and $C/\mathfrak{a}C$ at the image of $A \setminus \mathfrak{a}$. The identification $G = F \otimes_B C$ proves faithful flatness of $F \rightarrow G$ directly. For every F -module M ,

$$M \otimes_F G \cong M \otimes_B C.$$

Restriction of scalars is exact, while the right-hand tensor functor is exact and detects the zero module. The left-hand functor therefore has those properties too. The target G is geometrically regular over k , since $A \rightarrow C$ is regular. Geometric regularity descent F.51 applies to these Noetherian k -algebras and makes F geometrically regular. These are all the fibre conditions for $A \rightarrow B$, completing the proof. $\square$

Lemma F.126: relative global complete intersection koszul

Adapted from the Stacks project, more-algebra.tex, lines 8995–9012.

For the Noetherian relative global complete intersection above, the equation list has zero positive Koszul homology.

Proof. At primes containing its ideal the conormal lifting proof makes the list regular, and the Koszul cone induction in the preceding equivalence proof makes its positive homology zero. Away from that ideal choose coefficients a_i with $\sum a_i f_i = 1$ in the local ring. Wedging with $\sum a_i e_i$ gives $dh + hd = 1$ by the alternating Leibniz rule, making the Koszul complex contractible there. Its homology thus vanishes at every prime, and local detection makes it zero. $\square$

Lemma F.127: relative regular immersion algebra

Adapted from the Stacks project, more-algebra.tex, lines 8601–8631.

Let $A \rightarrow B$, $A \rightarrow A'$, $J = (f_i) \subset B$, and suppose B/J is A -flat. Quasi-regularity and H_1 -regularity of (f_i) are preserved in $B' = B \otimes_A A'$.

Proof. For quasi-regularity the hypothesis is $\text{gr}_J B \cong (B/J)[T_1, \dots, T_r]$, with T_i represented by f_i . Each graded piece is therefore A -flat. Induction on n in $0 \rightarrow J^n/J^{n+1} \rightarrow B/J^{n+1} \rightarrow B/J^n \rightarrow 0$ makes every B/J^n flat, using the extension-of-flat-modules proof AG-CA-07, Theorem 6.1. Tensoring $0 \rightarrow J^n \rightarrow B \rightarrow B/J^n \rightarrow 0$ consequently stays exact on the left. Its kernel identifies $J^n \otimes_A A'$ with $(J')^n$, since products of the images of the f_i generate the latter ideal. Taking consecutive quotients proves $\text{gr}_{J'} B' = \text{gr}_J B \otimes_A A' = (B'/J')[T_i]$.

For H_1 -regularity set $Z = \ker(B^r \xrightarrow{(f_i)} J)$. The sequences $0 \rightarrow J \rightarrow B \rightarrow B/J \rightarrow 0$, $Z \rightarrow B^r \rightarrow J \rightarrow 0$, and $\bigwedge^2 B^r \rightarrow Z \rightarrow H_1 \rightarrow 0$ are exact. After tensoring the first stays injective on the left by quotient flatness, and the last two stay right exact. Thus $Z \otimes_A A'$ surjects onto the new cycle module, and its quotient by the images of the Koszul boundaries is a quotient of $H_1 \otimes_A A'$. If $H_1 = 0$, the new H_1 is zero. This proves the assertion over arbitrary rings without assuming B flat. $\square$

Lemma F.128: symmetric algebra smooth

Adapted from the Stacks project, more-algebra.tex, lines 2111–2136.

Let A be a ring. Let M be an A -module. Then $C = \text{Sym}_A^*(M)$ is smooth over A if and only if M is a finite projective A -module.

Proof. We first construct the algebra and prove the identities used in Proposition D.10. Form the tensor algebra

$$T_A(M) = \bigoplus_{d \geq 0} M^{\otimes_A d}, \quad M^{\otimes_A 0} = A,$$

with multiplication given by concatenating pure tensors. The tensor relations and associativity calculation in AG-RG-S01, Lemma 1.0 make this multiplication well defined and associative; pure tensors generate each summand. Quotient by the two-sided ideal generated by $m \otimes n - n \otimes m$ for $m, n \in M$. Adjacent interchanges then commute every pair of products of degree-one elements, so the quotient is the commutative algebra $C = \text{Sym}_A(M)$. The relations are homogeneous of degree two. Hence $C_0 = A$, $C_1 = M$, and every higher-degree piece is generated by products of degree-one elements. Projection onto degree zero is an A -algebra augmentation $\epsilon : C \rightarrow A$. Its kernel J satisfies $J/J^2 = M$.

An A -linear map $u : M \rightarrow D$ to a commutative A -algebra extends to the tensor algebra by

$$m_1 \otimes \cdots \otimes m_d \longmapsto u(m_1) \cdots u(m_d).$$

This respects all tensor relations and kills the commutators, so gives an A -algebra map $C \rightarrow D$. It is unique, because A and the degree-one copy of M generate C . Restriction to that copy is the inverse construction. This proves the universal property without any finiteness assumption.

Here are explicit base-change and direct-sum identities, for any A -algebra B and any A -module N :

$$\begin{aligned} B \otimes_A \text{Sym}_A(M) &\simeq \text{Sym}_B(B \otimes_A M), \\ \text{Sym}_A(M \oplus N) &\simeq \text{Sym}_A(M) \otimes_A \text{Sym}_A(N). \end{aligned}$$

For the first identity, send the degree-one generator $b \otimes m$ on the right to $b \otimes m$ on the left; this is a B -linear map, so extends by the just-proved property. In the other direction send m to $1 \otimes m$ in the right-hand algebra and use the coproduct formula for tensor products of commutative algebras in AG-RG-S01, Lemma 1.0 to combine this map with its structural map from B . The two composites fix B and every degree-one generator, so are identities. For the second identity, send (m, n) to $m \otimes 1 + 1 \otimes n$. Conversely the two inclusions into $M \oplus N$ give maps from both symmetric algebras; their commuting images define the tensor-product map. The composites fix M, N and A , hence are identities. These maps preserve degree. Choosing the images of a basis similarly gives mutually inverse maps

$$\text{Sym}_A(A^{\oplus n}) \simeq A[X_1, \dots, X_n].$$

Thus the base-change, direct-sum and free-module identifications in D.10, including reduction modulo its base ideal, have now been proved for arbitrary rings and modules.

Suppose C is smooth over A . By finite presentation it has a finite list of algebra generators $z_1, \dots, z_r$. Put $a_i = \epsilon(z_i)$ and $y_i = z_i - a_i$. The y_i generate J : for every polynomial expression $P(z)$, subtracting $P(a)$ is a sum of multiples of the $z_i - a_i$, as follows by changing one variable at a time and expanding its powers. Every element of J has $P(a) = 0$. Therefore the classes of the y_i generate $J/J^2 = M$, so M is finite. Choose a finite free surjection $F = A^{\oplus r'} \twoheadrightarrow M$, with kernel K .

For any A -module V , the square-zero algebra $A \oplus V$ has product $(a, v)(b, w) = (ab, aw + bv)$. The map

$$A \oplus F \longrightarrow A \oplus M$$

is surjective and its kernel $0 \oplus K$ has square zero. The degree-one map $m \mapsto (0, m)$ gives an A -algebra map $C \rightarrow A \oplus M$. Smoothness supplies a lift to $A \oplus F$: its formal lifting meaning is proved equivalent to Definition F.6 in AG-CA-17, Theorem 3.1 and Section 4. The lift sends each m to $(0, s(m))$, since its first coordinate is forced to be zero by reduction. Its restriction $s : M \rightarrow F$ is A -linear, and the composite $M \xrightarrow{s} F \rightarrow M$ is identity. Thus $F = s(M) \oplus K$: decompose f as $s(\bar{f}) + (f - s(\bar{f}))$, and apply $F \rightarrow M$ to check that the intersection is zero. This makes M a finite direct summand of a finite free module, hence finite projective.

Conversely suppose M is finite projective. Choose a decomposition $F = A^{\oplus n} = M \oplus K$, with inclusion $i : M \rightarrow F$ and projection $\pi : F \rightarrow M$. The kernel K is generated by the projections onto K of the n basis vectors of F . The universal property identifies C with $A[X_1, \dots, X_n]$ modulo the linear forms corresponding to K : maps out of this quotient are exactly linear maps $F \rightarrow D$ killing K , hence exactly linear maps $M \rightarrow D$. Explicitly the variables map to the images under π of the basis vectors; the map back is induced by i , followed by the quotient. Both composites fix their degree-one generators. A finite generating list of K therefore gives a finite relation list, proving finite presentation of C .

Given a square-zero test $D \rightarrow D/H$ and an A -algebra map $C \rightarrow D/H$, restrict it to an A -linear map $u : M \rightarrow D/H$. Choose in D lifts of the images of the n basis vectors under $u\pi : F \rightarrow D/H$. They give an A -linear map $v : F \rightarrow D$ whose reduction is $u\pi$. The restriction $vi : M \rightarrow D$ lifts u , so the universal property extends it to an A -algebra lift $C \rightarrow D$. This proves formal smoothness for every test. Finite presentation and the same proved formal/conormal comparison make C smooth. All the arguments include $M = 0$, empty basis and relation lists, and the zero ring. No Noetherian hypothesis is used. $\square$

Lemma F.129: syntomic lci

Adapted from the Stacks project, more-algebra.tex, lines 9014–9061.

Let $R \rightarrow S$ be a ring map with R Noetherian. The following are equivalent

1. $R \rightarrow S$ is syntomic (Algebra, Definition F.1), and
2. $R \rightarrow S$ is flat and a local complete intersection.

Proof. Assume (1). Then $R \rightarrow S$ is flat by definition. By Algebra, Lemma F.87 and Lemma F.114 we see that it suffices to show a relative global complete intersection is a local complete intersection homomorphism which is Lemma F.126.

Assume (2). A local complete intersection is of finite presentation because a Koszul-regular ideal is finitely generated. Let $R \rightarrow k$ be a map to a field. It suffices to show that $S' = S \otimes_R k$ is a local complete intersection over k , see Algebra, Definition F.2. Choose a prime $\mathfrak{q}' \subset S'$. Write $S = R[x_1, \dots, x_n]/I$. Then $S' = k[x_1, \dots, x_n]/I'$ where $I' \subset k[x_1, \dots, x_n]$ is the image of I . Let $\mathfrak{p}' \subset k[x_1, \dots, x_n]$, $\mathfrak{q} \subset S$, and $\mathfrak{p} \subset R[x_1, \dots, x_n]$ be the corresponding primes. By Definition F.95 exists an $g \in R[x_1, \dots, x_n]$, $g \notin \mathfrak{p}$ and $f_1, \dots, f_r \in R[x_1, \dots, x_n]_g$ which form a Koszul-regular sequence generating I_g . Since S and hence S_g is flat over R we see that the images $f'_1, \dots, f'_r$ in $k[x_1, \dots, x_n]_g$ form a H_1 -regular sequence generating I'_g , see Lemma F.127. Thus $f'_1, \dots, f'_r$ map to a regular sequence in $k[x_1, \dots, x_n]_{\mathfrak{p}'}$ generating $I'_{\mathfrak{p}'}$ by Lemma F.118. Applying Algebra, Lemma F.57 we conclude $S'_{g\mathfrak{q}'}$ for some $g' \in S'$, $g' \notin \mathfrak{q}'$ is a global complete intersection over k as desired. $\square$

Lemma F.130: transitive lci at end

Adapted from the Stacks project, more-algebra.tex, lines 9068–9115.

Let $A \rightarrow B \rightarrow C$ be ring maps. Assume $B \rightarrow C$ is a local complete intersection homomorphism. Choose a presentation $\alpha : A[x_s, s \in S] \rightarrow B$ with kernel I . Choose a presentation $\beta : B[y_1, \dots, y_m] \rightarrow C$ with kernel J . Let $\gamma : A[x_s, y_t] \rightarrow C$ be the induced presentation of C with kernel K . Then we get a canonical commutative diagram

$$\begin{aligned}
0 &\rightarrow \Omega_{A[x_s]/A} \otimes C \\
\Omega_{A[x_s]/A} \otimes C &\rightarrow \Omega_{A[x_s, y_t]/A} \otimes C \\
\Omega_{A[x_s, y_t]/A} \otimes C &\rightarrow \Omega_{B[y_t]/B} \otimes C \\
\Omega_{B[y_t]/B} \otimes C &\rightarrow 0 \\
0 &\rightarrow I/I^2 \otimes C \\
I/I^2 \otimes C &\rightarrow K/K^2 \\
I/I^2 \otimes C &\rightarrow \Omega_{A[x_s]/A} \otimes C \\
K/K^2 &\rightarrow J/J^2 \\
K/K^2 &\rightarrow \Omega_{A[x_s, y_t]/A} \otimes C \\
J/J^2 &\rightarrow 0 \\
J/J^2 &\rightarrow \Omega_{B[y_t]/B} \otimes C
\end{aligned}$$

with exact rows. In particular, the six term exact sequence of Algebra, Lemma F.42 can be completed with a zero on the left, i.e., the sequence

$$0 \rightarrow H_1(\mathrm{NL}_{B/A} \otimes_B C) \rightarrow H_1(L_{C/A}) \rightarrow H_1(L_{C/B}) \rightarrow \Omega_{B/A} \otimes_B C \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$$

is exact.

Proof. The only thing to prove is the injectivity of the map $I/I^2 \otimes C \rightarrow K/K^2$. The arbitrary-ring comparison F.94.A makes the kernel J of this chosen presentation Koszul-regular. Hence we have $IA[x_s, y_j] \cap K^2 = IK$ by Lemma F.101. This means that the kernel of $K/K^2 \rightarrow J/J^2$ is isomorphic to $IA[x_s, y_j]/IK$. Since $I/I^2 \otimes_B C = IA[x_s, y_j]/IK$ by right exactness of tensor product, this provides us with the desired injectivity of $I/I^2 \otimes_B C \rightarrow K/K^2$. $\square$

Lemma F.131: transitivity gamma

Adapted from the Stacks project, more-algebra.tex, lines 9208–9228.

For fields $K \subset L \subset M$ the sequence $0 \rightarrow H_1(L_{L/K}) \otimes_L M \rightarrow H_1(L_{M/K}) \rightarrow H_1(L_{M/L}) \rightarrow \Omega_{L/K} \otimes_L M \rightarrow \Omega_{M/K} \rightarrow \Omega_{M/L} \rightarrow 0$ is exact.

Proof. First suppose the last algebra is a global complete-intersection L -algebra C . Present $L = P/I$ with $P = K[X_s]$, allowing arbitrary variables, and $C = L[Y]/J$ with J a regular equation ideal. Let $Q = P[Y]$ and H be the kernel of $Q \rightarrow C$. The sequence of conormal modules is

$$0 \rightarrow (I/I^2) \otimes_L C \rightarrow H/H^2 \rightarrow J/J^2 \rightarrow 0.$$

For its left injectivity, the relation calculation already proved at F.101 gives $IQ \cap H^2 = IH$. Thus its left module is IQ/IH , exactly the kernel in the middle. Surjectivity on the right is lifting ideal representatives. Polynomial differentials give the split short exact sequence $0 \rightarrow \Omega_{P/K} \otimes C \rightarrow \Omega_{Q/K} \otimes C \rightarrow \Omega_{L[Y]/L} \otimes C \rightarrow 0$ on the free X and Y bases. Apply the full snake-lemma proof already written to these two short exact rows and their conormal differentials. Their kernels are the three first homology groups and their cokernels the three differential modules, so it gives precisely the six-term exact sequence with zero at its left.

Now express M as the filtered union of global complete-intersection L -subalgebras by the preceding monic-tower proof. The conormal-complex filtered-colimit calculation proved above identifies all three complexes at the limit. Filtered colimits are exact by common-stage representatives. Finally tensoring with the field M over L is exact, so the first kernel in the limit is $H_1(L_{L/K}) \otimes_L M$, rather than merely a homology group of an unproved derived base change. This proves every term and the initial injectivity. $\square$

Proposition F.132: Noetherian complete G ring

Adapted from the Stacks project, more-algebra.tex, lines 12965–12999.

A Noetherian complete local ring is a G-ring.

Proof. Let A be a Noetherian complete local ring. By Lemma F.99 it suffices to check that $B = A/\mathfrak{q}$ has geometrically regular formal fibres over the minimal prime (0) of B . Thus we may assume that A is a domain and it suffices to check the condition for the formal fibres over the minimal prime (0) of A . Let K be the fraction field of A .

We can choose a subring $A_0 \subset A$ which is a regular complete local ring such that A is finite over A_0 , see Algebra, Lemma F.25. Moreover, we may assume that A_0 is a power series ring over a field or a Cohen ring. By Lemma F.96 we see that it suffices to prove the result for A_0 .

Assume that A is a power series ring over a field or a Cohen ring. Since A is regular local, each $A_{\mathfrak{p}}$ is regular local by the written localization proof AG-CA-14, Theorem 3.2. Hence the completions $A_{\mathfrak{p}}^{\wedge}$ are regular, see Lemma AG-CA-19, Theorem 4.1. Hence the fibre $A_{\mathfrak{p}}^{\wedge} \otimes_A K$ is, as a localization of $A_{\mathfrak{p}}^{\wedge}$, also regular. Thus we are done if the characteristic of K is 0. The positive characteristic case is the case $A = k[[x_1, \dots, x_d]]$ which is a special case of Lemma F.110. $\square$

Proposition F.133: characterization geometrically regular

Adapted from the Stacks project, more-algebra.tex, lines 9306–9458.

For a characteristic- p field k and Noetherian local k -algebra $(A, \mathfrak{m}, K)$, the following are equivalent: geometric regularity; regularity after every finite subextension of $k^{1/p}/k$; regularity of A and injectivity of $H_1(L_{K/k}) \rightarrow \mathfrak{m}/\mathfrak{m}^2$; regularity of A and injectivity of $\Omega_{k/\mathbb{F}_p} \otimes_k K \rightarrow \Omega_{A/\mathbb{F}_p} \otimes_A K$.

Proof. First identify the last two injections. The general Jacobi–Zariski calculation F.42 gives a natural map $H = H_1(L_{K/k}) \rightarrow E = \mathfrak{m}/\mathfrak{m}^2$. For a quotient presentation over A this first homology is exactly the conormal module E , since the degree-zero relative polynomial module is zero. Absolute prime-field formal smoothness of K , AG-CA-19S, Lemma 1.1, makes $H_1(L_{K/\mathbb{F}_p}) = 0$. Thus its Jacobi–Zariski row is $0 \rightarrow E \rightarrow V \rightarrow \Omega_{K/\mathbb{F}_p} \rightarrow 0$, with $V = \Omega_{A/\mathbb{F}_p} \otimes_A K$. The field Jacobi–Zariski row F.131 is $0 \rightarrow H \rightarrow \Omega_{k/\mathbb{F}_p} \otimes_k K \rightarrow \Omega_{K/\mathbb{F}_p} \rightarrow \Omega_{K/k} \rightarrow 0$. Naturality gives a square into the first row and identity on $\Omega_{K/\mathbb{F}_p}$. A vector killed by the middle vertical map lies in the kernel of the bottom map to $\Omega_{K/\mathbb{F}_p}$, hence comes uniquely from H ; it is killed precisely when its image in E is zero. Conversely every such vector maps to zero in V . The kernels are therefore isomorphic, proving equivalence of the two asserted injectivities.

Assume A regular and $H \rightarrow E$ injective. For a finite purely inseparable k'/k , put $A' = A \otimes_k k'$. A k -basis of k' containing 1 makes A' finite free over A and makes $A \rightarrow A'$ injective. A finite algebra is integral by AG-CA-05, Theorem 1.2. Choose e with $(k')^{p^e} \subset k$. Expanding a finite tensor sum shows that every $t \in A'$ satisfies $t^{p^e} \in A$. For a prime $\mathfrak{q}$ over $\mathfrak{p}$, membership $t \in \mathfrak{q}$ is equivalent to $t^{p^e} \in \mathfrak{p}$. Thus there is at most one prime over each $\mathfrak{p}$, and lying over gives one. Every maximal ideal of A' contracts to $\mathfrak{m}$ by AG-CA-05, Theorem 3.3, so A' is local. Let $\mathfrak{m}'$ be its maximal ideal and K' its residue field.

The extension K'/K is finite: the surjection

$$A'/\mathfrak{m}A' = K \otimes_k k' \longrightarrow A'/\mathfrak{m}' = K'$$

has a source of finite K -dimension $[k' : k]$. Write $E' = \mathfrak{m}'/\mathfrak{m}'^2$ and $H' = H_1(L_{K'/k'})$. The quotient instances of F.42 give the relative rows $0 \rightarrow H \otimes_K K' \rightarrow E \otimes_K K' \rightarrow V_0 \rightarrow W \rightarrow 0$ and $H' \rightarrow E' \rightarrow V_0 \rightarrow W' \rightarrow 0$. Differential base change AG-CA-16, Theorem 2.1, identifies the common V_0 , with $W = \Omega_{K/k} \otimes_K K'$ and $W' = \Omega_{K'/k'}$. The map $\alpha : W \rightarrow W'$ is surjective because both maps from V_0 are surjective. Both k'/k and K'/K are now verified finite, so F.108, with both transcendence degrees zero, gives $\dim \ker \alpha - \dim \ker \beta + \dim \operatorname{coker} \beta = 0$ for $\beta : H \otimes K' \rightarrow H'$. These defects are finite. Also $H \otimes K'$ injects into the finite $E \otimes K'$, so H' is finite too.

Let L, L' be the images of $E \otimes K', E'$ in V_0 . Then $L \subset L'$ and $L'/L = \ker \alpha$. Exactness gives $\dim E' \leq \dim H' + \dim L'$. On the top row $\dim E = \dim H + \dim L$. Subtracting yields

$$\dim E' - \dim E \leq (\dim H' - \dim H) + \dim \ker \alpha = \dim \operatorname{coker} \beta - \dim \ker \beta + \dim \ker \alpha = 0.$$

The proved integral inclusion $A \subset A'$ preserves dimension by AG-CA-05, Theorem 5.1. Thus $\dim E' \leq \dim E = \dim A = \dim A'$. Lift a basis of E' to $\mathfrak{m}'$. Nakayama's lemma F.10 makes these lifts generators, so the proved height theorem AG-CA-11, Theorem 3.1, gives $\dim A' \leq \dim E'$. Equality proves A' regular. The finite purely inseparable criterion F.50 therefore proves geometric regularity.

It remains to derive the absolute injection from regularity after all finite subextensions of $k^{1/p}$. This includes A itself regular. Choose a finite list $a_1, \dots, a_n \in k$ with independent absolute differentials. The root-degree proof F.81 gives $k' = k(a_i^{1/p}) = k[T_i]/(T_i^p - a_i)$ with degree p^n . Its base change A' is regular by assumption and has the same dimension as A by the proved integral-inclusion argument. The monic equations form a regular list, their relative derivatives vanish, and their conormal module is free by F.74. Thus the complex for A'/A is $A'^n \xrightarrow{0} A'^n$. The explicit connecting-map construction in F.42 sends its i th homology basis vector to $da_i \otimes 1$, up to the common sign convention, and gives

$$A'^n \longrightarrow \Omega_{A/\mathbb{F}_p} \otimes_A A' \longrightarrow \Omega_{A'/\mathbb{F}_p} \longrightarrow A'^n \longrightarrow 0.$$

Here is why exactness survives tensoring with K' . Let T be the kernel of the last surjection. The row $0 \rightarrow T \rightarrow \Omega_{A'/\mathbb{F}_p} \rightarrow A'^n \rightarrow 0$ splits because its quotient is free, so its injection remains injective after tensoring. Tensoring $A'^n \rightarrow \Omega_{A/\mathbb{F}_p} \otimes_A A' \rightarrow T \rightarrow 0$ is right exact. Combining these two facts shows that $\delta : V \otimes_K K' \rightarrow V' = \Omega_{A'/\mathbb{F}_p} \otimes_{A'} K'$ has kernel spanned by the $da_i \otimes 1$ and cokernel K'^n .

Compare the two absolute rows $0 \rightarrow E \otimes K' \rightarrow V \otimes K' \rightarrow \Omega_{K/\mathbb{F}_p} \otimes K' \rightarrow 0$ and $0 \rightarrow E' \rightarrow V' \rightarrow \Omega_{K'/\mathbb{F}_p} \rightarrow 0$. The maps on the left have equally dimensional finite spaces, since both local rings are regular of equal dimension, so their kernel and cokernel dimensions agree. The map on the right likewise has equal finite kernel and cokernel dimensions: the field sequence F.131 identifies them with $H_1(L_{K'/K})$ and $\Omega_{K'/K}$, and Cartier equality F.98 for the already proved finite extension $K \subset K'$ gives equality. The full snake sequence F.84 for these rows gives

$$\dim \ker \delta - \dim \operatorname{coker} \delta = (\dim \ker \delta_E - \dim \operatorname{coker} \delta_E) + (\dim \ker \delta_K - \dim \operatorname{coker} \delta_K) = 0.$$

All its defect spaces are finite; for the middle kernel this also follows from its preceding span by the n displayed vectors. Its cokernel has dimension n , so its kernel has dimension n and those n spanning $da_i \otimes 1$ are independent. A basis of $\Omega_{k/\mathbb{F}_p}$ is given by the absolute p -basis differentials, and every relation uses finitely many of them. The preceding argument therefore proves the whole absolute injection. Finally geometric regularity implies the stated root-height-one tests by definition. This closes every implication and the two dimension chases explicitly. $\square$

Proposition F.134: finite type over G ring

Adapted from the Stacks project, more-algebra.tex, lines 13161–13250.

Let R be a G-ring. If $R \rightarrow S$ is essentially of finite type then S is a G-ring.

Proof. Since being a G-ring is a property of the local rings it is clear that a localization of a G-ring is a G-ring. Conversely, if every localization at a prime is a G-ring, then the ring is a G-ring. Thus it suffices to show that $S_{\mathfrak{q}}$ is a G-ring for every finite type R -algebra S and every prime $\mathfrak{q}$ of S . Writing S as a quotient of $R[x_1, \dots, x_n]$ we see from Lemma F.96 that it suffices to prove that $R[x_1, \dots, x_n]$ is a G-ring. By induction on n it suffices to prove that $R[x]$ is a G-ring. Let $\mathfrak{q} \subset R[x]$ be a maximal ideal. By Lemma F.100 it suffices to show that

$$R[x]_{\mathfrak{q}} \longrightarrow R[x]_{\mathfrak{q}}^{\wedge}$$

is regular. If $\mathfrak{q}$ lies over $\mathfrak{p} \subset R$, then we may replace R by $R_{\mathfrak{p}}$. Hence we may assume that R is a Noetherian local G-ring with maximal ideal $\mathfrak{m}$ and that $\mathfrak{q} \subset R[x]$ lies over $\mathfrak{m}$. Note that there is a unique prime $\mathfrak{q}' \subset R^\wedge[x]$ lying over $\mathfrak{q}$. Consider the diagram

$$\begin{array}{ccc} R[x]_{\mathfrak{q}}^\wedge & \rightarrow & (R^\wedge[x]_{\mathfrak{q}'}^\wedge)^\wedge \\ R[x]_{\mathfrak{q}} & \rightarrow & R^\wedge[x]_{\mathfrak{q}'} \\ R[x]_{\mathfrak{q}} & \rightarrow & R[x]_{\mathfrak{q}}^\wedge \\ R^\wedge[x]_{\mathfrak{q}'} & \rightarrow & (R^\wedge[x]_{\mathfrak{q}'}^\wedge)^\wedge \end{array}$$

Since R is a G-ring the lower horizontal arrow is regular (as a localization of a base change of the regular ring map $R \rightarrow R^\wedge$). Suppose we can prove the right vertical arrow is regular. Then it follows that the composition $R[x]_{\mathfrak{q}} \rightarrow (R^\wedge[x]_{\mathfrak{q}'}^\wedge)^\wedge$ is regular. The local map $R[x]_{\mathfrak{q}} \rightarrow R^\wedge[x]_{\mathfrak{q}'}$ is flat by base change and localization of completion. Its completed map is therefore faithfully flat by Lemma F.99A. Lemma F.125 now proves that the left vertical arrow is regular. Hence we see that we may assume R is a Noetherian complete local ring and $\mathfrak{q}$ a prime lying over the maximal ideal of R .

Let R be a Noetherian complete local ring and let $\mathfrak{q} \subset R[x]$ be a maximal ideal lying over the maximal ideal of R . Let $\mathfrak{r} \subset \mathfrak{q}$ be a prime ideal. We want to show that $R[x]_{\mathfrak{q}}^\wedge \otimes_{R[x]} \kappa(\mathfrak{r})$ is a geometrically regular algebra over $\kappa(\mathfrak{r})$. Set $\mathfrak{p} = R \cap \mathfrak{r}$. Then we can replace R by $R/\mathfrak{p}$ and $\mathfrak{q}$ and $\mathfrak{r}$ by their images in $R/\mathfrak{p}[x]$, see Lemma F.99. Hence we may assume that R is a domain and that $\mathfrak{r} \cap R = (0)$.

Choose the regular complete local subring $R_0 \subset R$ supplied by Lemma F.25, with R finite over R_0 . For the finite extension $R_0[x] \rightarrow R[x]$, write $\mathfrak{q}_0, \mathfrak{r}_0$ for the contractions of $\mathfrak{q}, \mathfrak{r}$. The contraction $\mathfrak{q}_0$ is maximal and lies over the maximal ideal of R_0 : lying over and the finite integral extension identify the contractions of the maximal ideals of the complete local rings. Moreover $\mathfrak{r}_0 \cap R_0 = (0)$. The individual-fibre assertion of Lemma F.96, proved by the finite-completion product decomposition and field base change, reduces this particular fibre to the corresponding fibre for R_0 . This use does not assume that $R_0[x]$ is already a G-ring.

We can therefore assume R regular and complete. If $\mathfrak{r} = (0)$, the desired fibre is

$$\widehat{R[x]_{\mathfrak{q}}} \otimes_{R[x]} \text{Frac}(R[x]),$$

which is geometrically regular by the second assertion of Lemma F.110, in every characteristic. If $\mathfrak{r} \neq (0)$, its extension to $K[x]$, $K = \text{Frac}(R)$, is generated by an irreducible polynomial, and $\kappa(\mathfrak{r})/K$ is finite. Apply the second assertion of Lemma F.97. Its actual proof extends this residue field to each finite purely inseparable test field, chooses a finite complete local domain inside that field, and identifies the tested completed fibre with factors of

$$(\widehat{B[x]_{\mathfrak{q}_i}} \otimes_B \text{Frac}(B)) / (x - a).$$

The parenthesized ring is regular by the first assertion of F.97; the extended derivation $\partial/\partial x$ sends $x - a$ to one, so the proved hypersurface argument makes each quotient regular. Thus F.97 proves geometric regularity of this nonzero-polynomial fibre in every characteristic. This quotient is not asserted to be a localization. The two cases exhaust the primes over (0) , proving the required regular completion map and hence polynomial permanence of G-rings. $\square$

Proposition F.135: ubiquity G ring

Adapted from the Stacks project, more-algebra.tex, lines 13261–13281.

Every ring essentially of finite type over $\mathbf{Z}$ is a G-ring.

Proof. The localization of $\mathbf{Z}$ at zero is the field $\mathbf{Q}$, whose completion map is the identity. At (p) it is a discrete valuation ring; its completion is again a discrete valuation ring, proved by Completion, Theorem 4.1 and the regular one-dimensional description in the earlier valuation-ring lesson. The completion map is flat by Completion, Theorem 3.2. Its closed fibre is $\mathbf{F}_p$. Its generic fibre is $\mathbf{Q}_p$ over $\mathbf{Q}$, hence is geometrically regular: in characteristic zero a finite field extension is separable, and tensoring a field with a finite separable extension is a finite product of separable field extensions, which are regular. Equivalently, the proved finite purely inseparable criterion makes every characteristic-zero extension geometrically regular, because its only purely inseparable test extension is the identity. This proves the generic-fibre assertion with the exact criterion written below. Thus $\mathbf{Z}$ is a G-ring. Proposition F.134, followed by localization, proves the assertion. This proof does not consume the source's broader complete-local-ring or Dedekind-ring statements. $\square$

Lemma F.136: Noetherian

Adapted from the Stacks project, topology.tex, lines 1304–1357.

Let X be a Noetherian topological space.

1. Any subset of X with the induced topology is Noetherian.
2. The space X has finitely many irreducible components.
3. Each irreducible component of X contains a nonempty open of X .

Proof. Let $T \subset X$ be a subset of X . Let $T_1 \supset T_2 \supset \dots$ be a descending chain of closed subsets of T . Write $T_i = T \cap Z_i$ with $Z_i \subset X$ closed. Consider the descending chain of closed subsets $Z_1 \supset Z_1 \cap Z_2 \supset Z_1 \cap Z_2 \cap Z_3 \dots$. This stabilizes by assumption and hence the original sequence of T_i stabilizes. Thus T is Noetherian.

Let A be the set of closed subsets of X which do not have finitely many irreducible components. Assume that A is not empty to arrive at a contradiction. The set A is partially ordered by inclusion: $Z \leq Z' \Leftrightarrow Z \subset Z'$ for $Z, Z' \in A$. By the descending chain condition we may find a minimal element of A , say Z . As Z is not a finite union of irreducible components, it is not irreducible. Hence we can write $Z = Z' \cup Z''$ and both are strictly smaller closed subsets. By construction $Z' = \bigcup Z'_i$ and $Z'' = \bigcup Z''_j$ are finite unions of their irreducible components (Lemma F.137). Hence $Z = \bigcup Z'_i \cup \bigcup Z''_j$ is a finite union of irreducible closed subsets. After removing redundant members of this expression, this will be the decomposition of Z into its irreducible components (Lemma F.138), a contradiction.

Let $Z \subset X$ be an irreducible component of X . Let $Z_1, \dots, Z_n$ be the other irreducible components of X . Consider $U = Z \setminus (Z_1 \cup \dots \cup Z_n)$. This is not empty since otherwise the irreducible space Z would be contained in one of the other Z_i . Because $X = Z \cup Z_1 \cup \dots \cup Z_n$ (see Lemma F.137), also $U = X \setminus (Z_1 \cup \dots \cup Z_n)$ and hence open in X . Thus Z contains a nonempty open of X . $\square$

Lemma F.137: irreducible

Adapted from the Stacks project, topology.tex, lines 931–968.

The closure of an irreducible subset is irreducible. Irreducible components are closed, every irreducible subset lies in one, and the components cover the space.

Proof. If its closure is the union of two relatively closed subsets, intersection with the original irreducible subset puts that subset in one of them. Taking closures puts the entire closure there. Maximality then makes any component equal to its closure. For existence, order irreducible supersets of a fixed irreducible subset by inclusion. The union of a chain is irreducible: two nonempty relatively open subsets of the union meet some two members of the chain; a common larger member meets

both, and irreducibility of that member forces their intersection to be nonempty. Thus every chain has an upper bound, and Zorn's lemma gives a maximal member. Apply this to each singleton for the covering assertion. $\square$

Lemma F.138: pick irreducible components

Adapted from the Stacks project, topology.tex, lines 970–995.

If a space is a finite union of irreducible closed subsets, none contained in the union of the others, then those subsets are exactly its irreducible components.

Proof. Intersect a component with each member of the finite cover. This expresses it as a finite union of closed subsets. Repeatedly applying irreducibility puts it in one member, and maximality makes it equal to that member. Conversely any member lies in a component by the preceding existence proof. That component is a member by the first argument. Inclusion in a different member would make the original member redundant, contrary to the assumption. It is therefore itself a component. $\square$

6. Exact earlier proof dependencies

The statements in Sections 1–5 specify the scope actually used in approximation. The following earlier proofs supply their elementary algebra inputs. [Descent and Zariski Main](#), Lemma D2.2, Theorem C4.3 and Corollary B1.2, precedes the henselian section argument. The broader formal-smoothness/regularity equivalence is not consumed: the explicit completion arguments F.99A, F.110 and F.132–F.135 prove the required G-ring inputs. This dependency list points to the proofs of the results it names; results that it does not name are not covered by it. Finite type and finite presentation on every compatible affine pair, used in the target-descent step of AG-GS-05, Lemma 7.6, have their complete morphism argument in [AG-RG-S04, Lemma A1.8](#), using the preceding finite-generator and finite-relation algebra patching proof. In the application of AG-GS-05, Theorem 7.7, the fixed-character-group splitting scheme is globally a disjoint union on the splitting cover, so this application uses that case of Lemma 7.6.

Exact earlier programme proofs used

These locators refer to written programme proofs. They do not claim that an unused, broader source statement has been proved.

Consumed source label	Earlier programme proof
definition regular	AG-CA-14, Section 1, definition
definition regular local	AG-CA-14, Section 1, definition
definition standard smooth	AG-CA-17, Section 4, equations (6) and (7)
lemma Artin Rees	AG-CA-03, Theorem 5.1
lemma Noetherian irreducible components	AG-CA-03, Proposition 2.2
lemma Noetherian topology	AG-CA-03, Proposition 2.2
lemma Zariski topology	AG-CA-01, Theorem 1.2 and Proposition 2.1
lemma artinian finite length	AG-CA-03, Theorem 4.2
lemma characterize henselian	AG-CA-20, Theorem 1.2 and Theorem 2.2
lemma characterize projective	AG-CA-06S, Section 1
lemma cokernel flat	AG-CA-08, Lemma 5.1
lemma compose formally smooth	AG-CA-17, Theorem 1.2
lemma differentials base change	AG-CA-16, Theorem 2.1

Consumed source label	Earlier programme proof
lemma differentials finitely presented	AG-CA-16, Theorem 5.1, finite polynomial presentation
lemma dimension base fibre equals total	AG-CA-11, Theorem 5.1
lemma dimension base fibre total	AG-CA-11, Theorem 5.1
lemma dimension prime polynomial ring	AG-CA-09, Theorem 4.3
lemma disjoint implies product	AG-CA-01, Theorem 5.2 and Corollary 5.3
lemma etale at prime	AG-CA-17, Theorem 7.2 and Corollary 7.3
lemma exact sequence differentials	AG-CA-16, Theorem 3.1
lemma faithfully flat universally injective	AG-CA-08, Theorem 2.2
lemma finite finite fibres	AG-CA-05, Theorem 3.4
lemma finite projective	AG-CA-07, Theorem 5.3
lemma finite transitive	AG-CA-05, Theorem 1.2
lemma flat	AG-CA-07, Theorem 2.1
lemma flat base change	AG-CA-07, Proposition 3.2
lemma flat eq	AG-CA-07, Theorem 5.1
lemma flat going down	AG-CA-07, Theorem 6.3
lemma flat tor zero	AG-CA-07, Theorem 2.1
lemma formally etale etale	AG-CA-17, Theorem 2.2 and Proposition 6.1
lemma integral no inclusion	AG-CA-05, Theorem 3.3
lemma intersect powers ideal module zero	AG-CA-03, Theorem 6.1
lemma lci CM	AG-CA-12, Corollary 6.2
lemma maximal chain CM	AG-CA-12, Lemma 5.4 and Theorem 5.5
lemma quasi compact	AG-CA-01, Proposition 2.3
lemma rank omega	AG-CA-18, Proposition 3.1
lemma reformulate CM	AG-CA-12, Theorem 4.2 and Corollary 4.3
lemma regular domain	AG-CA-12, Theorem 6.1
lemma regular goes up	AG-CA-14, Proposition 3.3
lemma regular ring CM	AG-CA-12, Theorem 6.1
lemma regular sequence powers	AG-CA-12, Proposition 1.3
lemma spec localization	AG-CA-02, Theorem 3.1
lemma variant local criterion flatness	AG-CA-08, Corollary 4.4 and equation (6)
proposition CM module	AG-CA-12, Theorem 4.2 and Corollary 4.3
proposition characterize formally smooth	AG-CA-17, Theorem 3.1
proposition smooth formally smooth	AG-CA-17, Theorem 3.1 and the split conormal definition of smoothness in this draft
lemma completion regular	AG-CA-19, Theorem 4.1
complete-local simple-root lifting	AG-CA-19, Theorem 5.1
finite residue fields in finite-type fibres	AG-CA-06, Theorem 2.1
minimal resolutions and Tor detection	AG-CA-13, Theorem 2.2
regular Noetherian rings are normal	AG-CA-15, Corollary 4.5
arbitrary fields lift over their prime field	AG-CA-19S, Lemma 1.1
complete local coefficient rings	AG-CA-19S, Theorem 5.1
complete local power-series presentations	AG-CA-19S, Theorem 6.1

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